Continuous Time Financial Economics

Part 1 of 2

2025

Raman Uppal

EDHEC Business School

Core Course
Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics
Core Course

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Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 0

Syllabus and How This Course Is Organized

Raman Uppal

EDHEC Business School

0.1 Instructor for the Course and Contact Information

Instructor	Raman UPPAL, Professor of Finance, EDHEC Business School
Office hours	Please send email to set up an appointment
Email	Raman.Uppal@edhec.edu
Subject line	CTFE

0.2 Course Description

- This course builds on material covered in the course on **Financial Economics**, which is set in discrete time.
- This course develops the economic and mathematical foundations of asset pricing and portfolio selection in *continuous time*.
- The course starts by reviewing basic financial economics and introducing the mathematical tools required for continuous time finance.
- The course then applies these tools to the
  - ⋆ valuing of financial derivatives (on equities, currencies, forwards, etc.)
  - ⋆ modeling term structure of interest rates and valuing interest-rate options
  - ⋆ dynamic portfolio choice, and
  - ⋆ dynamic asset pricing in partial and general equilibrium.

0.3 Course Prerequisites

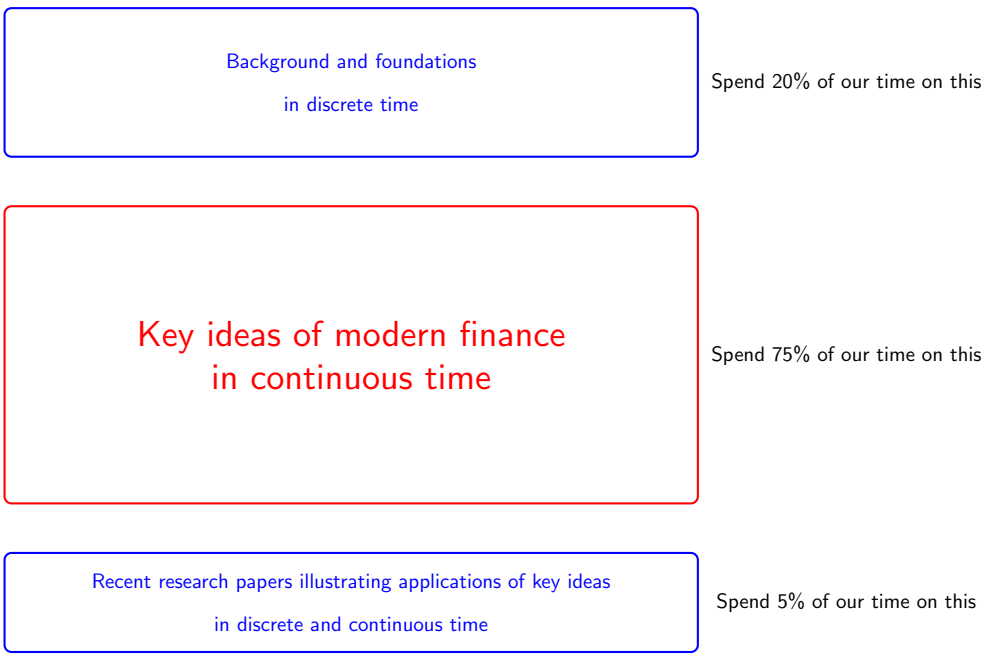
The prerequisites for the course are

- Solid understanding of material covered in course on **Financial Economics**. This means that you should understand very well the material in:
  1. Chapters 1–11 of Back (2017);
  2. Chapters 1–4 of Dumas and Luciano (2017); and
  3. Chapters 1–9 of Campbell (2018).
- Solid understanding of economics, probability, and calculus.
- Participate actively and be well-prepared for class.
- Do not sit in class without a **name card**
  (make one if you forget to bring your name card)

0.4 Course Objective

The objective of the course is to:

- Give you a **deep** understanding of the **key ideas** of modern finance;
- Provide you with the mathematical tools and theoretical foundations of financial economics that will help you
  - ⋆ understand the academic literature in finance
  - ⋆ solve problems in financial economics, and
  - ⋆ write your dissertation.
- Give you a good sense of the existing literature
  - ⋆ foundational research, which develops the key ideas;
  - ⋆ more recent research, which applies the key ideas to interesting problems.



0.5 Topics and Papers Covered in the Course

- **First half of the course**

1. Overview of Asset Pricing & Asset Pricing in Discrete Time
2. Mathematics for Continuous Time Finance
3. Stochastic Discount Factor and Returns
4. Change of Measure/Numeraire and Change of Drift
5. The Black-Scholes Model of Option Pricing
6. Extensions of Black-Scholes Model to
   - Dividends, Non-constant Interest Rates, Non-constant Volatility, and
   - Exchange Options, Options on Forwards & Foreign Currency
7. Summary of Part 1 of the course

- **Second half of the course**

8. Valuation of Fixed-Income Securities
9. Terms Structure of Interest Rates and Interest Rate Derivatives
10. Portfolio Selection: Mean-Variance and Two-Date Models
11. Portfolio Selection Using Martingale Methods
12. Portfolio Selection Using Dynamic Programming
13. Asset Pricing in Representative-Agent General-Equilibrium Economies
14. Summary of Part 2 of the course.

Papers that we study in Part 1 of the course

Paper	Google cite count (as of 2025/02/17)
Chapter 1	
• Cox and Ross (1976) The valuation of options for alternative stochastic processes	4,927
• Cox, Ross, and Rubinstein (1979) Option pricing: A simplified approach	10,942
Chapter 2	
• Merton (1990, Chapter 3) *Continuous time finance*	4,204
Chapter 3	
• Harrison and Kreps (1979) Martingales and arbitrage in multiperiod securities markets	5,779
Chapter 4	
• Geman, El Karoui, and Rochet (1995) Changes of numeraire, changes of probability measure and option pricing	1,027
Chapter 5	
• Black and Scholes (1973) The pricing of options and corporate liabilities	48,776
Chapter 6	
• Merton (1973) The theory of rational option pricing	17,338
• Black (1976) The pricing of commodity contracts	4,660
• Margrabe (1978) The value of an option to exchange one asset for another	2,636
• Garman and Kohlhagen (1983) Foreign currency option values	1,497

- Note that a Google Scholar cite count of 250 is good and a count of 500 is very good.

0.6 Course Method

- Class lectures.
- Practice problems to test your understanding of the material covered in class.
- Homework exercises to extend the material covered in class.
  - ⋆ It is **very important** that you do the homework exercises;
  - ⋆ If you do not do the homeworks, you will learn nothing from this course.
- In addition to doing the assigned homework problems, you should read at least one of the two books for the course, Back (2017) and Dumas and Luciano (2017), and do the exercises in these books.
- Readings from the academic literature.

0.7 Course Assessment

- First assignment to be returned on 7 April 2025.
  Second assignment to be returned on 2 June 2025.
- Please email homeworks to julie.smadja@edhec.edu.
- Homeworks should be typed (preferably in LaTeX).
- The assignments will make up 100% (50% each) of the course mark.
- Penalties for late submission of assignments
  - ⋆ Assignments submitted after the deadline will not be accepted, and a grade of 0 will be assigned.
  - ⋆ If neither assignment is submitted, the student will be deemed to have failed the course and will not be allowed to take the comprehensive exam.
- *The final exam for this course is online on 28 June 2025.*

0.8 Textbooks for the Course

The material that we will cover in the course can be found in the two books listed below.

1. Back, K. 2017. *Asset pricing and portfolio choice theory.* Second edition. NY: Oxford University Press.
   - We will cover the material in Chapters 12–18 of this book.
2. Dumas, B., and E. Luciano. 2017. *The economics of continuous-time finance.* Cambridge, MA: MIT Press.
   - We will cover the material in Chapters 5–12 and 16 of this book.

0.9 Other Textbooks (listed alphabetically by author name)

1. Back, K. 2005. *A course in derivative securities: Introduction to theory and computation.* NY: Springer.
2. Campbell, J. Y. 2018. *Finance decisions and markets: A course in asset pricing.* Princeton University Press.
3. Campbell, J. Y., A. W. Lo, and A. C. MacKinlay. 1997. *The econometrics of financial markets.* Princeton, New Jersey: Princeton University Press.
4. Cochrane, J. H. 2005. *Asset pricing.* Revised. Princeton, NJ: Princeton University Press.
5. Cvitanic, J., and F. Zapatero. 2004. *Introduction to the economics and mathematics of financial markets.* First. MIT Press.
6. Duffie, D. 2001. *Dynamic asset pricing theory.* Princeton, NJ: Princeton University Press.
7. Ferson, W. 2019. *Empirical asset pricing: Models and methods.* MIT Press.
8. Gollier, C. 2001. *The economics of risk and time.* Cambridge, MA: MIT Press.
9. Hull, J. C. 2017. *Options, futures, and other derivatives.* Upper Saddle River, NJ: Prentice Hall.
10. Ingersoll, J. E., Jr. 1987. *Theory of financial decision making.* Maryland: Rowman & Littlefield.
11. LeRoy, S. F., and J. Werner. 2001. *Principles of financial economics.* Cambridge, UK: Cambridge University Press.
12. Merton, R. C. 1990. *Continuous time finance.* Cambridge MA: Basil Blackwell.
13. Munk, C. 2013. *Financial asset pricing theory.* Oxford.
14. Pennacchi, G. 2008. *Theory of asset pricing.* MA: Addison-Wesley.

0.10 Review Articles

1. Sundaresan, S. M. 2000. Continuous-time methods in finance: A review and an assessment. *Journal of Finance* 55 (4): 1569–1622.
2. Campbell, J. Y. 2000. Asset pricing at the millennium. *Journal of Finance* 55 (4): 1515–67.
3. Brunnermeier, M. K., and Y. Sannikov. 2016. Macro, money, and finance: A continuous-time approach. In *Handbook of Macroeconomics,* vol. 2B. 1497-1545: Elsevier.

- There are several books and journals that specialize in publishing review articles. So, when starting your research, you may wish to first look for review papers in your area of interest.
  - ⋆ The books are typically called "*Handbook of . . .*".
  - ⋆ The journals have names such as:
    - "*Annual Review of Economics*"
    - "*Annual Review of Financial Economics*"
    - "*Foundations and Trends in Finance*"
- There is also a journal that specializes in publishing critiques of research articles, called the *Critical Finance Review*.

Sample articles from "Foundations and Trends in Finance" (sorted by year)

- Aragon, G. O., and W. E. Ferson. 2007. Portfolio performance evaluation. *Foundations and Trends in Finance* 2 (2): 83–190
- Mehra, R. 2007. The equity premium puzzle: A review. *Foundations and Trends in Finance* 2 (1): 1–81
- Brav, A., W. Jiang, and H. Kim. 2010. Hedge fund activism: A review. *Foundations and Trends in Finance* 4 (3): 185–246
- Gibson, R., F.-S. Lhabitant, and D. Talay. 2010. Modeling the term structure of interest rates: A review of the literature. *Foundations and Trends in Finance* 5 (1–2): 1–156
- Shefrin, H. 2010. Behavioralizing finance. *Foundations and Trends in Finance* 4 (1-2): 1–184
- Ang, A., W. N. Goetzmann, and S. M. Schaefer. 2011. The efficient market theory and evidence: Implications for active investment management. *Foundations and Trends in Finance* 5 (3): 157–242
- Cochrane, J. H. 2012. Continuous-time linear models. *Foundations and Trends in Finance* 6 (3): 165–219
- Agarwal, V., K. A. Mullally, and N. Y. Naik. 2015. The economics and finance of hedge funds: A review of the academic literature. *Foundations and Trends in Finance* 10 (1): 1–111

Sample articles from "Annual Review of Financial Economics" (by year)

- Aït-Sahalia, Y. 2009. Estimating and testing continuous-time models in finance: The role of transition densities. *Annual Review of Financial Economics* 1 (1): 341–359
- Bodie, Z., J. Detemple, and M. Rindisbacher. 2009. Life-cycle finance and the design of pension plans. *Annual Review of Financial Economics* 1 (1): 249–286
- Demirgüç-Kunt, A., and R. Levine. 2009. Finance and inequality: Theory and evidence. *Annual Review of Financial Economics* 1 (1): 287–318
- Jarrow, R. A. 2009. Credit risk models. *Annual Review of Financial Economics* 1 (1): 37–68
- Tufano, P. 2009. Consumer finance. *Annual Review of Financial Economics* 1 (1): 227–247
- Ferson, W. E. 2010. Investment performance evaluation. *Annual Review of Financial Economics* 2 (1): 207–234
- Frydman, C., and D. Jenter. 2010. CEO compensation. *Annual Review of Financial Economics* 2 (1): 75–102
- Jagannathan, R., E. Schaumburg, and G. Zhou. 2010. Cross-sectional asset pricing tests. *Annual Review of Financial Economics* 2 (1): 49–74
- Spatt, C. S. 2010. An informal perspective on the economics and regulation of securities markets. *Annual Review of Financial Economics* 2 (1): 127–143

- Wachter, J. A. 2010. Asset allocation. *Annual Review of Financial Economics* 2 (1): 175–206
- Burnside, C., M. Eichenbaum, and S. Rebelo. 2011. Carry trade and momentum in currency markets. *Annual Review of Financial Economics* 3 (1): 511–535
- Jegadeesh, N., and S. Titman. 2011. Momentum. *Annual Review of Financial Economics* 3 (1): 493–509
- Laeven, L. 2011. Banking crises: A review. *Annual Review of Financial Economics* 3 (1): 17–40
- Musto, D. K. 2011. The economics of mutual funds. *Annual Review of Financial Economics* 3 (1): 159–172
- Wermers, R. 2011. Performance measurement of mutual funds, hedge funds, and institutional accounts. *Annual Review of Financial Economics* 3 (1): 537–574
- Litzenberger, R., J. Castura, and R. Gorelick. 2012. The impacts of automation and high frequency trading on market quality. *Annual Review of Financial Economics* 4 (1): 59–98
- Strahan, P. E. 2013. Too big to fail: causes, consequences, and policy responses. *Annual Review of Financial Economics* 5 (1): 43–61
- Madhavan, A. 2014. Exchange-traded funds: An overview of institutions, trading, and impacts. *Annual Review of Financial Economics* 6 (1): 311–341

- Brav, A., W. Jiang, and H. Kim. 2015. Recent advances in research on hedge fund activism: Value creation and identification. *Annual Review of Financial Economics* 7 (1): 579–595
- Choi, J. J. 2015. Contributions to defined contribution pension plans. *Annual Review of Financial Economics* 7 (1): 161–178
- Hirshleifer, D. 2015. Behavioral finance. *Annual Review of Financial Economics* 7 (1): 133–159
- Bekaert, G., C. R. Harvey, A. Kiguel, and X. Wang. 2016. Globalization and asset returns. *Annual Review of Financial Economics* 8 (1): 221–288
- Ben-David, I., F. Franzoni, and R. Moussawi. 2017. Exchange-traded funds. *Annual Review of Financial Economics* 9 (1): 169–189
- Berk, J. B., and J. H. van Binsbergen. 2017. Mutual funds in equilibrium. *Annual Review of Financial Economics* 9 (1): 147–167
- Zhou, G. 2018. Measuring investor sentiment. *Annual Review of Financial Economics* 10 (1): 239–259
- Allen, F., J. Qian, and M. Qian. 2019. A review of China's institutions. *Annual Review of Financial Economics* 11 (1): 39–64
- Korteweg, A. 2019. Risk adjustment in private equity returns. *Annual Review of Financial Economics* 11 (1): 131–152

0.11 My Teaching Style

- Focus on *applying theory* to problems in financial economics
- Rigorous and technical
- Interactive and informal
  - ⋆ Do not be nervous when I call on you
  - ⋆ Feel free to ask for help from other students
  - ⋆ Objective of this is **not** to make you uncomfortable

▶ **My job**

- Provide *structure* for thinking about finance theory in a *systematic* manner.
- Provide you with the opportunity to learn this material.
- Office hours: By appointment and by email Raman.Uppal@edhec.edu

0.12 Your "Style" in Class . . .

- **Do not be shy** about asking questions
  - ⋆ if you knew everything you would not be here
  - ⋆ be honest with yourself about what you do/don't know
- You are **strongly encouraged to make mistakes** in responding to questions
- Feel free to interrupt and share your insights/experiences/funny thoughts
  It is important to have a good time in class.
- Be nice to other students—be patient when they ask questions
- Complain—when you disagree with something I say/do;
- Please do not let electronic devices (phones, laptops, etc.) distract you.

0.13 The Good News . . .

- We have two great books:
  - ⋆ Back (2017)
  - ⋆ Dumas and Luciano (2017)

0.14 The Bad News . . .

- YOU have to work through the book.
- Each chapter of Kerry Back's book has a discussion of the existing literature for that topic. You are strongly encouraged to read these papers:
  - ⋆ I will be covering material from many of these papers in class, but will not mention each paper explicitly in the lectures.
- I also do not intend to cover each lecture slide; many of the slides are for your information, but will not be discussed during the lecture.

0.15 Bottom line

- The course is a fantastic opportunity to learn modern finance.
- It is up to you to make this most of this opportunity.
- What you get out of this course will depend very much on the effort you put into the course.

Your Choice: Beauty, fragrance, or nectar?







Understanding finance is easy . . . just need to engage



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Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 1

Unifying View of Asset Pricing Problems and Methods

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EDHEC Business School

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1.1 Plan for the Day

Today

- **Main questions for the day:**
  - ⋆ Unifying view of asset pricing problems:
    – Valuation of redundant and primitive assets, hedging, portfolio selection
  - ⋆ Unifying view of asset pricing methods:
    – Stochastic discount factor, risk-neutral probability, change of measure

For Next Time

- Review the material covered in class.
- Read K. Back, Chapters 2 and 3; Dumas & Luciano, Chapter 2.
- Do the practice problems.

1.2 Main Points for Today

1. Unifying view of asset pricing problems:
   - Valuation and hedging of redundant claims
   - Selection of optimal portfolios
   - Valuation of primitive securities (stocks and bonds)
2. Unifying view of asset-pricing methods/concepts in a two-date setting:
   - Replication.
   - Hedging, and then discounting the hedged payoff at risk-free rate.
   - Risk-neutral valuation.
   - Arrow-Debreu state securities.
   - Change of numeraire and martingale pricing.
   - Stochastic discount factor or state price density.
   - Factor models of returns, such as the CAPM.

- First chapter, develop the main ideas in asset pricing in a two-date setting;
- Second chapter, develop the math for continuous time;
- Third chapter, apply the math to the main ideas in continuous time.

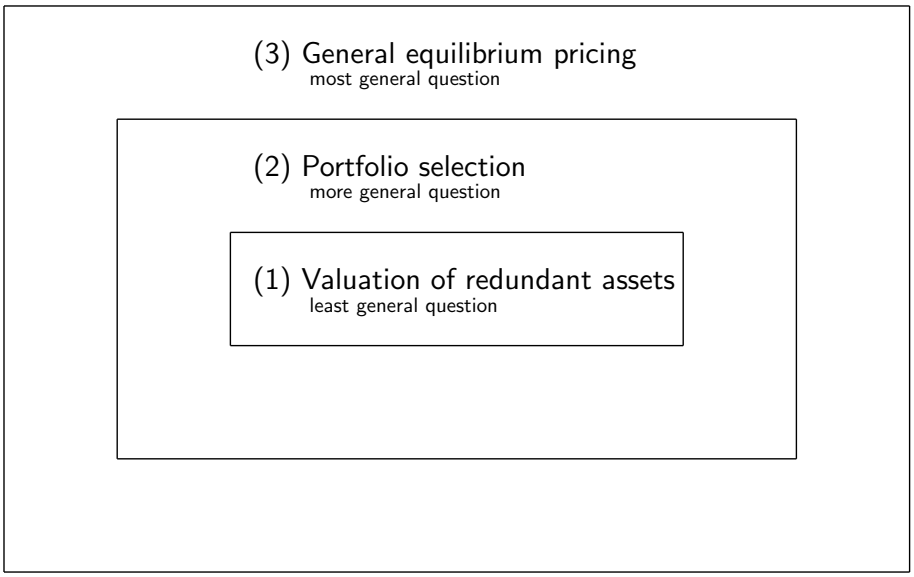
1.3 Overview of Asset Pricing Problems

1.3.1 There are Three Main Questions in Asset Pricing

The course will focus on answering these three questions:

1. How to value (and hedge) redundant securities? No-arbitrage valuation
2. How to determine optimal portfolio policies of investors? Portfolio selection
3. How to determine the values of stocks and bonds in general equilibrium with investors who are optimizing their consumption and portfolio policies?

These three questions are nested.



1. How to value redundant securities? No-arbitrage valuation
   Least general question:
   - *we are given prices of fundamental assets and the desired cashflows;*
   - *need to determine only the replicating portfolio strategy.*

2. How to determine optimal portfolio policies of investors? Portfolio selection
   More general than the previous question:
   - *we are given prices of the fundamental assets;*
   - *need to determine both the desired cashflows and the portfolio strategy generating these cashflows.*

3. How to determine the values of stocks and bonds in general equilibrium with investors who are optimizing their consumption and portfolio policies?
   Most general question:
   - *need to determine prices of fundamental assets, desired cashflows, and the portfolio strategy that replicates these cashflows.*

Today and the Future

- All asset pricing problems involve (at least) two dates:
  - ⋆ today, t;
  - ⋆ some future date, T.



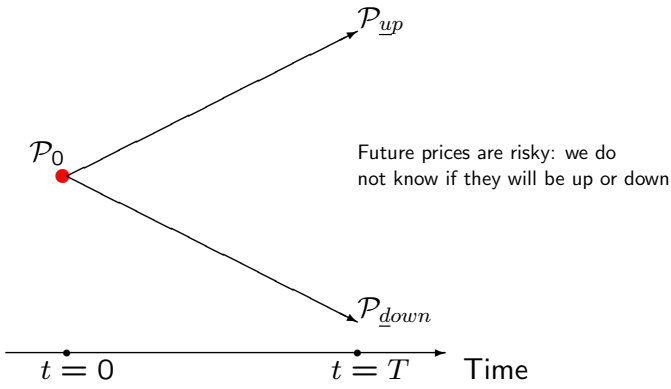
- You are usually given information about some date in the future (T), and you need to determine something today (t).
- **Examples**
  1. Given payoff of a derivative at T, find value of option today.
  2. Given your payoff/position at T, find the hedging strategy today.
  3. Given some objective function at T (say maximize expected-return-and-risk tradeoff), find portfolio strategy today.

1.3.2 The Two Things that Matter in Valuation: Time and Risk

Because all problems involve today and the future, there are only two things that one needs to account for when finding the current price, $\mathcal{P}_0$, of an asset:

1. Adjusting for the time value of the cashflows; and

2. Adjusting for the risk of cashflows.



1.3.3 Adjusting Cashflows for Time and Risk Using SDF or RNP

- We can adjust for time and risk in a single step, via
  - ⋆ the Stochastic Discount Factor (SDF), or
  - ⋆ via Arrow-Debreu State Prices, or
  - ⋆ via some variation of these constructs.

- Alternatively, we can adjust for time and risk in two distinct steps:
  1. Adjusting cashflows for time is done using
     - ⋆ the risk-free rate, or
     - ⋆ the value of a bank-deposit (or money-market) account.
  2. Adjusting cashflows for risk is done
     - ⋆ using the risk premium, which is the traditional method, or
     - ⋆ by changing the probabilities
       - ∗ from physical (objective) probabilities: $\mathbb{P}$-measure;
       - ∗ to probabilities that have been adjusted for risk: $\mathbb{Q}$-measure.
- Probabilities that have been adjusted for risk are called "risk-neutral probabilities" abbreviated as RNP.
- The risk-neutral probabilities adjust for risk by by putting more weight on bad outcomes, relative to the physical probabilities.

1.3.4 How Do We Obtain SDF and RNP?

- We get the SDF and RNP from either the prices of primitive securities, if these prices are given to us
  - ⋆ this is called "no-arbitrage pricing", where we use the prices of primitive securities to obtain the SDF or RNP;
  - ⋆ **Example**: Given the prices of the stock and bonds, *and* the dynamics of these prices, find the price of a European call option.
- Or, we get the SDF and RNP from the equilibrium conditions resulting from the optimizing actions of individual investors.
  - ⋆ This is called "equilibrium pricing", where we equate aggregate demand and supply of assets to obtain the SDF and RNP.
  - ⋆ **Example**: Given the process for the aggregate endowment, the SDF for a representative investor is her marginal-rate of substitution ($\tilde{m}$) between current (c_0) and future consumption (c_1): $\tilde{m} = \frac{\partial U(c_0,c_1)/\partial c_1}{\partial U(c_0,c_1)/\partial c_0}$.

1.3.5 Complete and Incomplete Financial Markets

There are two possible settings in which these three questions can be asked.

1. Financial markets are complete, and

2. Financial markets are incomplete.

Q. What does it mean for financial markets to be complete?

A. When one can use available securities to generate any pattern of cashflows, then financial markets are said to be complete.

- The desired pattern of cashflows can be generated through:
  1. Static trading (trade only at date $t = 0$); or,
  2. Dynamic trading (trade at date $t_0, t_1, \ldots, T$).

In this course, we will focus mostly on complete financial markets.

1.3.6 How Restrictive Is It to Assume Markets are Complete?

- How many securities do you need to complete the market in the following three figures:

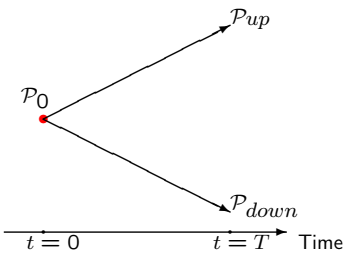


Figure A

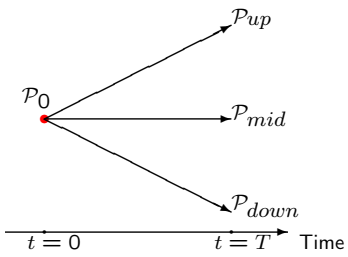


Figure B

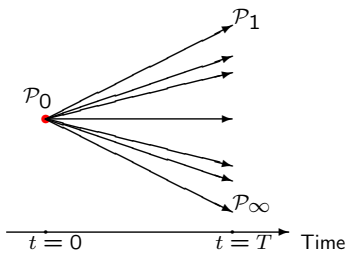
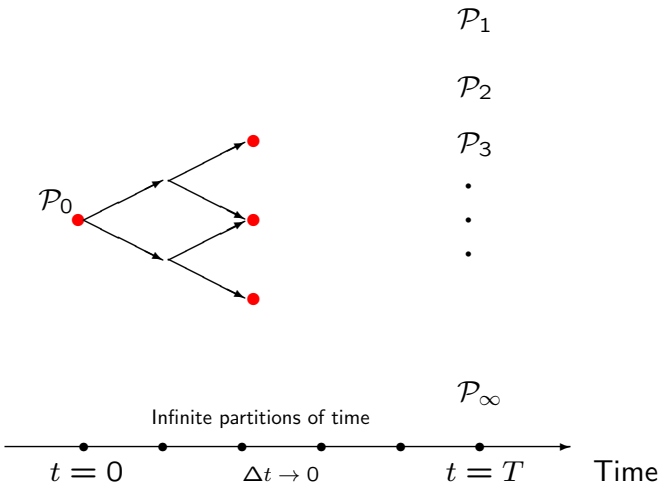


Figure C

Q. How do we fix this problem?

A. Think . . . the answer received a Nobel prize.

- Allow for dynamic (continuous) trading.
- In order to have smooth (continuous) arrival of information, best to think of each small interval of time to consist of a binomial model.



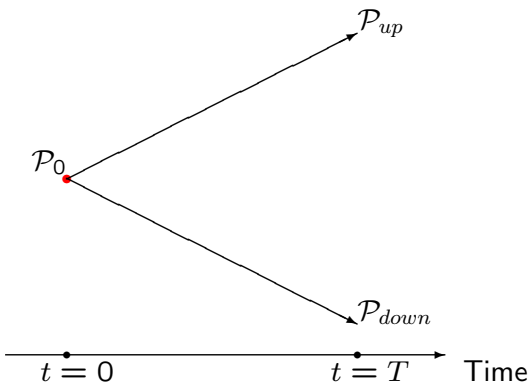
Allow for dynamic (continuous) trading between $t = 0$ and T

1.3.7 Complete Markets, SDF and RNP

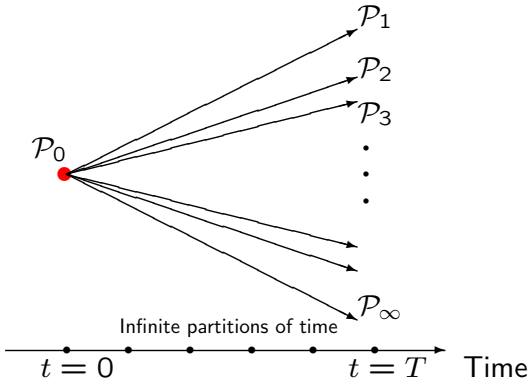
- If financial markets are complete with no arbitrage opportunities, then:
  - ⋆ SDF is unique and strictly positive;
  - ⋆ There is a unique RNP that corresponds to this SDF.
- If financial markets are incomplete, then:
  - ⋆ SDF is **not** unique (there can be more than one SDF);
  - ⋆ Different SDFs define different RNPs.
- Therefore, in markets that are incomplete, to choose one SDF from the many available, we need some additional assumption, which is usually ad-hoc.
  - ⋆ This is the reason why, for most of the course, we will consider only the setting where markets are complete.
  - ⋆ For each topic, I will explain solution in complete markets, and then describe various approaches for addressing that topic in incomplete markets.

1.4 Plan for Explaining the Course Material

1. Chapter 1:
Start with a one-period (two-date), binomial (two-state), numerical example.



2. Extend model to infinite dates (continuous time) and infinite states.



- Chapter 2: Continuous time mathematics
- Chapters 3 and 4: SDF and RNP in continuous time
- Chapter 5: Use the maths to get Black-Scholes call option pricing formula
- Chapter 6: Extend Black-Scholes option pricing formula to value other contracts
- Chapter 7: Summary of first part of the course

- Then, in the second part of the course:
  - ⋆ Chapter 8: Valuation of Fixed-Income Securities
  - ⋆ Chapter 9: Term Structure of Interest Rates and Interest Rate Derivatives
  - ⋆ Chapter 10: Portfolio Selection: Mean-Variance and Two-Date Models
  - ⋆ Chapter 11: Portfolio Selection Using Martingale Methods
  - ⋆ Chapter 12: Portfolio Selection Using Dynamic Programming
  - ⋆ Chapter 13: Asset Pricing in Representative-Agent General-Equilibrium Economies
  - ⋆ Chapter 14: Summary of second part of the course.

3. Basically, I use the following teaching method:

- Start by explaining each idea in discrete time
  - ⋆ Often, using a numerical example.
- Then, explain the same idea in continuous time.
  - ⋆ Often, using the setting of "option pricing"
- It is important to keep in mind this is **not** a course about "option pricing." However, we often use the context of option pricing, because option pricing is easy to understand, and is nested in the bigger problems of portfolio selection and general equilibrium asset pricing.

4. Bottom line:

- Less important (initially) to understand the math.
- More important to focus on understanding each idea.

1.5 Key insight about the entire course

- The only things that matter (in finance) are: time and risk.
- The SDF adjusts payoffs for both time and risk.
- The SDF is fully characterized by its mean and its volatility.
  - ⋆ The mean is given by the (negative of) the risk-free interest rate.
  - ⋆ The volatility is given by the (negative of) the Sharpe ratio.
- For most of the course,
  - ⋆ we take SDF as given and study how to use it to solve financial problems.
- In the last part of the course,
  - ⋆ we will study how to obtain the SDF; i.e. risk-free rate and Sharpe ratio.

1.6 Nobel Laureates Whose Work is Discussed in Course

Topic	Winner (year prize was received)
State prices	Arrow (1972), Debreu (1983), Dybvig (2022)
Stochastic discount factor	Hansen (2013)
Options	Merton (1997), Scholes (1997)
Portfolio choice	Markowitz (1990), Tobin (1981), Samuelson (1970), Merton (1997)
Behavioral finance	Kahneman (2002), Thaler (2017)
Pricing of equities	Sharpe (1997), Fama (2013), Shiller (2013)
Asset pricing in general equilibrium	Lucas (1995)

- UBS has a nice page describing the work of the Nobel laureates in Economics

  https://www.ubs.com/microsites/nobel-perspectives/en/laureates.html

- The work of each Nobel laureate is phrased in terms of a question.
- These questions provide an excellent starting point for your dissertations.

1.7 What the Course Will Not Cover

- The way theoretical models are developed is to:
  1. Write the "first-generation" theoretical model.
  2. Test the implications of the "first-generation" model empirically.
  3. If empirical evidence rejects the model, then extend the model by generalizing its assumptions to obtain "second-generation" model.
  4. Test implications of "second-generation" model empirically.
  5. Extend the model ...
  6. ...
- The course will not cover the most recent ("second-generation") models.
- The course will, however, prepare you to understand the most-recent models.

- **Example:**

1. Write "first-generation" model to value options:
   Black and Scholes (1973).
2. Test implications of the "first-generation" model empirically:
   Bates (1996) shows Black-Scholes prices a bit different from market prices.
3. Extend model to get "second-generation" model:
   - ⋆ account for stochastic volatility: Heston (1993).
   - ⋆ account for jumps (extreme moves) in returns and return volatility:
     Naik (1993); Das and Sundaram (1999).
   - ⋆ account for costly trading, liquidity risk, margin constraints:
     Edirisinghe, Naik, and Uppal (1993), Naik and Uppal (1994).
4. Test implications of the "second-generation" model empirically:
   Bakshi, Cao, and Chen (1997).
5. Extend the model ...
6. ...

End of
Overview of Asset Pricing Problems

1.8 Asset Pricing in a Two-Date Binomial Model

- In order to get a firm grip on ideas, we will start with the valuation of a call option in a two-date binomial model.

- Once we understand the key ideas in the familiar world of binomial option pricing, we will examine the same ideas in a discrete-time multiperiod model without restricting the claim to be a call option.

1.8.1 Eight Methods for Valuing Redundant Claims

- There are many ways to value a redundant contract, such as an option.

- All eight methods give the same answer, and will help us to
  - ⋆ Understand the key concepts;
  - ⋆ Understand the relation between the key concepts;
  - ⋆ Understand the terminology used in asset pricing.

- This material is *fundamental* to modern finance, so it is important that you understand the underlying logic.
  - ⋆ Throughout our discussion below, you should be asking one question.

Q. What is the common logic underlying the eight valuation methods?

1. Find a portfolio that replicates the payoff to the option;
   then, the value of the option must equal the value of the replicating portfolio.

2. Find a portfolio that hedges the payoff to the option;
   then, use the value of the hedged payoff to infer the value of the option.
   This is a re-interpretation of the replication method.

3. Find the risk-adjusted probabilities;
   then, use the probabilities to find risk-adjusted expected payoff to the option.

4. Find Arrow-Debreu state price today for receiving 1 dollar in a particular state (up or down) tomorrow; then, use state prices to find value of option.

5. Find the martingale value of the option, which is in terms of a bank deposit;
   then, use this to find the value of the option in terms of dollars.

6. Find the martingale value of the option, now in terms of the stock;
   then, use this to find the value of the option in terms of dollars.

7. Find the CAPM discount rate at which one can discount the expected payoff to the option; then, use this discount rate to find value of option.

8. Find the stochastic discount factor (SDF); then, use the SDF to find value of option.

Example 1.1 (Binomial option pricing for a call on stock)
We wish to find the value of a call option on one share of a non-dividend paying stock. We will assume that there is only one period to go before expiration of the option. Let HC denote your "home currency" (USD, GBP, JPY, . . .).

- Assume that:
  - ⋆ The strike price of the option is HC/share 1,050.
  - ⋆ The risk-free return is 1% per period.
  - ⋆ We denote the price at t of the risk-free bank deposit (money-market account) by $\mathcal{P}_t^D$.
  - ⋆ So, $\mathcal{P}_0^D = 1$ and $\mathcal{P}_1^D = 1.01$; that is, $R_f = \mathcal{P}_1^D/\mathcal{P}_0^D = 1.01$.
  - ⋆ The current (time-0) stock price, $\mathcal{P}_0^S$, is HC/share 1,000.
  - ⋆ At $t = 1$, the stock price can move either to $\mathcal{P}_u^S = 1100$ or to $\mathcal{P}_d^S = 950$.
- We wish to find the value of the call option at date 0, denoted $\mathcal{P}_0^C$.

- The payoff from the call in the up and down states is:
  - ⋆ in the up state: if $\mathcal{P}_1^S = \mathcal{P}_u^S = 1100$, then $\mathcal{P}_u^C = \text{HC}\,50$
  - ⋆ in the down state: if $\mathcal{P}_1^S = \mathcal{P}_d^S = 950$, then $\mathcal{P}_d^C = \text{HC}\,0$.
- Given the information above, how do we find the call price today, $\mathcal{P}_0^C$?

$$\mathcal{P}_0^S = 1000 \begin{cases} \mathcal{P}_u^S = 1100 \\ \mathcal{P}_d^S = 950 \end{cases} \qquad \mathcal{P}_0^C = ? \begin{cases} \mathcal{P}_u^C = 50 \\ \mathcal{P}_d^C = 0 \end{cases}$$

- Observe that the *delta* of the option at $t = 1$ is:

$$\text{delta} = \frac{\text{change in value of } \mathcal{P}_1^C}{\text{change in } \mathcal{P}_1^S} = \frac{50 - 0}{1100 - 950} = \frac{1}{3}.$$

- Assume that the physical (actual) probability of an up and down move is $\frac{1}{2}$.

Guide to Notation

- We will denote the price at date 0 by $\mathcal{P}_0$.
  - ⋆ Price of the *stock*: $\mathcal{P}_0^S$
  - ⋆ Price of the bank *deposit*: $\mathcal{P}_0^D$
  - ⋆ Price of the *call* option: $\mathcal{P}_0^C$
  - ⋆ Price of *Arrow*-Debreu claim that pays one unit in state $k = \{u, d\}$: $\mathcal{P}_0^{A_k}$
- The price at date 1, $\mathcal{P}_1$, will depend on whether the state is up (u) or down (d):
  - ⋆ Price of the stock at date 1: $\mathcal{P}_u^S$ or $\mathcal{P}_d^S$
  - ⋆ Price of the risk-free bank deposit: $\mathcal{P}_u^D = \mathcal{P}_d^D = \mathcal{P}_1^D$ . . . because risk free, independent of state
  - ⋆ Price of the call option: $\mathcal{P}_u^C$ or $\mathcal{P}_d^C$.
  - ⋆ Price of Arrow-Debreu claim
    - ∗ For the state-u claim: $\mathcal{P}_u^{Au} = 1$ and $\mathcal{P}_d^{Au} = 0$; . . . pays 1 in state u and 0 otherwise
    - ∗ For the state-d claim: $\mathcal{P}_u^{Ad} = 0$ and $\mathcal{P}_d^{Ad} = 1$. . . . pays 1 in state d and 0 otherwise
- The gross risk-free rate of return is: $\frac{\mathcal{P}_1^D}{\mathcal{P}_0^D} = R_f = 1 + r_f[\text{discrete}] = \exp(r_f[\text{continuous}])$.

Approach 1: Valuation via replication

- Let the replicating portfolio consist of a in a bank deposit, and b shares of the stock.
- The *replicating equations* that show how much to invest in the bank deposit, a, and the stock, b, to match the payoff to the option at $t = 1$ are:

$$\begin{aligned} a\,\mathcal{P}_1^D + b\,\mathcal{P}_u^S &= \mathcal{P}_u^C \quad \text{. . . replicate payoff in the "up" state} \\ a\,\mathcal{P}_1^D + b\,\mathcal{P}_d^S &= \mathcal{P}_d^C \quad \text{. . . replicate payoff in the "down" state} \end{aligned}$$

- Solving the two equations above, we get:

$$b = \frac{\mathcal{P}_u^C - \mathcal{P}_d^C}{\mathcal{P}_u^S - \mathcal{P}_d^S}, \quad \text{and} \quad a = \frac{\mathcal{P}_u^C - b\,\mathcal{P}_u^S}{\mathcal{P}_1^D} = \frac{\mathcal{P}_d^C - b\,\mathcal{P}_d^S}{\mathcal{P}_1^D} = \frac{\mathcal{P}_d^C\,\mathcal{P}_u^S - \mathcal{P}_u^C\,\mathcal{P}_d^S}{R_f(\mathcal{P}_u^S - \mathcal{P}_d^S)}.$$

- The cost this portfolio at $t = 0$ is: $a\,\mathcal{P}_0^D + b\,\mathcal{P}_0^S$.
- So, the value of the option, which equals cost of the replicating portfolio is:

$$\mathcal{P}_0^C = a\,\mathcal{P}_0^D + b\,\mathcal{P}_0^S = a + b\,\mathcal{P}_0^S \quad \text{. . . recall that } \mathcal{P}_0^D = 1.$$

Now, we verify the above using our numerical example.

- We have already computed: $b = \frac{50-0}{1100-950} = \frac{1}{3}$.
- And, $a = \frac{\mathcal{P}_u^C - b\mathcal{P}_u^S}{\mathcal{P}_1^D} = \frac{50-\frac{1}{3}1100}{1.01} = -313.53$.
- We can verify that 1/3 in the stock contract and a bank deposit with initial "deposit" of -313.531 replicates the payoff to the option:
  - ⋆ if $\mathcal{P}_1^S = 1100$, value of replicating portfolio is: $-313.53 \times 1.01 + (1/3) \times 1100 = 50$.
  - ⋆ if $\mathcal{P}_1^S = 950$, value of replicating portfolio is: $-313.53 \times 1.01 + (1/3) \times 950 = 0$.
- The total cost at $t = 0$ of this strategy is:

$$a + b\,\mathcal{P}_0^S = -313.53 + (1/3) \times 1000 = \text{HC}\,19.80. \tag{1.1}$$

- Hence, *in the absence of arbitrage*, value of the call is $\mathcal{P}_0^C = \text{HC}\ 19.80.$

Approach 2: Valuation via hedging

- Under this method, we need to choose b so that the payoff to the option is hedged; that is, the payoffs to the option in up and down states are equal:

  $$\mathcal{P}_u^C - b\,\mathcal{P}_u^S = \mathcal{P}_d^C - b\,\mathcal{P}_d^S$$

  As before, solving this gives: $b = \frac{\mathcal{P}_u^C - \mathcal{P}_d^C}{\mathcal{P}_u^S - \mathcal{P}_d^S}$.
- If you have a hedged (that is, riskless) payoff in the future, the present value of the hedged position can be obtained by discounting at the riskless rate:

  $$\text{PV of hedged position} = \frac{\mathcal{P}_u^C - b\,\mathcal{P}_u^S}{R_f} = a \quad \ldots \text{same as in Approach 1.}$$

- The hedged position contains the call option *and* a short position in ($b = \frac{1}{3}$) shares; that is, $\mathcal{P}_0^C - b\,\mathcal{P}_0^S$. So, the value of the option must equal

  $$\mathcal{P}_0^C = \text{PV of hedged position} + b\,\mathcal{P}_0^S = a + b\,\mathcal{P}_0^S \quad \ldots \text{same as in Approach 1.}$$

Now, we verify the above using our numerical example.

- We have already computed the (delta) hedge ratio: $b = \frac{50-0}{1100-950} = \frac{1}{3}$.
- We verify that this hedges the payoff from the option:
  - ⋆ if $\mathcal{P}_1^S = 1100$, value of hedged position is: $50 - (1/3) \times 1100 = -\text{HC}\ 316.667$.
  - ⋆ if $\mathcal{P}_1^S = \ 950$, value of hedged position is: $0 - (1/3) \times \ 950 = -\text{HC}\ 316.667$.
- Because the hedged portfolio gives us -316.667 in both states of the world, the cost of this portfolio today should be the present value of -316.667:

  PV of hedged position $= \frac{-316.667}{1.01} = -\text{HC}\ 313.53 = a$.
- The hedged position consists of the call option and short $(1/3)$ shares of the stock; therefore, the value of the option alone should be:

$$\begin{aligned}\mathcal{P}_0^C &= \text{PV of hedged position} + b\,\mathcal{P}_0^S \\ &= -313.53 + \frac{1}{3}1000 = \text{HC}\ 19.80. \quad \text{. . . same as in Approach 1.}\end{aligned}$$

Approach 3: Valuation using risk-neutral probabilities

- Risk-neutral probabilities are probabilities under which the stock earns a risk-free rate of return. Denoting by $\text{prob}_u^{\mathbb{Q}}$ and $\text{prob}_d^{\mathbb{Q}} = 1 - \text{prob}_u^{\mathbb{Q}}$ the risk-adjusted probabilities of the up and down move of the stock price:

$$\text{prob}_u^{\mathbb{Q}}\mathcal{P}_u^S + (1 - \text{prob}_u^{\mathbb{Q}})\mathcal{P}_d^S = \mathcal{P}_0^S R_f.$$

- Solving the above equation gives: $\text{prob}_u^{\mathbb{Q}} = \frac{\mathcal{P}_0^S R_f - \mathcal{P}_d^S}{\mathcal{P}_u^S - \mathcal{P}_d^S}$.
- So, the risk-adjusted expected future value of the option is:

$$\mathbb{E}^{\mathbb{Q}}[\mathcal{P}_1^C] = \text{prob}_u^{\mathbb{Q}}\,\mathcal{P}_u^C + \text{prob}_d^{\mathbb{Q}}\,\mathcal{P}_d^C.$$

- The current value of option is obtained by discounting at the riskless rate:

$$\mathcal{P}_0^C = \frac{\text{prob}_u^{\mathbb{Q}}\,\mathcal{P}_u^C + \text{prob}_d^{\mathbb{Q}}\,\mathcal{P}_d^C}{R_f} = \frac{\mathbb{E}^{\mathbb{Q}}[\mathcal{P}_1^C]}{R_f}.$$

Now, we verify the above using our numerical example.

- Under the risk-neutral probabilities, the stock should earn the risk-free return:

$$\text{prob}^{\mathbb{Q}} \underbrace{1100}_{\mathcal{P}_u^S} + (1 - \text{prob}^{\mathbb{Q}}) \underbrace{950}_{\mathcal{P}_d^S} = \underbrace{1010}_{\mathcal{P}_0^S \times (R_f)}$$

- This implies that: $\text{prob}_u^{\mathbb{Q}} = \frac{1010-950}{1100-950} = 0.4$; $\text{prob}_d^{\mathbb{Q}} = 0.6$.
(Recall, that the actual probability of an up move was 0.5.)

- Thus, the risk-adjusted expected value of the payoff to the call option is:

$$\mathbb{E}^{\mathbb{Q}}[\mathcal{P}_1^C] = (0.4 \times \underbrace{50}_{\mathcal{P}_u^C}) + (0.6 \times \underbrace{0}_{\mathcal{P}_d^C}) = \text{HC } 20.$$

- Hence, the value of the call today is the present value of the payoff, which can be discounted using the risk-free rate because the expected cashflow is already adjusted for risk.

$$\mathcal{P}_0^C = \frac{\mathbb{E}^{\mathbb{Q}}[\mathcal{P}_1^C]}{R_f} = \frac{20}{1.01} = \text{HC } 19.80.$$

Approach 4: Valuation using Arrow-Debreu state prices

- An *Arrow-Debreu state price* is the price today of a security that pays one dollar in a particular state in the future, and zero in all other states.
$\mathcal{P}_0^{A_u}$: Arrow-Debreu state security that pays 1 unit in the up state;
$\mathcal{P}_0^{A_d}$ Arrow-Debreu state security that pays 1 unit in the down state.

- The price of Arrow-Debreu securities can be identified using the prices of the stock and bank deposit. (The prices of Arrow-Debreu securities can also be identified using any of the three methods we have discussed so far.)

- The price of the stock must satisfy:

$$\mathcal{P}_0^S = \mathcal{P}_0^{A_u}\mathcal{P}_u^S + \mathcal{P}_0^{A_d}\mathcal{P}_d^S. \tag{1.2}$$

and the price of the bond must satisfy:

$$\begin{aligned} \mathcal{P}_0^D &= \mathcal{P}_0^{A_u}\mathcal{P}_u^D + \mathcal{P}_0^{A_d}\mathcal{P}_d^D \\ 1 &= \mathcal{P}_0^{A_u}\mathcal{P}_1^D + \mathcal{P}_0^{A_d}\mathcal{P}_1^D. \end{aligned} \tag{1.3}$$

- Solving (1.2) and (1.3) for the two unknowns, $\mathcal{P}_0^{A_u}$ and $\mathcal{P}_0^{A_d}$, we get:

$$\mathcal{P}_0^{A_u} = \frac{\mathcal{P}_0^S - \frac{\mathcal{P}_d^S}{\mathcal{P}_1^D}}{\mathcal{P}_u^S - \mathcal{P}_d^S} = \frac{\text{prob}_u^{\mathbb{Q}}}{R_f} = \frac{0.4}{1.01};$$
$$\mathcal{P}_0^{A_d} = \frac{\frac{\mathcal{P}_u^S}{\mathcal{P}_1^D} - \mathcal{P}_0^S}{\mathcal{P}_u^S - \mathcal{P}_d^S} = \frac{\text{prob}_d^{\mathbb{Q}}}{R_f} = \frac{0.6}{1.01}.$$

- Therefore, the price of the call option will be

$$\mathcal{P}_0^C = \left(\mathcal{P}_0^{A_u} \times \mathcal{P}_u^C\right) + \left(\mathcal{P}_0^{A_d} \times \mathcal{P}_d^C\right), \tag{1.4}$$

  which we can verify numerically is

$$\mathcal{P}_0^C = \left(\frac{0.4}{1.01} \times 50\right) + \left(\frac{0.6}{1.01} \times 0\right) = \frac{20}{1.01} = \text{HC}\, 19.80.$$

Approach 5: Martingale pricing using the bank-deposit as the numeraire

- A *martingale* is a variable whose current value is equal to its expected value:

$$X_t = \mathbb{E}_t\left(X_{t+1}\right).$$

- In finance, to have the price of an asset today be equal to its expected price, we need to adjust payoffs for *risk* and for *time*:
  - ⋆ We can adjust for risk by using risk-adjusted probabilities.
  - ⋆ We adjust for time by counting in units of the bank deposit.
    This is called a change of numeraire, and we will use this trick frequently.

- Let us denote the value of a bank deposit at t by $\mathcal{P}_t^D$.
  - ⋆ Recall our convention that $\mathcal{P}_0^D = 1$.
  - ⋆ Then, the one-period risk-free gross return is, $\mathcal{P}_1^D/\mathcal{P}_0^D = R_f$.

- Now, I claim that:
  the price of every asset in terms of the bank deposit is a martingale, when expectations are taken with respect to the risk-neutral ($\mathbb{Q}$) probabilities, instead of the physical probabilities $\mathbb{P}$:

$$\frac{\mathcal{P}_0^C}{\mathcal{P}_0^D} = \mathbb{E}_0^{\mathbb{Q}}\left[\frac{\mathcal{P}_1^C}{\mathcal{P}_1^D}\right] = \left(\text{prob}_u^{\mathbb{Q}} \times \frac{\mathcal{P}_u^C}{\mathcal{P}_1^D}\right) + \left(\text{prob}_d^{\mathbb{Q}} \times \frac{\mathcal{P}_d^C}{\mathcal{P}_1^D}\right).$$

- We now verify this numerically:

$$\begin{aligned}
\frac{\mathcal{P}_0^C}{1} &= \left(\text{prob}_u^{\mathbb{Q}} \times \frac{\mathcal{P}_u^C}{\mathcal{P}_1^D}\right) + \left(\text{prob}_d^{\mathbb{Q}} \times \frac{\mathcal{P}_d^C}{\mathcal{P}_1^D}\right) \\
&= \frac{1}{\mathcal{P}_1^D}\Big([0.4 \times 50] + [0.6 \times 0]\Big) \\
&= \frac{1}{1.01}20 \\
\mathcal{P}_0^C &= \text{HC}\,19.80.
\end{aligned}$$

Approach 6: Martingale pricing using the stock as the numeraire

- Instead of using the bank deposit as the numeraire, we can also use the stock as the numeraire (or any other asset), which can be very convenient.

- The claim is: the price of every asset in terms of the stock is a martingale, when expectations are taken with respect to the appropriate probabilities $\mathbb{S}$:

$$\frac{\mathcal{P}_0^C}{\mathcal{P}_0^S} = \mathbb{E}_0^{\mathbb{S}}\left[\frac{\mathcal{P}_1^C}{\mathcal{P}_1^S}\right] = \left(\text{prob}_u^{\mathbb{S}} \times \frac{\mathcal{P}_u^C}{\mathcal{P}_u^S}\right) + \left(\text{prob}_d^{\mathbb{S}} \times \frac{\mathcal{P}_d^C}{\mathcal{P}_d^S}\right).$$

- By equating $\mathcal{P}_0^C$ in the equation above to that in (1.4), reproduced below,

$$\mathcal{P}_0^C = \left(\mathcal{P}_0^{A_u} \times \mathcal{P}_u^C\right) + \left(\mathcal{P}_0^{A_d} \times \mathcal{P}_d^C\right), \tag{1.4}$$

  we see that the probabilities are (the Arrow-Debreu state price times the gross return on the numeraire):

$$\text{prob}_u^{\mathbb{S}} = \mathcal{P}_0^{A_u}\frac{\mathcal{P}_u^S}{\mathcal{P}_0^S} = \frac{\text{prob}_u^{\mathbb{Q}}}{R_f}\frac{\mathcal{P}_u^S}{\mathcal{P}_0^S}; \qquad \text{prob}_d^{\mathbb{S}} = \mathcal{P}_0^{A_d}\frac{\mathcal{P}_d^S}{\mathcal{P}_0^S} = \frac{\text{prob}_d^{\mathbb{Q}}}{R_f}\frac{\mathcal{P}_d^S}{\mathcal{P}_0^S}.$$

- We now verify this numerically.
- We first find the appropriate probabilities:

$$\text{prob}_u^{\mathbb{S}} = \frac{\text{prob}_u^{\mathbb{Q}}}{R_f}\frac{\mathcal{P}_u^S}{\mathcal{P}_0^S} = \frac{0.4}{1.01}\frac{1100}{1000} = 0.435644;$$

$$\text{prob}_d^{\mathbb{S}} = \frac{\text{prob}_d^{\mathbb{Q}}}{R_f}\frac{\mathcal{P}_d^S}{\mathcal{P}_0^S} = \frac{0.6}{1.01}\frac{950}{1000} = 0.564356.$$

- Next, we use these probabilities to write the price of the option, in terms of the stock, as a martingale:

$$\begin{aligned}\frac{\mathcal{P}_0^C}{\mathcal{P}_0^S} &= \left(\text{prob}_u^{\mathbb{S}} \times \frac{\mathcal{P}_u^C}{\mathcal{P}_u^S}\right) + \left(\text{prob}_d^{\mathbb{S}} \times \frac{\mathcal{P}_d^C}{\mathcal{P}_d^S}\right)\\ &= \left(0.435644 \times \frac{50}{1100}\right) + \left(0.564356 \times \frac{0}{950}\right)\\ &= 0.01980\end{aligned}$$

$$\begin{aligned}\mathcal{P}_0^C &= 0.01980 \times 1000\\ &= \text{HC}\, 19.80.\end{aligned}$$

Test Your Understanding 1.1 (Binomial option pricing for a call on stock)
We wish to find the value of a call option on one share of a non-dividend paying stock. We will assume that there is only one period to go before expiration of the option.

- Assume that:
  - ⋆ The strike price of the call option is HC/share 100.
  - ⋆ The risk-free return is 2% per period. So, $\mathcal{P}_0^D = 1$ and $\mathcal{P}_1^D = 1.02$; that is, $R_f = \mathcal{P}_1^D/\mathcal{P}_0^D = 1.02$.
  - ⋆ The current (time-0) stock price, $\mathcal{P}_0^S$, is HC 100 per share.
  - ⋆ At $t = 1$, the stock price can move either to $\mathcal{P}_u^S = 130$ or to $\mathcal{P}_d^S = 90$.
- We wish to find the value of the call option at date 0, denoted $\mathcal{P}_0^C$ using each of the methods that we have discussed above.

- The payoff from the call in the up and down states is:
  - ⋆ in the up state: if $\mathcal{P}_1^S = \mathcal{P}_u^S = 130$, then $\mathcal{P}_u^C = \text{HC } 30$
  - ⋆ in the down state: if $\mathcal{P}_1^S = \mathcal{P}_d^S = 90$, then $\mathcal{P}_d^C = \text{HC } 0$.
- We can summarize the above information as follows:

$$\mathcal{P}_0^S = 100 \begin{cases} \mathcal{P}_u^S = 130 \\ \mathcal{P}_d^S = 90 \end{cases} \qquad \mathcal{P}_0^C = ? \begin{cases} \mathcal{P}_u^C = 30 \\ \mathcal{P}_d^C = 0 \end{cases}$$

- At $t = 1$, we can calculate that delta $= \frac{\text{change in value of } \mathcal{P}_1^C}{\text{change in } \mathcal{P}_1^S} = \frac{30-0}{130-90} = \frac{3}{4}$.
- Assume that the physical (actual) probability of an up and down move is $\frac{1}{2}$.
- Think about why the risk-neutral and actual probabilities are not the same. Think about why the risk-neutral probability of the up-state is smaller than the actual probability.

Space for your work.

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1.8.2 Summary of Our Results

- Suppose the investor has available a bank deposit (money-market account) earning a rate of interest, R_f. In our example, $R_f = (1+r) = \mathcal{P}_1^D/\mathcal{P}_0^D$, but if interest were compounded continuously at the rate r_{cts}, then $R_f = e^{r_{cts}\,T}$.

- In the absence of arbitrage (this is what drives everything), it must be that:

$$\frac{\mathcal{P}_u^S}{\mathcal{P}_0^S} > R_f > \frac{\mathcal{P}_d^S}{\mathcal{P}_0^S} \qquad \Longleftrightarrow \qquad \frac{\mathcal{P}_u^S}{\mathcal{P}_0^S} > \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D} > \frac{\mathcal{P}_d^S}{\mathcal{P}_0^S}. \tag{1.5}$$

  That is,

  - ⋆ return on stock in up state is greater than return on the bank deposit; and,
  - ⋆ return on stock in down state is less than the return on the bank deposit.

- Then, we have the following results.

1. In the absence of arbitrage, there exist positive Arrow-Debreu state prices $\mathcal{P}_0^{A_k}$, such that the price today of any other security is the sum, across the states of the world, of its price in that state times the Arrow-Debreu state price:

$$\mathcal{P}_0 = \left[\mathcal{P}_u \times \mathcal{P}_0^{A_u}\right] + \left[\mathcal{P}_d \times \mathcal{P}_0^{A_d}\right].$$

2. Above, the valuation is described in state prices.
We can describe it also in terms of risk-neutral probabilities.

The price of any asset today is its expected future value using risk-neutral probabilities $\text{prob}_k^{\mathbb{Q}}$, discounted back to today using the risk-free rate:

$$\mathcal{P}_0 = \frac{1}{R_f}\left(\left[\mathcal{P}_u \times \text{prob}_u^{\mathbb{Q}}\right] + \left[\mathcal{P}_d \times \text{prob}_d^{\mathbb{Q}}\right]\right),$$

where $\dfrac{\text{prob}_k^{\mathbb{Q}}}{R_f} = \mathcal{P}_0^{A_k}$ . . . adjustment for risk done by $\text{prob}_k^{\mathbb{Q}}$; adjustment for time by $\frac{1}{R_f}$. (1.6)

3. Valuation using risk-neutral probabilities can be described in another way.

The ratio of an asset's price to the price of the bank deposit today is a martingale (equals the expected value of the same ratio at date 1), when we compute the expectations using the risk-neutral probabilities:

$$\frac{\mathcal{P}_0}{\mathcal{P}_0^D} = \text{prob}_u^{\mathbb{Q}} \frac{\mathcal{P}_u}{\mathcal{P}_1^D} + \text{prob}_d^{\mathbb{Q}} \frac{\mathcal{P}_d}{\mathcal{P}_1^D},$$

where $\mathcal{P}_0^D = 1$; $R_f = \mathcal{P}_1^D / \mathcal{P}_0^D = \mathcal{P}_1^D$; and

$$\text{prob}_k^{\mathbb{Q}} = \mathcal{P}_0^{A_k} \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D} = \mathcal{P}_0^{A_k} R_f.$$

4. In fact, instead of using the bank deposit as the numeraire, we can use some other asset as the numeraire. We have the general result that:

The ratio of an asset's price to the price of the numeraire today is a martingale (equals the expected value of the same ratio at date 1), when we compute the expectations using the appropriate probabilities:

$$\frac{\text{AssetPrice}_0}{\text{NumerairePrice}_0} = \mathbb{E}_0^{\text{Numeraire}} \left[\frac{\text{AssetPrice}_1}{\text{NumerairePrice}_1} \right].$$

For example, the price of any generic asset $\mathcal{P}^{gen}$:

$$\frac{\mathcal{P}_0^{gen}}{\mathcal{P}_0^S} = \text{prob}_u^{\mathbb{S}} \frac{\mathcal{P}_u^{gen}}{\mathcal{P}_u^S} + \text{prob}_d^{\mathbb{S}} \frac{\mathcal{P}_d^{gen}}{\mathcal{P}_d^S}, \quad \text{where} \quad \text{prob}_k^{\mathbb{S}} = \mathcal{P}_0^{A_k} \frac{\mathcal{P}_k^S}{\mathcal{P}_0^S}.$$

⋆ Note: if we change numeraire, we need to also change probability measure.

5. Finally, note that the valuation problem described in the previous four points, in terms of

- state prices,
- risk-neutral probabilities, or
- change of numeraire,

is identical to

- the portfolio problem
  of finding the strategy to replicate the payoff to the option;
- the risk-management problem
  of finding the delta to hedge the payoff to the option.

Thus, if we know how to solve one problem (say, valuation), we can infer from that the solution to the other problems.

1.9 Valuation using the CAPM

- The tradition approach for valuing a risky payoff is to divide it by a discount rate that has been appropriately adjusted for risk.
- One model used to adjust for risk is the Capital Asset Pricing Model (CAPM).
- In this section, we revisit our example to see how to find the value of the option using the CAPM, and examine whether it gives us the same answer.

1.9.1 From replication to CAPM

- There are many ways of showing that the various approaches that we have developed to find the value of the option are consistent also with the CAPM, which states that, for any asset j:

$$\mathbb{E}^{\mathbb{P}}[R^j] - R_f = \beta_j\Big(\mathbb{E}^{\mathbb{P}}[R^M] - R_f\Big),$$

where $\beta_j = \frac{\mathsf{Cov}(R^M, R^j)}{\mathsf{Var}(R^M)}$.

- We wish to show this is true also for the expected return on the call option:

$$\mathbb{E}^{\mathbb{P}}[R^C] - R_f = \beta_C\Big(\mathbb{E}^{\mathbb{P}}[R^M] - R_f\Big),$$

where $\beta_C = \frac{\mathsf{Cov}(R^M, R^C)}{\mathsf{Var}(R^M)}$.

- Note that in the CAPM, all expectations are under the physical probability measure $\mathbb{P}$, where the probabilities are $\text{prob}_u^{\mathbb{P}}$ and $\text{prob}_d^{\mathbb{P}} = 1 - \text{prob}_u^{\mathbb{P}}$.

- Below, we will illustrate the equivalence with the CAPM using as our starting point the replication approach.
- The actual derivation is straightforward.
- The only thing one has to be careful about is units:
  - ⋆ all the approaches that we looked at so far determined the dollar value of the option,
  - ⋆ but the CAPM is expressed in terms of returns.
- Therefore, we need to convert all expressions that are in dollar terms in to ones that are in terms of gross returns.

Notation

- In particular, we will use the following gross returns:
  - ⋆ The risk-free rate of return: R_f.
  - ⋆ The return on the market: R^M.
  - ⋆ The return on the stock: R^S,
    which in the up state is $R^S_u = \frac{\mathcal{P}^S_u}{\mathcal{P}^S_0}$ and in the down state is $R^S_d = \frac{\mathcal{P}^S_d}{\mathcal{P}^S_0}$.
  - ⋆ The return on the call option: R^C,
    which in the up state is $R^C_u = \frac{\mathcal{P}^C_u}{\mathcal{P}^C_0}$ and in the down state is $R^C_d = \frac{\mathcal{P}^C_d}{\mathcal{P}^C_0}$.

- Thus, the expected gross returns are:

$$\mu_S = \text{prob}^{\mathbb{P}}_u \frac{\mathcal{P}^S_u}{\mathcal{P}^S_0} + \text{prob}^{\mathbb{P}}_d \frac{\mathcal{P}^S_d}{\mathcal{P}^S_0} \tag{1.7}$$

$$\mu_C = \text{prob}^{\mathbb{P}}_u \frac{\mathcal{P}^C_u}{\mathcal{P}^S_0} + \text{prob}^{\mathbb{P}}_d \frac{\mathcal{P}^C_d}{\mathcal{P}^S_0} \tag{1.8}$$

- Similarly, noting that the variance of a Bernoulli random variable X is

$$\text{Var}(X) = \text{prob}^{\mathbb{P}}_u \, \text{prob}^{\mathbb{P}}_d \left(X_u - X_d\right)^2$$

the volatilities of gross returns for the stock and call, respectively are:

$$\sigma_S = \sqrt{\text{Var}\left(\frac{\mathcal{P}^S_1}{\mathcal{P}^S_0}\right)} = \sqrt{\text{prob}^{\mathbb{P}}_u \, \text{prob}^{\mathbb{P}}_d} \left(\frac{\mathcal{P}^S_u}{\mathcal{P}^S_0} - \frac{\mathcal{P}^S_d}{\mathcal{P}^S_0}\right) \tag{1.9}$$

$$\sigma_C = \sqrt{\text{Var}\left(\frac{\mathcal{P}^C_1}{\mathcal{P}^C_0}\right)} = \sqrt{\text{prob}^{\mathbb{P}}_u \, \text{prob}^{\mathbb{P}}_d} \left(\frac{\mathcal{P}^C_u}{\mathcal{P}^C_0} - \frac{\mathcal{P}^C_d}{\mathcal{P}^C_0}\right). \tag{1.10}$$

- Recall that we had already defined the "delta" of the option as

$$b = \frac{\text{change in } \mathcal{P}^C}{\text{change in } \mathcal{P}^S} = \frac{\mathcal{P}_u^C - \mathcal{P}_d^C}{\mathcal{P}_u^S - \mathcal{P}_d^S}.$$

- Now we define useful quantity, the elasticity of the option:

$$\mathcal{E} = \frac{\text{percentage change in } \mathcal{P}^C}{\text{percentage change in } \mathcal{P}^S} \quad \ldots \text{ definition of elasticity} \tag{1.11}$$

$$= \frac{\frac{\text{change in } \mathcal{P}^C}{\mathcal{P}_0^C}}{\frac{\text{change in } \mathcal{P}^S}{\mathcal{P}_0^S}} = \frac{\frac{\mathcal{P}_u^C - \mathcal{P}_d^C}{\mathcal{P}_0^C}}{\frac{\mathcal{P}_u^S - \mathcal{P}_d^S}{\mathcal{P}_0^S}} = \frac{\frac{\mathcal{P}_u^C}{\mathcal{P}_0^C} - \frac{\mathcal{P}_d^C}{\mathcal{P}_0^C}}{\frac{\mathcal{P}_u^S}{\mathcal{P}_0^S} - \frac{\mathcal{P}_d^S}{\mathcal{P}_0^S}} = \frac{\sigma_C}{\sigma_S} \quad \ldots \text{ in terms of volatilities} \tag{1.12}$$

$$= \frac{\mathcal{P}_0^S}{\mathcal{P}_0^C} \frac{\text{change in } \mathcal{P}^C}{\text{change in } \mathcal{P}^S} = \frac{\mathcal{P}_0^S}{\mathcal{P}_0^C} b \quad \ldots \text{ in terms of delta} \tag{1.13}$$

- From (1.12) and (1.13), we see that the delta can be expressed as:

$$b = \frac{\sigma_C}{\sigma_S} \frac{\mathcal{P}_0^C}{\mathcal{P}_0^S}. \quad \ldots \text{ we will use this below} \tag{1.14}$$

- We start with the replicating equations that we had before:

- The *replicating equations* that show how much to invest in the bank deposit, a, and the stock, b, to match the payoff to the option at $t = 1$:

$$a\,\mathcal{P}_1^D + b\,\mathcal{P}_u^S = \mathcal{P}_u^C \quad \ldots \text{ replicate payoff in the "up" state} \tag{1.15}$$
$$a\,\mathcal{P}_1^D + b\,\mathcal{P}_d^S = \mathcal{P}_d^C \quad \ldots \text{ replicate payoff in the "down" state} \tag{1.16}$$

had allowed us to demonstrate that the value of the option at $t = 0$ is:

$$\mathcal{P}_0^C = a + b\,\mathcal{P}_0^S. \tag{1.17}$$

- We rewrite (1.15) and (1.16) as:

$$a\,\mathcal{P}_1^D = \mathcal{P}_u^C - b\,\mathcal{P}_u^S \tag{1.18}$$
$$a\,\mathcal{P}_1^D = \mathcal{P}_d^C - b\,\mathcal{P}_d^S. \tag{1.19}$$

- Now, we multiply both sides of equation (1.18) by $\text{prob}_u^{\mathbb{P}}$ and both sides of equation (1.19) by $\text{prob}_d^{\mathbb{P}} = 1 - \text{prob}_u^{\mathbb{P}}$, and add them up:

$$a\,\mathcal{P}_1^D = \text{prob}_u^{\mathbb{P}}\big(\mathcal{P}_u^C - b\,\mathcal{P}_u^S\big) + \text{prob}_d^{\mathbb{P}}\big(\mathcal{P}_d^C - b\,\mathcal{P}_d^S\big)$$

$$\big(\mathcal{P}_0^C - b\,\mathcal{P}_0^S\big)\mathcal{P}_1^D = \text{prob}_u^{\mathbb{P}}\big(\mathcal{P}_u^C - b\,\mathcal{P}_u^S\big) + \text{prob}_d^{\mathbb{P}}\big(\mathcal{P}_d^C - b\,\mathcal{P}_d^S\big) \quad \ldots \text{from (1.17), } a = \mathcal{P}_0^C - b\,\mathcal{P}_0^S$$

$$\big(\mathcal{P}_0^C - b\,\mathcal{P}_0^S\big)\mathcal{P}_1^D = \big(\text{prob}_u^{\mathbb{P}}\mathcal{P}_u^C + \text{prob}_d^{\mathbb{P}}\mathcal{P}_d^C\big) - b\,\big(\text{prob}_u^{\mathbb{P}}\mathcal{P}_u^S + \text{prob}_d^{\mathbb{P}}\mathcal{P}_d^S\big) \quad \ldots \text{collecting } \mathcal{P}^S \text{ and } \mathcal{P}^C \text{ terms}$$

$$\big(\mathcal{P}_0^C - \frac{\sigma_C}{\sigma_S}\frac{\mathcal{P}_0^C}{\mathcal{P}_0^S}\mathcal{P}_0^S\big)\mathcal{P}_1^D = \big(\text{prob}_u^{\mathbb{P}}\mathcal{P}_u^C + \text{prob}_d^{\mathbb{P}}\mathcal{P}_d^C\big) - \frac{\sigma_C}{\sigma_S}\frac{\mathcal{P}_0^C}{\mathcal{P}_0^S}\big(\text{prob}_u^{\mathbb{P}}\mathcal{P}_u^S + \text{prob}_d^{\mathbb{P}}\mathcal{P}_d^S\big) \quad \ldots \text{from (1.14) } b = \frac{\sigma_C}{\sigma_S}\frac{\mathcal{P}_0^C}{\mathcal{P}_0^S}$$

$$\big(\mathcal{P}_0^C - \frac{\sigma_C}{\sigma_S}\mathcal{P}_0^C\big)\mathcal{P}_1^D = \big(\text{prob}_u^{\mathbb{P}}\mathcal{P}_u^C + \text{prob}_d^{\mathbb{P}}\mathcal{P}_d^C\big) - \frac{\sigma_C}{\sigma_S}\frac{\mathcal{P}_0^C}{\mathcal{P}_0^S}\big(\text{prob}_u^{\mathbb{P}}\mathcal{P}_u^S + \text{prob}_d^{\mathbb{P}}\mathcal{P}_d^S\big) \quad \ldots \text{canceling } \mathcal{P}_0^S \text{ in num. and den.}$$

$$\big(1 - \frac{\sigma_C}{\sigma_S}\big)\mathcal{P}_1^D = \big(\text{prob}_u^{\mathbb{P}}\frac{\mathcal{P}_u^C}{\mathcal{P}_0^C} + \text{prob}_d^{\mathbb{P}}\frac{\mathcal{P}_d^C}{\mathcal{P}_0^C}\big) - \frac{\sigma_C}{\sigma_S}\big(\text{prob}_u^{\mathbb{P}}\frac{\mathcal{P}_u^S}{\mathcal{P}_0^S} + \text{prob}_d^{\mathbb{P}}\frac{\mathcal{P}_d^S}{\mathcal{P}_0^S}\big) \quad \ldots \text{dividing both sides by } \mathcal{P}_0^C$$

- We are ready now to express our result in terms of returns.
- We start by rewriting the last equation we had on the previous slide:

$$\Big(1 - \frac{\sigma_C}{\sigma_S}\Big)\mathcal{P}_1^D = \Big(\text{prob}_u^{\mathbb{P}}\frac{\mathcal{P}_u^C}{\mathcal{P}_0^C} + \text{prob}_d^{\mathbb{P}}\frac{\mathcal{P}_d^C}{\mathcal{P}_0^C}\Big) - \frac{\sigma_C}{\sigma_S}\Big(\text{prob}_u^{\mathbb{P}}\frac{\mathcal{P}_u^S}{\mathcal{P}_0^S} + \text{prob}_d^{\mathbb{P}}\frac{\mathcal{P}_d^S}{\mathcal{P}_0^S}\Big)$$

$$\Big(1 - \frac{\sigma_C}{\sigma_S}\Big)R_f = \mu_C - \frac{\sigma_C}{\sigma_S}\mu_S \quad \ldots \text{recall } \mathcal{P}_1^D = R_f\text{; using definitions of } \mu_S \text{ and } \mu_C$$

$$\mu_C - R_f = \frac{\sigma_C}{\sigma_S}\big(\mu_S - R_f\big) = \mathcal{E}\big(\mu_S - R_f\big). \tag{1.20}$$

- We start by rewriting the last equation on the previous slide:

$$\mu_C - R_f = \frac{\sigma_C}{\sigma_S}\Big(\mu_S - R_f\Big) = \mathcal{E}\Big(\mu_S - R_f\Big). \tag{1.20}$$

- The equation above can be written in different ways, each leading to an interesting insight:

$$\mu_C = \mathcal{E}\mu_S + (1-\mathcal{E})R_f \quad \ldots \text{ call is a portfolio of stock and risk-free asset with weights } \mathcal{E} \text{ and } 1-\mathcal{E}$$

$$\mu_C - R_f = \underbrace{\frac{\mu_S - R_f}{\sigma_S}}_{\text{price of risk}} \underbrace{\sigma_C}_{\text{quantity of risk}} \quad \ldots \text{ risk premium on call is [price of risk] } \times \text{ [quantity of risk]}$$

$$\frac{\mu_C - R_f}{\sigma_C} = \frac{\mu_S - R_f}{\sigma_S} \quad \ldots \text{ Sharpe ratios of the option and stock are equal}$$

- Most importantly, equation (1.20) also allows us to show that the expected return on the call option satisfies the CAPM. Rewriting (1.20):

$$\mu_C - R_f = \mathcal{E}\Big(\mu_S - R_f\Big) \tag{1.20}$$

- and noting from the CAPM that the risk-premium on the stock is

$$\mu_S - R_f = \beta_S\, \mathbb{E}^{\mathbb{P}}\Big(\mu_M - R_f\Big), \tag{1.21}$$

which can be substituted into the right-hand side of (1.20) to get

$$\mu_C - R_f = \mathcal{E}\,\beta_S\, \mathbb{E}^{\mathbb{P}}\Big(\mu_M - R_f\Big) \tag{1.22}$$

which, because of my claim (proved on the next slide) that $\beta_C = \mathcal{E}\,\beta_S$, implies that the expected return on the option satisfies the CAPM:

$$\mu_C - R_f = \beta_C\, \mathbb{E}^{\mathbb{P}}\Big(\mu_M - R_f\Big). \qquad \ldots \text{CAPM} \tag{1.23}$$

1.9.2 Proof that $\beta_C = \mathcal{E}\,\beta_S$

- For Bernoulli variables X and Y where X_u and Y_u occur with probability $\text{prob}_u^{\mathbb{P}}$ and X_d and Y_d occur with probability $\text{prob}_d^{\mathbb{P}} = 1 - \text{prob}_u^{\mathbb{P}}$,

$$\text{Cov}(X, Y) = \text{prob}_u^{\mathbb{P}}\,\text{prob}_d^{\mathbb{P}}(X_u - X_d)(Y_u - Y_d).$$

- Therefore, for the covariance between the return on the market and the stock

$$\begin{aligned}\text{Cov}(R^M, R^S) &= \text{prob}_u^{\mathbb{P}}\,\text{prob}_d^{\mathbb{P}}\left(R_u^M - R_d^M\right)(R_u^S - R_d^S) \\ &= \text{prob}_u^{\mathbb{P}}\,\text{prob}_d^{\mathbb{P}}(R_u^M - R_d^M)\left(\frac{\mathcal{P}_u^S}{\mathcal{P}_0^S} - \frac{\mathcal{P}_d^S}{\mathcal{P}_0^S}\right) \\ &= \frac{1}{\mathcal{P}_0^S}\,\text{prob}_u^{\mathbb{P}}\,\text{prob}_d^{\mathbb{P}}\,(R_u^M - R_d^M)\left(\mathcal{P}_u^S - \mathcal{P}_d^S\right).\end{aligned}$$

- For the covariance between the return on the market and the call

$$\begin{aligned}\text{Cov}(R^M, R^C) &= \text{prob}_u^{\mathbb{P}}\,\text{prob}_d^{\mathbb{P}}\left(R_u^M - R_d^M\right)(R_u^C - R_d^C) \\ &= \text{prob}_u^{\mathbb{P}}\,\text{prob}_d^{\mathbb{P}}(R_u^M - R_d^M)\left(\frac{\mathcal{P}_u^C}{\mathcal{P}_0^C} - \frac{\mathcal{P}_d^C}{\mathcal{P}_0^C}\right) \\ &= \frac{1}{\mathcal{P}_0^C}\,\text{prob}_u^{\mathbb{P}}\,\text{prob}_d^{\mathbb{P}}\,(R_u^M - R_d^M)\left(\mathcal{P}_u^C - \mathcal{P}_d^C\right).\end{aligned}$$

- But, substituting from the replicating equations, which are rewritten below,

$$a\,\mathcal{P}_1^D + b\,\mathcal{P}_u^S = \mathcal{P}_u^C \qquad (1.15)$$
$$a\,\mathcal{P}_1^D + b\,\mathcal{P}_d^S = \mathcal{P}_d^C. \qquad (1.16)$$

- And, dividing and multiplying by $\mathcal{P}_0^S$, we get

$$\begin{aligned}\text{Cov}(R^M, R^C) &= \frac{b\,\mathcal{P}_0^S}{\mathcal{P}_0^C}\frac{1}{\mathcal{P}_0^S}\,\text{prob}_u^{\mathbb{P}}\,\text{prob}_d^{\mathbb{P}}\,(R_u^M - R_d^M)\left(\mathcal{P}_u^S - \mathcal{P}_d^S\right) \\ &= \frac{b\,\mathcal{P}_0^S}{\mathcal{P}_0^C}\text{Cov}(R^M, R^S) \\ &= \mathcal{E}\,\text{Cov}(R^M, R^S). \qquad \text{... using (1.13): } \mathcal{E} = \frac{b\,\mathcal{P}_0^S}{\mathcal{P}_0^C}\end{aligned}$$

- Dividing both sides by $\text{Var}(R^M)$, we get the result we wanted to show:

$$\text{Cov}(R^M, R^C) = \mathcal{E}\,\text{Cov}(R^M, R^S)$$

$$\frac{\text{Cov}(R^M, R^C)}{\text{Var}(R^M)} = \mathcal{E}\,\frac{\text{Cov}(R^M, R^S)}{\text{Var}(R^M)}$$

$$\beta_C = \mathcal{E}\,\beta_S.$$

Test Your Understanding 1.2 (Binomial option pricing using the CAPM)

- Let us continue with Example 1.1 on Page 77 and find the value of the call option using the CAPM.

- To use the CAPM to discount the expected payoff to the call option (with the expectation under the physical probabilities), we need the following steps:

  Step 1: Find the expected return on the stock.

  Step 2: Find the weights in the bank deposit and the stock that comprise the call option.

  Step 3: Use these weights to find the expected return on the call option.

  Step 4: Use the expected return on the call option to discount the expected payoff of the call option.

Space for your work.

Space for your work.

Space for your work.

Space for your work.

1.10 Stochastic Discount Factor (SDF) in Binomial Model

Definition of SDF

- We now express our results in yet another way, which is useful for relating no-arbitrage valuation to equilibrium valuation, and also for empirical work.
- Start with expressions for asset prices in terms of Arrow-Debreu state prices:

$$\mathcal{P}_0^C = \mathcal{P}_0^{Au}\,\mathcal{P}_u^C + \mathcal{P}_0^{Ad}\,\mathcal{P}_d^C, \tag{1.24}$$

$$\mathcal{P}_0^S = \mathcal{P}_0^{Au}\,\mathcal{P}_u^S + \mathcal{P}_0^{Ad}\,\mathcal{P}_d^S, \tag{1.25}$$

$$1 = \mathcal{P}_0^{Au}\,\mathcal{P}_1^D + \mathcal{P}_0^{Ad}\,\mathcal{P}_1^D \qquad \ldots \text{recall that } \mathcal{P}_0^D = 1\ . \tag{1.26}$$

- Let $\text{prob}_u^{\mathbb{P}}$, $\text{prob}_d^{\mathbb{P}}$ denote actual (physical) probabilities of up and down states.
- Define the stochastic discount factor (state price per probability):

$$m_u = \frac{\mathcal{P}_0^{Au}}{\text{prob}_u^{\mathbb{P}}} \quad \text{and} \quad m_d = \frac{\mathcal{P}_0^{Ad}}{\text{prob}_d^{\mathbb{P}}}. \tag{1.27}$$

1.10.1 SDF and Asset Prices

- We can then rewrite Equations (1.24)–(1.26) in terms of expectations with respect to the physical probabilities:

$$\mathcal{P}_0^C = \text{prob}_u^{\mathbb{P}}\left(m_u\,\mathcal{P}_u^C\right) + \text{prob}_d^{\mathbb{P}}\left(m_d\,\mathcal{P}_d^C\right) = \mathbb{E}^{\mathbb{P}}\left[m_1\,\mathcal{P}_1^C\right], \tag{1.28}$$

$$\mathcal{P}_0^S = \text{prob}_u^{\mathbb{P}}\left(m_u\,\mathcal{P}_u^S\right) + \text{prob}_d^{\mathbb{P}}\left(m_d\,\mathcal{P}_d^S\right) = \mathbb{E}^{\mathbb{P}}\left[m_1\,\mathcal{P}_1^S\right], \tag{1.29}$$

$$\mathcal{P}_0^D = \text{prob}_u^{\mathbb{P}}\left(m_u\,\mathcal{P}_1^D\right) + \text{prob}_d^{\mathbb{P}}\left(m_d\,\mathcal{P}_1^D\right) = \mathbb{E}^{\mathbb{P}}\left[m_1\,\mathcal{P}_1^D\right]. \tag{1.30}$$

- Obviously, in the expressions above, the adjustment for risk and time is being done in one shot by the stochastic discount factor, m_k.
- Equations (1.28)–(1.30) show that
  - ⋆ even though asset prices are not martingales,
  - ⋆ *after* multiplication by m, asset prices become martingales.

- Equations (1.28)–(1.30), suggest the following general principle:

> If there are no arbitrage opportunities, there exists for each date T a positive random variable m_T such that the value at date 0 of a non-dividend-paying security with price $\mathcal{P}^S$ is:
>
> $$\mathcal{P}_0^S = \mathbb{E}^{\mathbb{P}}\left[m_T\,\mathcal{P}_T^S\right], \tag{1.31}$$
>
> where the random variable m_T is called the stochastic discount factor (SDF), which is the state price per unit probability, and $\mathbb{E}^{\mathbb{P}}$ denotes the expectation of the random variable $m_T\,\mathcal{P}_T^S$ under the physical measure $\mathbb{P}$.

- The stochastic discount factor is called by a variety of names:
  - ⋆ state price per probability,
  - ⋆ state price density,
  - ⋆ pricing kernel, and
  - ⋆ marginal rate of substitution.

Example 1.2 (Continuing with binomial option pricing for a call on stock)

- Let us continue with Example 1.1 on Page 77 and find the value of the call option using the SDF.
- Recall that we had found the prices of Arrow-Debreu state prices:

$$\mathcal{P}_0^{A_u} = \frac{0.4}{1.01};$$
$$\mathcal{P}_0^{A_d} = \frac{0.6}{1.01}.$$

- Now, using our definition for the SDF m_k in (1.27), we get that

$$m_u = \frac{\mathcal{P}_0^{A_u}}{\text{prob}_u^{\mathbb{P}}} = \frac{0.4}{1.01} \times \frac{1}{\text{prob}_u^{\mathbb{P}}};$$
$$m_d = \frac{\mathcal{P}_0^{A_d}}{\text{prob}_d^{\mathbb{P}}} = \frac{0.6}{1.01} \times \frac{1}{\text{prob}_d^{\mathbb{P}}}.$$

- Finally, using the result in Equation (1.28), reproduced below, we have

$$\text{Price of call option} = \mathbb{E}^{\mathbb{P}}\left[m_1\,\mathcal{P}_1^C\right] \tag{1.28}$$

$$\begin{aligned}
\mathcal{P}_0^C &= \text{prob}_u^{\mathbb{P}}\, m_u\, \mathcal{P}_u^C + \text{prob}_d^{\mathbb{P}}\, m_d\, \mathcal{P}_d^C \\
&= \text{prob}_u^{\mathbb{P}} \left(\frac{\mathcal{P}_0^{A_u}}{\text{prob}_u^{\mathbb{P}}}\right) \mathcal{P}_u^C + \text{prob}_d^{\mathbb{P}} \left(\frac{\mathcal{P}_0^{A_d}}{\text{prob}_d^{\mathbb{P}}}\right) \mathcal{P}_d^C \\
&= \mathcal{P}_0^{A_u}\, \mathcal{P}_u^C + \mathcal{P}_0^{A_d}\, \mathcal{P}_d^C \\
&= \frac{0.4}{1.01}\,50 + \frac{0.6}{1.01}\,0 \\
&= \text{HC } 19.80.
\end{aligned}$$

- Note we did not need to know the physical probabilities, $\text{prob}_u^{\mathbb{P}}$ and $\text{prob}_d^{\mathbb{P}}$.
  - ⋆ This is because the SDF m_k also depends on the physical probabilities, and when we take expectations under $\mathbb{P}$, the $\text{prob}_k^{\mathbb{P}}$ terms cancel out.

1.10.2 Relation between SDF and Probabilities

- Substituting the results from (1.6) on page 106

$$\mathcal{P}_0^{A_u} = \frac{\text{prob}_u^{\mathbb{Q}}}{R_f} = \text{prob}_u^{\mathbb{Q}} \frac{\mathcal{P}_0^D}{\mathcal{P}_1^D} \qquad \text{and} \qquad \mathcal{P}_0^{A_d} = \frac{\text{prob}_d^{\mathbb{Q}}}{R_f} = \text{prob}_d^{\mathbb{Q}} \frac{\mathcal{P}_0^D}{\mathcal{P}_1^D} \tag{1.6}$$

into the definition of the SDF in (1.27) on page 129

$$m_u = \frac{\mathcal{P}_0^{A_u}}{\text{prob}_u^{\mathbb{P}}} \qquad \text{and} \qquad m_d = \frac{\mathcal{P}_0^{A_d}}{\text{prob}_d^{\mathbb{P}}}, \tag{1.27}$$

we get that

$$m_u = \frac{\text{prob}_u^{\mathbb{Q}}}{\text{prob}_u^{\mathbb{P}}} \frac{\mathcal{P}_0^D}{\mathcal{P}_1^D} \qquad \text{and} \qquad m_d = \frac{\text{prob}_d^{\mathbb{Q}}}{\text{prob}_d^{\mathbb{P}}} \frac{\mathcal{P}_0^D}{\mathcal{P}_1^D}, \tag{1.32}$$

which shows how m_u and m_d adjust for risk and for time.

1.10.3 Relation between SDF and Ratio of Probabilities

- We start by rewriting the last expression on the previous slide

$$m_u = \frac{\text{prob}_u^{\mathbb{Q}}}{\text{prob}_u^{\mathbb{P}}} \frac{\mathcal{P}_0^D}{\mathcal{P}_1^D} \qquad \text{and} \qquad m_d = \frac{\text{prob}_d^{\mathbb{Q}}}{\text{prob}_d^{\mathbb{P}}} \frac{\mathcal{P}_0^D}{\mathcal{P}_1^D}. \tag{1.32}$$

- These expressions can be written as

$$\frac{\text{prob}_k^{\mathbb{Q}}}{\text{prob}_k^{\mathbb{P}}} = m_k \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}, \tag{1.33}$$

  which says that: the ratio of $\text{prob}^{\mathbb{Q}}$ to $\text{prob}^{\mathbb{P}}$ for state k is equal to the discount factor for state k multiplied by the gross return on the numeraire.

- We will study the analog of this "ratio of probabilities" in continuous time, where it is referred to as the Radon-Nikodym derivative.

1.10.4 Relation between SDF and Numeraire-Specific Probabilities

- We start by rewriting the last expression on the previous slide

$$\frac{\text{prob}_k^{\mathbb{Q}}}{\text{prob}_k^{\mathbb{P}}} = m_k \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}, \tag{1.33}$$

- This can be re-arranged to get:

$$\text{prob}_k^{\mathbb{Q}} = m_k \, \text{prob}_k^{\mathbb{P}} \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}. \tag{1.34}$$

- Similarly, the probabilities corresponding to the stock-numeraire are:

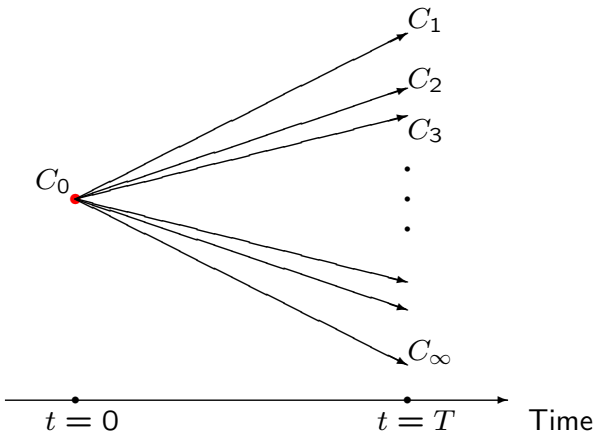
$$\text{prob}_k^{\mathbb{S}} = m_k \, \text{prob}_k^{\mathbb{P}} \frac{\mathcal{P}_k^S}{\mathcal{P}_0^S}. \tag{1.35}$$

- **Conclusion:** To convert from SDF to probabilities corresponding to different numeraires, we multiply three quantities:
  1. the physical probability of that state;
  2. the value of the SDF m_T in that state; and,
  3. the gross return of the numeraire in that state.

End of
Asset Pricing in a Binomial Model

1.11 SDF and RNP with a Continuum of States

- We now generalize the concepts of state prices and probabilities corresponding to different measures in a more general framework than the binomial model:
  – instead of two states, we allow for a continuum of states,
  – but, for now, we still allow for only two dates, 0 and $T = 1$.



- Equations (1.28)–(1.30), and the general version in (1.31), suggest that, if markets are complete, the following principle applies even with a continuum of states.

- If there are no arbitrage opportunities, there exists for each date T a positive random variable m_T such that the value at date 0 of a non-dividend-paying security with price $\mathcal{P}^S$ is:

$$\mathcal{P}_0^S = \mathbb{E}^{\mathbb{P}}\left[m_T\,\mathcal{P}_T^S\right], \tag{1.36}$$

  where the random variable m_T is the stochastic discount factor.

1.11.1 Converting From SDF to Risk-Neutral Probabilities (RNP)

- To convert from SDF to probabilities corresponding to different numeraires, we follow the same method as before, which is to multiply three quantities:
  1. the physical probability of that state;
  2. the value of the SDF m_T in that state; and,
  3. the gross return of the numeraire in that state.
- For example, from (1.35) on page 136: $\text{prob}_u^{\mathbb{S}} = \underbrace{\text{prob}_u^{\mathbb{P}}}\ \underbrace{m_u}\ \underbrace{\mathcal{P}_u^S/\mathcal{P}_0^S}$.
- If there are a continuum of states, then usually the probability of any particular state is 0.
- However, we can still add up these probabilities to define the probability of a particular event, A, where an event is a **set of states of the world**. For example, an event could be all the states when $\mathcal{P}_T^S > K$.

- To do this, let 1_A denote an indicator function, that is, the random variable that takes the value 1 when the event occurs and 0 otherwise.
- Then the probability of A, using the stock as the numeraire, is

$$\text{prob}^{\mathbb{S}}(A) = \mathbb{E}^{\mathbb{P}}\left[1_A\, m_1\, \frac{\mathcal{P}_1^S}{\mathcal{P}_0^S}\right]. \tag{1.37}$$

- Similarly, the probability of A, using the bond as the numeraire, would be

$$\text{prob}^{\mathbb{Q}}(A) = \mathbb{E}^{\mathbb{P}}\left[1_A\, m_1\, \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}\right]. \tag{1.38}$$

- We start by reproducing (1.37) from the previous slide:

$$\text{prob}^{\mathbb{S}}(A) = \mathbb{E}^{\mathbb{P}}\Big[1_A\, m_1\, \frac{\mathcal{P}_1^S}{\mathcal{P}_0^S}\Big]. \tag{1.37}$$

- To show that the above is a *probability measure*, we need to show that it has the following two properties:
  1. It is positive. This is indeed the case because all the quantities on the right-hand side of (1.37) are positive.
  2. If A denotes the set of all states of the world, then the probability over all outcomes equals one.
- Below, we show below that this is indeed the case.

- **Proof** that if A denotes the set of all states of the world, then the probability over all outcomes equals one.
- Applying the definition in (1.37) to the set of all events, we get:

$$\begin{aligned}
\text{prob}^{\mathbb{S}}\Big([\text{all events}]\Big) &= \mathbb{E}^{\mathbb{P}}\Big[1_{[\text{all events}]}\, m_1\, \frac{\mathcal{P}_1^S}{\mathcal{P}_0^S}\Big] && \ldots \text{from the definition in equation (1.37)} \\
&= \mathbb{E}^{\mathbb{P}}\Big[m_1\, \frac{\mathcal{P}_1^S}{\mathcal{P}_0^S}\Big] && \ldots \text{because } 1_{\text{all events}} = 1 \\
&= 1,
\end{aligned}$$

where the last line follows from equation (1.31) on page 131,

$$\mathcal{P}_0^S = \mathbb{E}^{\mathbb{P}}\left[m_1\, \mathcal{P}_1^S\right],$$

which, upon dividing both sides by $\mathcal{P}_0^S$, gives

$$1 = \mathbb{E}^{\mathbb{P}}\left[m_1\, \frac{\mathcal{P}_1^S}{\mathcal{P}_0^S}\right].$$

1.11.2 Relation between models with two vs. continuum of states

- The probability of event A defined on the previous page

$$\text{prob}^{\mathbb{S}}(A) = \mathbb{E}^{\mathbb{P}}\Big[1_A\, m_1\, \frac{\mathcal{P}_1^S}{\mathcal{P}_0^S}\Big], \tag{1.37}$$

corresponds exactly to the expression in (1.35) on Page 136 for the binomial model, which is reproduced below:

$$\text{prob}_u^{\mathbb{S}} = \text{prob}_u^{\mathbb{P}}\, m_u\, \frac{\mathcal{P}_u^S}{\mathcal{P}_0^S}; \quad \text{and} \quad \text{prob}_d^{\mathbb{S}} = \text{prob}_d^{\mathbb{P}}\, m_d\, \frac{\mathcal{P}_d^S}{\mathcal{P}_0^S}. \tag{1.35}$$

- There are three terms on the right-hand side of Equations (1.37) and (1.35).
  - ⋆ In Eq. (1.37), the expected value (under $\mathbb{P}$) of the indicator function 1_A corresponds to the $\text{prob}_k^{\mathbb{P}}$ term in (1.35); and
  - ⋆ the other two quantities (m_1 and the gross return on the numeraire) are the same in Equations (1.37) and (1.35).

- If instead we use the **bank-deposit as the numeraire**, we get:

$$\text{prob}^{\mathbb{Q}}(A) = \mathbb{E}^{\mathbb{P}}\Big[1_A\, m_1\, \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}\Big] = \mathbb{E}^{\mathbb{P}}\Big[1_A\, m_1\, R_f\Big], \tag{1.39}$$

which is called the risk-neutral probability measure (RNP), denoted by $\mathbb{Q}$.

- The RNP measure is also called the equivalent martingale measure, often abbreviated as EMM.

- If you look carefully at (1.39), you will see that it corresponds exactly to the expression in (1.34) on Page 136, which is reproduced below:

$$\text{prob}_u^{\mathbb{Q}} = \text{prob}_u^{\mathbb{P}}\, m_u\, \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}; \quad \text{and} \quad \text{prob}_d^{\mathbb{Q}} = \text{prob}_d^{\mathbb{P}}\, m_d\, \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}. \tag{1.34}$$

- To keep track of which probability we are using when computing expectations, we will use a superscript that indicates our choice.
  - ⋆ For example, to compute the expectation of the random variable X using the stock as the numeraire we have

$$\mathbb{E}^{\mathbb{S}}[X_1] = \mathbb{E}^{\mathbb{P}}\left[X_1\, m_1 \frac{\mathcal{P}_1^S}{\mathcal{P}_0^S}\right]. \tag{1.40}$$

- In our course, we will typically use one of three measures:
  - ⋆ The physical (objective or subjective) measure $\mathbb{P}$;
  - ⋆ The risk-neutral measure $\mathbb{Q}$, where the bank deposit is the numeraire;
  - ⋆ The measure $\mathbb{S}$, where the stock is the numeraire.
- A fourth measure is the "forward measure" $\mathbb{F}$, which corresponds to the numeraire being a discount bond (as opposed to the bank deposit); we will not need to use this in our course.

1.11.3 Martingale Pricing With a Continuum of States

- In the binomial model, we had shown that the ratio of the price of any (non-dividend paying) asset to the price of the numeraire asset is a **martingale** when we use the probability measure corresponding to the numeraire.
- The same result holds true even with a continuum of states.
- Denoting by $\mathbb{E}_t^{\mathbb{S}}$ the expectation conditional on information at $t < T$ and using $\mathcal{P}^S$ as the numeraire, the price of a non-dividend paying asset $\mathcal{P}^Y$ is:

$$\frac{\mathcal{P}_t^Y}{\mathcal{P}_t^S} = \mathbb{E}_t^{\mathbb{S}}\left[\frac{\mathcal{P}_T^Y}{\mathcal{P}_T^S}\right], \quad \text{which implies that the price of } Y \text{ is:} \tag{1.41}$$

$$\mathcal{P}_t^Y = \mathcal{P}_t^S\, \mathbb{E}_t^{\mathbb{S}}\left[\frac{\mathcal{P}_T^Y}{\mathcal{P}_T^S}\right]. \quad \text{Fundamental pricing formula} \tag{1.42}$$

- If the risk-free bank deposit is the numeraire, then we have:

$$\frac{\mathcal{P}_t^Y}{\mathcal{P}_t^D} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{\mathcal{P}_T^Y}{\mathcal{P}_T^D}\right]$$

$$\mathcal{P}_t^Y = \mathcal{P}_t^D\,\mathbb{E}_t^{\mathbb{Q}}\left[\frac{\mathcal{P}_T^Y}{\mathcal{P}_T^D}\right]$$

$$= e^{r\,t}\,\mathbb{E}_t^{\mathbb{Q}}\left[\frac{\mathcal{P}_T^Y}{e^{r\,T}}\right] \qquad \ldots \text{ assuming continuous compounding: } \mathcal{P}_0^D = 1;\ \mathcal{P}_t^D = e^{r\,t};\ \mathcal{P}_T^D = e^{r\,T}$$

$$\boxed{\mathcal{P}_t^Y = e^{-r\,(T-t)}\,\mathbb{E}_t^{\mathbb{Q}}\left[\mathcal{P}_T^Y\right], \qquad \ldots \text{ risk-neutral pricing}} \tag{1.43}$$

which says that the price today, $\mathcal{P}_t^Y$, is the expected price $\mathcal{P}_T^Y$,

- ⋆ with the expectation computed under the risk-neutral probability measure,
- ⋆ and then discounted at the risk-free rate for the remaining time $(T-t)$.

1.12 What Happens When Markets are Incomplete?

- So far, we assumed that markets were complete:
  - ⋆ that is, there existed enough primitive assets so that we could replicate exactly the payoff of the claim we wished to value.
  - ⋆ For example, in the binomial model, the stock and the bank deposit allowed us to replicate exactly the payoff to the call option in the up and down states.
- Consider now a "trinomial" model in which the stock price takes three possible values: $\mathcal{P}_u^S > \mathcal{P}_m^S > \mathcal{P}_d^S$.
- We continue to make the no-arbitrage assumption in (1.5).

- State prices must satisfy the following equations:

$$\mathcal{P}_0^S = \mathcal{P}_0^{A_u}\mathcal{P}_u^S + \mathcal{P}_0^{A_m}\mathcal{P}_m^S + \mathcal{P}_0^{A_d}\mathcal{P}_d^S, \tag{1.44}$$

$$1 = \mathcal{P}_0^{A_u}\mathcal{P}_1^D + \mathcal{P}_0^{A_m}\mathcal{P}_1^D + \mathcal{P}_0^{A_d}\mathcal{P}_1^D. \tag{1.45}$$

- In the binomial model, these equations could be solved for $\mathcal{P}_0^{A_u}$ and $\mathcal{P}_0^{A_d}$.

- However, in the trinomial model, we have only two equations but three state prices that are unknown; thus, **there exist many solutions**.

- Given any particular solution $\mathcal{P}_0^{A_u}$, $\mathcal{P}_0^{A_m}$, and $\mathcal{P}_0^{A_d}$ of (1.44) and (1.45), we can define the risk-neutral probabilities as before—for example, $\text{prob}_u^{\mathbb{Q}} = R_f\,\mathcal{P}_0^{A_u}$.

- Similarly, we can value calls and puts.

- But, the option values we get depend on the particular solution we choose for $\mathcal{P}_0^{A_u}$, $\mathcal{P}_0^{A_m}$, and $\mathcal{P}_0^{A_d}$.

- The reason that there are many arbitrage-free values for a call option is that a call cannot be replicated exactly in a trinomial model using only the stock and the risk-free asset.

- To see this, consider a portfolio of a units of the bank deposit and b shares of the stock. The value of this portfolio at $t = 1$ will be

$$a\,\mathcal{P}_1^D + b\,\mathcal{P}_i^S, \quad \text{where} \quad i = \{u, m, d\}.$$

- To replicate the call, we need a and b to satisfy the following three equations:

$$a\,\mathcal{P}_1^D + b\,\mathcal{P}_u^S = \max\left(0, \mathcal{P}_u^S - K\right), \tag{1.46}$$

$$a\,\mathcal{P}_1^D + b\,\mathcal{P}_m^S = \max\left(0, \mathcal{P}_m^S - K\right), \tag{1.47}$$

$$a\,\mathcal{P}_1^D + b\,\mathcal{P}_d^S = \max\left(0, \mathcal{P}_d^S - K\right). \tag{1.48}$$

- There are three equations but only two unknowns a and b, and so for any strike price K between S_d and S_u, the system has no solution.
- When there are state-contingent claims that cannot be replicated by trading in the fundamental assets (the stock and bond), we call the market incomplete.
- To value derivative securities in this situation, we have to select some particular solution $\mathcal{P}_0^{Au}$, $\mathcal{P}_0^{Am}$, and $\mathcal{P}_0^{Ad}$ of (1.44) and (1.45) and assume that the market uses that solution for valuation.
- Alternatively, we need to assume that the market uses a particular set of risk-neutral probabilities for valuation.
- This type of valuation is called equilibrium valuation, as opposed to no-arbitrage valuation, because to give a foundation to our particular choice of risk-neutral probabilities we have to assume something about the preferences and endowments of investors and the production possibilities.
- We will consider this kind of setting later in the course.

1.13 What are Other Limitations of the Binomial Model?

- The binomial model we described relies on a number of assumptions:
  1. Only two states at each future date; that is,
     the stock can take on only two possible values in the future.
  2. Stock does not pay any dividends.
  3. Only two trading dates.
  4. Existence of a risk-free bank deposit or interest rate.

     .....
  5. No frictions, such as transactions costs and taxes.
  6. No constraints on borrowing or short selling.
  7. No arbitrage opportunities.

- As we will see in the next few classes, many of these restrictions can be relaxed.
  1. We will extend the model to allow for an infinite number of states.
  2. It will be straightforward to allow for "dividends."
  3. We will extend the model to an infinite number of trading dates.
  4. We will not require the existence of a constant risk-free interest rate.

     .....
  5. It is more difficult to allow for frictions, such as transaction costs and taxes (Bensaid, Lesne, Pages, and Scheinkman 1992; Edirisinghe, Naik, and Uppal 1993; Jouini and Kallal 1995).
  6. It is more difficult to allow for constraints on borrowing or short selling (Naik and Uppal 1994; Broadie, Cvitanić, and Soner 1998).
  7. It would not make sense to have a model with infinite arbitrage profits; but it is possible to extend the model to allow for small arbitrage profits (Cochrane and Saa-Requejo 2000).

1.14 Summary

1. In complete markets, redundant securities can be valued in a straightforward manner, using a variety of equivalent methods:
   - 1.1. Find a portfolio that *replicates* the payoff to the option; then, the value of the option must equal the value of the replicating portfolio.
   - 1.2. Find a portfolio that *hedges* the payoff to the option; then, use the value of the hedged payoff to infer the value of the option.
   - 1.3. Find the *risk-adjusted probabilities*, $\text{prob}^{\mathbb{Q}}$; then, use the probabilities to find risk-adjusted expected payoff to the option and discount the expected payoff at the risk-free rate.
   - 1.4. Find *Arrow-Debreu state price* for receiving 1 dollar in up (or down) state, $\mathcal{P}_0^{A_k} = \frac{\text{prob}_k^{\mathbb{Q}}}{R_f}$; then, use the state prices to find the value of the option.
   - 1.5. Find *martingale* value of option in terms of bank deposit, $\mathcal{P}^D$, using probabilities $\text{prob}_k^{\mathbb{Q}}$; then, use this to find value of option in terms of dollars. Note $\text{prob}_k^{\mathbb{Q}} = \mathcal{P}_0^{A_k} \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}$.
   - 1.6. Find *martingale* value of option in terms of the stock price S, using the probabilities $\text{prob}_k^{\mathbb{S}}$; then, use this to find the option value in terms of dollars. Note that $\text{prob}_k^{\mathbb{S}} = \mathcal{P}_0^{A_k} \frac{\mathcal{P}_k^S}{\mathcal{P}_0^S}$.

2. The stochastic discount factor m_T turns asset prices into martingales:

$$\mathcal{P}_0 = \mathbb{E}^{\mathbb{P}}\left[m_T \times \mathcal{P}_T\right], \quad \text{where} \quad m_k = \frac{\mathcal{P}_0^{A_k}}{\text{prob}_k^{\mathbb{P}}}.$$

3. One can recover the risk-neutral probabilities (RNP) from the SDF as follows:

$$\text{prob}_u^{\mathbb{Q}} = \text{prob}_u^{\mathbb{P}}\, m_u \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}; \quad \text{and} \quad \text{prob}_d^{\mathbb{Q}} = \text{prob}_d^{\mathbb{P}}\, m_d \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}.$$

4. The results for the two-state binomial model extend in a straightforward manner to a model with a continuum of states.

5. The key insight of the material we have covered is that all the valuation methods discussed above rely on the same principle:

- use the information contained in prices of fundamental assets, such as bonds, stocks, foreign currency;
- to find the value of derivative (redundant) contracts, such as forwards, futures, options.

6. The second key insight of the material we have covered is that exactly the same method can be used to study all the problems that we have considered:

- Valuing redundant assets
- Hedging risky payoffs
- Choosing a portfolio to "replicate" a given payoff.

7. In terms of the (very) big picture, we have demonstrated above that

- if you know state prices,
- then it is straightforward to solve the problems of valuation, hedging (risk management), and portfolio choice in discrete time.

8. In the first part of the course we will see how, given the SDF, it is straightforward to solve the problems of valuation, hedging (risk management), and portfolio choice in continuous time.

9. In the second part of the course, we will see how to determine the SDF in continuous time.

10. Finally, the material in this chapter illustrates the power of the simple binomial model. When doing research,

- it is often best to start with this kind of a simple model, and
- often, a binomial model with only three dates, $t = \{0, 1, 2\}$, is sufficient to get the main insights that one would get from a more general model that is much more difficult to solve.

1.15 Suggestions for Further Readings

1.15.1 Source material for the lecture notes

- The binomial model for option pricing, and the concept of risk-neutral probabilities, is from Cox and Ross (1976) and Cox, Ross, and Rubinstein (1979).

- Van der Hoek and Elliott (2006) is an entire book devoted to the pricing options in the binomial model. See also the classic books Cox and Rubinstein (1985) and Jarrow and Rudd (1983).

- A book that covers a large amount of material on the valuation of various derivatives is Sundaram and Das (2015).

- The pricing of options in terms of different numeraires in the binomial model is discussed in Back (2005), which is his book on the valuation of derivatives, different from the book for our course.
  - ⋆ The numeraire approach was originally developed in Geman, El Karoui, and Rochet (1995).
  - ⋆ An entire book on asset pricing that relies on the numeraire approach is Vecer (2011).

- The CAPM-approach to the pricing of options in discrete time is described in the book McDonald (2013). The continuous-time derivation appears in Black and Scholes (1973).
  - ⋆ Scholes (1998) describes the discussion that took place around the decision on whether or not to use the CAPM when developing the Black-Scholes-Merton model.

1.15.2 Potential applications and extensions of lecture notes

- The relation of the binomial model to continuous-time models (including those with jumps) is explained in Cox and Ross (1976).

- The slim volume by Dixit (1993) has a very nice exposition connecting results in a discrete-time binomial setting to continuous time.

- Nelson and Ramaswamy (1990) extend the IID binomial model to allow for mean-reversion.

- He (1990) extends the single-risky asset binomial model to allow for *multiple* risky assets, which is useful if one is working on a problem with many risky assets.

- Application of the binomial model to other assets:
  - ⋆ Valuation of options contingent on interest rates (Ho and Lee 1986).
  - ⋆ Valuation of exotic options (Rubinstein 1999).
  - ⋆ Valuation of real options (Dixit and Pindyck 1993).

- There are two books that are set entirely in discrete time and do an excellent job of covering many of the topics that we will be covering, beyond the valuation of derivatives:
  - ⋆ Shreve (2005), which is the counterpart to the book covering the same material in continuous time, Shreve (2004).
  - ⋆ Pliska (1997); note, however, that this book uses non-standard notation so you need to exert some extra effort when reading it (but the payoff justifies the extra effort).

1.15.3 Ph.D. elective courses offered on related topics

- Rama Cont taught a course on *Monte Carlo Methods* in Finance in 2010 and 2013; Marcel Rindisbacher taught this course in 2017.
- Tim Bollerslev taught a course on *Volatility Modelling* in 2010, 2012, 2016.
- Jerome Detemple taught a course on *Derivative Securities* in 2011.
- Mikhail Chernov taught a course on *Empirical Option Pricing* in 2012 and 2014.
- Riccardo Rebonato taught a course on *Term Structure Modelling in the P and Q Measures* in 2017.

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Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 2

Maths for Continuous Time Finance

Raman Uppal

EDHEC Business School

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2.1 Plan for the Day

Last Time

- Asset pricing in a discrete time (two date) model.

Today

- **Develop mathematical model for stock prices and derivatives on stocks**
  - ⋆ Brownian motion and Itô process – to model stock prices
  - ⋆ Itô's formula – how value of option changes with change in stock price

For Next Time

- Review the material covered in class.
- Read Back, Chapter 12 and Dumas & Luciano, Chapters 5 and 7.
- Do the practice problems.

2.2 Main Points for Today

1. Brownian motion
   - Quadratic variation
2. Local martingales and Martingale Representation Theorem
3. Itô processes and Itô's formula for one and many variables
4. Geometric Brownian Motion (a special case of an Itô process)
5. Transformations of models (from many Brownian motions to one)

- This chapter: develop the math for continuous time finance;
- Next chapter: apply this math to ideas of the last class (SDF, RNP, etc.);
- Rest of course: apply the main ideas (SDF, RNP) to solve finance problems.

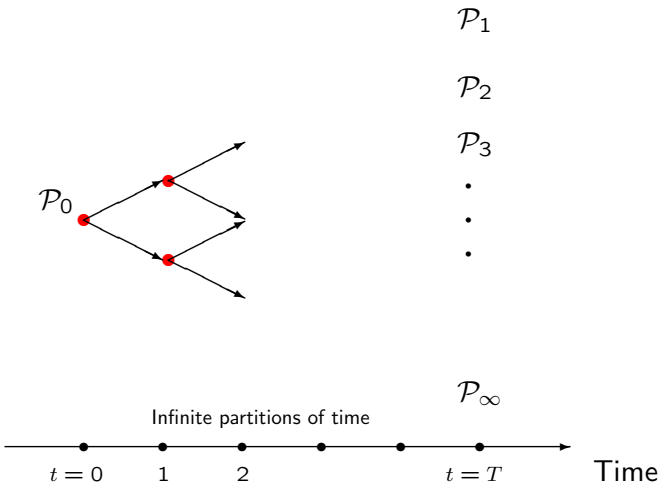
2.3 What We Wish to Accomplish in this Chapter

- Recall from the binomial model that to find the value of an option:
  1. We modeled stock prices as a Binomial Process
  2. We computed the delta of the option; that is, we computed how the value of the option changes as the stock price changes.

- In this chapter, we wish to develop the exact counterparts to these two steps, but in a setting where time is continuous:
  1. We want to model stock prices in continuous time
     - ⋆ The stock price needs to have a stochastic (random) component, which we will model using a Brownian Motion.
     - ⋆ The stock price also needs to have a deterministic component, which will be a standard process.
     - ⋆ Together, the deterministic component and the Brownian Motion, are called an Itô Process.
  2. To compute the delta of the option, that is, how the value of the option changes as the stock price changes, we will need to develop calculus that applies to functions of stochastic processes; this is called Itô's formula.

2.4 Motivation: Why Use Continuous Time

Wish to model a world where information arrival is continuous.



Advantages of using continuous-time models

1. The possibility of dynamic trading implies that we need only a few securities to complete the market; this is because information arrival is now continuous.

2. The "instantaneous" or local analysis makes models appear to be "mean-variance" models, which then facilitates getting closed-form solutions.

3. There are a large number of mathematical results available for continuous-time stochastic processes that you will be able to use.

4. For many markets, data is now collected at a high-frequency level, so continuous time models may be more appropriate.

Disadvantages of using continuous-time models

1. The mathematics is a bit more difficult (initially).

2. If data is discrete, you will need to reconcile discretely-sampled data with a model set in continuous time.

2.5 Preliminaries: Some Basic Results from Statistics

- Recall from statistics that the variance of a random variable x is:

$$\text{Variance}[x] = \mathbb{E}\Big[x^2\Big] - \Big(\mathbb{E}[x]\Big)^2 \quad \ldots \text{ [mean of the square] } - \text{ [square of the mean]}$$

$$\text{Volatility}[x] = \sqrt{\text{Variance}[x]} \quad \ldots \text{ by definition, volatility is the square root of variance}$$

- The covariance of two random variables x and y, is:

$$\text{Covariance}[x, y] = \mathbb{E}\Big[x\, y\Big] - \Big(\mathbb{E}[x]\Big)\Big(\mathbb{E}[y]\Big) \quad \ldots \text{ [mean of the product] } - \text{ [product of the means]}$$

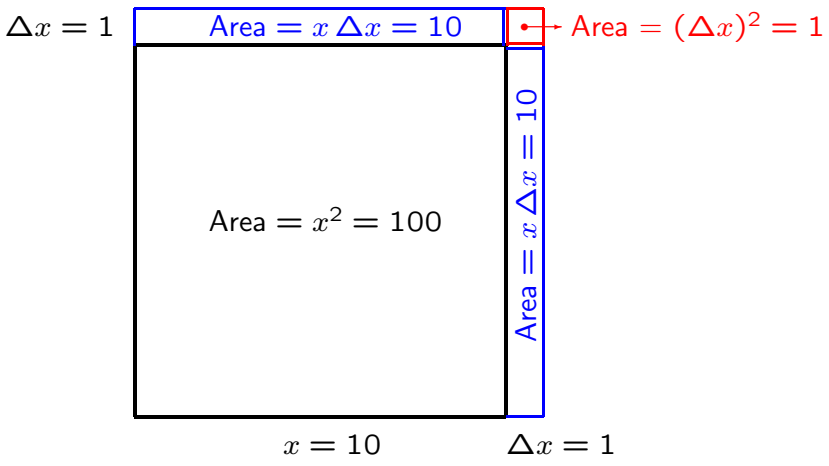
$$\text{Correlation}[x, y] = \frac{\text{Covariance}[x, y]}{\text{Volatility}[x]\,\text{Volatility}[y]}. \quad \ldots \text{ by definition of correlation}$$

2.6 An Example to Motivate Stochastic Calculus

- Compute the area of a square with side: $x = 10$.
  - ⋆ Area $= x^2 = 100$.
- Next, find the change in area if we increase the side from 10 to 11.
  - ⋆ Change in Area $= 121 - 100 = 21$.
- Now, use calculus to find the change in the area of a square:
  - ⋆ Change in Area $= \frac{d\,\text{Area}}{dx} = \frac{d\,(x^2)}{dx} = 2x = 20$.
- Observe that the two answers are not the same: $21 \neq 20$
  - ⋆ Why?
  - ⋆ How can we get the two answers to be the same?

- Because we are looking at a change that is not small, to get the correct change in area using calculus, we need to include the second order term:

$$\begin{aligned}\text{Change in Area} &= \frac{d\,(x^2)}{dx} + \frac{1}{2}\frac{d^2(x^2)}{dx^2} \\ &= \quad 2x \quad + \frac{1}{2}\,2 = 21.\end{aligned}$$



2.7 Brownian Motion

Definition 2.1 (Brownian Motion)
A Brownian motion, $B(t)$, is a

- **stochastic process** (a variable that changes randomly over time)
- that evolves **continuously** in time (so, no jumps),
- but is **non-differentiable** almost everywhere, and
- conditional on the information up to the current date t, its change, $B(u) - B(t)$, over any time period, $u - t > 0$, is
  1. Normally distributed, with
  2. **mean** zero (so, it is a martingale) and
  3. **variance** equal to the length of the time period, $u - t$, so, volatility is equal to $\sqrt{u-t}$.

2.7.1 Properties of a Brownian motion

- $B(t)$ is the level of the Brownian motion at time t.
- $B(0) = 0$ (where $B(0)$ is the level of the Brownian motion at $t = 0$); Just a convention that says that the Brownian motion starts at zero.
- $B(u) - B(t)$ for $u > t$, is the change in the Brownian motion over the time interval $u - t$.
- In the limit, as $(u - t) = dt \to 0$, the change in the Brownian motion $B(u) - B(t) = dB$.

- From the properties of the Brownian motion on the previous page:
  1. dB is Normally distributed, where $dB = \Big[B(u) - B(t)\Big]$ with $(u - t) \to 0$
  2. The (conditional) mean of dB is zero:

     $$\mathbb{E}_t[dB] = 0.$$

  3. The (conditional) variance of dB is equal to the time elapsed, dt:

     $$\text{Var}_t(dB) = \mathbb{E}_t\Big[(dB)^2\Big] - \Big(\mathbb{E}_t[dB]\Big)^2 = \mathbb{E}_t\Big[(dB)^2\Big] = dt;$$

     or equivalently, its volatility (standard deviation) is:

     $$\text{Volatility}_t(dB) = \sqrt{dt}.$$

- *All our results will be a consequence of these three properties.*

2.7.2 Quadratic Variation of a Brownian Motion

- A distinctive property of a Brownian motion is that it jiggles rapidly, moving up and down in an erratic (random) way.
- This is considered to be a reasonable model for the behavior of a stock price.
- The plot of a Brownian motion is much less smooth than that of standard functions; this is captured by the concept of **quadratic variation**.
- Consider a discrete partition of time:

  $$0 = t_0 < t_1 < t_2 < \cdots < t_N = T$$

  of the time interval $[0, T]$.

- Calculate the sum of the **squared changes** of the Brownian motion over these time intervals:

  $$\sum_{i=1}^{N} \left[\Delta B(t_i)\right]^2$$

  where $\Delta B(t_i)$ denotes the change $B(t_i) - B(t_{i-1})$.
- The limit of this sum, as we get the partitions get finer and finer (that is, as we let the interval $t_i - t_{i-1}$ go to zero), is called the **quadratic variation**.
- For a Brownian motion, the quadratic variation over the interval $[0, T]$ is T.
- In contrast, continuously differentiable functions of time have a quadratic variation of 0, as we show on the next slide.

Example 2.1 (Quadratic variation of a continuously differentiable function)

- Consider a linear function $X(t) = a\,t$, for some constant a,
  and let $t_i - t_{i-1} = \Delta t = T/N$ for each i (that is, all the time intervals are of equal length).

- Then
  $$\Delta X(t_i) = \Delta(a\,t_i) = a\Delta t_i = a(t_i - t_{i-1}) = a\frac{T}{N},$$
  so,
  $$\Big[\Delta X(t_i)\Big]^2 = \left(a\frac{T}{N}\right)^2,$$
  and so,
  $$\sum_{i=1}^{N}[\Delta X(t_i)]^2 = \sum_{i=1}^{N}\left(a\frac{T}{N}\right)^2 = N\left(a\frac{T}{N}\right)^2 = a^2\frac{T^2}{N},$$
  which in the limit, as N gets large, is equal to 0:
  $$\lim_{N\to\infty} a^2\frac{T^2}{N} = 0.$$

- Essentially, the same argument shows that the quadratic variation of any continuously differentiable function is zero, because such a function is approximately linear at each point.

Total variation

- A concept related to quadratic variation is "total variation", where total variation is defined as the sum of the absolute changes:

$$\lim_{N\to\infty}\sum_{i=1}^{N}\Big|\Delta B(t_i)\Big| = \infty.$$

- For processes with non-zero quadratic variation, **total variation is infinite**. Thus, Brownian motions have a total variation that is infinite.

- In contrast for functions that are continuously differentiable, and have a **quadratic variation of zero**, the **total variation is finite**.

2.7.3 Why Are We Interested in Brownian Motions

- We are interested in continuous martingales.
  - ⋆ Asset prices involve martingales (random variables whose expected change is zero).
  - ⋆ Continuous processes (variables whose paths are continuous functions of time) are mathematically tractable.
- A Brownian motion is an example of a continuous martingale; that is, a martingale with continuous paths.
- Every continuous martingale is a transformation of a Brownian motion:

  Levy's Theorem: A continuous martingale is a Brownian motion if and only if its quadratic variation over each time interval $[0, T]$ is T.
- And, continuous martingales can be constructed as "stochastic integrals" with respect to Brownian motions, as shown in the next section.

2.8 Itô Integral and Local Martingale

- The basis of continuous time finance is Itô's concept of a stochastic integral.
- A stochastic process $\theta(t, \omega)$ is said to be measurable if each sample path (that is, the trajectory for each ω) is a measurable function.
  - ⋆ In plain English, "measurable function" means a function whose value is known.
- A stochastic process θ is said to be adapted to B if θ_t depends only on the history of B through t.
  - ⋆ An adapted process is a process that does *not* look into the future.
- A stochastic process is said to be jointly measurable in (t, ω) if the integral itself is a random variable (that is, if the integral is measurable in ω).

- If θ is a stochastic process adapted to the information with respect to which B is a Brownian motion, is jointly measurable in (t, ω), and satisfies $\int_0^T \theta^2(t)dt < \infty$, with probability 1, and if M_0 is a constant, then one can define the stochastic process

$$M_t = M_0 + \underbrace{\int_0^t \theta_s \, dB_s}_{\text{stochastic integral}} , \qquad \text{for } t \in [0, T]. \tag{2.1}$$

- For each t, the stochastic integral can be approximated as

$$\int_0^t \theta_s \, dB_s \approx \sum_{i=1}^{N} \theta_{t_{i-1}} \left(B_{t_i} - B_{t_{i-1}} \right), \quad \ldots \text{important to note that it is } \theta(t_{i-1}), \text{ not } \theta(t_i)$$

where $0 = t_0 < t_1 \cdots < t_N = T$ and $t_i - t_{i-1}$ going to zero as $N \to \infty$.

2.8.1 Local Martingale

- The stochastic process in Equation (2.1)

$$M_t = M_0 + \underbrace{\int_0^t \theta_s \, dB_s}_{\text{stochastic integral}} , \qquad \text{for } t \in [0, T]. \tag{2.1}$$

is a local martingale, which is a class of stochastic processes that includes martingales.

- That is, all martingales are local martingales (but all local martingales may not be martingales). We will discuss this in greater detail below.

- The stochastic process M_t in (2.1) has continuous paths and quadratic variation equal to:

$$\int_0^t \theta_s^2 \, ds.$$

2.8.2 Differential Form of Stochastic Integral

- The stochastic process in Equation (2.1)

$$M_t = M_0 + \int_0^t \theta_s \, dB_s, \qquad \text{for } t \in [0, T], \tag{2.1}$$

  can also be written in differential form

$$dM_t = \theta_t \, dB_t. \tag{2.2}$$

- Heuristically, one can go from the differential form in (2.2) plus the initial condition M_0, to the integral form in (2.1) by "summing" the changes dM:

$$M_t = M_0 + \int_0^t dM_s = M_0 + \int_0^t \theta_s \, dB_s.$$

2.8.3 Local Martingales and Doubling Strategies

- A local martingale M_t is a stochastic processes whose each independent increment, dM has a mean of zero; that is, locally (at each instant) M is a martingale.

- However, because there are an infinite number of instants in any time interval, being a local martingale at each instant does not necessarily imply that M is a martingale over the entire time interval.

- For an example of a process that is a local martingale but not a martingale, consider the following wealth process in discrete time.
  - ⋆ Consider an infinite sequence of dates $0 = t_0 < t_1 < t_2 < \cdots < T$ within some interval $[0, T]$ for a finite T.
  - ⋆ Suppose a casino allows you to gamble on the toss of a fair coin at each date t_n, with you winning if the coin turns up heads.
  - ⋆ Suppose initial stake is $\$1$, and the stake doubles at each successive round.

- Suppose the game ends the first time you get heads. Then, the claim is that you will always win $\$1$ in this game.
  - ⋆ If you get heads on the first coin flip, then you make $\$1$.
  - ⋆ If you get heads on the second coin flip, then you lose $\$1$ on the first flip, but win $\$2$ on the second coin flip, so overall you make $\$1$.
  - ⋆ If you get heads on the third coin flip, then you lose $\$1$ on the first flip, you lose $\$2$ on the second flip, but you win $\$4$ on the third flip, so overall you make $\$1$.
- Each individual coin toss is a fair game; that is, the expected gain is 0, or equivalently, your wealth process is a local martingale.
- But, the overall game is *not* fair, because you always win $\$1$, and so your wealth on date T is: $\mathbb{E}[W_T] = W_0 + 1 \neq W_0$.

  Thus, your wealth process is not a martingale because the expected value of the change in your wealth, $\mathbb{E}\Big[W_T - W_0\Big] \neq 0$.

- To ensure that the processes we study in Finance are martingales, we need to rule out these kind of "doubling strategies."
  - ⋆ One way to rule out these strategies is to impose a **lower bound on wealth**, which then rules out an infinite credit line (which is what you would need at the casino for the doubling strategy).
    - ∗ If W is a local martingale and bounded below, then it can be shown that $\mathbb{E}_t[W_u] \leq W_t$ for $u \geq t$.
  - ⋆ Another alternative is to impose the condition that θ (interpreted as the **amount invested in the gamble**) is not too large; that is, it is bounded:

$$\mathbb{E}\left[\int_t^u \theta_s^2 \, ds\right] < \infty. \tag{2.3}$$

2.8.4 Martingale Representation Theorem

- Consider a local martingale M that depends only on a Brownian motion B.
- That is, for each t, M_t depends only on the history of B through t.
- This is expressed by saying that M is adapted to B.
- Any such local martingale has the property that we can always find a θ_s so that

$$M_t = M_0 + \int_0^t \theta_s \, dB_s.$$

- Thus, all local martingales adapted to a Brownian motion are stochastic integrals with respect to the Brownian motion.

- This is a spanning property: a Brownian motion spans all local martingales adapted to it in the sense that

$$dM = \theta \, dB.$$

- This is called the Martingale Representation Theorem.
- The theorem extends to vectors of Brownian motions:
  If M is a local martingale adapted to a vector $(B_1, B_2, \ldots, B_N)$ of Brownian motions, then there exist stochastic processes θ_n such that

$$dM = \sum_{n=1}^{N} \theta_n \, dB_n.$$

- The spanning property is analogous to the spanning property of a stock in a binomial model. Market completeness properties follow from the martingale representation theory, and it is this property that underlies the Black-Scholes option pricing formula and much of continuous time finance.

2.9 Itô Process

Definition 2.2 (Itô Process)
The **sum** of an ordinary integral and a stochastic integral is an Itô process.

$$X(T) = X(0) + \underbrace{\int_0^T \mu(t)dt}_{\text{ordinary integral}} + \underbrace{\int_0^T \sigma(t)\, dB(t)}_{\text{stochastic integral}}, \text{ for any } T > 0, \tag{2.4}$$

where $B(t)$ is a Brownian motion, and μ and σ can also be random processes that satisfy some regularity conditions and $\mu(t)$ and $\sigma(t)$ are known at t.

- Recall that the **stochastic integral** is to be interpreted as

$$\int_0^T \sigma(t)dB(t) \approx \sum_{i=1}^{N} \sigma(t_{i-1})\Delta B(t_i)$$

  where $0 = t_0 < t_1 \cdots < t_N = T$ and the time intervals $t_i - t_{i-1}$ are small.
- The **expected value** of the stochastic integral is zero, so it is a martingale.
  The expected value of the ordinary integral, and hence the Itô process, is not zero.

Itô Process in Differential Form

- **In differential form**, an Itô process X changes over time as

$$dX(t) = \mu(t)dt + \sigma(t)dB(t) \tag{2.5}$$

- We can go from the differential form to the integral form by summing the changes over time:

$$X(T) = X(0) + \int_0^T \mu(t)dt + \underbrace{\int_0^T \sigma(t)\, dB(t)}_{\text{stochastic integral}}, \text{ for any } T > 0.$$

2.9.1 Properties of an Itô Process

- An Itô process

$$dX(t) = \mu(t)dt + \sigma(t)dB(t)$$

  - ⋆ evolves continuously over time
  - ⋆ $\mu(t)$ is called the drift of X at t;
  - ⋆ $\sigma(t)$ is called the diffusion coefficient of X at t.

- The (conditional) expected change in X is: $\mathbb{E}_t[dX(t)] = \mu(t)\,dt$.

- Clearly, the Itô process $dX(t)$ is a local martingale only if $\mu = 0$.

- This fact, along with Itô's formula that we discuss below, leads to the partial differential equations (PDEs) that appear in Finance—for example, the fundamental PDE for valuing derivative securities and the Hamilton-Bellman-Jacobi equations for optimal portfolio choice.

2.9.2 Rules for Working with Itô Process

- Throughout the course, we will use the following rules when computing variances and covariances of Itô processes that are driven by Brownian motions dB_x and dB_y.

$$(dt)^2 = 0, \quad \ldots \text{ because } dt \text{ is "small," so } (dt)^2 \to 0 \tag{2.6}$$

$$(dt)\,(dB_x) = 0, \quad \ldots dB_x \text{ is of order } \sqrt{dt}\text{, so } (dt)\,(dB_x) = (dt)\,(\sqrt{dt}) = (dt)^{3/2} \to 0 \tag{2.7}$$

$$(dB_x)^2 = dt, \quad \ldots (dB_x)^2 \text{ is of order dt from property of Brownians} \tag{2.8}$$

$$(dB_x)(dB_y) = \rho(t)\,dt, \quad \ldots \text{where } \rho(t) \text{ is the correlation between the two Brownians.} \tag{2.9}$$

2.9.3 Mean of an Itô Process $dX(t)$

- For an Itô process X

$$dX(t) = \mu(t)dt + \sigma(t)dB(t)$$

- The mean of the change in X is

$$\begin{aligned}\mathbb{E}_t\Big[dX(t)\Big] &= \mathbb{E}_t\Big[\mu(t)dt + \sigma(t)dB(t)\Big] \\ &= \mathbb{E}_t\Big[\mu(t)dt\Big] + \mathbb{E}_t\Big[\sigma(t)dB(t)\Big] \\ &= \mu(t)dt \qquad \ldots \text{because } \mathbb{E}_t[\sigma(t)dB(t)] = \sigma(t)\mathbb{E}_t[dB(t)] = 0.\end{aligned} \tag{2.10}$$

- The mean change in X per unit time is

$$\frac{1}{dt}\mathbb{E}_t\Big[dX(t)\Big] = \mu(t).$$

- The term $\mu(t)dt$ is called the *deterministic* component of dX.

2.9.4 Variance of $dX(t) = \mu(t)dt + \sigma(t)dB(t)$

$$\begin{aligned}\text{Variance}_t[dX] &= \mathbb{E}_t\Big[(dX)^2\Big] - \big(\mathbb{E}_t[dX]\big)^2 && \ldots \text{Variance[Y]} = \mathbb{E}\Big[Y^2\Big] - \big(\mathbb{E}[Y]\big)^2 \\ &= \mathbb{E}_t\Big[(dX)^2\Big] - \Big(\mathbb{E}_t\Big[\mu\,dt + \sigma\,dB\Big]\Big)^2 && \ldots \text{substituting definition of } dX(t) \\ &= \mathbb{E}_t\Big[(dX)^2\Big] - \Big(\mu\,dt\Big)^2 && \ldots \text{because } \mathbb{E}_t[dB] = 0 \\ &= \mathbb{E}_t\Big[(dX)^2\Big] && \ldots \text{because } dt \text{ is "small," so } (dt)^2 \to 0 \\ &= \mathbb{E}_t\Big[(\mu\,dt + \sigma\,dB)^2\Big] && \ldots \text{substituting definition of } dX(t) \\ &= \mathbb{E}_t\Big[(\mu^2(dt)^2 + 2\mu\,\sigma(dt)(dB) + \sigma^2(dB)^2)\Big] && \ldots (a+b)^2 = a^2 + 2ab + b^2 \\ &= \mathbb{E}_t\Big[(\mu^2(dt)^2 + 2\mu\sigma(dt)(\sqrt{dt}) + \sigma^2 dt)\Big] && \ldots \text{because } dB \text{ is of order } \sqrt{(dt)};\ (dB)^2 = dt \\ &= \mathbb{E}_t\Big[(\mu^2(dt)^2 + 2\mu\sigma(dt)^{3/2} + \sigma^2 dt)\Big] && \ldots \text{because } (dt)(\sqrt{dt}) = (dt)^{3/2} \\ &= \mathbb{E}_t\Big[\sigma^2 dt\Big] && \ldots dt \text{ is "small," so } (dt)^2 \to 0,\ (dt)^{3/2} \to 0 \\ &= \sigma^2 dt.\end{aligned}$$

2.9.5 Quadratic Variation and Variance of an Itô Process

- For an Itô process $dX(t) = \mu(t)dt + \sigma(t)dB(t)$ its quadratic variation is

$$\lim_{N\to\infty} \sum_{i=1}^{N} [\Delta X(t_i)]^2 = \int_0^T \sigma^2(t)dt,$$

- If the expected quadratic variation over $[0, T]$ is finite, $\mathbb{E}_t\left[\int_0^T \sigma^2(t)dt\right] < \infty$, then the variance of the date-T value of the Itô process X is:

$$\text{var}[X(t)] = \mathbb{E}_t\left[\int_0^T \sigma^2(t)dt\right].$$

2.10 Itô's Formula (often called "Itô's Lemma")

- Recall some basic facts of **ordinary calculus**.
  - ⋆ Suppose $y = y(x)$ and $x = x(t)$, then

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = y'\Big(x(t)\Big)\, x'(t).$$

  - ⋆ Over a time period $[0, T]$, this implies that:

$$\begin{aligned} y(T) &= y(0) + \int_0^T \frac{dy}{dt}\, dt \\ &= y(0) + \int_0^T y'\Big(x(t)\Big)\, x'(t)\, dt \\ &= y(0) + \int_0^T y'(x(t))\, dx \qquad \ldots \text{because } dx = x'(t)\, dt. \end{aligned} \tag{2.11}$$

- Now, consider a function of a Brownian motion.
  - ⋆ Suppose $B(t)$ is a Brownian motion, and $Y(t) = Y(B(t))$, where Y is a twice-continuously differentiable function.
  - ⋆ Then,

$$\begin{aligned} Y(T) &= Y(0) + \int_0^T Y'(B(t))\, dB(t) + \frac{1}{2}\int_0^T Y''(B(t))\, \Big(dB(t)\Big)^2 \\ &= Y(0) + \int_0^T Y'(B(t))\, dB(t) + \frac{1}{2}\int_0^T Y''(B(t))\, dt. \end{aligned} \tag{2.12}$$

  - ⋆ In differential form, the above would be written as:

$$dY(t) = Y'(B(t))\, dB(t) + \frac{1}{2} Y''(B(t))\, dt. \tag{2.13}$$

- Why do we have an **extra term** for differentials of functions of Brownian motions?

2.10.1 Understanding the Extra Term in Itô's Formula

- We start with ordinary calculus, and consider again the function $y = y\Big(x(t)\Big)$.
- Let $u > t$, and recall that the derivative defines a linear approximation of the change in y over this time period; that is, setting $\Delta x = x(u) - x(t)$, and $\Delta y = y(u) - y(t)$, we have the approximation

$$\Delta y \approx y'(x(t))\, \Delta x.$$

- A better approximation is given by the **second-order** Taylor expansion:

$$\Delta y \approx y'\Big(x(t)\Big)\, \Delta x + \frac{1}{2} y''\Big(x(t)\Big)\, [\Delta x]^2.$$

- The reason why the second-order term does not appear in standard calculus is that, when we consider very small time periods, the second-order term becomes very small, and in the limit, as $dt \to 0$, the term $[\Delta x]^2 \to 0$.
- *Recall our motivating example of finding the change in the area of a square.*

- However, for the function of Brownians, consider the second-order Taylor series expansion of $Y = Y\Big(B(t)\Big)$:

$$\Delta Y \approx Y'\Big(B(t)\Big)\,\Delta B + \frac{1}{2}Y''\Big(B(t)\Big)\,[\Delta B]^2.$$

- Now, as we reduce the time intervals, the extra term does not disappear because we know that for Brownian motions:

$$\lim_{N\to\infty}\sum_{i=1}^{N}\Big[\Delta B(t_i)\Big]^2 = T,$$

whereas for the continuously differentiable function, $x(t)$, this limit is zero.

- Another way of seeing this is to interpret $[\Delta B(t_i)]^2$ as $(dB)^2$, which we have agreed is equal to dt (and not zero).

2.10.2 Itô's Formula For a Single Itô Process

- Consider a function $C = g(t, X(t))$, where X is an Itô process given by:

$$dX(t) = \mu_x(t)\,dt + \sigma_x(t)\,dB_x(t).$$

- Then, Itô's formula tells us that:

$$\begin{aligned}
C(T) &= C(0) + \int_0^T \frac{\partial g}{\partial t}\,dt + \int_0^T \frac{\partial g}{\partial x}\,dX(t) + \frac{1}{2}\int_0^T \frac{\partial^2 g}{\partial x^2}\,[dX(t)]^2 \qquad (2.14)\\
&= C(0) + \int_0^T \frac{\partial g}{\partial t}\,dt + \int_0^T \frac{\partial g}{\partial x}\,[\mu_x(t)\,dt + \sigma_x(t)\,dB_x(t)] + \frac{1}{2}\int_0^T \frac{\partial^2 g}{\partial x^2}\,[\sigma_x(t)\,dB_x(t)]^2\\
&= C(0) + \int_0^T \frac{\partial g}{\partial t}\,dt + \int_0^T \frac{\partial g}{\partial x}\,[\mu_x(t)\,dt + \sigma_x(t)\,dB_x(t)] + \frac{1}{2}\int_0^T \frac{\partial^2 g}{\partial x^2}\,[\sigma_x(t)]^2\,dt\\
&= C(0) + \int_0^T \left(\frac{\partial g}{\partial t} + \frac{\partial g}{\partial x}\mu_x(t) + \frac{1}{2}\frac{\partial^2 g}{\partial x^2}\,[\sigma_x(t)]^2\right)dt + \int_0^T \frac{\partial g}{\partial x}\,\sigma_x(t)\,dB_x(t)
\end{aligned}$$

- Instead of writing Itô's formula in its integral form, as on the previous page:

$$C(T) \;=\; C(0) + \int_0^T \left(\frac{\partial g}{\partial t} + \frac{\partial g}{\partial x}\mu_x(t) + \frac{1}{2}\frac{\partial^2 g}{\partial x^2}\,[\sigma_x(t)]^2 \right) dt + \int_0^T \frac{\partial g}{\partial x}\,\sigma_x(t)\,dB_x(t),$$

we can write Itô's formula in its "differential" form, which is easier to remember:

$$\begin{aligned} dC \;&=\; g_t\,dt + g_x\,dX + \frac{1}{2}g_{xx}\,(dX)^2 \\ &=\; \left(g_t + g_x\mu_x(t) + \frac{1}{2}g_{xx}\,[\sigma_x(t)]^2 \right) dt + g_x\,\sigma_x(t)\,dB_x, \end{aligned} \tag{2.15}$$

where g_t refers to $\partial g/\partial t$, g_x refers to $\partial g/\partial x$, and g_{xx} refers to $\partial^2 g/\partial x^2$.

Test Your Understanding 2.1 (Itô's formula applied to exponentials)
If $C = g(X) = e^X$, then find dC.

Space for your work.

Test Your Understanding 2.2 (Itô's formula applied to logarithms)
If $C = \log(X)$, then find dC.

Space for your work.

Test Your Understanding 2.3 (Application of Itô's Formula)
Let $dX = \mu\, dt + \sigma\, dB$. Consider a function $Y = X^{\alpha}$. Compute dY.

Space for your work.

Test Your Understanding 2.4 (Application of Itô's Formula)
Let $dX = \mu\, dt + \sigma\, dB$. Consider a function $Y = e^{\alpha X - \beta t}$. Find the process followed by $Y(t) = Y(t, X(t))$.

- Recall from regular calculus the following partial derivatives:

$$\frac{\partial Y}{\partial t} = -\beta\, Y; \qquad \frac{\partial Y}{\partial X} = \alpha\, Y; \qquad \frac{\partial^2 Y}{\partial X^2} = \alpha^2 Y.$$

Space for your work.

2.10.3 Itô's Formula for Multiple Itô Processes

- Consider two Itô processes:

$$dX(t) = \mu_x(t)\,dt + \sigma_x(t)\,dB_x(t), \tag{2.16}$$
$$dY(t) = \mu_y(t)\,dt + \sigma_y(t)\,dB_y(t), \tag{2.17}$$

  where B_x and B_y can be different Brownian motions.

- We already know that if $u > t$, then the changes $B_x(u) - B_x(t)$ and $B_y(u) - B_y(t)$ are Normally distributed with mean 0 and variance $u - t$.

- The relation between the two Brownian motions is given by their covariance or correlation. There will exist a (possibly random) process ρ such that the covariance between these two Normally distributed random variables, given information at t, is:

$$\mathbb{E}_t\left[\int_0^u \rho(s)ds\right].$$

- The process ρ is called the **correlation coefficient** of the two Brownian motions because when ρ is constant, the correlation of the changes $B_x(u) - B_x(t)$ and $B_y(u) - B_y(t)$ is:

$$\begin{aligned}
\text{correlation} &= \frac{\text{covariance}}{\sqrt{\text{variance of } B_x}\ \sqrt{\text{variance of } B_y}} \\
&= \frac{\int_t^u \rho(s)ds}{\sqrt{u-t}\ \sqrt{u-t}} \\
&= \frac{\rho(u-t)}{(u-t)} \\
&= \rho.
\end{aligned}$$

- Moreover, given increasingly fine partitions $0 = t_0 < t_1 < \cdots < t_N = T$ of an interval $[0, T]$, we will have

$$\sum_{i=1}^{N} \Delta B_x(t_i) \times \Delta B_y(t_i) = \int_0^T \rho(t)dt,$$

and

$$\sum_{i=1}^{N} \Delta X(t_i) \times \Delta Y(t_i) = \int_0^T \sigma_x(t)\,\sigma_y(t)\,\rho(t)dt. \tag{2.18}$$

where we have used, in addition to the rules in (2.6)–(2.8), the new rule:

$$(dB_x)(dB_y) = \rho(t)\,dt. \tag{2.19}$$

- Thus, the covariance between $dB_x(t)$ and $dB_y(t)$ is:

$$\begin{aligned}
&\text{Cov}[dB_x(t), dB_y(t)] \\
&= \mathbb{E}\Big[dB_x(t) \times dB_y(t)\Big] - \Big(\mathbb{E}[dB_x(t)] \times \mathbb{E}[dB_y(t)]\Big) && \text{... from definition of covariance on page 179} \\
&= \mathbb{E}\Big[dB_x(t) \times dB_y(t)\Big] && \text{... because } \mathbb{E}[dB_x(t)] = \mathbb{E}[dB_y(t)] = 0 \\
&= \rho(t)\, dt. && \text{... from (2.19) on page 222}
\end{aligned}$$

- Similarly, the correlation between $dB_x(t)$ and $dB_y(t)$ is:

$$\begin{aligned}
&\text{Correlation}[dB_x(t), dB_y(t)] \\
&= \frac{\text{Cov}[dB_x(t), dB_y(t)]}{\text{Volatility}[dB_x(t)]\,\text{Volatility}[dB_y(t)]} && \text{... from definition of correlation on page 179} \\
&= \frac{\rho(t)\, dt}{\sqrt{dt} \times \sqrt{dt}} && \text{... substituting our results for covariance and volatility} \\
&= \rho(t). && \text{... canceling } dt \text{ term in numerator and denominator}
\end{aligned}$$

- Based on the results on the previous slide, we have that if:

$$dX(t) = \mu_x(t)\, dt + \sigma_x(t)\, dB_x(t), \tag{2.16}$$
$$dY(t) = \mu_y(t)\, dt + \sigma_y(t)\, dB_y(t), \tag{2.17}$$

- Then:

$$\begin{aligned}
&dX(t) \times dY(t) \\
&= \Big[\mu_x(t)\, dt + \sigma_x(t)\, dB_x(t)\Big] \times \Big[\mu_y(t)\, dt + \sigma_y(t)\, dB_y(t)\Big] && \text{... using definitions of } dX \text{ and } dY \\
&= \sigma_x(t)\, \sigma_y(t)\, dB_x(t)\, dB_y(t) && \text{... all the other terms go to zero} \\
&= \sigma_x(t)\, \sigma_y(t)\, \rho(t)\, dt. && \text{... using the rule: } dB_x\, dB_y = \rho(t) dt
\end{aligned}$$

- Thus, the covariance between $dX(t)$ and $dY(t)$ is:

$$\begin{aligned}
&\text{Cov}[dX(t), dY(t)] \\
&= \mathbb{E}\Big[dX(t) \times dY(t)\Big] - \Big(\mathbb{E}[dX(t)] \times \mathbb{E}[dY(t)]\Big) && \text{. . . from definition of covariance on page 179} \\
&= \mathbb{E}\Big[dX(t) \times dY(t)\Big] && \text{. . . because } \mathbb{E}[dX(t)] \times \mathbb{E}[dY(t)] = \ldots dt \times \ldots dt = \ldots (dt)^2 = 0 \\
&= \sigma_x(t)\, \sigma_y(t)\, \rho(t)\, dt. && \text{. . . because the other cross terms are all zero}
\end{aligned}$$

- Similarly, the correlation between $dX(t)$ and $dY(t)$ is:

$$\begin{aligned}
&\text{Correlation}[dX(t), dY(t)] \\
&= \frac{\text{Cov}[dX(t), dY(t)]}{\text{Volatility[dX(t)] Volatility[dY(t)]}} && \text{. . . from definition of correlation on page 179} \\
&= \frac{\sigma_x(t)\, \sigma_y(t)\, \rho(t)\, dt}{\sigma_x(t) \sigma_y(t)\, dt} && \text{. . . substituting our results for covariance and volatility} \\
&= \rho(t). && \text{. . . canceling terms in numerator and denominator}
\end{aligned}$$

- The most general case of Itô's formula that we will use is for a function of time and two stochastic processes, $C(t) = g\Big(t, X(t), Y(t)\Big)$, where X and Y are Itô processes as in (2.16) and (2.17). Then, Itô's formula is:

$$\begin{aligned}
C(T) \;=\; & C(0) + \int_0^T \frac{\partial g}{\partial t}\, dt + \int_0^T \frac{\partial g}{\partial x}\, dX(t) + \int_0^T \frac{\partial g}{\partial y}\, dY(t) \\
& + \frac{1}{2}\int_0^T \frac{\partial^2 g}{\partial x^2}\, [dX(t)]^2 + \frac{1}{2}\int_0^T \frac{\partial^2 g}{\partial y^2}\, [dY(t)]^2 + \int_0^T \frac{\partial^2 g}{\partial x\, \partial y}\, [dX(t)]\, [dY(t)] \\
= \; & C(0) + \int_0^T \left(\frac{\partial g}{\partial t} + \frac{\partial g}{\partial x}\mu_x(t) + \frac{\partial g}{\partial y}\mu_y(t) \right. \\
& \left. + \frac{1}{2}\frac{\partial^2 g}{\partial x^2}\, [\sigma_x(t)]^2 + \frac{1}{2}\frac{\partial^2 g}{\partial y^2}\, [\sigma_y(t)]^2 + \frac{\partial^2 g}{\partial x\, \partial y}\, \sigma_x(t)\sigma_y(t)\rho(t) \right) dt \\
& + \int_0^T \frac{\partial g}{\partial x}\, \sigma_x(t)\, dB_x(t) + \int_0^T \frac{\partial g}{\partial y}\, \sigma_y(t)\, dB_y(t),
\end{aligned}$$

- and in its "differential" form:

$$
\begin{aligned}
dC &= g_t\,dt + g_x\,dX + g_y\,dY + \frac{1}{2}g_{xx}\,(dX)^2 + \frac{1}{2}g_{yy}\,(dY)^2 + g_{xy}\,(dX)\,(dY) \\
&= \Big(g_t + g_x\,\mu_x(t) + g_y\,\mu_y(t) \\
&\quad + \frac{1}{2}g_{xx}\,[\sigma_x(t)]^2 + \frac{1}{2}g_{yy}\,[\sigma_y(t)]^2 + g_{xy}\,\sigma_x(t)\,\sigma_y(t)\,\rho(t)\Big)\,dt \\
&\quad + g_x\,\sigma_x(t)\,dB_x \\
&\quad + g_y\,\sigma_y(t)\,dB_y.
\end{aligned}
\tag{2.20}
$$

- Note that
  - ⋆ dt multiplies (i) first derivatives and (ii) second derivatives;
  - ⋆ Brownians (dB_x and dB_y) multiply only the first derivatives.

Test Your Understanding 2.5 (Itô's formula applied to products)
If $C = g(X, Y) = X\,Y$, then find dC.

Space for your work.

Test Your Understanding 2.6 (Volatility of a product of stochastic processes)

- In the equation

$$dS = \mu\, dt + \sigma\, dB$$

where B is a Brownian motion, we say that σ is the **volatility** of S.

- Suppose

$$\begin{aligned} \frac{dX}{X} &= \mu_x\, dt + \sigma_x\, dB_x, \\ \frac{dY}{Y} &= \mu_y\, dt + \sigma_y\, dB_y, \end{aligned}$$

where B_x and B_y can be different Brownian motions with correlation ρ, and $\{\mu_x, \mu_y, \sigma_x, \sigma_y, \rho\}$ can be quite general processes.

- If $C = X\,Y$, then what is the volatility of dC/C?

Space for your work.

Test Your Understanding 2.7 (Itô's formula applied to ratios)
If $C = \frac{Y}{X}$, then find dC.

Space for your work.

Test Your Understanding 2.8 (Volatility of a ratio of stochastic processes)

- Suppose

$$\frac{dX}{X} = \mu_x\,dt + \sigma_x\,dB_x,$$
$$\frac{dY}{Y} = \mu_y\,dt + \sigma_y\,dB_y,$$

  where B_x and B_y can be different Brownian motions with correlation ρ, and $\{\mu_x, \mu_y, \sigma_x, \sigma_y, \rho\}$ can be quite general processes.

- If $C = Y/X$, then what is the volatility of dC/C?

Space for your work.

Space for your work.

Test Your Understanding 2.9 (Itô's formula for compounding/discounting)
Suppose that $Y(t) = \exp\left(\int_0^t q(s)ds\right)$ for some (possibly random) process q and define $C = X\,Y$ for any Itô process X. Then, find dC.

Space for your work.

2.11 Special Case: Geometric Brownian Motion

- If μ and σ are constants, and B is a Brownian motion, then the following process is called a geometric Brownian motion:

$$\frac{dS}{S} = \mu\, dt + \sigma\, dB \quad \ldots \quad \text{geometric Brownian motion.} \tag{2.21}$$

- The solution to the stochastic differential equation (SDE) in (2.21) is:

$$S(t) = S(0) \exp\left(\mu\, t - \frac{1}{2}\sigma^2\, t + \sigma\, B(t)\right). \tag{2.22}$$

- We show this below.

- Let us apply Itô's formula to the function $Y = \log S = \ln S$.

$$\begin{aligned}
dY &= \frac{\partial Y}{\partial S}(dS) + \frac{1}{2}\frac{\partial^2 Y}{\partial S^2}(dS)^2 && \ldots \text{applying Itô's formula to } Y \\
&= \frac{\partial \log S}{\partial S}\, dS + \frac{1}{2}\frac{\partial^2 \log S}{\partial S^2}\,(dS)^2 && \ldots \text{substituting the function } Y = \log S \\
&= \frac{dS}{S} - \frac{1}{2}\left(\frac{dS}{S}\right)^2 && \ldots \partial(\log S) = \tfrac{1}{S}; \partial^2(\log S) = \tfrac{-1}{S^2} \\
&= (\mu\, dt + \sigma\, dB) - \frac{1}{2}(\mu\, dt + \sigma\, dB)^2 && \ldots \text{substituting the given process for } \tfrac{dS}{S} \\
d\log S(t) &= (\mu - \frac{1}{2}\sigma^2)dt + \sigma dB(t) && \ldots \text{collecting the } dt \text{ and } dB \text{ terms} \\
\log S(t) - \log S(0) &= \left(\mu\, t - \frac{1}{2}\sigma^2\, t + \sigma\, B(t)\right) && \ldots \text{integrating both sides} \\
\log S(t) &= \log S(0) + \left(\mu\, t - \frac{1}{2}\sigma^2\, t + \sigma\, B(t)\right) && \ldots \text{adding } \log S(0) \text{ to both sides} \\
S(t) &= S(0) \exp\left(\mu\, t - \frac{1}{2}\sigma^2\, t + \sigma\, B(t)\right) && \ldots \text{taking the exponential of both sides}
\end{aligned}$$

Test Your Understanding 2.10 (Itô's formula with geometric Brownian motion)
Find dC if $C = g(S,t)$ and $\frac{dS}{S} = \mu\,dt + \sigma\,dB$.

Space for your work.

Test Your Understanding 2.11 (Itô's formula for one geometric Brownian motion)

- Suppose that a stock price S, follows a geometric Brownian motion:

$$dS = \mu\, S\, dt + \sigma\, S\, dB, \quad \text{where } \mu \text{ and } \sigma \text{ are constants,}$$

  and where B is a Brownian motion under the actual probability measure.

- Consider a function $Y = f(S)$.

- Compute dY for the following functions, and explain if the process for Y is a geometric Brownian motion:

  1. $Y = S^{\alpha}$, where α is a constant.
  2. $Y = \log(S)$.
  3. $Y = \exp(S)$.

Space for your work.

Space for your work.

Space for your work.

Test Your Understanding 2.12 (Itô's formula – two geometric Brownian motions)
Find dC if $C = g(t, S_x, S_y)$ and

$$\frac{dS_x}{S_x} = \mu_x\,dt + \sigma_x\,dB_x,$$

$$\frac{dS_y}{S_y} = \mu_y\,dt + \sigma_y\,dB_y.$$

Space for your work.

Space for your work.

2.12 Transformations of Models

- Suppose that you are given an Itô process

$$dY = \alpha\, dt + \sum_{k=1}^{K} \sigma_k\, dB_k, \quad \text{for } K > 1 \text{ independent Brownian motions } B_k.$$

- One can transform this so that the stochastic part depends only on a single Brownian motion.

- To do this, define a new stochastic process $\widehat{B}$ by $\widehat{B}_0 = 0$, and

$$d\widehat{B} = \frac{1}{\sqrt{\sum_{k=1}^{K} \sigma_k^2}} \left(\sum_{k=1}^{K} \sigma_k\, dB_k \right).$$

- We reproduce the expression for the new stochastic process $\hat{B}$:

$$d\hat{B} = \frac{1}{\sqrt{\sum_{k=1}^{K} \sigma_k^2}} \left(\sum_{k=1}^{K} \sigma_k \, dB_k \right).$$

- Then, observe that $d\hat{B}$ has no drift (that is, its mean equals zero), and

$$\left(d\hat{B}\right)^2 = \frac{1}{\sum_{k=1}^{K} \sigma_k^2} \left(\sum_{k=1}^{K} \sigma_k^2 \, dt \right) = dt,$$

so Levy's theorem implies that $\hat{B}$ is a Brownian motion, and that

$$dY = \alpha \, dt + \hat{\sigma} \, d\hat{B}, \quad \text{where} \quad \hat{\sigma} = \sqrt{\sum_{k=1}^{K} \sigma_k^2}.$$

2.13 Summary

1. A stochastic process is a variable that changes randomly over time.

2. A Brownian motion is a stochastic process that

- evolves continuously in time (no jumps),
- whose change over time is
  - ⋆ Normally distributed,
  - ⋆ with mean zero, and
  - ⋆ variance equal to the length of time,
- whose quadratic variation over $[0, T]$ is T, and whose total variation over time is infinite.

3. Every continuous martingale is a transformation of a Brownian motion.

4. An Itô process is a variable X that changes over time as

$$dX(t) = \mu(t)dt + \sigma(t)dB(t)$$

where $B(t)$ is a Brownian motion, and μ and σ can also be random processes that satisfy some regularity conditions, and $\mu(t)$ and $\sigma(t)$ are known at t.

- Itô processes satisfy the following "rules":

$$\begin{aligned} (dt)^2 &= 0, \\ (dt)\,(dB) &= 0, \\ (dB)^2 &= dt \\ (dB_x)(dB_y) &= \rho\, dt \end{aligned}$$

  . . . where ρ is the correlation between the two Brownians.

- A geometric Brownian motion is an Itô process in which the drift μ and volatility σ are constant.

- An Itô process $dX(t)$ is a martingale only if it has zero drift: $\mu = 0$.

5. Itô's formula tells us that for

- If $C = g(t, X(t))$, and X is an Itô process given by

$$dX(t) = \mu_x(t)\, dt + \sigma_x(t)\, dB_x(t), \tag{2.23}$$

then
$$\begin{aligned} dC &= g_t\, dt + g_x\, dX + \frac{1}{2} g_{xx}\, (dX)^2 \\ &= \left(g_t + g_x \mu_x(t) + \frac{1}{2} g_{xx}\, [\sigma_x(t)]^2 \right) dt + g_x\, \sigma_x(t)\, dB_x. \end{aligned}$$

- If $C(t) = g(t, X(t), Y(t))$, and X and Y are Itô processes given in (2.23) and (2.24)

$$dX(t) = \mu_x(t)\,dt + \sigma_x(t)\,dB_x(t), \tag{2.23}$$
$$dY(t) = \mu_y(t)\,dt + \sigma_y(t)\,dB_y(t), \tag{2.24}$$

where B_x & B_y can be different Brownian motions, and $(dB_x)(dB_y) = \rho\,dt$, then:

$$\begin{aligned} dC &= g_t\,dt + g_x\,dX + g_y\,dY + \frac{1}{2}g_{xx}\,(dX)^2 + \frac{1}{2}g_{yy}\,(dY)^2 + g_{xy}\,(dX)\,(dY) \\ &= \left(g_t + g_x\mu_x(t) + g_y\mu_y(t) + \frac{1}{2}g_{xx}\sigma_x^2(t) + \frac{1}{2}g_{yy}\sigma_y^2(t) + g_{xy}\sigma_x(t)\sigma_y(t)\rho_{xy}(t)\right)dt \\ &\quad + g_x\sigma_x(t)dB_x + g_y\sigma_y(t)dB_y. \end{aligned}$$

6. Itô's formula for log, product, ratio, and compounding/discounting:

- If $C = \log(X)$, then

$$dC = \frac{dX}{X} - \frac{1}{2}\left(\frac{dX}{X}\right)^2. \tag{2.25}$$

- If $C = g(X, Y) = X\,Y$, then

$$\frac{dC}{C} = \frac{dX}{X} + \frac{dY}{Y} + \left(\frac{dX}{X}\right)\left(\frac{dY}{Y}\right). \tag{2.26}$$

- If $C = \frac{Y}{X}$, then

$$\frac{dC}{C} = -\frac{dX}{X} + \frac{dY}{Y} + \left(\frac{dX}{X}\right)^2 - \left(\frac{dX}{X}\right)\left(\frac{dY}{Y}\right). \tag{2.27}$$

- If $Y(t) = \exp\left(\int_0^t q(s)ds\right)$ and $C = X\,Y$ for any Itô process X, then

$$\frac{dC}{C} = \underbrace{\frac{dX}{X}}_{\text{capital gains}} + \underbrace{q(t)dt}_{\text{dividends}}. \tag{2.28}$$

We will be making frequent use of these results, so it is vital to understand them well.

7. Suppose

$$\frac{dX}{X} = \mu_x\,dt + \sigma_x\,dB_x, \qquad \text{and} \qquad \frac{dY}{Y} = \mu_y\,dt + \sigma_y\,dB_y,$$

where B_x and B_y are different Brownian motions with correlation ρ.

- The **volatility** of XY per unit time is

$$\sqrt{\sigma_x^2 + \sigma_y^2 + 2\rho\,\sigma_x\,\sigma_y}, \qquad \text{and} \tag{2.29}$$

- The **volatility** of X/Y per unit time is

$$\sqrt{\sigma_x^2 + \sigma_y^2 - 2\rho\,\sigma_x\,\sigma_y}. \tag{2.30}$$

We will be making frequent use of these results, so it is vital to understand them well.

2.14 Suggestions for Further Readings

2.14.1 Source material for the lecture notes

- These notes are based on the book by Back (2017). You should also look at the book by Dumas and Luciano (2017).

- There are a large number of books that develop the mathematics for continuous time finance. If you want to look at where this all started from, I would recommend:
  - ⋆ Merton (1975), which is a description (without any equations) of the beauty of continuous-time mathematics, and
  - ⋆ Merton (1990, Chapter 3), which develops the continuous-time equations starting from the corresponding discrete-time equations.

- As mentioned in the previous chapter, the relation of the binomial model to continuous-time models is explained in Cox and Ross (1976).

- The slim volume by Dixit (1993) has a very nice exposition connecting results in a discrete-time binomial setting to continuous time.

- You can also read the book Shreve (2004), which is the continuous-time counterpart to the book set in discrete time, Shreve (2005).

- There are a very large number of books on continuous-time stochastic processes, too many to list here, so please talk to me if you wish to explore a particular topic.

2.14.2 Potential applications and extensions of lecture notes

- A good starting point is the review paper by Sundaresan (2000), who lists the following applications of continuous-time models:

  1. Valuation of options and other derivatives
  2. Modeling of the term structure of interest rates
  3. Design of optimal asset-allocation and portfolio-choice strategies
  4. Pricing of primitive assets (stocks and bonds)
  5. Determining default risk and credit spreads
  6. Understanding effect of capital-market frictions such as: transactions costs, asymmetric information, restricted market participation, taxes, etc.
  7. International finance and exchange rate dynamics
  8. Real options.

- In the rest of our course, we will study the applications of this material to the first four items on the list above:
  1. Valuation, hedging, and the design of of contracts contingent on stock prices, forward prices, foreign exchange rates, etc. (Part 1).
  2. Modeling of the term structure and valuation of contracts contingent on interest rates or bond prices (Part 2).
  3. Design of dynamic asset-allocation and portfolio strategies (Part 2).
  4. Valuation of primitive assets, such as stocks and bonds (Part 2).

2.14.3 Ph.D. elective courses offered on related topics

- Rama Cont taught a course on *Monte Carlo Methods* in Finance in 2010 and 2013; Marcel Rindisbacher taught this course in 2017.
- Jerome Detemple taught a course on *Derivative Securities* in 2011.
- Yacine Aït-Sahalia taught a course on *Financial Econometrics of Jumps* in 2011.
- Mikhail Chernov taught a course on *Empirical Option Pricing* in 2012 and 2014.
- Federico Bandi taught a course on *The Econometrics of Continuous-Time Models* in 2012 and 2015, and a course on *Estimation of Continuous-Time Models* in 2013.
- Torben Andersen taught a course on *High-Frequency Asset Pricing* in 2013 and 2015.
- Peter Christoffersen taught a course on *Risk Management and Extreme Risk* in 2011, 2013, and 2015.

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Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 3

Stochastic Discount Factor and Returns in Continuous Time

Raman Uppal

EDHEC Business School

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3.1 Plan for the Day

Last Time

- Maths for continuous time finance.

Today

- **Understand the SDF in continuous time:**
  - ⋆ Stochastic discount factor and its properties in continuous time.

For Next Time

- Review the material covered in class.
- Read Back, Chapter 13 and Dumas & Luciano, Chapter 8.
- Do the practice problems.

3.2 Main Points for Today

1. Stochastic discount factor and its properties in continuous time.
   - Beta pricing in continuous time.

- Last chapter, developed maths for continuous time setting;
- This chapter, apply the maths for continuous time to the concepts of SDF developed in the first class;
- Next chapter, study change of measure and numeraire in continuous time.
- From then onwards (for the rest of the course), apply concepts of SDF and RNP to various problems in finance.

3.3 SDF, Returns, and Excess Returns in Discrete Time

- So far, our discussion has been in terms of asset prices.
- However, traditionally (after the development of the CAPM in the 1960s), results in finance are often stated in terms of returns.
- In this section, we show that all the traditional results that you are familiar with in terms of returns (and many new results, too) can be expressed also in terms of the SDF.
- Every asset-pricing model/theory implies a particular SDF:

  ⋆ So, all we need to know about any asset-pricing model is the particular SDF implied by it;

  ⋆ Conversely, knowing the SDF fully characterizes the asset-pricing model it comes from.

3.3.1 SDF and Expected Gross Returns

- Let us start with our basic expression on Page 131 of Chapter 1 for the price at date $t = 0$ of a non-dividend paying asset, $\mathcal{P}^{S_i}$, in terms of the SDF at $t = 1$, m_1, which we reproduce below:

$$\mathcal{P}_0^{S_i} = \mathbb{E}^{\mathbb{P}}\left[m_1 \, \mathcal{P}_1^{S_i}\right]. \tag{1.31}$$

- If $\mathcal{P}_0^{S_i} > 0$, then we can divide the above equation by $\mathcal{P}_0^{S_i}$ to get:

$$1 = \mathbb{E}^{\mathbb{P}}\left[m_1 \frac{\mathcal{P}_1^{S_i}}{\mathcal{P}_0^{S_i}}\right] = \mathbb{E}^{\mathbb{P}}\left[m_1 \, R_i\right], \qquad \text{gross return } R_i = \mathcal{P}_1^{S_i}/\mathcal{P}_0^{S_i}. \tag{3.1}$$

- Equation (3.1) says that after we adjust for time and risk (via m_1), the expected gross return on asset R_i is 1 (and so, the net return is zero).

3.3.2 SDF and Expected Excess Returns

- In finance, we often study excess returns. So, next we identify the relation between the SDF and excess returns.
- We start by reproducing the last equation on the previous slide:

$$1 = \mathbb{E}^{\mathbb{P}}\left[m_1\, R_i\right] \quad \text{and similarly:} \quad 1 = \mathbb{E}^{\mathbb{P}}\left[m_1\, R_j\right]. \tag{3.1}$$

- Taking the difference of the two expressions above, observe that we can write (3.1) also in terms of excess returns:

$$0 = \mathbb{E}^{\mathbb{P}}\left[m_1\ (R_i - R_j)\right], \quad \text{where } R_i \text{ is the return on asset } i. \tag{3.2}$$

- The expressions above in (3.1) and (3.2) are true for any asset or *portfolio* of assets.
- Below, we will look at the special cases of
  - ⋆ the return on the risk-free asset (R_f), and
  - ⋆ the return on the market portfolio (R_{mkt}).

3.3.3 SDF and the Risk-free Rate

- Recall from basic statistics that: $\text{Cov}[x, y] = \mathbb{E}[x\, y] - \mathbb{E}[x]\,\mathbb{E}[y]$, which implies that: $\mathbb{E}[x\, y] = \text{Cov}[x, y] + \mathbb{E}[x]\,\mathbb{E}[y]$.
- Starting with equation (3.1), and writing m_1 as m:

$$1 = \mathbb{E}^{\mathbb{P}}\left[m\, R_i\right], \tag{3.1}$$

$$1 = \text{Cov}[m, R_i] + \mathbb{E}^{\mathbb{P}}[m]\,\mathbb{E}^{\mathbb{P}}[R_i]$$

$$\mathbb{E}^{\mathbb{P}}[R_i]\,\mathbb{E}^{\mathbb{P}}[m] = 1 - \text{Cov}[m, R_i]$$

$$\mathbb{E}^{\mathbb{P}}[R_i] = \frac{1}{\mathbb{E}^{\mathbb{P}}[m]} - \frac{1}{\mathbb{E}^{\mathbb{P}}[m]}\text{Cov}[m, R_i]. \tag{3.3}$$

- Consider the special case of the risk-free asset, so $R_i = R_f$, and $\text{Cov}[m, R_f] = 0$, which implies that

$$\mathbb{E}[R_f] = R_f = \frac{1}{\mathbb{E}^{\mathbb{P}}[m]} \quad \ldots \text{ useful result that will appear frequently.} \tag{3.4}$$

Example 3.1 (Continuing with binomial option pricing for a call on stock)

- Let us continue with Example 1.1 on Page 77 and find the risk-free return and the return on the stock in terms of the SDF, as given below:

$$R_f = \frac{1}{\mathbb{E}^{\mathbb{P}}[m]} \tag{3.4}$$

- In Example 1.2, we already calculated:

$$m_u = \frac{\mathcal{P}_0^{Au}}{\text{prob}_u^{\mathbb{P}}} = \frac{0.4}{1.01} \times \frac{1}{\text{prob}_u^{\mathbb{P}}}; \quad m_d = \frac{\mathcal{P}_0^{Ad}}{\text{prob}_d^{\mathbb{P}}} = \frac{0.6}{1.01} \times \frac{1}{\text{prob}_d^{\mathbb{P}}}.$$

- Therefore,

$$\mathbb{E}^{\mathbb{P}}[m] = \text{prob}_u^{\mathbb{P}} m_u + \text{prob}_d^{\mathbb{P}} m_d = \text{prob}_u^{\mathbb{P}} \left(\frac{0.4}{1.01} \times \frac{1}{\text{prob}_u^{\mathbb{P}}} \right) + \text{prob}_d^{\mathbb{P}} \left(\frac{0.6}{1.01} \times \frac{1}{\text{prob}_d^{\mathbb{P}}} \right)$$

$$= \left(\frac{0.4}{1.01} \right) + \left(\frac{0.6}{1.01} \right) = \frac{1}{1.01}.$$

- So, we confirm that

$$R_f = \frac{1}{\mathbb{E}^{\mathbb{P}}[m]} = 1.01.$$

3.3.4 SDF, Risk Premia, and Covariance

- Substituting $R_f = \frac{1}{\mathbb{E}^{\mathbb{P}}[m]}$ in the result $\mathbb{E}[R_i] = \frac{1}{\mathbb{E}^{\mathbb{P}}[m]} - \frac{1}{\mathbb{E}^{\mathbb{P}}[m]}\text{Cov}[m, R_i]$ gives:

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_f - R_f \text{Cov}[m, R_i]$$

$$\boxed{\mathbb{E}^{\mathbb{P}}[R_i] - R_f = -R_f \text{Cov}[m, R_i] \quad \text{... useful result that will appear frequently.}} \tag{3.5}$$

- In words, the above equation says that:
  the risk-premium on asset i over and above the risk-free rate R_f,
  is negatively related to the covariance of the return on asset i with the SDF.

- The reason for the negative relation with the covariance of the return on asset i with the SDF m is that when wealth decreases, the SDF increases.

  ⋆ Thus, assets that pay off in states of the world where wealth is low (and SDF is high), need to pay only a small risk-premium (because they have a higher payoff in states where we value this payoff much more).

- Now, we confirm the following result: $\mathbb{E}^{\mathbb{P}}[R_i] = R_f - R_f \text{Cov}[m, R_i]$ for the binomial setting that we looked at earlier.

- Let us start by finding a nice expression and interpretation for $\text{Cov}[m, R_i]$:

$$
\begin{aligned}
\text{Cov}[m, R_i] &= \mathbb{E}^{\mathbb{P}}[m\, R_i] - \mathbb{E}^{\mathbb{P}}[m]\, \mathbb{E}^{\mathbb{P}}[R_i] && \text{. . . using definition of Cov} \\
&= \text{prob}_u^{\mathbb{P}} m_u\, R_u + \text{prob}_d^{\mathbb{P}} m_d\, R_d - \frac{1}{R_f}\mathbb{E}^{\mathbb{P}}[R_i] && \text{. . . taking expectations, } \mathbb{E}^{\mathbb{P}}[m\, R_i] \\
&= \text{prob}_u^{\mathbb{P}}\Big(\frac{\mathcal{P}_0^{A_u}}{\text{prob}_u^{\mathbb{P}}}\Big) R_u + \text{prob}_d^{\mathbb{P}}\Big(\frac{\mathcal{P}_0^{A_d}}{\text{prob}_d^{\mathbb{P}}}\Big) R_d - \frac{1}{R_f}\mathbb{E}^{\mathbb{P}}[R_i] && \text{. . . using definition of } m_k = \frac{\mathcal{P}_0^{A_k}}{\text{prob}_k^{\mathbb{P}}} \\
&= \mathcal{P}_0^{A_u}\, R_u + \mathcal{P}_0^{A_d}\, R_d - \frac{1}{R_f}\mathbb{E}^{\mathbb{P}}[R_i] && \text{. . . cancel } \text{prob}_k^{\mathbb{P}} \text{ from num. \& denom.} \\
&= \Big(\frac{\text{prob}_u^{\mathbb{Q}}}{R_f}\Big)\, R_u + \Big(\frac{\text{prob}_d^{\mathbb{Q}}}{R_f}\Big)\, R_d - \frac{1}{R_f}\mathbb{E}^{\mathbb{P}}[R_i] && \text{. . . using definition of } \mathcal{P}_0^{A_k} = \frac{\text{prob}_k^{\mathbb{Q}}}{R_f} \\
&= \frac{1}{R_f}\Big(\text{prob}_u^{\mathbb{Q}}\, R_u + \text{prob}_d^{\mathbb{Q}}\, R_d\Big) - \frac{1}{R_f}\mathbb{E}^{\mathbb{P}}[R_i] && \text{. . . collecting terms} \\
&= -\frac{1}{R_f}\Big(\mathbb{E}^{\mathbb{P}}[R_i] - \mathbb{E}^{\mathbb{Q}}[R_i]\Big) && \text{. . . collecting terms, and rearranging}
\end{aligned}
$$

- We start by reproducing the last line from the last previous page:

$$
\begin{aligned}
\text{Cov}[m, R_i] &= -\frac{1}{R_f}\Big(\mathbb{E}^{\mathbb{P}}[R_i] - \mathbb{E}^{\mathbb{Q}}[R_i]\Big) \\
&= -\frac{1}{R_f}\Big(\mathbb{E}^{\mathbb{P}}[R_i] - R_f\Big) && \text{. . . under } \mathbb{Q}\text{, expected return on every asset is } R_f \\
-R_f\, \text{Cov}[m, R_i] &= \Big(\mathbb{E}^{\mathbb{P}}[R_i] - R_f\Big) && \text{. . . multiplying both sides by } -R_f \\
R_f - R_f\, \text{Cov}[m, R_i] &= R_f + \Big(\mathbb{E}^{\mathbb{P}}[R_i] - R_f\Big) && \text{. . . adding to both sides } R_f \\
&= \mathbb{E}^{\mathbb{P}}[R_i], && \text{. . . canceling } R_f \text{ and } -R_f \text{ on right-hand side}
\end{aligned}
\tag{3.5}
$$

which is the result we wanted to prove.

3.4 SDF and Factor Beta-Pricing Models in Discrete Time

3.4.1 Single-factor beta pricing model

- A single-factor beta-pricing model with factor f is said to exist if there are constants R_z and λ such that for each return R_i

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \lambda \underbrace{\frac{\mathrm{Cov}[f, R_i]}{\mathrm{Var}[f]}}_{beta}. \tag{3.6}$$

- Note that if a risk-free asset exists, then $\mathrm{Cov}[f, R_f] = 0$ and so setting $R_i = R_f$ in the equation above, gives us $R_z = R_f$.

- More generally, R_z (where the "z" stands for "*zero* beta") is the expected value of the return satisfying $\mathrm{Cov}[f, R_z] = 0$.

- In the absence of a risk-free asset, different beta-pricing models can have different expected zero-beta returns (that is, each model may imply a different return for the asset with a beta of zero).

3.4.2 Multifactor beta pricing model

- It is straightforward to extend Equation (3.6) to a multi-factor setting.

- A multi-factor beta-pricing model with factors $F = (f_1, f_2, \ldots, f_K)^\top$ is said to exist if there are constants R_z and $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_K)^\top$ such that for each return R_i

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \lambda^\top \Sigma_F^{-1} \mathrm{Cov}[F, R_i], \tag{3.7}$$

  where

  - $^\top$ denotes the transpose operator;
  - Σ_F is the invertible covariance matrix of the vector F;
  - $\mathrm{Cov}[F, R_i]$ is the column vector of dimension K with elements $\mathrm{Cov}[f, R_i]$;
  - $\Sigma_F^{-1} \mathrm{Cov}[F, R_i]$ is the vector of multiple regression betas of R_i on F;
  - λ are the factor risk premia.

3.4.3 Beta-Pricing Models in Terms of Covariances

- One can always express beta-pricing models in terms of covariances instead of betas:

- For instance, we can write the single-factor model given in (3.6) as

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \psi \,\mathsf{Cov}[f, R_i], \quad \text{where} \quad \psi = \frac{\lambda}{\mathsf{Var}[f]}. \tag{3.8}$$

- Similarly, we can write the multifactor model given in (3.7) as:

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \psi^{\top} \mathsf{Cov}[F, R_i], \quad \text{where} \quad \psi = \Sigma_F^{-1}\lambda. \tag{3.9}$$

3.4.4 Number of Factors is Not Uniquely Determined

- The number of factors in a beta-pricing model is not uniquely determined.

- Given a K factor model, one can always use $\lambda^{\top} \Sigma_F^{-1} F$ as a single factor.

- One can also use the SDF m as the single systematic factor, provided $\mathbb{E}[m] \neq 0$, as we showed in (3.3), which is reproduced below:

$$\mathbb{E}^{\mathbb{P}}[R_i] = \frac{1}{\mathbb{E}^{\mathbb{P}}[m]} - \frac{1}{\mathbb{E}^{\mathbb{P}}[m]}\mathsf{Cov}[m, R_i], \tag{3.3}$$

  with $1/\mathbb{E}[m] = R_f$ if there exists a risk-free asset.

3.4.5 Single-factor Models with Returns as Factors

- If a factor is a return, then its factor risk premium is its ordinary risk premium, treating R_z as a proxy for the risk-free return.

- To see this, suppose that there is a single-factor beta-pricing model with the factor being the return R_*. Then,

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \lambda \frac{\mathsf{Cov}[R_*, R_i]}{\mathsf{Var}[R_*]}. \tag{3.10}$$

- But, the above equation must be true for $R_i = R_*$

$$\mathbb{E}^{\mathbb{P}}[R_*] = R_z + \lambda \frac{\mathsf{Cov}[R_*, R_*]}{\mathsf{Var}[R_*]} = R_z + \lambda \frac{\mathsf{Var}[R_*]}{\mathsf{Var}[R_*]} = R_z + \lambda,$$

and so

$$\lambda = \mathbb{E}^{\mathbb{P}}[R_*] - R_z. \tag{3.11}$$

- Substituting $\lambda = \mathbb{E}^{\mathbb{P}}[R_*] - R_z$ from (3.11) into (3.10) leads to:

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \left(\mathbb{E}^{\mathbb{P}}[R_*] - R_z\right) \frac{\mathsf{Cov}[R_*, R_i]}{\mathsf{Var}[R_*]}. \tag{3.12}$$

- Thus, there is a beta-pricing model with a return as a single factor if and only if the return is on mean-variance frontier and is not equal to the risk-free rate, if one exists, and otherwise not equal to global minimum-variance portfolio.

- The Capital Asset Pricing Model (CAPM) states that (3.12) is true for the factor return being the return on the market portfolio, $R_* = R_{mkt}$:

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \left(\mathbb{E}^{\mathbb{P}}[R_{mkt}] - R_z\right) \frac{\mathsf{Cov}[R_{mkt}, R_i]}{\mathsf{Var}[R_{mkt}]}, \tag{3.13}$$

which is equivalent to the market portfolio being on the mean-variance frontier and not equal to the risk-free rate, if one exists, and otherwise not equal to global minimum-variance portfolio.

3.4.6 SDF and Beta Pricing

- There is a beta-pricing model with respect to some factors with the expected zero-beta return being non-zero if and only if there is an SDF m such that:
  1. $\mathbb{E}[m] \neq 0$; and,
  2. m is an affine function of the factors ($m = a + b\, f$).
- This is true for both single- and multi-factor models.
- In a single-factor model such as (3.8), reproduced below:

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \psi\, \mathsf{Cov}[f, R_i], \quad \text{where} \quad \psi = \frac{\lambda}{\mathsf{Var}[f]}, R_z \neq 0, \tag{3.8}$$

  the following is an SDF:

$$m = \frac{1}{R_z} - \frac{1}{R_z}\psi \left(f - \mathbb{E}^{\mathbb{P}}[f]\right). \tag{3.14}$$

Proof that m in Equation (3.14) is a valid SDF

- Suppose that there is a one-factor beta-pricing model

$$\mathbb{E}^{\mathbb{P}}[R_i] = R_z + \psi\, \mathsf{Cov}[f, R_i] \tag{3.8}$$

  and we now wish to show that the following is a valid SDF:

$$m = \frac{1}{R_z} - \frac{1}{R_z}\psi \left(f - \mathbb{E}^{\mathbb{P}}[f]\right). \tag{3.14}$$

- If m is a valid SDF, then it must be that $\mathbb{E}^{\mathbb{P}}[m\, R_i] = 1$.

- Start by multiplying m in (3.14) by R_i:

$$m\,R_i = \frac{R_i}{R_z} - \frac{1}{R_z}\left(\psi\,f\,R_i - \mathbb{E}^{\mathbb{P}}[\psi\,f]R_i\right). \tag{3.15}$$

- Now take expectations, and then simplify:

$$\begin{aligned}
\mathbb{E}^{\mathbb{P}}[m\,R_i] &= \frac{\mathbb{E}^{\mathbb{P}}[R_i]}{R_z} - \frac{1}{R_z}\Big(\mathbb{E}^{\mathbb{P}}[\underbrace{\psi\,f}_{x}\,\underbrace{R_i}_{y}] - \mathbb{E}^{\mathbb{P}}[\underbrace{\psi\,f}_{x}]\mathbb{E}^{\mathbb{P}}[\underbrace{R_i}_{y}]\Big) \\
&= \frac{\mathbb{E}^{\mathbb{P}}[R_i]}{R_z} - \frac{1}{R_z}\mathsf{Cov}[\psi\,f, R_i] && \ldots \text{because } \mathsf{Cov}[x,y] = \mathbb{E}[x\,y] - \mathbb{E}[x]\,\mathbb{E}[y] \\
&= \frac{1}{R_z}\left(\mathbb{E}^{\mathbb{P}}[R_i] - \psi\,\mathsf{Cov}[f, R_i]\right) && \ldots \text{because } \psi \text{ is a constant} \\
&= \frac{1}{R_z}\left(R_z\right) && \ldots \text{from } \mathbb{E}^{\mathbb{P}}[R_i] = R_z + \psi\,\mathsf{Cov}[f, R_i] \text{ in (3.8)} \\
&= 1.
\end{aligned}$$

3.4.7 SDF in the Capital Asset Pricing Model (CAPM)

- For example, if the CAPM is true, then substituting

$$f = R_{mkt} \quad \text{and} \quad \psi = \frac{\mathbb{E}^{\mathbb{P}}[R_{mkt}] - R_z}{\mathsf{Var}[R_{mkt}]}$$

in (3.14), which is reproduced below

$$m = \frac{1}{R_z} - \frac{1}{R_z}\psi\left(f - \mathbb{E}^{\mathbb{P}}[f]\right), \tag{3.14}$$

gives us the following SDF for the CAPM:

$$m = \frac{1}{R_z} - \frac{1}{R_z}\underbrace{\left(\frac{\mathbb{E}^{\mathbb{P}}[R_{mkt}] - R_z}{\mathsf{Var}[R_{mkt}]}\right)}_{\psi}\left(R_{mkt} - \mathbb{E}^{\mathbb{P}}[R_{mkt}]\right). \tag{3.16}$$

END OF DISCUSSION IN DISCRETE TIME.

3.5 Stochastic Discount Factor (SDF) in Continuous Time

- Let $\mathcal{P}^D(t)$ denote the price of the bank deposit, which follows

$$\frac{d\mathcal{P}_t^D}{\mathcal{P}_t^D} = r_t\,dt, \quad \text{for some } r_t, \text{ which could be a stochastic process.} \tag{3.17}$$

- Let $\mathcal{P}^{S_n}$ be the price of a non-dividend paying asset (stock),

$$\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} = \mu_t\,dt + \sigma_t\,dB^{\mathbb{P}},$$

where $B^{\mathbb{P}}$ is a Brownian motion under the physical probability measure and μ_t and σ_t are quite general random processes.

3.5.1 Definition of SDF in Continuous Time

Definition 3.1 (SDF)
We define M to be an SDF process if

1. $M_0 = 1$;
2. $M_t > 0$ for all t with probability 1; and,
3. $M\,\mathcal{P}^D$ and $M\,\mathcal{P}^{S_n}$ are local martingales for $n = 1, 2, \ldots, N$.

- If there is only a **single** risky asset, i.e. $n = 1$, then we will show below that a stochastic process M is an SDF process if and only if $M_0 = 1$ and

$$\frac{dM_t}{M_t} = -r_t\, dt - \lambda_t\, dB_t^{\mathbb{P}}, \tag{3.18}$$

where

$$\lambda_t = \frac{\mu_t - r_t}{\sigma_t}, \tag{3.19}$$

is the Sharpe ratio of the risky asset.

- Observe that the SDF is fully described by
  1. the risk-free interest rate, r_t, and
  2. the Sharpe ratio, λ_t.

- For the general case with many risky assets, we will show below that a stochastic process M is an SDF process if and only if $M_0 = 1$ and

$$\boxed{\frac{dM_t}{M_t} = -r_t\, dt - \lambda_t^\top\, dB_t^{\mathbb{P}},} \tag{3.20}$$

where the stochastic process λ_t satisfies

$$\sigma_t\, \lambda_t = \mu_t - r_t \mathbf{1}. \tag{3.21}$$

- The vector λ is called the vector of market prices of risk.
- If there are as many risky assets as Brownian motions ($N = K$), then markets are complete, and given our assumption that there are no redundant assets, the volatility matrix σ_t is non-singular so

$$\lambda_t = \sigma_t^{-1}(\mu_t - r_t \mathbf{1}), \qquad \ldots \text{multiplying both sides of (3.21) by } \sigma_t^{-1} \tag{3.22}$$

and there is at most one SDF process.

3.5.2 Drift of SDF Process in Continuous Time

- The analysis of SDF processes in continuous time is very similar to that in a single-period model.
- In a single-period model, we have $\mathbb{E}^{\mathbb{P}}[m] = 1/R_f$.
- The analogous condition in continuous time is that the drift of dM/M is $-r$.
- This follows from the definition that $Y \equiv M\,\mathcal{P}^D$ is a local martingale.
- Applying Itô's formula to $Y \equiv M\,\mathcal{P}^D$ gives:

$$\frac{dY}{Y} = \frac{dM}{M} + \frac{d\mathcal{P}^D}{\mathcal{P}^D} = \frac{dM}{M} + r_t\,dt.$$

- For Y to be a local martingale, we need $\mathbb{E}^{\mathbb{P}}\left[\frac{dY}{Y}\right] = 0$, which then implies

$$\mathbb{E}^{\mathbb{P}}\left[\frac{dM}{M}\right] = -r_t\,dt.$$

3.5.3 Beta Pricing in Continuous Time

- Recall that in a single-period model, there is a beta-pricing model using any SDF as the factor:

$$\mathbb{E}^{\mathbb{P}}[R_n] - R_f = -R_f\,\mathrm{Cov}[m, R_n].$$

- The analog of the above result in continuous time follows from the condition that $M\,\mathcal{P}^{S_n}$ is a local martingale, as is shown below.

- To see this, apply Itô's formula to $Y \equiv M\,\mathcal{P}^{S_n}$ to obtain:

$$\frac{dY}{Y} = \frac{dM}{M} + \frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} + \left(\frac{dM}{M}\right)\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right).$$

- For Y to be a local martingale, we need $\mathbb{E}^{\mathbb{P}}\left[\frac{dY}{Y}\right] = 0$, which then implies:

$$\begin{aligned} 0 &= \mathbb{E}^{\mathbb{P}}\left[\frac{dY}{Y}\right] \\ &= \mathbb{E}^{\mathbb{P}}\left[\frac{dM}{M}\right] + \mathbb{E}^{\mathbb{P}}\left[\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right] + \left(\frac{dM}{M}\right)\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right) \\ &= -r_t\,dt + \mu_{n,t}\,dt + \left(\frac{dM}{M}\right)\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right) \end{aligned}$$

$$(\mu_{n,t} - r_t)\,dt = -\left(\frac{dM}{M}\right)\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right) \tag{3.23}$$

$$\boxed{(\mu_{n,t} - r_t)\,dt = -\mathrm{Cov}\left[\frac{dM}{M}, \frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right].} \tag{3.24}$$

- To see the interpretation of λ_t as the vector of market prices of risk, note that substituting the expression for $\frac{dM_t}{M_t}$ from (3.20), which is reproduced below,

$$\frac{dM_t}{M_t} = -r_t\,dt - \lambda_t^{\top}\,dB_t^{\mathbb{P}}, \tag{3.20}$$

into the result for the risk premium in (3.23) derived on the previous page

$$(\mu_{n,t} - r_t)\,dt = -\left(\frac{dM}{M}\right)\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right), \tag{3.23}$$

we get

$$\underbrace{(\mu_{n,t} - r_t)}_{\text{risk premium}}\,dt = \lambda_t^{\top}\underbrace{\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dB_t^{\mathbb{P}})}_{\text{covariances}}, \quad \ldots \text{ substituting } dM_t/M_t \text{ from (3.20) and dropping terms} \to 0$$

which says that the risk premium on the left-hand side is a linear combination of the conditional covariances of the rates of return with the changes dB_k of the Brownian motions, with λ_k being the price of risk for factor $dB_k^{\mathbb{P}}$.

3.5.4 Volatility of SDF Process in Continuous Time

- Note that using Equation (3.20), reproduced below,

$$\frac{dM_t}{M_t} = -r_t\, dt - \lambda_t^\top\, dB_t^{\mathbb{P}}, \tag{3.20}$$

the expression for a single asset n

$$(\mu_{n,t} - r_t)\, dt = -\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(\frac{dM}{M}\right)$$

can be rewritten in vector form for all N assets as

$$(\mu_t - r_t\,\mathbf{1})\, dt = (\sigma_t\, dB_t^{\mathbb{P}})(dB_t^{\mathbb{P}})^\top \lambda_t = \sigma_t\, \lambda_t\, dt,$$

which verifies the volatility of the SDF process given in Equation (3.21) on Page 294; that is,

$$\mu_t - r_t\mathbf{1} = \sigma_t\, \lambda_t \tag{3.21}$$
$$\lambda_t = \sigma_t^{-1}(\mu_t - r_t\mathbf{1}).$$

3.6 Summary

1. SDF

   A stochastic process M is an SDF process if and only if $M_0 = 1$ and

$$\frac{dM_t}{M_t} = -r_t\, dt - \lambda_t^\top\, dB_t^{\mathbb{P}}, \quad \text{where} \quad \lambda_t = \sigma_t^{-1}(\mu_t - r_t\mathbf{1}).$$

2. The relation between the risk-free rate and the SDF is:

$$\mathbb{E}^{\mathbb{P}}\left[\frac{dM}{M}\right] = -r_t\, dt, \quad \text{and}$$

3. The relation between the risk premium and the SDF is:

$$(\mu_{n,t} - r_t)\, dt = -\text{Cov}\left[\frac{dM}{M}, \frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right].$$

3.7 Suggestions for Further Readings

3.7.1 Source material for the lecture notes

- This material, in discrete time and in continuous time, is from Back (2017, Chapters 3, 8, 13). See also Dumas and Luciano (2017).
- The origin of the SDF is not clear; given that it is simply the marginal utility of consumption, one can consider that it originated in microeconomics.
  - ⋆ In Finance, the first discussion is probably in Rubinstein (1976).
  - ⋆ For a nice historical account of the development of research in the areas of investments and asset pricing, see the book by Rubinstein (2006).
- Dybvig and Ross (1985) develop the "*fundamental theorem of asset pricing*," which establishes the equivalence of: (1) absence of arbitrage opportunities, (2) existence of a strictly positive SDF, and (3) existence of an optimum for an agent with strictly monotone utility.

3.7.2 Potential applications and extensions of lecture notes

- For the classic papers on SDF, see Hansen and Jagannathan (1991, 1997).
- For a comparison of the SDF and beta-pricing approaches, see Jagannathan and Wang (2002).
- A largely descriptive account of asset pricing in terms of the SDF is presented in Cochrane and Culp (2003).
- The book by Cochrane (2005), views asset pricing through the lens of the SDF and provides a tight link between the *theory* of asset pricing and the *estimation* of asset-pricing models.
- Using the SDF to empirically assess asset-pricing models is discussed in Nagel and Singleton (2011).
- Dello-Preite, Uppal, Zaffaroni, and Zviadadze (2023) use an SDF based on the Arbitrage Pricing Theory (APT) to show the dominant role of unsystematic risk relative to systematic risk for pricing the cross section of stock returns.

- For applications of the SDF to:
  - ⋆ Asset pricing: see Campbell and Cochrane (1999) and Bansal and Yaron (2004), and the review papers by Campbell (2000), Back (2010), Mehra (2012), and Breeden, Litzenberger, and Jia (2015).
  - ⋆ International finance: Sandulescu, Trojani, and Vedolin (2018).
  - ⋆ Behavioral finance: the book by Shefrin (2008) and the review paper by Shefrin (2010).
  - ⋆ Macrofinance: see review paper by Brunnermeier and Sannikov (2016).
  - ⋆ Effect of financial intermediation on asset prices and the consequences for financial crises: Brunnermeier and Sannikov (2014) and He and Krishnamurthy (2013, 2018).

- Following the award of the Nobel prize to Fama, Hansen, and Shiller, Campbell (2014) explains the contribution of their work in terms of the SDF.
- For *empirical* work using the SDF, see
  - ⋆ Almeida and Garcia (2012, 2013)
  - ⋆ Kozak, Nagel, and Santosh (2018, 2020),
  - ⋆ Ghosh, Julliard, and Taylor (2017, 2019),
  - ⋆ Korsaye, Quaini, and Trojani (2019),
  - ⋆ Chen, Pelger, and Zhu (2019), and
  - ⋆ Dello-Preite, Uppal, Zaffaroni, and Zviadadze (2023).

- Most modern research papers in the area of asset pricing, both theoretical and empirical, are stated in terms of the SDF.
- In our course, almost all our results will be framed in terms of the SDF.
  - ⋆ First, we will take the SDF as given, and use it to:
    - ∗ Derive the value of contracts contingent on stock prices, forward rates, and exchange rates (in Part 1 of the course);
    - ∗ Derive the value of contracts contingent on bond prices and interest rates (in Part 2 of the course);
    - ∗ Derive the optimal portfolio policies (in Part 2 of the course).
  - ⋆ Then, we derive the SDF itself, making assumptions on primitives of the economy (the last chapter in Part 2 of the course).

3.7.3 Ph.D. elective courses offered on related topics

- Pietro Veronesi taught
  - ⋆ *Recent Developments in Portfolio Allocation and Macro-Finance Model* in 2009, and
  - ⋆ *Macroeconomic and Political Uncertainty and the Dynamics of Asset Prices* in 2012, 2014, and 2016.
  - ⋆ He will teach a course on *Fixed Income Securities* in 2021.
- Yacine Aït-Sahalia taught *Financial Econometrics of Jumps* in 2011.
- Jerome Detemple taught a course on *Derivative Securities* in 2011.
- Ravi Bansal taught a course on *Long-Run Risks in Asset Prices* in 2011.
- Mikhail Chernov taught *Empirical Option Pricing* in 2012 and 2014.
- Federico Bandi taught *The Econometrics of Continuous-Time Models* in 2012 and 2015, and *Estimation of Continuous-Time Models* in 2013.
- Torben Andersen taught *High-Frequency Asset Pricing* in 2013 and 2015.

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Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 4

Change of Measure/Numeraire and Change of Drift in Continuous Time

Raman Uppal

EDHEC Business School

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4.1 Plan for the Day

Last Time

- Stochastic Discount Factor (SDF) and returns in continuous time.

Today

- **Understanding change of measures in continuous time:**
  - ⋆ Changing measure/numeraire and drift in continuous time.
  - ⋆ Tail probabilities of a geometric Brownian motion

For Next Time

- Review the material covered in class.
- Read Back, Chapters 15–16 and Dumas & Luciano, Chapter 8.
- Do the practice problems.

4.2 Main Points for Today

1. Change of measure, drift, and numeraire in continuous time.
   - Girsanov's Theorem.
2. Tail probabilities of geometric Brownian motion.

- Last chapter, we studied the properties of SDF in a continuous time setting;
- This chapter, we apply the maths for continuous time to the concepts of change of measure and change of numeraire developed in the first class;
- From next time onwards (for the rest of the course), we will apply concepts of SDF and change of measure to various problems in finance.

4.3 Change of Measure and Change of Drift

- When we change probability measures,
  - ⋆ a process that was a Brownian motion under the old measure
  - ⋆ may not be a Brownian motion under the new measure; that is, its expected change may no longer be zero.
- However, even under the new measure, Itô processes will continue to be Itô processes
  - ⋆ with the same diffusion coefficient (and hence, also the same covariances),
  - ⋆ but a different drift (because the expected change is no longer zero).
- Thus, changing probability measures changes the likelihood of various paths, but it will not affect how each particular path jiggles, which is determined by the diffusion coefficient of the Itô process.

4.3.1 Risk-Neutral Probabilities in Discrete Time

- Recall, from Chapter 1, the results for the change of measure:
  - ⋆ For the two-date binomial model, Equation (1.34) on Page 136 of Ch. 1:

$$\text{prob}_k^{\mathbb{Q}} = \text{prob}_k^{\mathbb{P}}\, m_k\, \frac{\mathcal{P}_T^D}{\mathcal{P}_0^D} = \text{prob}_k^{\mathbb{P}}\, m_k\, \mathcal{P}_T^D \qquad \ldots \text{recall that } \mathcal{P}_0^D = 1. \quad (1.34)$$

  - ⋆ For the two-date model with a continuum of states, Equation (1.39) on Page 146 of Ch. 1:

$$\text{prob}^{\mathbb{Q}}(A) = \mathbb{E}^{\mathbb{P}}\left[1_A\, m_T\, \frac{\mathcal{P}_T^D}{\mathcal{P}_0^D}\right] = \mathbb{E}^{\mathbb{P}}\left[1_A\, m_T\, \mathcal{P}_T^D\right]. \quad (1.39)$$

  - ⋆ We now wish to obtain the risk-neutral measure for a model set in continuous time and with a continuum of states.

4.3.2 Radon-Nikodym Derivative and Change of Measure in Continuous Time

- Let M be an SDF process and assume $M\,\mathcal{P}^D$ is a $\mathbb{P}$-martingale, instead of just a local martingale, so $\mathbb{E}^{\mathbb{P}}[M_T\,\mathcal{P}^D_T] = \mathcal{P}^D_0 = 1$, with $T < \infty$.
- Define $\xi_t \equiv M_t\,\frac{\mathcal{P}^D_t}{\mathcal{P}^D_0} = M_t\,\mathcal{P}^D_t$ for each $t < T$.
- Define for each event A that is distinguishable at date T:

$$\underbrace{\mathbb{Q}(A)}_{\text{usual notation}} = \underbrace{\text{prob}^{\mathbb{Q}}(A)}_{\text{our notation}} = \mathbb{E}^{\mathbb{P}}[\xi_T\,1_A]. \tag{4.1}$$

- The construction (4.1) defines ξ_T as the Radon-Nikodym derivative $d\mathbb{Q}/d\mathbb{P}$ or, in terms of the notation we are using, $d\text{prob}^{\mathbb{Q}}/d\text{prob}^{\mathbb{P}}$:

$$\xi_T = \frac{d\mathbb{Q}}{d\mathbb{P}} = \frac{d\,\text{prob}^{\mathbb{Q}}}{d\,\text{prob}^{\mathbb{P}}}.$$

- It follows from (4.1), which is reproduced below,

$$\underbrace{\mathbb{Q}(A)}_{\text{usual notation}} = \underbrace{\text{prob}^{\mathbb{Q}}(A)}_{\text{our notation}} = \mathbb{E}^{\mathbb{P}}[\xi_T\,1_A]. \tag{4.1}$$

  and our definition of ξ_T, also reproduced below,

$$\xi_T = \frac{d\mathbb{Q}}{d\mathbb{P}} = \frac{d\,\text{prob}^{\mathbb{Q}}}{d\,\text{prob}^{\mathbb{P}}}, \tag{4.2}$$

  that for any random variable X depending on date-T information:

$$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[\xi_T\,X]. \tag{4.3}$$

- Note that each SDF process M defines a different risk-neutral probability.

- To understand (4.1) and (4.2): From Equation (1.33) on page 135 of Chapter 1 for the binomial model, we see that

$$\frac{\text{prob}_k^{\mathbb{Q}}}{\text{prob}_k^{\mathbb{P}}} = m_k \, \mathcal{P}_T^D = \xi_T, \tag{1.33}$$

that is, $\xi_T = m_k \mathcal{P}_T^D = \text{prob}^{\mathbb{Q}}/\text{prob}^{\mathbb{P}}$, rather than $d\,\text{prob}^{\mathbb{Q}}/d\,\text{prob}^{\mathbb{P}}$, because the binomial distribution is discrete.

- To understand (4.3): Note that in the binomial setting,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[X] &= \mathbb{E}^{\mathbb{P}}[\xi_T \, X] \\
&= \text{prob}_u^{\mathbb{P}} \, \xi_u \, X_u + \text{prob}_d^{\mathbb{P}} \, \xi_d \, X_d \\
&= \text{prob}_u^{\mathbb{P}} \, \frac{\text{prob}_u^{\mathbb{Q}}}{\text{prob}_u^{\mathbb{P}}} \, X_u + \text{prob}_d^{\mathbb{P}} \, \frac{\text{prob}_d^{\mathbb{Q}}}{\text{prob}_d^{\mathbb{P}}} \, X_d \\
&= \text{prob}_u^{\mathbb{Q}} \, X_u + \text{prob}_d^{\mathbb{Q}} \, X_d.
\end{aligned}$$

4.3.3 Girsanov's Theorem

This theorem states that, in continuous time, changing the probability measure changes only the drift, not volatility or correlation of the process.

- Girsanov's theorem states that if
  1. $B^{\mathbb{P}}$ is a vector of $\mathbb{P}$-Brownian motions;
  2. ξ is a strictly positive $\mathbb{P}$-martingale;
  3. $\mathbb{Q}$ is a probability measure satisfying (4.1): $\mathbb{Q}(A) = \text{prob}^{\mathbb{Q}}(A) = \mathbb{E}^{\mathbb{P}}[\xi_T \, 1_A]$;
  4. $d\xi/\xi = -\lambda^\top dB^{\mathbb{P}}$ for a vector $\lambda = (\lambda_1, \ldots, \lambda_N)^\top$ of stochastic processes,

  then

$$B_t^{\mathbb{Q}} = B_t^{\mathbb{P}} + \int_0^t \lambda_u \, du, \tag{4.4}$$

  or equivalently: $$dB_t^{\mathbb{Q}} = dB_t^{\mathbb{P}} + \lambda_t \, dt, \tag{4.5}$$

  defines a vector of independent $\mathbb{Q}$-Brownian motions.

- The main implication of Girsanov's theorem is that changing measures changes drifts but does not affect volatilities or correlations.

- To see this, let Y be an Itô process with

$$dY = \mu\, dt + \sum_{n=1}^{N} \sigma_n\, dB_{n,t}^{\mathbb{P}} \tag{4.6}$$

and recall equation (4.5), which is reproduced below:

$$dB_t^{\mathbb{Q}} = dB_t^{\mathbb{P}} + \lambda_t\, dt, \tag{4.5}$$

which implies that for the nth Brownian motion:

$$dB_{n,t}^{\mathbb{P}} = dB_{n,t}^{\mathbb{Q}} - \lambda_{n,t}\, dt. \tag{4.7}$$

- Substituting $dB_{n,t}^{\mathbb{P}} = dB_{n,t}^{\mathbb{Q}} - \lambda_{n,t}\, dt$ into equation (4.6), leads to:

$$\begin{aligned} dY &= \mu\, dt + \sum_{n=1}^{N} \sigma_n \left(dB_{n,t}^{\mathbb{Q}} - \lambda_{n,t}\, dt \right) \\ &= \underbrace{\left(\mu - \sum_{n=1}^{N} \sigma_n\, \lambda_{n,t} \right)}_{\text{drift}} dt + \underbrace{\sum_{n=1}^{N} \sigma_n}_{\text{volatility}}\, dB_{n,t}^{\mathbb{Q}}, \qquad \text{... collecting "}dt\text{" terms} \end{aligned} \tag{4.8}$$

we see that

⋆ only the drift has changed (from μ to $\mu - \sum_{n=1}^{N} \sigma_n\, \lambda_{n,t}$)

⋆ but the volatility coefficients are still the same (σ_n).

4.3.4 Understanding Girsanov's Theorem

- Girsanov's Theorem tells us
  - ⋆ how to change the probability measure (or numeraire)
  - ⋆ when we change the drift of a process.

- On the next few pages, we provide the intuition underlying Girsanov's Theorem.

- In the section below,
  - ⋆ instead of considering two Brownians $B_t^{\mathbb{P}}$ and $B_t^{\mathbb{Q}}$,
  - ⋆ we will consider two generic Brownian motions denoted B_t and B_t^*,
  - ⋆ with the associated probability measures denoted prob and prob*.

- Let λ be a constant (at least for now, to keep things simple) and let B be a Brownian motion under some measure prob.

- Let

$$\begin{aligned} & B_t^* = B_t + \lambda\, t \\ \text{or equivalently:} \quad & dB_t^* = dB_t + \lambda\, dt \end{aligned} \tag{4.9}$$

- So, we have obtained the process B^* by changing the drift of the process B.

- Girsanov's Theorem tells us how to change the probability measure so that
  - ⋆ drift of $B^* = 0$;
  - ⋆ implying that B^* is a martingale;
  - ⋆ hence, from Levy's Theorem, B^* is a Brownian motion.

Girsanov's Theorem: A Discrete-Time Exposition

- Below, we explain the meaning of Girsanov's Theorem in discrete time.
- Consider discrete time periods of length Δt.
- Approximate ΔB by a binomial process that can take the values

  $+\sqrt{\Delta t}$ with $\text{prob}_u = 1/2$;

  $-\sqrt{\Delta t}$ with $\text{prob}_d = 1/2$.
- Therefore, just like for a true Brownian motion,

$$\mathbb{E}[\Delta B] = 0$$

$$\text{Var}[\Delta B] = \Delta t.$$

- Recall from (4.9) on Page 324 that $dB^* = \lambda dt + dB$, which implies that
  - ⋆ in the up state: $\Delta B^* = \lambda \Delta t + \sqrt{\Delta t}$;
  - ⋆ in the down state: $\Delta B^* = \lambda \Delta t - \sqrt{\Delta t}$.
- We now wish to find new probabilities prob_u^* and prob_d^* so that B^* has a zero drift under the probability measure prob*. Let the new probabilities be:

$$\text{prob}_u^* = \frac{1}{2}\left(1 - \lambda\sqrt{\Delta t}\right);$$

$$\text{prob}_d^* = \frac{1}{2}\left(1 + \lambda\sqrt{\Delta t}\right).$$

- Under the new probability measure prob*:

$$\begin{aligned}\mathbb{E}^*[\Delta B^*] &= \text{prob}_u^*\left(\lambda\Delta t + \sqrt{\Delta t}\right) + \text{prob}_d^*\left(\lambda\Delta t - \sqrt{\Delta t}\right)\\ &= \frac{1}{2}\left(1 - \lambda\sqrt{\Delta t}\right)\left(\lambda\Delta t + \sqrt{\Delta t}\right) + \frac{1}{2}\left(1 + \lambda\sqrt{\Delta t}\right)\left(\lambda\Delta t - \sqrt{\Delta t}\right)\\ &= 0.\end{aligned}$$

- Therefore, B^* is a martingale under the measure prob*.

Girsanov's Theorem: Ratio of Probabilities of Paths

- Changing the probabilities of each "branch" of the binomial tree in this way implies that the probability of a path through the tree is changed as follows.
- The probability of a path is the product of the probabilities of the branches.
- Therefore, the ratio of the path probabilities is:

$$\frac{\text{prob}^*(\text{path through time } t)}{\text{prob}(\text{path through time } t)} = \frac{\text{prob}^*(\text{path through time } t-\Delta t)}{\text{prob}(\text{path through time } t-\Delta t)} \times \frac{\text{prob}^*(\text{branch at } t)}{\text{prob}(\text{branch at } t)} \tag{4.10}$$

- Recall our definition of $B(t)$, which is
  - $\star$ $+\sqrt{\Delta t}$ in the up state, and
  - $\star$ $-\sqrt{\Delta t}$ in the down state.
- This definition of $B(t)$ implies that the ratio of probabilities is:

$$\frac{\text{prob}^*(\text{up branch at } t)}{\text{prob}(\text{up branch at } t)} = \frac{\frac{1}{2}(1-\lambda\sqrt{\Delta t})}{\frac{1}{2}} = 1-\lambda(+\sqrt{\Delta t}) = 1-\lambda\Delta B(t) \tag{4.11}$$

$$\frac{\text{prob}^*(\text{down branch at } t)}{\text{prob}(\text{down branch at } t)} = \frac{\frac{1}{2}(1+\lambda\sqrt{\Delta t})}{\frac{1}{2}} = 1-\lambda(-\sqrt{\Delta t}) = 1-\lambda\Delta B(t) \tag{4.12}$$

- Therefore, substituting (4.11) and (4.12) in (4.10), we get

$$\frac{\text{prob}^*(\text{path through time } t)}{\text{prob}(\text{path through time } t)} = \frac{\text{prob}^*(\text{path through time } t-\Delta t)}{\text{prob}(\text{path through time } t-\Delta t)} \times \Big(1-\lambda\Delta B(t)\Big) \tag{4.13}$$

- We start by reproducing (4.13)

$$\frac{\text{prob}^*(\text{path through time } t)}{\text{prob}(\text{path through time } t)} = \frac{\text{prob}^*(\text{path through time } t-\Delta t)}{\text{prob}(\text{path through time } t-\Delta t)} \times \Big(1-\lambda\Delta B(t)\Big). \tag{4.13}$$

- Defining $\xi(t)$ as the ratio of probabilities through time t, we can then write (4.13) in terms of $\xi(t)$:

$$\begin{aligned}
\xi(t) &= \xi(t-\Delta t) \times \Big[1-\lambda\Delta B(t)\Big] \\
&= \xi(t-\Delta t) - \xi(t-\Delta t)\lambda\Delta B(t) \qquad \ldots \text{expanding the square bracket} \\
\frac{\xi(t)-\xi(t-\Delta t)}{\xi(t-\Delta t)} &= -\lambda\Delta B(t) \qquad \ldots \text{percentage change in } \xi \\
\frac{d\xi(t)}{\xi(t)} &= -\lambda\, dB(t) \qquad \ldots \text{in continuous time.}
\end{aligned} \tag{4.14}$$

  which gives us exactly the expression in Item 4 on Page 320.

- We start by reproducing the last equation on the previous page:

$$\frac{d\xi(t)}{\xi(t)} = -\lambda\, dB(t). \tag{4.14}$$

- Note that $\xi(0) = 0$, so ξ is a Brownian motion with the explicit solution of (4.14) being:

$$\xi(t) = \exp\left[-\frac{1}{2}\lambda^2 t - \lambda B(t)\right] \quad \text{if } \lambda \text{ is constant,} \tag{4.15}$$

  and, if λ is not constant, then we would get

$$\xi(t) = \exp\left[-\frac{1}{2}\int_0^t \lambda^2(u)\, du - \int_0^t \lambda(u)\, dB(u)\right]. \tag{4.16}$$

- Therefore, (4.15) or (4.16) define the ratio of path probabilities prob*/prob, such that B^* is a martingale under prob*.

Girsanov's Theorem Restated

- Girsanov's Theorem basically says: Because B^* is continuous and its quadratic variation through each date t is equal to t (because the addition of $\lambda\, t$ to B does not alter the quadratic variation of B), Levy's Theorem implies that B^* must be a Brownian motion with respect to the measure prob*.

- In the formal statement of Girsanov's Theorem, there is no reference to ratio of path probabilities because individual paths have zero probability under prob or prob*.

- Instead, the theorem states that B^* is converted to a Brownian motion by multiplying the probability of any *event* (*set* of paths) by the conditional expectation of ξ, given the event; that is,

  $$\text{prob}^*(A) = \mathbb{E}\left[1_A\, \xi(T)\right]$$

  and for any random variable X whose mean exists, we have

  $$\mathbb{E}^*[X] = \mathbb{E}[\xi(T)\, X].$$

Girsanov's Theorem: Reversing the Process

- We can also reverse the above process:
  - ⋆ if we are given the random variable ξ defining the change of measure,
  - ⋆ we can then find the appropriate change in the drift of a Brownian motion

- This is useful, for example, to find the drift of volatility or the interest rate, which are not prices, and so we cannot use the change of measure interpretation.

- To do this, define $\xi_t = \mathbb{E}_t[\xi_T]$.
- Equations (4.15) and (4.16) on Page 330 imply that

$$\frac{d\xi(t)}{\xi(t)} = -\lambda\, dB(t).$$

- Consequently, multiplying both sides by $-dB(t)$, we get

$$-dB(t) \times \frac{d\xi(t)}{\xi(t)} = -dB(t) \times \Big(-\lambda\, dB(t)\Big) = \lambda\, dt.$$

- Then, the definition below, gives us a Brownian motion B^* relative to the measure prob*:

$$dB^*(t) = dB(t) - dB(t)\frac{d\xi(t)}{\xi(t)} = dB(t) + \lambda\, dt.$$

- That is, the drift of B under the measure prob* is: $dB(t)\frac{d\xi(t)}{\xi(t)}$.

4.4 Change of Drift and Change of Numeraire

- We now wish to understand how the distribution of an asset price (that is, its probability measure) changes as we change the numeraire.
- Let $\mathcal{P}^S$ be the price of a non-dividend paying asset (stock), and $r(t)$ the instantaneous risk-free rate at t, with $\mathcal{P}^D(t)$ the price of the bank deposit:

$$\frac{d\mathcal{P}^S}{\mathcal{P}^S} = \mu\, dt + \sigma\, dB^{\mathbb{P}}, \tag{4.17}$$

$$\frac{d\mathcal{P}^D}{\mathcal{P}^D} = r\, dt, \tag{4.18}$$

  where $B^{\mathbb{P}}$ is a Brownian motion under the physical probability measure $\mathbb{P}$ and μ and σ are quite general random processes.

- We study the dynamics of the asset price $\mathcal{P}^S$ under two probability measures:
  - ⋆ The bank deposit $\mathcal{P}^D$ is the numeraire;
  - ⋆ The asset $\mathcal{P}^S$ is the numeraire.

- In each case, we follow the same line of attack:

  Step 1:
  Recall that the ratio of the asset price to the numeraire asset price must be a martingale;

  Step 2:
  Use Itô's formula to compute the drift of this ratio;

  Step 3:
  Identify the drift of $d\mathcal{P}^S/\mathcal{P}^S$ under the new measure using the fact that a martingale has zero drift.

- We will use these results for deriving the Black-Scholes price for European options.

4.4.1 Money-Market Account as Numeraire (Risk-Neutral Probability)

- If the bank deposit is the numeraire,
  then the process $Y \equiv \mathcal{P}^S/\mathcal{P}^D$ is a prob$^{\mathbb{Q}}$-martingale.
- Using Itô's formula for ratios (see Slide 256 of Chapter 2)

$$\begin{aligned}
\frac{dY}{Y} &= \frac{d\mathcal{P}^S}{\mathcal{P}^S} - \frac{d\mathcal{P}^D}{\mathcal{P}^D} && \text{... only two terms because } \mathcal{P}^D \text{ is risk-free and so has only } dt \text{ terms} \\
&= \Big[\mu\, dt + \sigma\, dB^{\mathbb{P}}\Big] - r\, dt && \text{... substituting for the return processes} \\
&= \Big[\mu\ - r\Big]\, dt + \sigma\, dB^{\mathbb{P}} && \text{... collecting } dt \text{ and } dB^{\mathbb{P}} \text{ terms} \\
&= \sigma \left[\frac{\mu\ - r}{\sigma} dt + dB^{\mathbb{P}}\right] && \text{... dividing and multiplying by } \sigma \\
&= \sigma\, dB^{\mathbb{Q}}, && \text{... from the definition of } dB^{\mathbb{Q}}, \text{ which is given below}
\end{aligned}$$

where we define the Itô process $B^{\mathbb{Q}}$ by $B_0^{\mathbb{Q}} = 0$ and

$$dB^{\mathbb{Q}} = dB^{\mathbb{P}} + \frac{\mu\ - r}{\sigma} dt. \tag{4.19}$$

- For $Y \equiv \mathcal{P}^S/\mathcal{P}^D$ to be a prob$^{\mathbb{Q}}$-martingale, it must be that $B^{\mathbb{Q}}$ is a local-martingale under prob$^{\mathbb{Q}}$. It then follows from Levy's theorem that $B^{\mathbb{Q}}$ is a prob$^{\mathbb{Q}}$-Brownian motion.
- Substituting (4.19), reproduced below:

$$dB^{\mathbb{Q}} = dB^{\mathbb{P}} + \frac{\mu - r}{\sigma} dt \quad \Longleftrightarrow \quad dB^{\mathbb{P}} = dB^{\mathbb{Q}} - \frac{\mu - r}{\sigma} dt \tag{4.19}$$

into (4.17), which is:

$$\begin{aligned} \frac{d\mathcal{P}^S}{\mathcal{P}^S} &= \mu\, dt + \sigma\, dB^{\mathbb{P}} && \text{. . . this was given to us} \\ &= \mu\, dt + \sigma \left(dB^{\mathbb{Q}} - \frac{\mu - r}{\sigma} dt \right) && \text{. . . substituting for } dB^{\mathbb{P}} \text{ from (4.19)} \\ &= \mu\, dt + \sigma\, dB^{\mathbb{Q}} - \mu dt + r dt. && \text{. . . canceling the } \mu\, dt \text{ and } -\mu\, dt \text{ terms} \\ &= r\, dt + \sigma\, dB^{\mathbb{Q}}. && \text{. . . note that the drift is now the risk-free rate} \end{aligned} \tag{4.17}$$

- Thus, the process for $d\mathcal{P}^S/\mathcal{P}^S$ under the risk-neutral prob$^{\mathbb{Q}}$ measure is:

$$\frac{d\mathcal{P}^S}{\mathcal{P}^S} = r\, dt + \sigma\, dB^{\mathbb{Q}}, \tag{4.20}$$

that we will write in the form in which we will need to use it later

$$\begin{aligned} d \log \mathcal{P}^S &= \frac{d\mathcal{P}^S}{\mathcal{P}^S} - \frac{1}{2}\sigma^2 dt && \text{. . . using Itô's formula applied to logs, as on page 256} \\ &= \left(r - \frac{1}{2}\sigma^2 \right) dt + \sigma\, dB^{\mathbb{Q}}. \end{aligned} \tag{4.21}$$

4.4.2 Risky Asset as the Numeraire

- If now the risky asset is the numeraire (instead of the risk-free asset), then the process $Y \equiv \mathcal{P}^D/\mathcal{P}^S$ is a prob$^{\mathbb{S}}$-martingale.
- Using Itô's formula for ratios (see Slide 256 of Chapter 2)

$$\begin{aligned}\frac{dY}{Y} &= \frac{d\mathcal{P}^D}{\mathcal{P}^D} - \frac{d\mathcal{P}^S}{\mathcal{P}^S} + \left(\frac{d\mathcal{P}^S}{\mathcal{P}^S}\right)^2 \\ &= r\,dt - \left[\mu\,dt + \sigma\,dB^{\mathbb{P}}\right] + \sigma^2\,dt \\ &= -\left[\mu - r - \sigma^2\right]dt - \sigma\,dB^{\mathbb{P}} \\ &= -\sigma\left[\frac{\mu - r - \sigma^2}{\sigma}dt + dB^{\mathbb{P}}\right] \\ &= -\sigma\,dB^{\mathbb{S}},\end{aligned}$$

where we define the Itô process $B^{\mathbb{S}}$ by $B_0^{\mathbb{S}} = 0$ and

$$dB^{\mathbb{S}} = dB^{\mathbb{P}} + \frac{\mu - r - \sigma^2}{\sigma}dt. \tag{4.22}$$

- For $Y \equiv \mathcal{P}^D/\mathcal{P}^S$ to be a prob$^{\mathbb{S}}$-martingale, it must be that $B^{\mathbb{S}}$ is a local-martingale under prob$^{\mathbb{S}}$. It then follows from Levy's theorem that $B^{\mathbb{S}}$ is a prob$^{\mathbb{S}}$-Brownian motion.
- Substituting (4.22), reproduced below:

$$dB^{\mathbb{S}} = dB^{\mathbb{P}} + \frac{\mu - r - \sigma^2}{\sigma}dt \quad \Longleftrightarrow \quad dB^{\mathbb{P}} = dB^{\mathbb{S}} - \frac{\mu - r - \sigma^2}{\sigma}dt \tag{4.22}$$

into (4.17), also reproduced below, leads to

$$\begin{aligned}\frac{d\mathcal{P}^S}{\mathcal{P}^S} &= \mu\,dt + \sigma\,dB^{\mathbb{P}} \qquad (4.17)\\ &= \mu\,dt + \sigma\left(dB^{\mathbb{S}} - \frac{\mu - r - \sigma^2}{\sigma}dt\right) \\ &= \mu\,dt + \sigma\,dB^{\mathbb{S}} - \mu\,dt + r\,dt + \sigma^2 dt \\ &= \left(r + \sigma^2\right)dt + \sigma\,dB^{\mathbb{S}}.\end{aligned}$$

- This gives us the process for $d\mathcal{P}^S/\mathcal{P}^S$ under the prob$^{\mathbb{S}}$ measure:

$$\frac{d\mathcal{P}^S}{\mathcal{P}^S} = \left(r + \sigma^2\right) dt + \sigma\, dB^{\mathbb{S}}, \tag{4.23}$$

that we will write in the form in which we will need to use it later

$$\begin{aligned} d\log \mathcal{P}^S &= \frac{d\mathcal{P}^S}{\mathcal{P}^S} - \frac{1}{2}\sigma^2 dt \qquad \text{. . . using Itô's formula applied to logs, as on page 256} \\ &= \left(r + \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{S}}. \end{aligned} \tag{4.24}$$

4.5 Tail Probabilities of Geometric Brownian Motions

- Thus, the process for the risky asset under the two numeraires is:

$$d\log \mathcal{P}^S = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{Q}} \qquad \text{. . . with bank-deposit as numeraire,}$$

$$d\log \mathcal{P}^S = \left(r + \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{S}} \qquad \text{. . . with stock as numeraire.}$$

- Because we want to do our next set of calculations only once, we write these two processes using common notation:

$$d\log \mathcal{P}^S = \alpha\, dt + \sigma\, dB, \tag{4.25}$$

where α takes on different values, $(r - \frac{1}{2}\sigma^2)$ or $(r + \frac{1}{2}\sigma^2)$, depending on the whether the Brownian dB is under $\mathbb{Q}$ or $\mathbb{S}$.

- We will assume in the rest of this section that α and σ are constants.

- The essential calculation in pricing options is to compute $\text{prob}(\mathcal{P}_T^S > K)$,
  - ⋆ for a constant K (strike price of an option), and
  - ⋆ where prob denotes the probabilities associated with a particular numeraire at date 0 (the date we are pricing the option); for example, $\text{prob}^{\mathbb{Q}}$ and $\text{prob}^{\mathbb{S}}$.

- Equation (4.25), reproduced from the previous page,

$$d \log \mathcal{P}^S = \alpha \, dt + \sigma \, dB, \tag{4.25}$$

gives us, upon integration:

$$\log \mathcal{P}_T^S = \log \mathcal{P}_0^S + \alpha \, T + \sigma \, B_T,$$

which we will use on the next page.

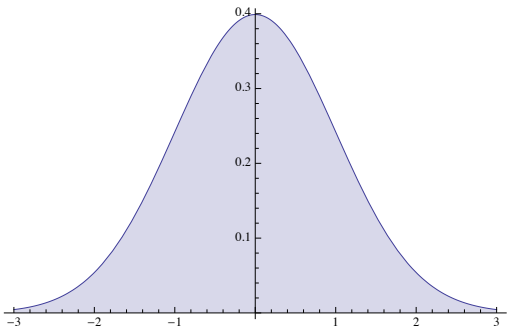
- In order to compute $\text{prob}(\mathcal{P}_T^S > K)$, note that:

$$\begin{aligned}
\mathcal{P}_T^S > K &\iff \log \mathcal{P}_T^S > \log K && \text{. . . taking logs on both sides} \\
&\iff \log \mathcal{P}_0^S + \alpha \, T + \sigma \, B_T > \log K && \text{. . . Using the result: } \log \mathcal{P}_T^S = \log \mathcal{P}_0^S + \alpha \, T + \sigma \, B_T \\
&\iff \sigma \, B_T > \log K - \log \mathcal{P}_0^S - \alpha \, T && \text{. . . Isolating the term with the random variable, } B_T \\
&\iff \frac{B_T}{\sqrt{T}} > \frac{\log K - \log \mathcal{P}_0^S - \alpha \, T}{\sigma \sqrt{T}} && \text{. . . dividing by } \sigma/\sqrt{T} \text{ to make variance equal to 1} \\
&\iff -\frac{B_T}{\sqrt{T}} < \frac{\log \mathcal{P}_0^S - \log K + \alpha \, T}{\sigma \sqrt{T}} && \text{. . . multiplying by } -1 \text{ to change direction of inequality} \\
&\iff -\frac{B_T}{\sqrt{T}} < \frac{\log \left(\frac{\mathcal{P}_0^S}{K} \right) + \alpha \, T}{\sigma \sqrt{T}} && \text{. . . collecting log terms}
\end{aligned} \tag{4.26}$$

- Equation (4.26), reproduced below, tells us that:

$$\mathcal{P}_T^S > K \iff -\frac{B_T}{\sqrt{T}} < \frac{\log\left(\frac{\mathcal{P}_0^S}{K}\right) + \alpha T}{\sigma\sqrt{T}}. \tag{4.26}$$

- The random variable $-\frac{B_T}{\sqrt{T}}$ has a standard Normal distribution—with mean zero and variance equal to one.



Probability Distribution Function

for a Normal Distribution with $\mu = 0$ and $\sigma = 1$.

Test Your Understanding 4.1 (Normal Distribution)

Suppose d is variable with a standard Normal distribution.

1. What is $N(0)$?
2. What is $N(-\infty)$?
3. What is $N(+\infty)$?
4. What is $N(1.64)$?
5. What is $N(1.96)$?

Space for your work.

- We reproduce yet again the equation

$$\mathcal{P}_T^S > K \quad \Longleftrightarrow \quad -\frac{B_T}{\sqrt{T}} < \frac{\log\left(\frac{\mathcal{P}_0^S}{K}\right) + \alpha\, T}{\sigma\sqrt{T}}. \tag{4.26}$$

- Let $N(d)$ denote probability that a standard Normal variable is less than d.

- Then, Equation (4.26) above tells us that

$$\text{prob}\left(\mathcal{P}_T^S > K\right) \;=\; N(d), \quad \text{where} \quad d = \frac{\log\left(\frac{\mathcal{P}_0^S}{K}\right) + \alpha\, T}{\sigma\sqrt{T}}, \tag{4.27}$$

$$\text{prob}\left(\mathcal{P}_T^S < K\right) \;=\; 1 - N(d) = N(-d). \tag{4.28}$$

- The value of α we use, will depend on our choice of numeraire; that is, whether we wish to compute $\text{prob}^{\mathbb{Q}}\left(\mathcal{P}_T^S > K\right)$ or $\text{prob}^{\mathbb{S}}\left(\mathcal{P}_T^S > K\right)$.

4.5.1 Deriving $\text{prob}^{\mathbb{S}}(\mathcal{P}_T^S > K) = N(d_1)$

- We start with the equation on the previous page:

$$\text{prob}\left(\mathcal{P}_T^S > K\right) = N(d), \quad \text{where} \quad d = \frac{\log\left(\frac{\mathcal{P}_0^S}{K}\right) + \alpha T}{\sigma\sqrt{T}}, \tag{4.27}$$

- For the case where the numeraire is the underlying risky asset, we have that

$$\text{prob}^{\mathbb{S}}\left(\mathcal{P}_T^S > K\right) = N(d_1), \tag{4.29}$$

$$\text{where} \quad d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{K}\right) + (r + \frac{1}{2}\sigma^2)\,T}{\sigma\sqrt{T}},$$

$$= \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-rT}K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}. \tag{4.30}$$

4.5.2 Deriving $\text{prob}^{\mathbb{Q}}(\mathcal{P}_T^S > K) = N(d_2)$

- We repeat the analysis with the numeraire being the bank deposit.

$$\text{prob}\left(\mathcal{P}_T^S > K\right) = N(d), \quad \text{where} \quad d = \frac{\log\left(\frac{\mathcal{P}_0^S}{K}\right) + \alpha T}{\sigma\sqrt{T}}. \tag{4.27}$$

- Then, for the case where the numeraire is the underlying bank deposit,

$$\text{prob}^{\mathbb{Q}}\left(\mathcal{P}_T^S > K\right) = N(d_2), \tag{4.31}$$

$$\text{where} \quad d_2 = \frac{\log\left(\frac{\mathcal{P}_0^S}{K}\right) + (r - \frac{1}{2}\sigma^2)\,T}{\sigma\sqrt{T}},$$

$$= \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-rT}K}\right) - \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \tag{4.32}$$

$$= d_1 - \sigma\sqrt{T}. \tag{4.33}$$

4.6 Summary

1. Change of Measure

If $\xi_t = M_t \mathcal{P}_t^D$, for each $t < T$,
then: $\text{prob}^{\mathbb{Q}}(A) = \mathbb{E}^{\mathbb{P}}[\xi_T \, 1_A]$,

so that for any random variable X depending on date-T information:

$$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[\xi_T \, X].$$

2. Girsanov's theorem states that

- if:
  - ⋆ $B^{\mathbb{P}}$ is a vector of $\mathbb{P}$-Brownian motions;
  - ⋆ ξ is a strictly positive $\mathbb{P}$-martingale;
  - ⋆ $\mathbb{Q}$ is a probability measure satisfying $\text{prob}^{\mathbb{Q}}(A) = \mathbb{E}^{\mathbb{P}}[\xi_T \, 1_A]$;
  - ⋆ $d\xi/\xi = \lambda^\top dB^{\mathbb{P}}$ for a vector $\lambda = (\lambda_1, \ldots, \lambda_N)^\top$ of stochastic processes,
- then the following defines a vector of independent $\mathbb{Q}$-Brownian motions:

$$dB_t^{\mathbb{Q}} = dB_t^{\mathbb{P}} + \lambda_t \, dt.$$

- Girsanov's theorem tells us that changing measures
  - ⋆ changes drifts,
  - ⋆ but does not change volatilities of correlations.

3. Price process for risky asset under different measures
If the price process for the risky asset under $\mathbb{P}$ is

$$\frac{d\mathcal{P}^S}{\mathcal{P}^S} = \mu\, dt + \sigma\, dB^{\mathbb{P}},$$

or equivalently: $\quad d\log \mathcal{P}^S = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{P}},$

- then under the probability measure $\mathbb{S}$ the price process is:

$$d\log \mathcal{P}^S = \left(r + \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{S}} \qquad \ldots \mu \text{ has been replaced by } r + \sigma^2.$$

- and, under the probability measure $\mathbb{Q}$ the price process is:

$$d\log \mathcal{P}^S = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{Q}}, \qquad \ldots \mu \text{ has been replaced by } r,$$

4. Tail probability for geometric Brownian motion with stock as numeraire.
Given the process

$$d\log \mathcal{P}^S = \left(r + \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{S}},$$

we have that

$$\text{prob}^{\mathbb{S}}\left(\mathcal{P}_T^S > K\right) = N(d_1), \qquad d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-rT}K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}.$$

5. Tail probability for geometric Brownian motion with bank deposit as numeraire.
Given the process

$$d \log \mathcal{P}^S = \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma\, dB^{\mathbb{Q}},$$

we have that

$$\text{prob}^{\mathbb{Q}}\left(\mathcal{P}_T^S > K\right) = N(d_2), \qquad d_2 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-rT}K}\right) - \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}.$$

4.7 Suggestions for Further Readings

4.7.1 Source material for the lecture notes

- Much of this material is from Back (2005), which is his book on derivatives; in particular, Section 4.3.4 of our notes is from Back (2005, Appendix B.1).
- See also Back (2017, Chapter 15) and Dumas and Luciano (2017, Chapter 8).
- The original article with Girsanov's theorem is Girsanov (1960).
- Harrison and Kreps (1979) develop further the concept of a risk-neutral probability measure.
- The change of numeraire approach is from Geman, El Karoui, and Rochet (1995) and further developed in Schroder (1999).

4.7.2 Potential applications and extensions of lecture notes

- For a textbook treatment of valuation using different numeraires, see Vecer (2011), who uses this approach:
  - ⋆ in three kinds of models—binomial, diffusion, and jump models
  - ⋆ to value various options—European options on equities and interest-rates, American options, barrier options, lookback options, quantos, etc.
- The books by Back (2005) and Bingham and Kiesel (2013) provide a risk-neutral approach to the valuation and hedging of contingent contracts.

- Breeden and Litzenberger (1978) explain how one can use option prices to obtain Arrow-Debreu state prices.
- For a review paper on using options to extract the risk-neutral probability, see Figlewski (2018).
- For papers on the topic of "recovering" the physical measure $\mathbb{P}$ from the risk-neutral measure $\mathbb{Q}$, see Ross (2015), Borovička, Hansen, and Scheinkman (2016), and Schneider and Trojani (2019).
- The SDF and RNP approaches are not very different, so for further applications and extensions, please see the references cited in Chapter 3.7.2 on page 302.

4.7.3 Ph.D. elective courses offered on related topics

- Pietro Veronesi taught
  - ⋆ *Recent Developments in Portfolio Allocation and Macro-Finance Model* in 2009, and
  - ⋆ *Macroeconomic and Political Uncertainty and the Dynamics of Asset Prices* in 2012, 2014, and 2016.
  - ⋆ He taught a course on *Fixed Income Securities* in 2021.
- Yacine Aït-Sahalia taught *Financial Econometrics of Jumps* in 2011.
- Jerome Detemple taught a course on *Derivative Securities* in 2011.
- Ravi Bansal taught a course on *Long-Run Risks in Asset Prices* in 2011.
- Mikhail Chernov taught *Empirical Option Pricing* in 2012 and 2014.
- Federico Bandi taught *The Econometrics of Continuous-Time Models* in 2012 and 2015, and *Estimation of Continuous-Time Models* in 2013.
- Torben Andersen taught *High-Frequency Asset Pricing* in 2013 and 2015.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 5

Black-Scholes Option Pricing Model

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EDHEC Business School

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5.1 Plan for the Day

Last Time

- Change of numeraire/measure and drift in continuous time.

Today

- **Understand the Black-Scholes-Merton model of option pricing:**
  - ⋆ Valuing call and put options.
  - ⋆ Greeks and hedging of options.

For Next Time

- Review the material covered in class.
- Read Back, Chapters 15–16 and Dumas & Luciano, Chapter 6.
- Do the practice problems.

5.2 Main Points for Today

1. The stochastic discount factor (SDF) in the option-pricing model
2. Valuing
   - digital options and share digital options
   - call and put options on stock (using the prices of digitals and share digitals)
3. Hedging of options, and option greeks
4. Implied volatility

- In the last class, we developed the ideas of SDF and RNP in continuous time.
- In today's class, we use these ideas to price/hedge/replicate options on stock.
- In the next class, we will use the same methods to value many other options.

5.3 The Black-Scholes Option-Pricing Model

5.3.1 Assumptions of the Black-Scholes Model

- Let $\mathcal{P}^S$ be the price of a non-dividend paying asset (stock)

$$\frac{d\mathcal{P}^S}{\mathcal{P}^S} = \mu_t\, dt + \sigma_t\, dB^{\mathbb{P}}, \tag{5.1}$$

  where B is a single Brownian motion under the physical probability measure $\mathbb{P}$ and μ and σ are quite general random processes.

- Let $r(t)$ denote the instantaneous risk-free rate at t, with $\mathcal{P}^D(t)$ the price of the bank deposit, where

$$\mathcal{P}^D(t) = \exp\left(\int_0^t r(u)du\right)$$
$$\frac{d\mathcal{P}^D}{\mathcal{P}^D} = r_t\, dt.$$

- For the rest of this lecture, we assume that
  - ⋆ the risk-free rate r is constant, and
  - ⋆ the volatility σ is constant.
- These assumptions, along with the assumption that $M\mathcal{P}^D$ is a martingale, imply that the market is complete for payoffs that depend only on the history of $\mathcal{P}^S$.
- Later, we will see how to extend the model to allow for:
  - ⋆ Non-constant interest rate;,
  - ⋆ Non-constant volatility of the risky asset; and,
  - ⋆ Dividends paid by the risky asset;

5.3.2 European Call Option in Terms of Digital and Share Digitals

- We wish to find the price today ($t = 0$), of a call option that has a payoff at a future date T of $\max(0, \mathcal{P}_T^S - K)$, where K denotes the strike price.

- Let A denote the set of states of the world such that $\mathcal{P}_T^S \geq K$, and let 1_A denote the random variable that equals 1 when $\mathcal{P}_T^S \geq K$.

- Then, one can write the payoff to the call option as

$$\max(0, \mathcal{P}_T^S - K) = \mathcal{P}_T^S \, 1_A - K \, 1_A. \tag{5.2}$$

- Reproducing the last equation on the previous slide:

$$\max(0, \mathcal{P}_T^S - K) = \mathcal{P}_T^S \, 1_A - K \, 1_A. \tag{5.2}$$

- The random variable $K \, 1_A$ is the payoff to what is called a digital or binary option, which pays K when $\mathcal{P}_T^S > K$, and zero otherwise.

- The random variable $\mathcal{P}_T^S \, 1_A$ is the payoff to what is called a share digital option, which pays one share worth $\mathcal{P}_T^S$ when $\mathcal{P}_T^S > K$, and zero otherwise.

- From (5.2), we see that a European call option is equivalent to
  - ⋆ a long position in a share digital, and
  - ⋆ a short position in a digital option.

5.3.3 Valuation of European Call in Terms of Digital and Share Digital

- We start by re-writing the payoff at maturity, T, to a European call option:

$$\max(0, \mathcal{P}_T^S - K) = \mathcal{P}_T^S \, 1_A - K \, 1_A. \tag{5.2}$$

- The price at $t = 0$ of this date-T payoff is

$$\mathcal{P}_0^C = \mathbb{E}^{\mathbb{P}} \left[M_T \, \mathcal{P}_T^S \, 1_A \right] - K \, \mathbb{E}^{\mathbb{P}} \left[M_T \, 1_A \right], \tag{5.3}$$

  where M_T is the SDF that adjusts for time and for risk of the payoffs.

- As we show below, it will be convenient to use
  - ⋆ the risky asset as the numeraire to value the first part of the payoff, and
  - ⋆ the bank-deposit as the numeraire to value the second part.

- Recall that $M_T \, \mathcal{P}_T^S / \mathcal{P}_0^S$ is the Radon-Nikodym derivative in the change of measure to $\text{prob}^{\mathbb{S}}$. Therefore,

$$\begin{aligned}
\mathbb{E}^{\mathbb{P}} \left[M_T \, \mathcal{P}_T^S \, 1_A \right] &= \mathcal{P}_0^S \, \mathbb{E}^{\mathbb{P}} \left[\frac{M_T \, \mathcal{P}_T^S}{\mathcal{P}_0^S} \, 1_A \right] && \text{. . . multiplying and dividing by } \mathcal{P}_0^S \\
&= \mathcal{P}_0^S \, \mathbb{E}^{\mathbb{S}} \left[1_A \right] && \text{. . . changing probability measure to } \mathbb{S} \\
&= \mathcal{P}_0^S \, \text{prob}^{\mathbb{S}}(A) && \text{. . . expected value of } 1_A \text{ is simply probability of } A \\
&= \mathcal{P}_0^S \, N(d_1), && \text{. . . we have shown earlier that } \text{prob}^{\mathbb{S}}(A) = N(d_1)
\end{aligned}$$

  where in the last class (Chapter 4), we already derived the expression for $\text{prob}^{\mathbb{S}}(A)$ in Equation (4.29) on Page 349, which we reproduce below:

$$\text{prob}^{\mathbb{S}} \left(\mathcal{P}_T^S > K \right) = N(d_1), \qquad d_1 = \frac{\log \left(\frac{\mathcal{P}_0^S}{e^{-rT} K} \right) + \frac{1}{2} \sigma^2 \, T}{\sigma \sqrt{T}}. \tag{4.29}$$

- Similarly, recall that $M_T\,\mathcal{P}_T^D/\mathcal{P}_0^D$ is the Radon-Nikodym derivative in the change of measure to $\text{prob}^{\mathbb{Q}}$. Therefore,

$$\begin{aligned}
\mathbb{E}^{\mathbb{P}}\left[M_T\,K\,1_A\right] &= \frac{\mathcal{P}_0^D}{\mathcal{P}_T^D}\,K\,\mathbb{E}^{\mathbb{P}}\left[\frac{M_T\,\mathcal{P}_T^D}{\mathcal{P}_0^D}\,1_A\right] && \ldots \text{ multiplying and dividing by } \mathcal{P}_0^D/\mathcal{P}_T^D \\
&= e^{-r\,T}\,K\,\mathbb{E}^{\mathbb{Q}}\left[1_A\right] && \ldots \text{ changing measure to } \mathbb{Q};\ \mathcal{P}_0^D = 1;\ 1/\mathcal{P}_T^D = e^{-r\,T} \\
&= e^{-r\,T}\,K\,\text{prob}^{\mathbb{Q}}(A) && \ldots \text{ expected value of } 1_A \text{ is simply probability of } A
\end{aligned}$$

where in the last class (Chapter 4) we already derived the expression for $\text{prob}^{\mathbb{Q}}(A)$ in Equation (4.31) on Page 350,

$$\text{prob}^{\mathbb{Q}}\left(\mathcal{P}_T^S > K\right) = N(d_2), \qquad d_2 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}\,K}\right) - \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}} \qquad (4.31)$$
$$= d_1 - \sigma\,\sqrt{T}.$$

- Substituting these results in (5.3), reproduced below, we get the Black-Scholes formula for the value of a European call option:

$$\begin{aligned}
\mathcal{P}_0^C &= \mathbb{E}^{\mathbb{P}}\left[M_T\,\mathcal{P}_T^S\,1_A\right] - K\,\mathbb{E}^{\mathbb{P}}\left[M_T\,1_A\right] && (5.3) \\
&= \mathcal{P}_0^S\,\text{prob}^{\mathbb{S}}(A) - e^{-r\,T}\,K\,\text{prob}^{\mathbb{Q}}(A) && (5.4) \\
&= \mathcal{P}_0^S\,N(d_1) - e^{-r\,T}\,K\,N(d_2), && (5.5)
\end{aligned}$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}\,K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}}; \qquad (5.6)$$

$$d_2 = d_1 - \sigma\,\sqrt{T}. \qquad (5.7)$$

5.3.4 Understanding the Black-Scholes Call Option Formula

$$\mathcal{P}_0^C = \mathcal{P}_0^S\, N(d_1) - e^{-r\,T}\, K\, N(d_2) \tag{5.5}$$

$$d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}\,K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}}; \qquad d_2 = d_1 - \sigma\,\sqrt{T}. \tag{5.6}$$

- In (5.5), the price today of a call option is the difference between the
  - ⋆ present value of what you will receive, $\mathcal{P}_0^S$,
    multiplied by the appropriate probability of getting this, $N(d_1) = \text{prob}^{\mathbb{S}}(\mathcal{P}_T^S \geq K)$, and;
  - ⋆ present value of what you will deliver, $e^{-r\,T}\,K$,
    multiplied by the appropriate probability of paying this, $N(d_2) = \text{prob}^{\mathbb{Q}}(\mathcal{P}_T^S \geq K)$.
- In (5.6), the argument of log has the ratio of the present values of the quantities to be received and paid, $\mathcal{P}_0^S/(e^{-r\,T}\,K)$; and,
- In (5.6), the difference between d_1 and d_2, $\sigma\,\sqrt{T}$, is the difference in the volatilities of the quantities to be received and paid.
- Many other options will have exactly this same interpretation.

5.3.5 Understanding the Term Missing in Black-Scholes Option Formula

- Just like the physical probabilities did not appear in the two-date binomial model of option pricing that we studied in Chapter 1, the expected return of the risky asset, μ, does *not* appear in the Black-Scholes prices of the option.
- This is because the effect of μ is reflected in the current asset price, $\mathcal{P}_0^S$.
- Another way of saying this is that when μ changes,
  - ⋆ then this will lead to a change in the distribution of $\mathcal{P}_T^S$;
  - ⋆ but, if $\mathcal{P}_0^S$ stays the same, it must imply that the value of M_T has decreased,
  - ⋆ otherwise it would not be possible to still have: $\mathcal{P}_0^S = \mathbb{E}\left[M_T\,\mathcal{P}_T^S\right]$.
  - ⋆ Intuitively, states in which $\mathcal{P}_T^S$ is larger must become cheaper; the exact change in M_T when μ changes is given by (3.18), reproduced below:

$$\frac{dM_t}{M_t} = -r_t\,dt - \left(\frac{\mu_t - r_t}{\sigma_t}\right) dB_t. \tag{3.18}$$

5.3.6 When Not to Use the Black-Scholes Formula

Under what circumstances or market conditions would it be inappropriate to use the Black-Scholes formula for valuing a particular option?

- The logic underlying the Black-Scholes-Merton formula is
  - ⋆ ability to replicate exactly the cashflow to contract we are valuing, and
  - ⋆ the no-arbitrage condition, which says that the value of the replicating portfolio and the contract must be the same.

- So, we cannot use the Black-Scholes-Merton model
  - ⋆ if cannot replicate cashflows to contract because market is incomplete:
    - ∗ the underlying is not traded (the option to make movie sequels);
    - ∗ cashflows are not geometric Brownian motion (stochastic volatility);
    - ∗ cashflows of the contract are not continuous ("big news");
  - ⋆ or, if cannot execute arbitrage because of market frictions when the price of the contract and its replicating portfolio deviate; for example, if there are
    - ∗ transactions costs, or
    - ∗ shortsale constraints.
- It is very important not to use the option pricing models in situations where the conditions underlying the model are not satisfied.

5.4 Option Greeks

We will cover this section in class only briefly—just for completeness.

- The partial derivatives of the option price with respect to the various inputs (parameters) are called the "Greeks."

With Respect To	With-Respect-To Symbol	Greek	Greek Symbol
Stock price	$\mathcal{P}_0^S$	delta	Δ
delta	Δ	gamma	Γ
$-$ Time to maturity	$-T$	theta	Θ
Volatility	σ	vega	$\mathcal{V}$
Interest rate	r	rho	ρ

- Each of these partial derivatives can be calculated by differentiating the Black-Scholes formula using the chain rule from (standard) differential calculus.

- Θ measures how the value of the option changes as t increases, which is equivalent to measuring the change in option value as T decreases.

5.4.1 Preliminaries for Calculating the Greeks

- The Normal distribution function $N(d)$ is:

$$N(d) = \int_{-\infty}^{d} n(x)\,dx = \frac{1}{\sqrt{2\,\pi}} \int_{-\infty}^{d} e^{-\frac{1}{2}x^2}\,dx.$$

- The derivative of the Normal distribution function $N(d)$, is the Normal density function $n(d)$, which is defined as:

$$N'(d) = n(d) = \frac{1}{\sqrt{2\,\pi}} e^{-\frac{1}{2}d^2}.$$

- One can verify the following two results, which are very useful for simplifying the derivation of the Greeks for Black-Scholes options.

$$\mathcal{P}_0^S\, n(d_1) = e^{-r\,T} K\, n(d_2), \quad \text{and} \tag{5.8}$$

$$\frac{\partial d_1}{\partial \mathcal{P}_0^S} = \frac{\partial d_2}{\partial \mathcal{P}_0^S} \qquad \ldots \text{because } d_2 = d_1 - \sigma\sqrt{T}, \tag{5.9}$$

5.4.2 Greeks for the Black-Scholes Call Option Price

- **Delta**
  - ⋆ Measures how the option value changes as the value of the underlying asset changes.

$$\begin{aligned}
\Delta &= \frac{\partial \mathcal{P}_0^C}{\partial \mathcal{P}_0^S} \quad \text{where} \quad \mathcal{P}_0^C = \mathcal{P}_0^S\, N(d_1) - e^{-r\,T} K\, N(d_2) \\
&= N(d_1) + \mathcal{P}_0^S\, n(d_1)\frac{\partial d_1}{\partial \mathcal{P}_0^S} - \underbrace{e^{-r\,T} K\, n(d_2)}_{=\mathcal{P}_0^S\, n(d_1)}\frac{\partial d_2}{\partial \mathcal{P}_0^S} \\
&= N(d_1) + \mathcal{P}_0^S\, n(d_1)\underbrace{\left(\frac{\partial d_1}{\partial \mathcal{P}_0^S} - \frac{\partial d_2}{\partial \mathcal{P}_0^S}\right)}_{=0} \\
&= N(d_1).
\end{aligned} \tag{5.10}$$

- **Gamma**
  - ⋆ Measures the change in the delta of the option as the value of the underlying changes.
  - ⋆ This is a measure of the "curvature" of the option.

$$\Gamma = \frac{\partial^2 \mathcal{P}_0^C}{\partial (\mathcal{P}_0^S)^2}$$

$$= \frac{\partial \Delta}{\partial \mathcal{P}_0^S}, \qquad \text{where} \quad \Delta = N(d_1) = \frac{\partial \mathcal{P}_0^C}{\partial \mathcal{P}_0^S}$$

$$= n(d_1) \frac{\partial d_1}{\partial \mathcal{P}_0^S} \quad \text{where} \quad d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{K}\right) + (r + \sigma^2/2)\, T}{\sigma \sqrt{T}}$$

$$= n(d_1) \frac{1}{\mathcal{P}_0^S \, \sigma \sqrt{T}}.$$

- **Theta**
  - ⋆ Measures the change in the value of the option as time t increases, or equivalently, as the time-to-maturity T decreases.

$$\Theta = \frac{\partial \mathcal{P}_0^C}{\partial (-T)} = -\frac{\partial \mathcal{P}_0^C}{\partial T} \quad \text{where} \quad \mathcal{P}_0^C = \mathcal{P}_0^S \, N(d_1) - e^{-r\,T} \, K \, N(d_2)$$

$$= -\mathcal{P}_0^S \, n(d_1) \frac{\partial d_1}{\partial T} + e^{-r\,T} \, K \, n(d_2) \frac{\partial d_2}{\partial T} - r \, e^{-r\,T} \, K \, N(d_2)$$

$$= \mathcal{P}_0^S \, n(d_1) \left(\frac{\partial d_2}{\partial T} - \frac{\partial d_1}{\partial T} \right) - r \, e^{-r\,T} \, K \, N(d_2)$$

$$= -\mathcal{P}_0^S \, n(d_1) \left(\frac{\sigma}{2\sqrt{T}} \right) - r \, e^{-r\,T} \, K \, N(d_2).$$

Aside: $d_2 = d_1 - \sigma \sqrt{T} \quad \Longrightarrow \quad d_2 - d_1 = -\sigma \sqrt{T}.$

Thus,

$$\frac{\partial}{\partial T} (d_2 - d_1) = -\frac{\sigma}{2\sqrt{T}}.$$

- **Vega**
  - ⋆ This measures the change in the value of the option as the volatility of the underlying changes.

$$\begin{aligned}\mathcal{V} &= \frac{\partial \mathcal{P}_0^C}{\partial \sigma} \qquad \text{where } \mathcal{P}_0^C = \mathcal{P}_0^S\, N(d_1) - e^{-r\,T}\, K\, N(d_2)\\ &= \mathcal{P}_0^S\, n(d_1)\, \frac{\partial d_1}{\partial \sigma} - e^{-r\,T}\, K\, n(d_2)\, \frac{\partial d_2}{\partial \sigma}\\ &= \mathcal{P}_0^S\, n(d_1)\, \left(\frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma}\right)\\ &= \mathcal{P}_0^S\, n(d_1)\, \sqrt{T}.\end{aligned}$$

Aside: $d_2 = d_1 - \sigma\,\sqrt{T} \Longrightarrow d_1 - d_2 = \sigma\,\sqrt{T}$.
Thus,

$$\frac{\partial}{\partial \sigma}\,(d_1 - d_2) = \sqrt{T}.$$

- **Rho**
  - ⋆ This measures the change in the value of the option as the interest rate changes.

$$\begin{aligned}\rho &= \frac{\partial \mathcal{P}_0^C}{\partial r} \qquad \text{where } \mathcal{P}_0^C = \mathcal{P}_0^S\, N(d_1) - e^{-r\,T}\, K\, N(d_2)\\ &= \mathcal{P}_0^S\, n(d_1) \frac{\partial d_1}{\partial r} - e^{-r\,T}\, K\, n(d_2) \frac{\partial d_2}{\partial r} + T\, e^{-r\,T}\, K\, N(d_2)\\ &= \mathcal{P}_0^S\, n(d_1)\, \left(\frac{\partial d_1}{\partial r} - \frac{\partial d_2}{\partial r}\right) + T\, e^{-r\,T}\, K\, N(d_2)\\ &= T\, e^{-r\,T}\, K\, N(d_2).\end{aligned}$$

Aside: $d_2 = d_1 - \sigma\,\sqrt{T} \Longrightarrow d_1 - d_2 = \sigma\,\sqrt{T}$.
Thus,

$$\frac{\partial}{\partial r}\,(d_1 - d_2) = 0.$$

5.4.3 Greeks for put options

- To find the delta and gamma of European put options one can use the results derived above for call options, along with put-call parity (where q is the dividend yield on the stock), which tells us the relation between call and put prices at date 0:

  Put Price = Call Price + $e^{-rT}K - e^{-qT}$ Stock Price.

- The put-call parity relation in terms of our notation:

  $\mathcal{P}^P = \mathcal{P}^C + e^{-rT}K - e^{-qT}\mathcal{P}^S$.

- Differentiating the above equation with respect to $\mathcal{P}^S$, we see that the delta of a put option is the delta of the call option minus e^{-qT}:

  $\frac{\partial \mathcal{P}^P}{\partial \mathcal{P}^S} = \frac{\partial \mathcal{P}^C}{\partial \mathcal{P}^S} + 0 - e^{-qT}$ (and, if $q = 0$, then: $\frac{\partial \mathcal{P}^P}{\partial \mathcal{P}^S} = \frac{\partial \mathcal{P}^C}{\partial \mathcal{P}^S} - 1$).

- And, from the previous result, we see that the gamma of a put option is the same as the gamma of a corresponding call option.

  $\frac{\partial^2 \mathcal{P}^P}{\partial \mathcal{P}^{S2}} = \frac{\partial^2 \mathcal{P}^C}{\partial \mathcal{P}^{S2}}$.

5.5 Delta Hedging

- In Equation (5.10) on Page 382, we have shown that the delta of a call option is equal to $\Delta = N(d_1)$.

- The delta of a call is the number of shares of the underlying asset that one needs to hold in order to replicate a call option using the underlying and risk-free assets.

  ⋆ Recall from Chapter 1, the relation between hedging and replicating.

- That is, it is the number of shares to hold at each instant t in order to obtain a portfolio worth $\max(0, \mathcal{P}_T^S - K)$ at date T.

- If one sells (writes) a call, then one needs to hedge this by replicating a "long call" position; thus, to hedge a short call position, one needs to buy delta shares. This is called delta hedging.

- These statements about replicating calls are verified below.

5.5.1 Return on Portfolio of Risky Asset and Bank Deposit

- Suppose one can invest in a bank deposit and one single risky asset.
- A portfolio with investment in these two assets is defined by the proportion π of wealth invested in the risky asset at each instant t.
- If no funds are invested or withdrawn from the portfolio during the time period $[0, T]$, then the wealth process, W, is defined by

$$\begin{aligned}
\frac{dW}{W} &= (1-\pi_t)\frac{d\mathcal{P}_t^D}{\mathcal{P}_t^D} + \pi_t \frac{d\mathcal{P}_t^S}{\mathcal{P}_t^S} && \ldots \text{return on portfolio is weighted return on investments} \\
&= (1-\pi_t)\, r\, dt + \pi_t \left(\mu\, dt + \sigma\, dB^{\mathbb{P}}\right) && \ldots \text{substituting for } d\mathcal{P}_t^D/\mathcal{P}_t^D \text{ and } d\mathcal{P}_t^S/\mathcal{P}_t^S \\
&= [\pi_t(\mu - r) + r]\, dt + \pi_t\, \sigma\, dB^{\mathbb{P}}. && \ldots \text{collecting "}dt\text{" and "}dB^{\mathbb{P}}\text{" terms}
\end{aligned} \tag{5.11}$$

- This is a self-financing portfolio (no inflows/outflows between 0 and T); we will see this again in the course, when we study portfolio choice.

- We reproduce the last equation:

$$\frac{dW}{W} = [\pi_t(\mu - r) + r]\, dt + \pi_t\, \sigma\, dB^{\mathbb{P}}. \tag{5.11}$$

- It is important to note that in (5.11) the portfolio returns are calculated based on the beginning-of-period wealth, π_{t-1}; that is, one needs to choose the portfolio before knowing the return on the risky asset, $d\mathcal{P}^S/\mathcal{P}^S$.
- Note that
  - ⋆ the expected return on the portfolio, $[(1-\pi_t)r + \pi_t\mu]$, is a weighted average of the expected rates of return on the two assets.
  - ⋆ the volatility of the portfolio, $\pi_t\,\sigma$, depends on the volatility of the risky asset, σ, and the proportion of wealth invested in this asset, π_t.
- We will now select the portfolio to replicate the payoff to a call option and also identify the strategy to hedge this (risk management).

5.5.2 Instantaneous Rate of Return of the Call Option

- Let $V(t, \mathcal{P}^S)$ denote value of call option at date $t \leq T$ when $\mathcal{P}_t^S = \mathcal{P}^S$.

- This value is given by the Black-Scholes formula in (5.5) on Page 374, but with T in the formula replaced by $T - t$, which is the time to maturity.

- Applying Itô's formula to $V(t, \mathcal{P}^S)$ and using subscripts to denote partial derivatives (that is, $V_t = \partial V(t, \mathcal{P}^S)/\partial t$; $V_{\mathcal{P}^S} = \partial V(t, \mathcal{P}^S)/\partial \mathcal{P}^S$), we have:

$$dV = V_t\, dt + V_{\mathcal{P}^S}\, d\mathcal{P}^S + \frac{1}{2} V_{\mathcal{P}^S \mathcal{P}^S} (d\mathcal{P}^S)^2$$

$$\frac{dV}{V} = \frac{V_t}{V} dt + \frac{V_{\mathcal{P}^S}}{V} d\mathcal{P}^S + \frac{1}{2} \frac{V_{\mathcal{P}^S \mathcal{P}^S}}{V} (d\mathcal{P}^S)^2 \qquad \text{. . . dividing both sides by } V$$

$$= \left(\frac{V_t}{V} + \mu\, \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V} + \frac{1}{2} \sigma^2\, (\mathcal{P}^S)^2 \frac{V_{\mathcal{P}^S \mathcal{P}^S}}{V} \right) dt + \sigma\, \mathcal{P}^S\, \frac{V_{\mathcal{P}^S}}{V} dB^{\mathbb{P}}. \tag{5.12}$$

  where the last line is obtained by substituting for $d\mathcal{P}^S$ from (5.1) on page 367.

- This is the instantaneous rate of return of the call option.

5.5.3 Portfolio that Replicates the Return of the Call Option

- To replicate the call, we need a portfolio that has the same return as the call:

$$\frac{dV}{V} = \left(\frac{V_t}{V} + \mu\, \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V} + \frac{1}{2} \sigma^2\, (\mathcal{P}^S)^2 \frac{V_{\mathcal{P}^S \mathcal{P}^S}}{V} \right) dt + \sigma \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V} dB^{\mathbb{P}}. \quad (5.12)$$

- The return on our portfolio, reproduced below, is;

$$\frac{dW}{W} = [\pi_t(\mu - r) + r]\, dt + \pi_t\, \sigma\, dB^{\mathbb{P}}. \tag{5.11}$$

- For the rate of return on our portfolio to replicate the rate of return on the call, we need to choose π_t so that the coefficients of dt and $dB^{\mathbb{P}}$ in (5.11) are equal to those in (5.12).

- To match the $dB^{\mathbb{P}}$ terms,

$$\pi_t \sigma = \sigma \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V}$$
$$\pi_t = \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V}$$
$$\boxed{\pi_t \frac{V}{\mathcal{P}^S} = V_{\mathcal{P}^S} = \frac{\partial V(t, \mathcal{P}^S)}{\partial \mathcal{P}^S} = \Delta.} \tag{5.13}$$

- Note that
  - ⋆ π is the proportion of the portfolio value invested in the risky asset;
  - ⋆ thus, πV is the total dollar value invested in the risky asset, and
  - ⋆ thus, $\pi V / \mathcal{P}^S$ is the number of shares held in the portfolio.
- Thus, as claimed, we can replicate a long call position by holding Δ shares.
- The amount invested in the bank deposit is $V - \Delta \mathcal{P}^S$.
  Because $V - \Delta \mathcal{P}^S < 0$, it means the replicating strategy requires borrowing.

5.6 The Fundamental Partial Differential Equation

- Continuing with our discussion in the previous section, in order to match the dt terms, given $\pi_t = \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V}$, we must have:

$$\frac{V_t}{V} + \mu \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V} + \frac{1}{2}\sigma^2 (\mathcal{P}^S)^2 \frac{V_{\mathcal{P}^S \mathcal{P}^S}}{V} = \pi_t(\mu - r) + r \quad \text{... equating "dt" terms}$$
$$\frac{V_t}{V} + \mu \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V} + \frac{1}{2}\sigma^2 (\mathcal{P}^S)^2 \frac{V_{\mathcal{P}^S \mathcal{P}^S}}{V} = \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V}(\mu - r) + r \quad \text{... substituting for } \pi_t$$
$$\frac{V_t}{V} + \frac{1}{2}\sigma^2 (\mathcal{P}^S)^2 \frac{V_{\mathcal{P}^S \mathcal{P}^S}}{V} = r - r \mathcal{P}^S \frac{V_{\mathcal{P}^S}}{V} \quad \text{... canceling common terms}$$

$$\boxed{V_t + r \mathcal{P}^S V_{\mathcal{P}^S} + \frac{1}{2}\sigma^2 (\mathcal{P}^S)^2 V_{\mathcal{P}^S \mathcal{P}^S} = r V \quad \text{... Fundamental PDE}} \tag{5.14}$$

- This is the differential equation that Black and Scholes (1973) solved (by transforming it to the transfer-of-heat equation).

- We reproduce the equation from the previous page:

$$V_t + r\,\mathcal{P}^S\,V_{\mathcal{P}^S} + \frac{1}{2}\sigma^2\,(\mathcal{P}^S)^2\,V_{\mathcal{P}^S\mathcal{P}^S} = r\,V. \tag{5.14}$$

  - ⋆ The left-hand side of the PDE contains the drift of the option value, but with μ replaced by r; that is, the drift under the risk-neutral measure.
  - ⋆ The right-hand side is the risk-free return on investing V.

- The following three statements are equivalent:

  1. The expected rate of return of V under a risk-neutral probability is r;
  2. $e^{-r\,t}\,V(t,\mathcal{P}_t^S) = V(t,\mathcal{P}_t^S)/\mathcal{P}_t^D$ is a local martingale under a risk-neutral measure (so, $\frac{d\,[e^{-r\,t}\,V(t,\mathcal{P}_t^S)]}{[e^{-r\,t}\,V(t,\mathcal{P}_t^S)]} = -r\,dt + \frac{dV}{V}$ has zero drift, which implies that $\mathbb{E}[\frac{dV}{V}] = r\,dt$);
  3. The fundamental PDE in (5.14) holds.

- We reproduce the fundamental PDE:

$$V_t + r\,\mathcal{P}^S\,V_{\mathcal{P}^S} + \frac{1}{2}\sigma^2\,(\mathcal{P}^S)^2\,V_{\mathcal{P}^S\mathcal{P}^S} = r\,V. \tag{5.14}$$

- It is called the fundamental PDE, because it holds for every derivative security.

- Different derivatives have different values only because of their boundary conditions.

  - ⋆ The boundary conditions are the intrinsic value at maturity, the optimality conditions for early exercise, etc.

- Some examples of boundary conditions are:

  - ⋆ Value of call option at maturity is $\max\left(0, \mathcal{P}_T^S - K\right)$.
  - ⋆ Value of put option at maturity is $\max\left(0, K - \mathcal{P}_T^S\right)$.
  - ⋆ Value of a (long) forward contract on the stock is $\mathcal{P}_T^S - K$.

Example: Valuation of a forward contract using the PDE

- Consider a forward contract on a non-dividend paying stock.
- Denote by $\mathcal{P}^{Fwd}(t,T)$ the value at t of a forward contract maturing at T.
- Let the maturity date of the forward be T, the delivery (strike) price K, so that the payoff of the forward contract at maturity is $\mathcal{P}^S(T) - K$.
  - ⋆ The strike being K means that the forward price is: $\mathcal{P}^{Fwd}(0,T) = K$.
- We wish to show, using the fundamental PDE developed above, that the value of the forward contract at date t is: $\mathcal{P}^S(t) - e^{-r\,(T-t)}\,K$.
- And, we wish to give some intuition for the expression for the value of the forward contract.

- Let us start with the intuition.
- From Covered Interest Parity, we know that the relation between the forward price and the spot price for a non-dividend paying asset can be written as:

$$\mathcal{P}^{Fwd}(t,T) = e^{r\,(T-t)}\mathcal{P}^S(t). \qquad \ldots \text{Forward price} = [\text{interest rate factor}] \times [\text{Spot price}]$$

- Noting that the value of a forward contract at t is the present value of the difference between the forward rate at t and the forward rate at 0:

$$\begin{aligned}
& e^{-r\,(T-t)} \times \left[\mathcal{P}^{Fwd}(t,T) - \mathcal{P}^{Fwd}(0,T)\right] \\
&= e^{-r\,(T-t)} \times \left[\mathcal{P}^{Fwd}(t,T) - K\right] && \ldots \text{substituting } \mathcal{P}^{Fwd}(0,T) = K \\
&= e^{-r\,(T-t)} \times \left[e^{r\,(T-t)}\mathcal{P}^S(t) - K\right] && \ldots \text{substituting } \mathcal{P}^{Fwd}(t,T) = e^{r\,(T-t)}\mathcal{P}^S(t) \\
&= \mathcal{P}^S(t) - e^{-r\,(T-t)}\,K && \ldots \text{expanding terms in square bracket and simplifying.}
\end{aligned}$$

- Now, we show that the fundamental PDE is satisfied by the value of the forward contract at date t, which we denote by: $V(t, \mathcal{P}^S) = \mathcal{P}^S(t) - e^{-r\,(T-t)}\,K$.

- We start by calculating the partial derivatives of $V(t, \mathcal{P}^S)$:

$$\frac{\partial V}{\partial t} = V_t = -r\,e^{-r\,(T-t)}\,K;$$
$$\frac{\partial V}{\partial \mathcal{P}^S} = V_{\mathcal{P}^S} = 1;$$
$$\frac{\partial^2 V}{\partial \mathcal{P}^S \mathcal{P}^S} = V_{\mathcal{P}^S\mathcal{P}^S} = 0.$$

Why is this zero? What does it tell us about the forward contract payoff?

- Substituting these partial derivatives into the left-hand side of the PDE:

$$\begin{aligned} V_t + V_{\mathcal{P}^S}\,r\,\mathcal{P}^S + (1/2)V_{\mathcal{P}^S\mathcal{P}^S}\,\sigma^2\,(\mathcal{P}^S)^2 &= -r\,e^{-r\,(T-t)}\,K + 1\,r\,\mathcal{P}^S + 0 \\ &= r\left(\mathcal{P}^S - e^{-r\,(T-t)}\,K\right) \\ &= r\,V, \end{aligned}$$

  which is equal to the right-hand side of the PDE. Done.

5.7 Hedging Option Positions

- To hedge a call option, one needs to short delta shares.
  - ⋆ Recall from the first class that this is the argument Black and Scholes used to construct a riskless position, and hence, to value the option.

- This hedge portfolio, consisting of a long position in the call option and a short position in delta shares (or short the call option and long delta shares), will not have any instantaneous exposure to the price of the underlying asset.

- To maintain a perfect hedge, this portfolio must be adjusted continuously because the delta changes when the price of the underlying changes and with the passage of time.

- If one cannot trade continuously, which is the case in the real world, then the hedge will not be perfect.

- We now study the two cases of rebalancing (1) continuously and (2) discretely.

5.7.1 Hedging with continuous rebalancing: Delta hedging

- Consider a European call option with maturity T, and let $\mathcal{P}^C(\mathcal{P}^S, t)$ denote the value of the option at date $t < T$ when the stock price is $\mathcal{P}^S$ at t.

- Consider a portfolio that is
  1. short one call option (that is, you have sold a call option),
  2. long Δ shares of the underlying asset, and
  3. (short) cash equal to $\mathcal{P}^C - \Delta\,\mathcal{P}^S$.

- The above portfolio has zero value at date t.

- The change in the value of the above portfolio over the instant dt is:

$$-d\mathcal{P}^C + \Delta\, d\mathcal{P}^S + q\,\Delta\,\mathcal{P}^S dt + (\mathcal{P}^C - \Delta\,\mathcal{P}^S) r\, dt, \tag{5.15}$$

  - ⋆ the first term reflects the change in the value of the option,
  - ⋆ the second term is the capital gain or loss on Δ shares of stock,
  - ⋆ the third term is the dividends (q) received on Δ shares of stock, and
  - ⋆ the fourth term is the interest expense on the short cash position.

- But, we know from Ito's formula applied to $\mathcal{P}^C(\mathcal{P}^S, t)$ that

$$d\mathcal{P}^C = \frac{\partial \mathcal{P}^C}{\partial t} dt + \frac{\partial \mathcal{P}^C}{\partial \mathcal{P}^S} d\mathcal{P}^S + \frac{1}{2} \frac{\partial^2 \mathcal{P}^C}{\partial (\mathcal{P}^S)^2} \left(d\mathcal{P}^S\right)^2 \tag{5.16}$$

- Substituting (5.16) into (5.15) shows that the change in value of portfolio is:

$$-\Big[\underbrace{\frac{\partial \mathcal{P}^C}{\partial t}dt+\frac{\partial \mathcal{P}^C}{\partial \mathcal{P}^S}d\mathcal{P}^S+\frac{1}{2}\frac{\partial^2 \mathcal{P}^C}{\partial(\mathcal{P}^S)^2}\left(d\mathcal{P}^S\right)^2}_{d\mathcal{P}^C}\Big]+\Delta\, d\mathcal{P}^S+q\,\Delta\,\mathcal{P}^S\,dt+(\mathcal{P}^C-\Delta\,\mathcal{P}^S)r\,dt,$$

- But, $\frac{\partial \mathcal{P}^C}{\partial \mathcal{P}^S}=\Delta$, so the above simplifies to (the terms in red have opposite sign, so cancel out):

$$-\left[\frac{\partial \mathcal{P}^C}{\partial t}dt+\frac{1}{2}\frac{\partial^2 \mathcal{P}^C}{\partial(\mathcal{P}^S)^2}\left(d\mathcal{P}^S\right)^2\right]+q\,\Delta\mathcal{P}^S\,dt+(\mathcal{P}^C-\Delta\mathcal{P}^S)r\,dt, \tag{5.17}$$

- In (5.17), the expression for the change in the price of the underlying ($d\mathcal{P}^S$) does not appear anywhere; thus, delta hedging eliminates exposure to the source of risk, the Brownian motion, dB.

- Setting equation (5.17) equal to 0, because if the portfolio has no risk and no initial investment then the change in its value should be 0, gives:

$$0=-\left[\frac{\partial \mathcal{P}^C}{\partial t}dt+\frac{1}{2}\frac{\partial^2 \mathcal{P}^C}{\partial(\mathcal{P}^S)^2}\left(d\mathcal{P}^S\right)^2\right]+q\,\Delta\mathcal{P}^S\,dt+(\mathcal{P}^C-\Delta\mathcal{P}^S)r\,dt, \tag{5.17}$$

  which, upon simplifying (substituting for $(d\mathcal{P}^S)^2$ and expanding brackets), leads to

$$0=-\frac{\partial \mathcal{P}^C}{\partial t}-\frac{1}{2}\frac{\partial^2 \mathcal{P}^C}{\partial(\mathcal{P}^S)^2}\left(\sigma\mathcal{P}^S\right)^2+q\,\Delta\mathcal{P}^S+r\,\mathcal{P}^C-r\,\Delta\mathcal{P}^S,$$

  and then rearranging terms gives

$$r\,\mathcal{P}^C=\frac{\partial \mathcal{P}^C}{\partial t}+(r-q)\mathcal{P}^S\Delta+\frac{1}{2}\sigma^2(\mathcal{P}^S)^2\frac{\partial^2 \mathcal{P}^C}{\partial(\mathcal{P}^S)^2},$$

  which is identical to the fundamental PDE in (5.14) on page 394, reproduced below:

$$r\,V=V_t+r\,\mathcal{P}^S\,V_{\mathcal{P}S}+\frac{1}{2}\sigma^2\,(\mathcal{P}^S)^2V_{\mathcal{P}S\mathcal{P}S}\quad\ldots\text{Fundamental PDE} \tag{5.14}$$

  once we set the dividend yield $q=0$, and note that $\mathcal{P}^C(\mathcal{P}^S,t)=V(\mathcal{P}^S,t)$ and $\Delta=V_{\mathcal{P}S}$.

- Deriving the fundamental PDE via delta hedging corresponds to the approach used by Black and Scholes originally.

5.7.2 Hedging with discrete rebalancing: Gamma hedging

- In the discussion above, we had assumed that it is possible to rebalance one's position at each instant in time (that is, continuously).
- In practice, one can rebalance (trade) only discretely.
- If only discrete rebalancing is possible, then the delta of the position will change during the time interval that one does not trade.
- One can improve the performance of the discretely-rebalanced hedge by holding a portfolio that is both delta neutral and gamma neutral.
  - ⋆ Delta neutral means no exposure to changes in price of underlying, $d\mathcal{P}^S$.
  - ⋆ Gamma neutral means that the delta of the position has no exposure to changes in the price of the underlying; that is, the second derivative of the portfolio value, with respect to the price of the underling (portfolio gamma) is zero.

- Gamma hedging is done by using an additional asset in the hedge portfolio.
- To see this, consider the following example.
- Suppose we have written (sold) a call option and we want to hedge both the delta and the gamma using the underlying asset and another option, for example, another call option with a different strike.
- Let Δ and Γ denote the delta and gamma of the written option.
- Let Δ' and Γ' denote the delta and gamma of the option used to hedge.
- Consider holding
  - ⋆ a shares of the stock, and
  - ⋆ b units of the option purchased to hedge the written call option

- Note that the delta of the stock is 1 ($\frac{d\mathcal{P}^S}{d\mathcal{P}^S} = 1$),
  so to obtain a portfolio that is delta neutral, we need:

$$0 = -\Delta + a\,1 + b\,\Delta' \quad \text{– [delta of option sold] + } a\times \text{ [delta of one share] + } b\times \text{ [delta of option bought]} \tag{5.18}$$

- The gamma of the stock is zero ($\frac{d^2\mathcal{P}^S}{d\mathcal{P}^{S2}} = 0$),
  so to obtain a portfolio that is gamma neutral, we need

$$\begin{aligned} 0 &= -\Gamma + a\,0 + b\,\Gamma' \quad \text{– [gamma of option sold] + } a\times \text{ [gamma of one share] + } b\times \text{ [gamma of option bought]} \\ &= -\Gamma + b\,\Gamma'. \end{aligned} \tag{5.19}$$

- From Equation (5.19) we see that for our portfolio to be gamma neutral,
  the number of the (second) option to hold is:

$$b = \frac{\Gamma}{\Gamma'}.$$

- Substituting this in (5.18), we see that for our portfolio to be delta-neutral,
  the number of shares to hold is:

$$a = \Delta - \frac{\Gamma}{\Gamma'}\Delta'.$$

5.8 American Options

- Let us assume that the risky asset does not pay dividends.
- Then, it is never optimal to exercise early an American call. . . . Why?
- However, it may be optimal to exercise early an American put. . . . Why?
- It will be optimal to exercise an American put if it is sufficiently in the money; that is, if $\mathcal{P}^S$ is sufficiently low relative to the strike, K.
  - ⋆ Early exercise of the put has the benefit that one can start earning early the interest on the strike received from exercising the option.
- We now study the problem of valuing an American put, whose payoff at exercise date t is: $\max(0, K - \mathcal{P}_t^S)$.
- Note that many "real options" are American; for example, the option to invest in a project, the option to export to a new country, etc.

- In general, one must consider American options in conjunction with exercise policies.
- For an American put, under the assumptions of the Black and Scholes model, it is sufficient to consider exercise boundaries; that is, continuous functions $f : [0, T] \to [0, \infty)$, with the interpretation that the option is exercised at the first time that $\mathcal{P}^S$ falls to the boundary; that is, the first time $\mathcal{P}^S_t \leq f(t)$.
- Because one would never exercise an out of the money put, we can assume that $f(t) \leq K$ for all t.
- Given an exercise boundary, let A denote the set of states of the world such that $\mathcal{P}^S_t \leq f(t)$, for some $t \leq T$.
- On the events A define $\tau = \min\{t \mid \mathcal{P}^S_t \leq f(t)\}$; that is, τ is the first time that $\mathcal{P}^S_t \leq f(t)$ (often referred to as a "stopping time").
- The American put with this exercise policy pays $K - \mathcal{P}^S_\tau$ at the random date τ if the event A occurs, and 0 otherwise.

- Under the Black-Scholes assumptions, there is a self-financing replicating strategy in the underlying asset and the bank-deposit with the same payoff. The cost at $t = 0$ of this portfolio is

$$\mathbb{E}^{\mathbb{Q}}\left[e^{-r\tau}(K - \mathcal{P}^S_\tau)1_A\right]. \tag{5.20}$$

- Under the Black-Scholes assumptions, there is an optimal exercise boundary denoted f^* that maximizes the value of the American put in (5.20).
- Denote this maximum option value by V_0.
- One would like to calculate V_0 and f^*, but there is no simple formula like the one for European call and put options.
- However, f^* is known to be an increasing continuously differentiable function with the property that $f^*_T = K$.
- The value of the American put can be calculated using numerical methods (for example, in a binomial model with multiple dates).

5.9 Value Matching and Smooth Pasting Conditions

- Consider an American put. Given an exercise boundary f, let A denote the exercise event, and τ the exercise time defined on A.

- For any t and $\mathcal{P}^S \geq f(t)$, define

$$V(t, \mathcal{P}^S; f) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(\tau - t)} \left(K - \mathcal{P}^S_\tau \right) 1_A \mid \mathcal{P}^S_t = \mathcal{P}^S, \mathcal{P}^S_u > f(u) \text{ for all } u < t \right]. \quad (5.21)$$

- This is the value of an American put exercised at the boundary f, conditional on it not having already been exercised by time t.

- Like any other asset value, $e^{-r\,t} V(t, \mathcal{P}^S_t; f)$ is a local martingale under the risk-neutral probability;

  ⋆ hence, prior to exercise, the fundamental PDE holds, which means that it holds in the region $\{(t, \mathcal{P}^S_t) \mid \mathcal{P}^S > f(t)\}$.

  ⋆ This is called the continuation region associated with f (the region where you do not exercise the option).

- The put value also satisfies $V(t, \mathcal{P}^S; f) = K - \mathcal{P}^S$ when $\mathcal{P}^S = f(t)$, which is called the value-matching condition.

- The condition $V(t, \mathcal{P}^S; f^*) \geq K - \mathcal{P}^S$ must also be satisfied, which says that the put must be worth at least its intrinsic value, which is clearly true.

- Finally, the optimal exercise boundary and associated with the value of the put $V(t, \mathcal{P}^S; f^*)$ must also satisfy the smooth pasting condition:

$$-1 = V_{\mathcal{P}^S}(t, \mathcal{P}^S; f^*) = \lim_{x \downarrow 0} \frac{V(t, \mathcal{P}^S + x; f^*) - V(t, \mathcal{P}^S; f^*)}{x},$$

  when $\mathcal{P}^S = f^*(t)$, and where the partial derivative $V_{\mathcal{P}^S}(t, \mathcal{P}^S; f^*)$ is the right-hand derivative.

- The smooth pasting condition implies that the boundary f^* is optimal, so
  - ⋆ the value of the put $V(t, \mathcal{P}^S; f^*)$ just above the exercise boundary f^*, and
  - ⋆ the value $K - \mathcal{P}^S$ of immediate exercise below the exercise boundary,

  "paste" together in a smooth (continuously differentiable) way at the boundary.

5.10 Summary

1. The Black-Scholes value of a European call option is

$$\begin{aligned} \mathcal{P}_0^C &= \mathbb{E}^{\mathbb{P}}\left[M_T\, \mathcal{P}_T^S\, 1_A\right] - K\, \mathbb{E}^{\mathbb{P}}\left[M_T\, 1_A\right] \\ &= \mathcal{P}_0^S\, \text{prob}^{\mathbb{S}}(A) - e^{-r\,T}\, K\, \text{prob}^{\mathbb{Q}}(A) \\ &= \mathcal{P}_0^S\, N(d_1) - e^{-r\,T}\, K\, N(d_2), \end{aligned}$$

   where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}\,K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}};$$

$$d_2 = d_1 - \sigma\,\sqrt{T}.$$

2. The price today of a European call option

$$\mathcal{P}_0^C = \mathcal{P}_0^S\, N(d_1) - e^{-r\,T}\, K\, N(d_2),$$
$$d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}\,K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}}; \qquad d_2 = d_1 - \sigma\,\sqrt{T},$$

can be interpreted as the difference between the

- present value of what you will receive, $\mathcal{P}_0^S$, multiplied by the appropriate probability of getting this, $N(d_1) = \text{prob}^{\mathbb{S}}(\mathcal{P}_T^S \geq K)$, and;
- present value of what you will deliver, $e^{-r\,T}\,K$, multiplied by the appropriate probability of paying this, $N(d_2) = \text{prob}^{\mathbb{Q}}(\mathcal{P}_T^S \geq K)$.
- In d_1, the argument of log has the ratio of the present values of the quantities to be received and paid, $\mathcal{P}_0^S/(e^{-r\,T}\,K)$; and,
- The difference between d_1 and d_2, which is $\sigma\,\sqrt{T}$, is the difference in the volatility of the quantities to be received and to be paid.

3. To interpret the Black-Scholes formula for a European call option in another way, recall that the forward price for a stock is:

$$\mathcal{P}_0^{Fwd}(T) \;=\; e^{r\,T} \times \mathcal{P}_0^S \qquad \Longleftrightarrow \qquad e^{-r\,T}\mathcal{P}_0^{Fwd}(T) = \mathcal{P}_0^S.$$

Now, we substitute this into the Black-Scholes formula:

$$\begin{aligned}\mathcal{P}_0^C &= \overbrace{\mathcal{P}_0^S}^{\text{PV of asset}}\; N(d_1) - \overbrace{e^{-r\,T}\,K}^{\text{PV of strike}}\; N(d_2)\\ &= e^{-r\,T}\mathcal{P}_0^{Fwd}(T)\, N(d_1) - e^{-r\,T}\, K\, N(d_2)\\ &= e^{-r\,T}\left[\mathcal{P}_0^{Fwd}(T)\, N(d_1) - K\, N(d_2)\right].\end{aligned}$$

Thus, the value of a European call option is the discounted value of the difference between

- the amount you will receive at T, the forward price, multiplied by the probability of receiving this under the appropriate probability; and,
- the amount you will pay at T, the strike, multiplied by the probability of receiving this under the appropriate probability;

Similarly, we can express d_1 and d_2 in terms of the forward price:

$$d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}}$$

$$= \frac{\log\left(\frac{e^{r\,T}\,\mathcal{P}_0^S}{K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}}$$

$$= \frac{\log\left(\frac{\mathcal{P}_0^{Fwd}(T)}{K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\,\sqrt{T}.$$

4. The fundamental PDE that all derivatives must satisfy (in the absence of dividends) is:

$$V_t + r\,\mathcal{P}^S\,V_{\mathcal{P}^S} + \frac{1}{2}\sigma^2\,(\mathcal{P}^S)^2\,V_{\mathcal{P}^S\mathcal{P}^S} \;= r\,V.$$

5. The expected return on the stock, μ,

- does not appear explicitly in the Black-Scholes formula, but
- its effect is present via the stock price (or via the SDF).

6. The "Greeks" are partial derivatives that measure how the value of an option changes as the parameters of the model change:

- delta $= \Delta$: change in the value of an option wrt value of underlying;
- gamma $= \Gamma$: change in the value of the delta wrt value of underlying;
- theta $= \theta$: change in the value of an option wrt time-to-maturity $(-T)$;
- vega $= \mathcal{V}$: change in the value of an option wrt volatility of underlying;
- rho $= \rho$: change in the value of an option wrt risk-free interest rate.

7. To hedge a written call or replicate an option (or a "book of options"), one needs to construct a self-financing portfolio of the underlying and the bank deposit with the same delta.

8. The logic underlying the Black-Scholes formula is

- ability to replicate exactly the cashflow to contract we are valuing, and
- the no-arbitrage condition, which says that the value of the replicating portfolio and the contract must be the same.

Thus, one should not use the Black-Scholes model to value an option

- if one cannot replicate the cashflows to an option; or,
- if one cannot execute the arbitrage when the price of the option and replicating portfolio are different.

5.11 Suggestions for Further Readings

5.11.1 Source material for the lecture notes

- The notes are based on Back (2017, Chapter 16). See also Dumas and Luciano (2017, Chapters 6 & 8).
- Back (2017, Pages 425–6) discusses the various different forms in which the fundamental partial differential equation can be expressed, including the heat equation. He also explains the relation of the partial differential equation to the Feynman-Kac theorem.
- Duffie (1988) also uses the Feynman-Kac representation to extend the Black and Scholes (1973) model to a setting where the risk-free interest rate is stochastic and markets are not complete.
- Breeden and Litzenberger (1978) explain how one can obtain prices of Arrow-Debreu securities (state prices) from the prices of options.

- The seminal papers on option valuation are Black and Scholes (1973) and Merton (1973).
- Black (1989), Scholes (1998), and Merton (1998, Section I) provide a historical account of the development of the Black-Scholes-Merton model of option pricing.
- Following the award of the Nobel prize to Merton and Scholes, the paper by Duffie (1998) explains the contribution of the work done by Black, Scholes, and Merton.
- Mehrling (2005) provides an excellent biography of Fischer Black that I strongly recommend to you.

- For a detailed discussion of American options, see Karatzas (1988), the book Detemple (2005), and the review paper Detemple (2014).
  - ⋆ Detemple taught a course on this material at EDHEC, so you can also see the video recordings.
- For a detailed discussion and deeper understanding of the value-matching and smooth-pasting conditions, see Dixit (1991), Dumas (1991), and the book Dixit (1993).

5.11.2 Potential applications and extensions of lecture notes

- There are many, many other areas of finance and economics to which the insights of contingent claim valuation/hedging/replication apply:
  - ⋆ Options on individual stocks, stock-index and mutual-fund portfolios, on bonds and other fixed-income securities, on currencies, commodities, real estate, publishing and movie rights; see Merton (1998, Section II).
  - ⋆ Corporate liabilities such as equity, debt, convertible bonds, and debentures; see Merton (1974) and the review paper by Sundaresan (2013).
  - ⋆ Volatility derivatives, such as variance swaps: see the review paper by Carr and Lee (2009).
  - ⋆ Credit risk: see the review paper by Jarrow (2009).
  - ⋆ Price-support guarantees, loan guarantees, and deposit insurance, prepayment options on mortgages; see the book Merton (1990).
  - ⋆ Real options to initiate or expand, to abandon or contract, and to wait, slow down, or speed up investments; for a starting point, see the books by Dixit and Pindyck (1993) and Trigeorgis (1996).

- For a more general discussion of the various applications of option pricing, see Merton (1998, Sections II & III) and the more recent review paper by Bodie (2019).
- Two recent talks by Merton (with links to the respective PDFs):
  - ⋆ Finance Science, Financial Innovation and Long-Term Asset Management
  - ⋆ On the Role of Financial Innovation and Derivative Markets in Global Economic Growth and Development

5.11.3 Ph.D. elective courses offered on related topics

- Rama Cont taught a course on *Monte Carlo Methods* in Finance in 2010 and 2013; Marcel Rindisbacher taught this course in 2017.
- Tim Bollerslev taught a course on *Volatility Modelling* in 2010, 2012, 2016.
- Yacine Aït-Sahalia taught *Financial Econometrics of Jumps* in 2011.
- Jerome Detemple taught a course on *Derivative Securities* in 2011.
- Mikhail Chernov taught *Empirical Option Pricing* in 2012 and 2014.
- Federico Bandi taught *The Econometrics of Continuous-Time Models* in 2012 and 2015, and *Estimation of Continuous-Time Models* in 2013.
- Torben Andersen taught *High-Frequency Asset Pricing* in 2013 and 2015.

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Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 6

Extensions of Black-Scholes Option Pricing Model

Raman Uppal

EDHEC Business School

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6.1 Plan for the Day

Last Time

- The Black-Scholes model of option pricing

Today

- **Extensions of the Black-Scholes model**
  - ⋆ Extend the model to allow for:
    Dividends, non-constant interest rate, non-constant volatility.
  - ⋆ Extend the model for options on equity in order to apply it to:
    Options on forwards, futures, and currency; exchange options, etc.

For Next Time

- Review the material covered in class.
- Read Back, Chapter 17 and Dumas & Luciano, Chapter 6.
- Do the practice problems.

6.2 Main Points for Today

- The main point we wish to make is that, once we understand the Black-Scholes model at a deep level, it is easy to generalize the model.
- This generalization allows us to
  1. Drop the assumptions that there are no dividends, and that the risk-free rate and volatility are constant.
  2. Value a number of different options (without much additional work):
     - ⋆ Exchange options, deferred-exchange options
     - ⋆ Options on forwards and futures
     - ⋆ Foreign-currency options
     - ⋆ Various exotic options

Bottom line: A single model, with the same underlying logic, can be used

- in a fairly general setting
- to value a wide variety of financial derivatives.

6.3 General Interpretation of the Black-Scholes Model

- Recall that the Black-Scholes price today ($t = 0$) for a call option maturing at T is

$$\text{Value of call option today} = \mathcal{P}_0^S \, N(d_1) - e^{-r\,T} \, K \, N(d_2) \tag{6.1}$$

$$\text{where} \qquad d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}} \tag{6.2}$$

$$d_2 = d_1 - \sigma\,\sqrt{T}. \tag{6.3}$$

General Interpretation of Each Term of the Option Pricing Model

- $\mathcal{P}_0^S$ is the present value of the asset that would be received if the option is exercised.
- $e^{-r\,T}\,K$ is the present value of the asset (cash) that is paid if the option is exercised.
- σ is the volatility of the ratio of the receivable to the payable; that is, the volatility of $\mathcal{P}^S/(e^{-r\,T}K)$.
- d_1 contains the log of the ratio of these present values + one-half $\sigma^2\,T$, with the sum divided by $\sigma\sqrt{T}$.
- d_2 is d_1 minus $\sigma\sqrt{T}$, which is the adjustment for the difference in volatility of the receivable (risky asset) and payable (cash).
- $N(d_1)$ is probability of option being exercised, but where numeraire is the underlying asset.
- $N(d_2)$ is the risk-neutral probability of the option being exercised; that is, the probability of option exercise where the numeraire is the risk-free asset, the bank deposit.
- All the models discussed below share this interpretation.

6.4 European Options on Dividend-Paying Assets

- So far, we considered options on risky assets that did not pay dividends.
- Suppose that the underlying risky asset pays dividends continuously at a rate proportional to the asset price; that is, the dividend rate at date t is $D_t = q\,\mathcal{P}_t^S$, for a constant q.
- The quantity $D_t/\mathcal{P}_t^S = q$ is called the "dividend yield."
- Thus, the price of the asset including dividends is $e^{q\,t}\,\mathcal{P}_t^S$, which is the value of a portfolio that starts with one share of the asset and reinvests the dividends.
- This means that if one starts with $e^{-q\,T}\,\mathcal{P}_t^S$ shares, then the value at date T will be $\mathcal{P}_T^S$, which is exactly what the option pays off at maturity.
- Therefore, if an asset pays a constant dividend yield q, we can apply Black-Scholes to value European calls and puts, simply by using for the initial asset price $e^{-q\,T}\,\mathcal{P}_t^S$ instead of $\mathcal{P}_t^S$.

Black-Scholes Value of Call On Dividend-Paying Asset

- The $t = 0$ value of a call on an asset with continuous dividend yield of q is:

$$\mathcal{P}_0^C = e^{-q\,T}\,\mathcal{P}_0^S N(d_1) - e^{-r\,T}\,K N(d_2)$$

$$\text{where} \qquad d_1 = \frac{\log\left(\frac{e^{-q\,T}\,\mathcal{P}_0^S}{e^{-r\,T}K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}}; \qquad d_2 = d_1 - \sigma\,\sqrt{T}.$$

- Note that our interpretation in terms of present value of receivable and payable does not change.

- In the presence of a continuous dividend yield of q:

  ⋆ The delta of the call is: $e^{-qT}N(d_1)$.

  ⋆ The drift of $d\mathcal{P}^S/\mathcal{P}^S$ under risk-neutral measure is: $(r-q)$ instead of r.

  ⋆ The fundamental PDE for the value $V(t,\mathcal{P}_t^S)$ of an option is:

$$V_t + (r-q)\,\mathcal{P}^S\,V_{\mathcal{P}^S} + \frac{1}{2}\sigma^2\,(\mathcal{P}^S)^2\,V_{\mathcal{P}^S\mathcal{P}^S} = r\,V.$$

Example 6.1 (Valuation of European call and put options)
Consider a stock with a dividend yield of 1% p.a. whose current price is \$100 and volatility is 35% per annum. Let the continuously-compounded risk-free interest rate be 2% per annum,

1. Calculate the price of a 3-month European call option on this stock if the strike price is \$90.

2. Calculate the price of a 3-month European put option on this stock if the strike price is \$90.

Space for your work.

6.5 Option to Exchange One Risky Asset for Another

- Consider two risky assets with prices $\mathcal{P}^{S_1}$ and $\mathcal{P}^{S_2}$, and a European option to deliver Asset 2 in exchange for receiving Asset 1 at date T.
- The value of this option at maturity is

$$\max\left(0, \mathcal{P}_T^{S_1} - \mathcal{P}_T^{S_2}\right).$$

- This can be viewed as
  - ⋆ a **call option** on the first asset with random strike price $\mathcal{P}_T^{S_2}$, or
  - ⋆ a **put option** on the second asset with random strike price $\mathcal{P}_T^{S_1}$.
- How should we find the value of this option?
  - ⋆ Margrabe (1978) provides the answer to this.

- Assume that the two assets pay constant dividend yields q_i and assume that their prices satisfy

$$\frac{d\mathcal{P}^{S_i}}{\mathcal{P}_i^{S_i}} = \mu_i\, dt + \sigma_i\, dB_i^{\mathbb{P}}, \qquad i = \{1, 2\},$$

where each $B_i^{\mathbb{P}}$ is a Brownian motion under the actual probability measure, and the correlation between them is ρ.

- As before, the drifts μ_i can be quite general random processes. We can also allow the σ_i and ρ to be general processes as long as

$$\sigma \equiv \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\,\sigma_2}, \qquad \text{. . . see equation (2.30) on page 257} \tag{6.4}$$

which is the volatility of $\mathcal{P}^{S_1}/\mathcal{P}^{S_2}$, is a constant.

- The assumption of a constant σ can be relaxed to allow σ to be time-varying, as long as it is non-random.

- To value the exchange option, we use as numeraire Asset 2.

- If Asset 2 is the numeraire, then:
  - ⋆ The value of the first asset in terms of the second asset is: $\mathcal{P}^{S_1}/\mathcal{P}^{S_2}$;
  - ⋆ The value of the second asset in terms of the second asset is $\mathcal{P}^{S_2}/\mathcal{P}^{S_2} = 1$.
  - ⋆ The value of the exchange option at maturity is:

$$\frac{1}{\mathcal{P}_T^{S_2}} \times \max\left(0, \mathcal{P}_T^{S_1} - \mathcal{P}_T^{S_2}\right) = \max\left(0, \frac{\mathcal{P}_T^{S_1}}{\mathcal{P}_T^{S_2}} - 1\right).$$

- But, the payoff above,

$$\max\left(0, \frac{\mathcal{P}_T^{S_1}}{\mathcal{P}_T^{S_2}} - 1\right),$$

can be interpreted as the value of a standard call option, with:

⋆ strike price being equal to 1;

⋆ "interest rate" being q_2 (note the price of the second asset is always 1 when Asset 2 is the numeraire, and so its dividend yield is risk-free);

⋆ dividend yield on the first asset being q_1, whose value in terms of the second asset is: $q_1\,\mathcal{P}_t^{S_1}/\mathcal{P}_t^{S_2}$;

⋆ volatility of the first asset in terms of the second asset is σ, which is given in (6.4), and is reproduced below:

$$\sigma \equiv \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\,\sigma_2}. \tag{6.4}$$

- Using these inputs, and our general interpretation of the Black-Scholes model, the value of a European exchange option to receive Asset 1 in exchange for Asset 2 at date T is given by Margrabe's formula:

$$\text{Value in units of } \mathcal{P}_0^{S_2} \;=\; e^{-q_1 T}\,\frac{\mathcal{P}_0^{S_1}}{\mathcal{P}_0^{S_2}}\,N(d_1) - e^{-q_2 T}\,\underbrace{1}_{=K}\,N(d_2),$$

so

$$\begin{aligned}
\text{Value in cash} &= \mathcal{P}_0^{S_2} \times \left[\text{Value in terms of } \mathcal{P}_0^{S_2}\right], \\
&= e^{-q_1 T}\,\mathcal{P}_0^{S_1}\,N(d_1) - e^{-q_2 T}\,\mathcal{P}_0^{S_2}\,N(d_2) \qquad (6.5) \\
\text{where} \qquad d_1 &= \frac{\log\left(\frac{e^{-q_1 T}\,\mathcal{P}_0^{S_1}}{e^{-q_2 T}\,\mathcal{P}_0^{S_2}}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \\
d_2 &= d_1 - \sigma\sqrt{T}, \quad \text{where } \sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}. \qquad (6.6)
\end{aligned}$$

6.6 Options on Forwards

- Black (1976) tells us how to value options on forwards.
- Consider a forward contract that matures at T' and a call or put option (on this forward) that matures at date $T < T'$.



- A call option on the forward implies that you have the option to go long the forward, with the forward price being the strike price of the option.
  - ⋆ That is, if you exercise the call option at T, you will then receive the underlying at date T' and you will also pay the strike price at T'.
- A put option on the forward gives you the option to go short the forward.

- We will assume that the price of the forward contract, $\mathcal{P}^{Fwd}(t, T')$, satisfies

$$\frac{d\mathcal{P}^{Fwd}}{\mathcal{P}^{Fwd}} = \mu\, dt + \sigma\, dB^{\mathbb{P}}, \tag{6.7}$$

  where $B^{\mathbb{P}}$ is a Brownian motion, μ can be quite a general random process, and σ is constant (but can be extended to be time-varying but non-random).
- We do not assume that there is a constant risk-free interest rate; instead, we assume there is a "discount bond" (zero-coupon bond) that pays $\$1$ at T'.

  The price of this zero-coupon discount bond at date t is denoted $\mathcal{P}^{Z}(t, T')$.
- Based on our general interpretation of the Black-Scholes model, the value of European call options on forward contracts is derived below.

- We derive the above result by interpreting the options on forwards in terms of Margrabe's formula for an exchange option.
- We start by noting that, if the call option on a forward is exercised at T, this results in a long position in the underlying with forward price K.
- The value of this on the settlement date T' is: $\left[\mathcal{P}^{Fwd}(T,T') - K\right]$.
- The present value of this at date T is: $\mathcal{P}^{Z}(T,T')\ \left[\mathcal{P}^{Fwd}(T,T') - K\right]$.

- Re-writing the last statement on the previous slide:

  The present value of the option payoff at date T is: $\mathcal{P}^{Z}(T,T')\ \left[\mathcal{P}^{Fwd}(T,T') - K\right]$.

- Now define

$$\mathcal{P}_T^{S_1} \equiv \mathcal{P}^{Z}(T,T')\,\mathcal{P}^{Fwd}(T,T'), \quad \text{and} \tag{6.8}$$

$$\mathcal{P}_T^{S_2} \equiv \mathcal{P}^{Z}(T,T')\,K. \tag{6.9}$$

- The payoff to the call option on the forward can therefore be written as

$$\text{Payoff} = \max\left(0, \mathcal{P}_T^{S_1} - \mathcal{P}_T^{S_2}\right). \tag{6.10}$$

- Finally, the volatility of $\frac{\mathcal{P}_t^{S_1}}{\mathcal{P}_t^{S_2}} = \frac{\mathcal{P}^{Z}(t,T')\,\mathcal{P}^{Fwd}(t,T')}{\mathcal{P}^{Z}(t,T')\,K} = \frac{\mathcal{P}^{Fwd}(t,T')}{K}$ is σ, the volatility of the forward in (6.7), because K is a constant; and $q_1 = q_2 = 0$.

- Based on the insights above, Black's formula for call options on the forward contract is:

$$\mathcal{P}_0^C = \mathcal{P}^Z(0,T')\,\mathcal{P}^{Fwd}(0,T')\,N(d_1) - \mathcal{P}^Z(0,T')\,K\,N(d_2) \tag{6.11}$$

$$= \mathcal{P}^Z(0,T')\left[\mathcal{P}^{Fwd}(0,T')\,N(d_1) - K\,N(d_2)\right], \tag{6.12}$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^{Fwd}(0,T')}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \tag{6.13}$$

$$d_2 = d_1 - \sigma\sqrt{T},$$

σ = volatility of the forward (*not* the volatility of the spot).

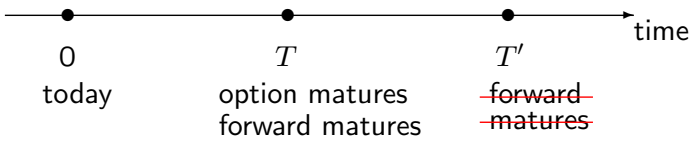
6.7 Option Price when Risk-free Rate is Not Constant

- The Merton (1973) model of option pricing is the same as Black-Scholes model, but
  - ⋆ it does not assume the existence of a constant risk-free rate;
  - ⋆ it also allows for arbitrary dividends.
- Instead, it assumes only the existence of a discount bond that matures on the same date as the option, T, and whose price at date t is denoted $\mathcal{P}^Z(t,T)$.
- And, instead of assuming that the volatility of the stock return is constant, it assumes that the volatility of the forward price is constant.

- Note that at maturity the price of the forward must be equal to the spot price. Thus we have: $\mathcal{P}^{Fwd}(T,T) = \mathcal{P}^S_T$.

  (We do not need the forward to actually be traded, because we can construct its price via synthetic replication.)

- Now, consider a call option on the forward, with the forward maturing at T also (instead of at T', as for Black's model for options on forwards on page 447).



- In terms of the notation of the previous section, we have $T' = T$, and hence for the price of the zero-coupon bond, we have:

$$\mathcal{P}^Z(T,T') = \mathcal{P}^Z(T,T) = 1.$$

- Therefore, the value of the call on the forward at its maturity date is the same as the value of the call on the stock:

$$\max\left(0, \mathcal{P}^S_T - K\right) = \max\left(0, \mathcal{P}^{Fwd}(T,T) - K\right) \qquad \ldots \text{ because } \mathcal{P}^S_T = \mathcal{P}^{Fwd}(T,T)$$

- Hence, the value at date 0 of the call on the stock must be the same as that of the call on the forward, which we have already priced in the last section.

- Assuming only the existence of a discount bond with price $\mathcal{P}^Z(0,T) = e^{-yT}$, where y is the yield on the bond, but *without* assuming a constant risk-free rate, Black's formula for call options on the asset, is:

$$\mathcal{P}_0^C = \mathcal{P}^Z(0,T)\left[\mathcal{P}^{Fwd}(0,T)\,N(d_1) - K\,N(d_2)\right], \tag{6.14}$$

$$= e^{-yT}\left[\mathcal{P}^{Fwd}(0,T)\,N(d_1) - K\,N(d_2)\right], \tag{6.15}$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^{Fwd}(0,T)}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \tag{6.16}$$

$$d_2 = d_1 - \sigma\sqrt{T},$$

$$\sigma = \text{volatility of the forward.}$$

- Note from Covered Interest Rate Parity, the price of the forward is

$$\mathcal{P}^{Fwd}(0,T) = \frac{e^{-qT}\,\mathcal{P}_0^S}{\mathcal{P}^Z(0,T)} \iff \mathcal{P}^Z(0,T)\mathcal{P}^{Fwd}(0,T) = e^{-qT}\,\mathcal{P}_0^S$$

- Substituting the above in Black's formula in (6.14), we get Merton's formula:

$$\mathcal{P}_0^C = e^{-qT}\,\mathcal{P}_0^S\,N(d_1) - \mathcal{P}^Z(0,T)\,K\,N(d_2), \tag{6.17}$$

where

$$d_1 = \frac{\log\left(\frac{e^{-qT}\,\mathcal{P}_0^S}{\mathcal{P}^Z(0,T)\,K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \tag{6.18}$$

$$d_2 = d_1 - \sigma\sqrt{T}, \tag{6.19}$$

and σ is the constant volatility of the forward price, which, from Covered Interest Rate Parity, we see is equal to the volatility of $\mathcal{P}^S/\mathcal{P}^Z$.

Volatility of the forward price

- One can compute the volatility of the forward price in terms of the volatilities and correlation of the stock and zero-coupon discount bond as follows.

- Assume that

$$\frac{d\mathcal{P}^S}{\mathcal{P}^S} = \mu_s\,dt + \sigma_s\,dB_s^{\mathbb{P}}, \quad \text{and} \quad \frac{d\mathcal{P}^Z}{\mathcal{P}^Z} = \mu_z\,dt + \sigma_z\,dB_z^{\mathbb{P}},$$

  where $B_s^{\mathbb{P}}$ and $B_z^{\mathbb{P}}$ are Brownians under $\mathbb{P}$ with correlation ρ.

- From Covered Interest Parity, for a dividend-paying asset

$$\mathcal{P}^{Fwd}(t,T) = e^{-q(T-t)}\frac{\mathcal{P}_t^S}{\mathcal{P}^Z(t,T)}.$$

- Then, from Itô's formula (using the rule for "ratios") we know that the volatility of $\mathcal{P}^{Fwd}(t,T)$ is: $\sigma = \sqrt{(\sigma_s^2 + \sigma_z^2 - 2\rho\,\sigma_s\,\sigma_z)}$.

6.8 Deferred-Exchange Options

- Observe that a call option on a forward can be viewed as an option to exchange K dollars for the underlying asset, with the exchange taking place at a later date, which is the maturity date of the forward.



- Therefore, it is an exchange option in which the exchange takes place at a fixed date after the option expires.

- Using this insight, we can extend Margrabe's formula to value options to exchange assets when the exchange takes place after the option matures

- These options are called deferred-exchange options.

- Consider two assets that pay constant dividend yields q_i with prices that satisfy

$$\frac{d\mathcal{P}^{S_i}}{\mathcal{P}^{S_i}} = \mu_i\, dt + \sigma_i\, dB_i^{\mathbb{P}},$$

  where each $B_i^{\mathbb{P}}$ is a Brownian motion under the actual probability measure, and the correlation between them is ρ.

- As before, the drifts μ_i can be quite general random processes.

- Similarly, σ_i and ρ are allowed to be general processes, as long as σ is a constant, where σ is the volatility of the ratio of the two asset prices and is given by

$$\sigma \equiv \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\,\sigma_2}.$$

- Consider an option maturing at T to exchange the second asset for the first asset at date T', where $T' > T$.

- To determine the value of the option at date T, suppose the option is exercised. Then, to unwind the position, one can sell forward the asset to be received and buy forward the asset to be delivered, with these forwards maturing at T'.

- Thus,

  ⋆ value of the payoff at date T' is: $\left[\mathcal{P}^{Fwd_1}(T,T') - \mathcal{P}^{Fwd_2}(T,T')\right]$;

  ⋆ present value of this at T is: $\mathcal{P}^{Z}(T,T')\left[\mathcal{P}^{Fwd_1}(T,T') - \mathcal{P}^{Fwd_2}(T,T')\right]$;

  ⋆ value of the option at T is:

$$\max\left(0, \mathcal{P}^{Z}(T,T')\,\mathcal{P}^{Fwd_1}(T,T') - \mathcal{P}^{Z}(T,T')\,\mathcal{P}^{Fwd_2}(T,T')\right).$$

- As we did before, let us define the prices of two non-dividend paying assets:

$$\mathcal{P}_t^{S_1^*} = \mathcal{P}^Z(t,T')\,\mathcal{P}^{Fwd_1}(t,T'), \quad \text{and} \tag{6.20}$$

$$\mathcal{P}_t^{S_2^*} = \mathcal{P}^Z(t,T')\,\mathcal{P}^{Fwd_2}(t,T'). \tag{6.21}$$

- Thus, the option to exchange the two assets at T' must have the same value as an option to exchange at date T the assets with prices $\mathcal{P}^{S_1^*}$ and $\mathcal{P}^{S_2^*}$.

- Recall from Covered Interest Parity that

$$\mathcal{P}_i^{fwd}(t,T') = \frac{e^{-q_i(T'-t)}}{\mathcal{P}^Z(t,T')}\mathcal{P}_t^{S_i}. \tag{6.22}$$

- Substituting (6.22) into (6.20), we get:

$$\begin{aligned}\mathcal{P}_t^{S_i^*} &= \mathcal{P}^Z(t,T')\,\mathcal{P}_i^{fwd}(t,T') \\ &= \mathcal{P}^Z(t,T')\,\frac{e^{-q_i(T'-t)}}{\mathcal{P}^Z(t,T')}\mathcal{P}_t^{S_i} \\ &= e^{-q_i(T'-t)}\mathcal{P}_t^{S_i}.\end{aligned}$$

- Observe that $\mathcal{P}_t^{S_i^*} = e^{-q_i(T'-t)}\mathcal{P}_t^{S_i}$,

- Thus, the volatility of the ratio $\mathcal{P}_t^{S_1^*}/\mathcal{P}_t^{S_2^*}$ is the same as the volatility of the ratio $\mathcal{P}_t^{S_1}/\mathcal{P}_t^{S_2}$, because $e^{-q_i(T'-t)}$ is a function only of time, and so is non-random.

- Therefore, we can use the formula of Margrabe (1978) to find the value of a deferred-exchange option,
  - ⋆ using the prices $\mathcal{P}_0^{S_i^*} = e^{-q_i T'}\mathcal{P}_0^{S_i}$ as the initial asset prices, and
  - ⋆ zero as their dividend yields:

- Thus, the value of a deferred exchange option is:

$$\begin{aligned}
\mathcal{P}_{\text{deferred-exch. option}} &= \mathcal{P}^Z(0,T')\left[\mathcal{P}^{Fwd_1}(0,T')N(d_1) - \mathcal{P}^{Fwd_2}(0,T')N(d_2)\right] \\
&= e^{-q_1 T'}\mathcal{P}_0^{S_1}N(d_1) - e^{-q_2 T'}\mathcal{P}_0^{S_2}N(d_2), \\
\text{where} \quad d_1 &= \frac{\log\left(\frac{e^{-q_1 T'}\mathcal{P}_0^{S_1}}{e^{-q_2 T'}\mathcal{P}_0^{S_2}}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \\
d_2 &= d_1 - \sigma\sqrt{T}, \\
\sigma &= \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\,\sigma_2}.
\end{aligned}$$

6.9 Currency Options

- We now see how to apply the Black-Scholes model to value currency options and options on foreign assets.
- In this section, we will use a * to indicate quantities for the foreign country.
- Let $\mathcal{P}_t^X$ denote the HC (home currency) price of one unit of FC (foreign currency) at date t.
  - ⋆ For example, if you live in France, then the spot rate for Japanese yen would be the price in Euro of one unit of Japanese yen: $\mathcal{P}^X = \text{EUR } 0.01/\text{JPY}$.
  - ⋆ Similarly, if $\mathcal{P}_t^{S^*}$ is the price of a stock in units of FC, then its home currency value is given by $\mathcal{P}_t^X \times \mathcal{P}_t^{S^*}$.

- We will continue to make the assumptions of the Black-Scholes model:
  - ⋆ There is a foreign asset with price $\mathcal{P}_t^{S^*}$ in terms of FC.
  - ⋆ It has a constant yield q and a constant volatility σ.
  - ⋆ The exchange rate has a constant volatility σ_x.
  - ⋆ The correlation between the foreign asset and the exchange rate is ρ.
  - ⋆ The constant domestic and foreign risk-free interest rates are r and r^*.
    Below, we will see that we do *not* need to assume risk-free constant interest rates.

6.9.1 Options on the Spot Exchange Rate

- A European call option on the exchange rate $\mathcal{P}^X$ pays at date T:

  $$\max\left(0, \mathcal{P}_T^X - K\right), \quad \text{where } K \text{ is the strike price in HC.}$$

- The underlying for this option is the foreign risk-free asset, which has a "yield" of r^*.
  - ⋆ Therefore, the cost at date 0 of obtaining one unit of FC at T is: $e^{-r^*T}\mathcal{P}_0^X$.
- Thus,
  - ⋆ the exchange rate is analogous to a stock price,
  - ⋆ with the foreign risk-free rate being its dividend yield.

- Thus, the value of options on foreign currency can be obtained using the Black-Scholes model with the following inputs:
  - ⋆ Initial asset price $= \mathcal{P}_0^X$;
  - ⋆ Dividend-yield $= r^*$;
  - ⋆ Risk-free rate $= r$;
  - ⋆ Volatility $= \sigma_x$; that is, the volatility of the exchange rate.
- This is the Garman and Kohlhagen (1983) formula for currency options.

6.9.2 Allowing for Nonconstant Home and Foreign Interest Rates

- The forward exchange rate at date 0, for a contract maturing at T, must be

$$\mathcal{P}^{Fwd}(0,T) = e^{(r-r^*)T}\mathcal{P}_0^X \quad \ldots \text{Covered Interest Parity.}$$

- We can write this in terms of the prices of HC and FC discount bonds,
  - ⋆ $\mathcal{P}^Z(0,T) = e^{-yT}$; and,
  - ⋆ $\mathcal{P}^{Z^*}(0,T) = e^{-y^*T}$

  where y and y^* are the yields on HC and FC bonds, respectively. Then,

$$\mathcal{P}^{Fwd}(0,T) = \frac{\mathcal{P}^{Z^*}(0,T)}{\mathcal{P}^Z(0,T)}\mathcal{P}_0^X \quad \ldots \text{Covered Interest Parity.}$$

- Substituting this result in Merton's formula, we get the value of currency options without assuming that home and foreign interest rates are constant:

$$\text{Call} = \mathcal{P}^{Z^*}(0,T)\,\mathcal{P}_0^X\,N(d_1) - \mathcal{P}^Z(0,T)\,K\,N(d_2), \tag{6.23}$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^{Z^*}(0,T)\,\mathcal{P}_0^X}{\mathcal{P}^Z(0,T)}\frac{1}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} = \frac{\log\left(\frac{\mathcal{P}^{Fwd}(0,T)}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \tag{6.24}$$

$$d_2 = d_1 - \sigma\sqrt{T}. \tag{6.25}$$

in which σ now denotes the constant volatility of the forward rate.

- This is the Grabbe (1983) formula for currency options.

Test Your Understanding 6.1 (Currency option prices in terms of forward rates) On the previous page, we have

$$\text{Call} = \mathcal{P}^{Z^*}(0,T)\,\mathcal{P}_0^X\,N(d_1) - \mathcal{P}^Z\,K\,N(d_2),$$

$$\text{Put} = \mathcal{P}^Z(0,T)\,K\,N(-d_2) - \mathcal{P}^{Z^*}(0,T)\,\mathcal{P}_0^X\,N(-d_1).$$

Explain how you can use Covered Interest Parity to express these option prices in terms of the forward exchange rate. Then, use your solution to provide an interpretation for the option prices.

Hint: Note that Covered Interest Rate Parity can be written as

$$\mathcal{P}^{Fwd}(0,T) = \frac{e^{r\,T}}{e^{r^*\,T}}\,\mathcal{P}_0^X = \frac{\mathcal{P}^{Z^*}(0,T)}{\mathcal{P}^Z(0,T)}\,\mathcal{P}_0^X.$$

Space for your work.

Space for your work.

6.10 Exotic Options

- So far, we have seen the valuation of a variety of options:
  - ⋆ exchange options, deferred exchange options, options on forwards, and currency options.
- The key insight from looking at these various options is that one can obtain their value using the insights from the Black-Scholes-Merton model.
- In this section, we show that one can obtain the value also of exotic options without doing any additional calculations.

- Suppose we wish to find the price of an exotic option that depends on the maximum of the prices of two risky assets, $\mathcal{P}^{S_1}(t)$ and $\mathcal{P}^{S_2}(t)$:

$$\max\left[\mathcal{P}^{S_1}(T), \mathcal{P}^{S_2}(T)\right].$$

- Consider a setting where the two risky assets pay constant dividend yields q_i, $i = \{1, 2\}$, and assume that the price of each of the risky assets satisfies:

$$\frac{dS_i}{S_i} = \mu_i\, dt + \sigma_i\, dB_i^{\mathbb{P}}, \qquad \text{for} \qquad i = \{1, 2\},$$

where each $B_i^{\mathbb{P}}$ is a Brownian motion under the actual (physical) probability measure, and the correlation between $dB_1^{\mathbb{P}}$ and $dB_2^{\mathbb{P}}$ is $\rho\, dt$.

- Furthermore, assume that μ_i and σ_i are constants.

- We have already studied the valuation of a European option to *exchange one risky asset for another*; that is, an option whose payoff is given by:

$$\max\left[\mathcal{P}^{S_1}(T) - \mathcal{P}^{S_2}(T), 0\right], \tag{6.26}$$

and we found that at $t = 0$:

$$\text{Value}_0^{\text{exchange option}} = e^{-q_1 T}\,\mathcal{P}^{S_1}(0)\,N(d_1) - e^{-q_2 T}\,\mathcal{P}^{S_2}(0)\,N(d_2), \tag{6.27}$$

where

$$d_1 = \frac{\log\left(\frac{e^{-q_1}\mathcal{P}^{S_1}(0)}{e^{-q_2}\mathcal{P}^{S_2}(0)}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}; \qquad d_2 = d_1 - \sigma\sqrt{T}; \qquad \text{and}$$
$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}.$$

- We wish to find the price of a European option whose payoff at maturity is the maximum of the two asset prices:

$$\max\left[\mathcal{P}^{S_1}(T), \mathcal{P}^{S_2}(T)\right]. \tag{6.28}$$

- Note that by subtracting and adding $\mathcal{P}^{S_2}(T)$ to the payoff in (6.28), you can express the payoff as:

$$\max\left[\mathcal{P}^{S_1}(T), \mathcal{P}^{S_2}(T)\right] = \underbrace{\max\left[\mathcal{P}^{S_1}(T) - \mathcal{P}^{S_2}(T), 0\right]}_{\text{first component}} + \underbrace{\mathcal{P}^{S_2}(T)}_{\text{second component}}.$$

- So, we need to find the value of the two components in the above equation.

- We start by reproducing the payoff that we wish to value:

$$\max\left[\mathcal{P}^{S_1}(T), \mathcal{P}^{S_2}(T)\right] = \underbrace{\max\left[\mathcal{P}^{S_1}(T) - \mathcal{P}^{S_2}(T), 0\right]}_{\text{first component}} + \underbrace{\mathcal{P}^{S_2}(T)}_{\text{second component}}.$$

- The present value of the first component is just the value of an exchange option, which we have already determined.
- The second component, the payoff $\mathcal{P}^{S_2}(T)$, is equal to delivering one share of Stock 2 at date T.
- If the stock did not pay any dividends, then to deliver one share at date T you would need to buy one share today, which has the present value $\mathcal{P}^{S_2}(0)$.
- However, if the share pays dividends, then we need to start out with an amount that is adjusted for the dividends paid from 0 to T.
  - ⋆ Thus, the payoff $\mathcal{P}^{S_2}(T)$ has a present value of $e^{-q_2 T}\mathcal{P}^{S_2}(0)$.

- Thus, the value today of the payoff $\max\left[\mathcal{P}^{S_1}(T), \mathcal{P}^{S_2}(T)\right]$ is the sum of the values of these two component payoffs:

$$\begin{aligned}
V_0^{\max} &= V_0^{\text{exch}} + V_0\left(\mathcal{P}^{S_2}(T)\right) \\
&= \left[e^{-q_1 T}\mathcal{P}^{S_1}(0)\, N(d_1) - e^{-q_2 T}\mathcal{P}^{S_2}(0)\, N(d_2)\right] + e^{-q_2 T}\mathcal{P}^{S_2}(0) \\
&= e^{-q_1 T}\mathcal{P}^{S_1}(0)\, N(d_1) + e^{-q_2 T}\mathcal{P}^{S_2}(0)\,[1 - N(d_2] \\
&= e^{-q_1 T}\mathcal{P}^{S_1}(0)\, N(d_1) + e^{-q_2 T}\mathcal{P}^{S_2}(0)\,[N(-d_2)]
\end{aligned}$$

where $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$,

$$d_1 = \frac{\log\left(\frac{e^{-q_1}\mathcal{P}^{S_1}(0)}{e^{-q_2}\mathcal{P}^{S_2}(0)}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} = \frac{\log\left(\frac{e^{-q_1}\mathcal{P}^{S_1}(0)}{e^{-q_2}\mathcal{P}^{S_2}(0)}\right)}{\sigma\sqrt{T}} + \frac{1}{2}\sigma\sqrt{T}; \quad \text{and}$$

$$d_2 = d_1 - \sigma\sqrt{T} = \frac{\log\left(\frac{e^{-q_1}\mathcal{P}^{S_1}(0)}{e^{-q_2}\mathcal{P}^{S_2}(0)}\right)}{\sigma\sqrt{T}} - \frac{1}{2}\sigma\sqrt{T}.$$

- In the example above, we looked at the valuation of a particular exotic option: the option on the maximum of the prices of two risky assets.

- Other exotic options that can be valued in the same way include:

  ⋆ Forward start option, which comes to life only at some future date T^* and has a life of τ years, starting from T^*, and the strike price K is not specified at the outset but is determined at T^* as equal to some proportion of the stock price at date T^*.

  ⋆ Chooser option, where the option holder has the right to decide by a specified date whether the option is a call option or a put option.

  ⋆ Compound option is an option on an option; that is, the underlying asset itself is an option. For example, a call option on a call option gives the holder the right to buy the call by paying K^*, which has the payoff:

$$\max\left[\mathcal{P}^C(\mathcal{P}^S, K, T-t) - K^*, 0\right]$$

  ⋆ Quanto option, which is a cross-currency option in which the option is written on a security that trades in one currency but the payoff is translated into a different currency in a prespecified manner.

  ⋆ Option on minimum of two assets: $\min\left[\mathcal{P}^{S_1}(T), \mathcal{P}^{S_2}(T)\right]$.

  ⋆ Option on maximum of two assets and cash : $\max\left[\mathcal{P}^{S_1}(T), \mathcal{P}^{S_2}(T), K\right]$.

- All the options listed above are path-independent options; that is, their payoff does not depend on the price path of the underlying asset.

- One can value also path-dependent options, but their valuation is often more complex, and for many of them a closed-form formula for the value does not exist, so valuation has to be done numerically. These include:

  ⋆ Barrier options;

  ⋆ Asian options;

  ⋆ Lookback options, etc.

6.11 Implied Volatility

- All of the inputs into the option pricing formula are observable, except for the volatility coefficient, σ.

- We can estimate volatility

  1. either from historical data,
  2. or from the prices of other options, by exploiting the one-to-one relation between the price given by the Black Scholes formula and the σ; the σ computed in this way is called implied volatility.

6.11.1 Non-constant volatility

- We derived the Black-Scholes option prices assuming that the natural logarithm of the stock price at maturity is normally distributed with a certain mean (depending on the numeraire) and variance equal to $\sigma^2 T$.

- But we do not need the volatility to be constant over time.

- The option pricing formulas are still valid if

$$\frac{d\mathcal{P}_t^S}{\mathcal{P}_t^S} = \mu_t \, dt + \sigma_t \, dB_t,$$

  where

  ⋆ σ_t is some non-random function of time,

  ⋆ while μ_t can be a quite general random process.

- In this case, the variance of $\log \mathcal{P}_T^S$ will be

$$\int_0^T \sigma_t^2 \, dt, \tag{6.29}$$

  which is essentially the sum of the instantaneous variances $\sigma^2 \, dt$.

- Thus, we need to replace $\sigma^2 \, T$ by the expression in (6.29) in the d_1 and d_2 terms of the option pricing formula.

- Or, we can replace σ by σ_{avg}, where

$$\sigma_{\text{avg}}^2 = \frac{1}{T} \int_0^T \sigma_t^2 \, dt. \tag{6.30}$$

6.11.2 Smiles and Smirks

- If we compute implied volatilities for options with the same maturity but different strikes, we obtain different implied volatilities rather than the same.

- If we plot the implied volatility against the strike, the pattern one normally sees for options on stocks and stock indexes is that
  - ⋆ the implied volatility declines as the strike increases, until the strike price is near the current value of the underlying (so the option is "at the money"),
  - ⋆ and then the implied volatility flattens out or increases slightly.
  - ⋆ This pattern looks like a twisted "smile" or a "smirk."

- This pattern of implied volatilities is inconsistent with the assumption of a lognormal risk-neutral distribution:
  - ⋆ it suggests that there are "fat tails" with the probability of extreme negative returns (disasters) being higher than the probability of extreme positive returns (that is, the return distribution is "skewed.")

- A simple way to modify the Black-Scholes-Merton model to match the market prices of options having different strikes and maturities is to assume that the volatility is a function $\sigma(t, \mathcal{P}_t^S)$, rather than a constant.
- This is called a local volatility model.
- Subject to $M\mathcal{P}^D$ and $M\mathcal{P}^S$ being martingales, the market is still complete, and the pricing formula for a call, given below, is still valid:

$$\mathcal{P}^C = e^{-q\,T}\,\mathcal{P}_0^S\,\text{prob}^{\mathbb{S}}(A) - e^{-y\,T}\,K\,\text{prob}^{\mathbb{Q}}(A).$$

- The only difference is that now one needs to compute $\text{prob}^{\mathbb{S}}(A)$ and $\text{prob}^{\mathbb{Q}}(A)$ numerically.

- There is a logical inconsistency in adopting this approach because
  - ⋆ on one hand, the model assumes that σ is constant,
  - ⋆ but, on the other hand, we change the function $\sigma(t, \mathcal{P}_t^S)$ as market prices change.
- To avoid this logical inconsistency, one can develop models that explicitly allow for stochastic volatility, such as Heston (1993).
- You can read about the stochastic-volatility model in Kerry Back's book.

6.12 Summary

1. The Black-Scholes formula for European call options can be extended to allow for dividends, and can then be interpreted as:

$$\text{Value of call option} = \overbrace{e^{-qT}\mathcal{P}_0^S}^{\text{PV of receipt}} N(d_1) - \overbrace{e^{-rT}K}^{\text{PV of payment}} N(d_2)$$

$$\text{where} \qquad d_1 = \frac{\log\left(\frac{e^{-qT}\mathcal{P}_0^S}{e^{-rT}K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}.$$

2. The option to exchange one risky asset for another can be valued by changing the numeraire to one of the risky assets.

Margrabe's formula for exchange options is:

$$\text{Value of exchange option} = e^{-q_1 T}\mathcal{P}_0^{S_1} N(d_1) - e^{-q_2 T}\mathcal{P}_0^{S_2} N(d_2)$$

$$\text{where} \qquad d_1 = \frac{\log\left(\frac{e^{-q_1 T}\mathcal{P}_0^{S_1}}{e^{-q_2 T}\mathcal{P}_0^{S_2}}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}.$$

where σ is the constant volatility of the ratio of the two asset prices.

3. Interpreting the option on forwards in terms of Margrabe's formula leads to Black's formula for the price of European options on forwards:

$$\text{Call Price} = \mathcal{P}^Z(0,T') \left[\mathcal{P}^{Fwd}(0,T')\, N(d_1) - K\, N(d_2)\right],$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^{Fwd}(0,T')}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T},$$

and σ is the volatility of the forward contract.

4. Using Covered Interest Parity and noting that at maturity the value of a forward contract and the spot are the same, we get Merton's formula for options, which does not require the risk-free rate to be constant:

$$\text{Call} = e^{-q\,T}\,\mathcal{P}_0^S\, N(d_1) - \mathcal{P}^Z(0,T)\,K\,N(d_2),$$

where

$$d_1 = \frac{\log\left(\frac{e^{-q\,T}\,\mathcal{P}_0^S}{\mathcal{P}^Z(0,T)\,K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T},$$

where σ is the constant volatility of the forward price.

5. Using Margrabe's formula and Covered Interest Parity, we obtain the the value of a deferred exchange option

$$\text{Value} = e^{-q_1 T'} \mathcal{P}_0^{S_1} N(d_1) - e^{-q_2 T'} \mathcal{P}_0^{S_2} N(d_2),$$

$$\text{where} \quad d_1 = \frac{\log\left(\frac{e^{-q_1 T'} \mathcal{P}_0^{S_1}}{e^{-q_2 T'} \mathcal{P}_0^{S_2}}\right) + \frac{1}{2}\sigma^2}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}.$$

where

- σ is the volatility of the ratio of the prices of the two assets being exchanged,
- the option matures at date T, and
- the exchange takes place at $T' > T$.

6. The value of options on foreign currency can be obtained using the Black-Scholes model with the following inputs:

- Initial asset price $= \mathcal{P}_0^X$;
- Dividend-yield $= r^*$; that is, the foreign interest rate;
- Risk-free rate $= r$;
- Volatility $= \sigma_x$; that is, the volatility of the spot exchange rate.

7. Using Covered Interest Parity and Merton's formula, we can obtain the value of currency options without assuming that domestic and foreign risk-free interest rates are constant:

$$\text{Call} = \mathcal{P}^{Z^*}(0,T)\,\mathcal{P}_0^X\,N(d_1) - \mathcal{P}^{Z}(0,T)\,K\,N(d_2),$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^{Z^*}(0,T)\,\mathcal{P}_0^X}{\mathcal{P}^{Z}(0,T)}\frac{1}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} = \frac{\log\left(\frac{\mathcal{P}^{Fwd}(0,T)}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}.$$

where σ now denotes the constant volatility of the forward exchange rate.

8. All of the inputs into the option pricing formula are observable, except for the volatility coefficient, σ.

- We can estimate volatility either
  - ⋆ from historical data (standard deviation), or
  - ⋆ from the prices of other options (implied volatility).

- The Black-Scholes formula is still valid even if σ_t is a not constant,
  - ⋆ as long as it is a deterministic (non-random) function of time.
  - ⋆ In this case, we need to replace in the formula σ by $\sigma^2_{\text{avg}} = \frac{1}{T}\int_0^T \sigma_t^2\,dt$.

9. If we compute implied volatilities for options with the same maturity but different strikes, we obtain different implied volatilities rather than the same.

- This suggests that the return distribution is fat tailed and skewed, rather than being normal, and that we should consider a model that allows for stochastic volatility.

6.13 Suggestions for Further Readings

6.13.1 Source material for the lecture notes

- This material is based on Back (2005, Ch. 6–8) and Back (2017, Ch. 17).
- The models we discussed in this chapter are from the following papers: Margrabe (1978), Black (1976), Merton (1973), Garman and Kohlhagen (1983), Grabbe (1983), and (Stulz 1982).
- As we will see in the second part of the course, similar arguments can be used also to value options on: interest rates, bond prices, and swaps.

6.13.2 Potential applications and extensions of lecture notes

- The book by Rubinstein (1999) does an excellent job of explaining how one can obtain, in a straightforward manner, the value of a large number of exotic options from the prices of standard options using replication arguments.
- The book by Vecer (2011) is written along the same lines.
- The same kind of analysis can be used to value options other than the ones discussed in the lecture notes, such as quantos, compound options, choosers, barrier options, and lookback options; see Back (2005, Chapter 8).

- **Jump processes** are discussed in Back (2017, Pages 375–376).
- There are several models of option pricing with jumps. For instance,
  - ⋆ Merton (1976) allows for jumps when pricing options.
  - ⋆ Bakshi, Cao, and Chen (1997) and Bates (1996, 2008) provide empirical tests of option pricing models with jumps.
  - ⋆ Das and Sundaram (1999) develop a general option pricing model with jumps in the level and volatility of the underlying asset price.
  - ⋆ Naik and Lee (1990) develop a general equilibrium model for pricing of options on the market when its return may have jumps.

6.13.3 Ph.D. elective courses offered on related topics

- Rama Cont taught a course on *Monte Carlo Methods* in Finance in 2010 and 2013; Marcel Rindisbacher taught this course in 2017.
- Tim Bollerslev taught a course on *Volatility Modelling* in 2010, 2012, 2016.
- Yacine Aït-Sahalia taught *Financial Econometrics of Jumps* in 2011.
- Jerome Detemple taught a course on *Derivative Securities* in 2011.
- Mikhail Chernov taught *Empirical Option Pricing* in 2012 and 2014.
- Federico Bandi taught *The Econometrics of Continuous-Time Models* in 2012 and 2015, and *Estimation of Continuous-Time Models* in 2013.
- Torben Andersen taught *High-Frequency Asset Pricing* in 2013 and 2015.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 7

Course Summary: Key Ideas Covered in Part 1

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EDHEC Business School

Chapter contents

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7.1 Unifying View of Asset Pricing Problems and Methods

1. The valuation of all payoffs relies on the principle of no-arbitrage.

2. There are only **two** things that matter when valuing (replicating) cashflows:

2.1. Adjusting for (matching) the **timing** of the cashflows; and

2.2. Adjusting for (matching) the **riskiness** of cashflows.

3. There are two possible settings in which valuation has to be done.

- The first is where financial markets are **complete**, and
- the second is where markets **incomplete**.

4. In complete markets, redundant securities can be valued in a straightforward manner, using a variety of equivalent methods:

4.1. Find a portfolio that *replicates* the payoff to the option; then, the value of the option must equal the value of the replicating portfolio.

4.2. Find a portfolio that *hedges* the payoff to the option; then, use the value of the hedged payoff to infer the value of the option.

4.3. Find the *risk-adjusted probabilities*, $\text{prob}^{\mathbb{Q}}$; then, use the probabilities to find risk-adjusted expected payoff to the option and discount the expected payoff at the risk-free rate.

4.4. Find *Arrow-Debreu state price* for receiving 1 dollar in up (or down) state, $\mathcal{P}_0^{A_k} = \frac{\text{prob}_k^{\mathbb{Q}}}{R_f}$; then, use the state prices to find the value of the option.

4.5. Find *martingale* value of option in terms of bank deposit, $\mathcal{P}^D$, using probabilities $\text{prob}_k^{\mathbb{Q}}$; then, use this to find value of option in terms of dollars. Note $\text{prob}_k^{\mathbb{Q}} = \mathcal{P}_0^{A_k} \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}$.

4.6. Find *martingale* value of option in terms of the stock price S, using the probabilities $\text{prob}_k^{\mathbb{S}}$; then, use this to find the option value in terms of dollars. Note that $\text{prob}_k^{\mathbb{S}} = \mathcal{P}_0^{A_k} \frac{\mathcal{P}_k^S}{\mathcal{P}_0^S}$.

5. The stochastic discount factor m_T turns asset prices into martingales:

$$\mathcal{P}_0 = \mathbb{E}^{\mathbb{P}}\left[m_T \times \mathcal{P}_T\right], \quad \text{where} \quad m_k = \frac{\mathcal{P}_0^{A_k}}{\text{prob}_k^{\mathbb{P}}}.$$

6. One can recover the risk-neutral probabilities (RNP) from the SDF as follows:

$$\text{prob}_u^{\mathbb{Q}} = \text{prob}_u^{\mathbb{P}}\, m_u\, \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}; \quad \text{and} \quad \text{prob}_d^{\mathbb{Q}} = \text{prob}_d^{\mathbb{P}}\, m_d\, \frac{\mathcal{P}_1^D}{\mathcal{P}_0^D}.$$

7. The key insight of the material we have covered is that all the valuation methods discussed above rely on the same principle:

- use the information contained in prices of fundamental assets, such as bonds, stocks, foreign currency;
- to find the value of derivative (redundant) contracts, such as forwards, futures, options.

8. The second key insight of the material we have covered is that exactly the same method can be used to study all the problems that we have considered:

- Valuing redundant assets
- Hedging risky payoffs
- Choosing a portfolio to "replicate" a given payoff.

9. In terms of the (very) big picture, we have demonstrated above that

- if you know state prices,
- then it is straightforward to solve the problems of valuation, hedging (risk management), and portfolio choice in discrete time.

7.2 Maths for Continuous Time Finance

1. A stochastic process is a variable that changes randomly over time.

2. A Brownian motion is a stochastic process that

- evolves continuously in time (no jumps),
- whose change over time is
  - ⋆ Normally distributed,
  - ⋆ with mean zero, and
  - ⋆ variance equal to the length of time,
- whose quadratic variation over $[0, T]$ is T, and whose total variation over time is infinite.

3. Every continuous martingale is a transformation of a Brownian motion.

4. An Itô process is a variable X that changes over time as

$$dX(t) = \mu(t)dt + \sigma(t)dB^{\mathbb{P}}(t)$$

where $B^{\mathbb{P}}(t)$ is a Brownian motion, and μ and σ can also be random processes that satisfy some regularity conditions, and $\mu(t)$ and $\sigma(t)$ are known at t.

- Itô processes satisfy the following "rules":

$$\begin{aligned} (dt)^2 &= 0, \\ (dt)\,(dB^{\mathbb{P}}) &= 0, \\ (dB^{\mathbb{P}})^2 &= dt \\ (dB_x^{\mathbb{P}})(dB_y^{\mathbb{P}}) &= \rho\, dt \end{aligned}$$

  . . . where ρ is the correlation between the two Brownians.

- A geometric Brownian motion is an Itô process in which the drift μ and volatility σ are constant.

- An Itô process $dX(t)$ is a martingale only if it has zero drift: $\mu = 0$.

5. Itô's formula tells us that for

- If $C = g(t, X(t))$, and X is an Itô process given by

$$dX(t) = \mu_x(t)\, dt + \sigma_x(t)\, dB_x^{\mathbb{P}}(t), \tag{7.1}$$

then
$$\begin{aligned} dC &= g_t\, dt + g_x\, dX + \frac{1}{2} g_{xx}\, (dX)^2 \\ &= \left(g_t + g_x \mu_x(t) + \frac{1}{2} g_{xx}\, [\sigma_x(t)]^2 \right) dt + g_x\, \sigma_x(t)\, dB_x^{\mathbb{P}}. \end{aligned}$$

- If $C(t) = g(t, X(t), Y(t))$, and X and Y are Itô processes given in (2.23) and (2.24)

$$dY(t) = \mu_y(t)\,dt + \sigma_y(t)\,dB_y(t), \tag{7.2}$$

where $B_x^{\mathbb{P}}$ & $B_y^{\mathbb{P}}$ can be different Brownian motions, and $(dB_x^{\mathbb{P}})(dB_y^{\mathbb{P}}) = \rho\,dt$, then:

$$\begin{aligned} dC &= g_t\,dt + g_x\,dX + g_y\,dY + \frac{1}{2}g_{xx}\,(dX)^2 + \frac{1}{2}g_{yy}\,(dY)^2 + g_{xy}\,(dX)\,(dY) \\ &= \left(g_t + g_x\mu_x(t) + g_y\mu_y(t) + \frac{1}{2}g_{xx}\sigma_x^2(t) + \frac{1}{2}g_{yy}\sigma_y^2(t) + g_{xy}\sigma_x(t)\sigma_y(t)\rho_{xy}(t)\right)dt \\ &\quad + g_x\sigma_x^{\mathbb{P}}(t)dB_x + g_y\sigma_y^{\mathbb{P}}(t)dB_y. \end{aligned}$$

6. Itô's formula for products, ratios, and compounding/discounting:

- If $C = \log(X)$, then

$$dC = \frac{dX}{X} - \frac{1}{2}\left(\frac{dX}{X}\right)^2. \tag{7.3}$$

- If $C = g(X, Y) = X\,Y$, then

$$\frac{dC}{C} = \frac{dX}{X} + \frac{dY}{Y} + \left(\frac{dX}{X}\right)\left(\frac{dY}{Y}\right). \tag{7.4}$$

- If $C = \frac{Y}{X}$, then

$$\frac{dC}{C} = -\frac{dX}{X} + \frac{dY}{Y} + \left(\frac{dX}{X}\right)^2 - \left(\frac{dX}{X}\right)\left(\frac{dY}{Y}\right). \tag{7.5}$$

- If $Y(t) = \exp\left(\int_0^t q(s)ds\right)$ and $C = X\,Y$ for any Itô process X, then

$$\frac{dC}{C} = \underbrace{\frac{dX}{X}}_{\text{capital gains}} + \underbrace{q(t)dt}_{\text{dividends}}. \tag{7.6}$$

7. Suppose

$$\frac{dX}{X} = \mu_x\, dt + \sigma_x\, dB_x^{\mathbb{P}}, \qquad \text{and} \qquad \frac{dY}{Y} = \mu_y\, dt + \sigma_y\, dB_y^{\mathbb{P}},$$

where $B_x^{\mathbb{P}}$ and $B_y^{\mathbb{P}}$ are different Brownian motions with correlation ρ.

- Then, the **volatility** of $X\,Y$ per unit time is

$$\sqrt{\sigma_x^2 + \sigma_y^2 + 2\rho\,\sigma_x\,\sigma_y}, \qquad \text{and} \tag{7.7}$$

- the **volatility** of X/Y per unit time is

$$\sqrt{\sigma_x^2 + \sigma_y^2 - 2\rho\,\sigma_x\,\sigma_y}. \tag{7.8}$$

7.3 Stochastic Discount Factor and Returns

1. SDF

A stochastic process M is an SDF process if and only if $M_0 = 1$ and

$$\frac{dM_t}{M_t} = -r_t\, dt - \lambda_t^\top\, dB_t^{\mathbb{P}}, \quad \text{where} \quad \lambda_t = \sigma_t^{-1}(\mu_t - r_t\mathbf{1}).$$

2. The relation between the risk-free rate and the SDF is:

$$\mathbb{E}\left[\frac{dM}{M}\right] = -r_t\, dt, \quad \text{and}$$

3. The relation between the risk premium and the SDF is:

$$(\mu_{n,t} - r_t)\, dt = -\text{Cov}\left[\frac{dM}{M}, \frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right].$$

7.4 Change of Measure, Drift, and Numeraire

1. Change of Measure

If $\xi_t = M_t \mathcal{P}_t^D$, for each $t < T$,
then: $\text{prob}^{\mathbb{Q}}(A) = \mathbb{E}^{\mathbb{P}}[\xi_T \, 1_A]$,

so that for any random variable X depending on date-T information:

$$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[\xi_T \, X].$$

2. Girsanov's theorem states that

- if:
  - $B^{\mathbb{P}}$ is a vector of $\mathbb{P}$-Brownian motions;
  - ξ is a strictly positive P-martingale;
  - $\mathbb{Q}$ is a probability measure satisfying $\text{prob}^{\mathbb{Q}}(A) = \mathbb{E}^{\mathbb{P}}[\xi_T \, 1_A]$;
  - $d\xi/\xi = \lambda^\top dB^{\mathbb{P}}$ for a vector $\lambda = (\lambda_1, \ldots, \lambda_N)^\top$ of stochastic processes,
- then the following defines a vector of independent $\mathbb{Q}$-Brownian motions:

$$dB_t^{\mathbb{Q}} = dB_t^{\mathbb{P}} + \lambda_t \, dt.$$

- Girsanov's theorem tells us that changing measures
  - changes drifts,
  - but does not change volatilities of correlations.

3. Price process for risky asset under different measures

If the price process for the risky asset under $\mathbb{P}$ is

$$\frac{d\mathcal{P}^S}{\mathcal{P}^S} = \mu\, dt + \sigma\, dB^{\mathbb{P}},$$

or equivalently: $\quad d\log \mathcal{P}^S = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{P}},$

- then, under the probability measure $\mathbb{Q}$ the price process is:

$$d\log \mathcal{P}^S = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{Q}}, \qquad \ldots \mu \text{ has been replaced by } r,$$

- and under the probability measure $\mathbb{S}$ the price process is:

$$d\log \mathcal{P}^S = \left(r + \frac{1}{2}\sigma^2\right) dt + \sigma\, dB^{\mathbb{S}} \qquad \ldots \mu \text{ has been replaced by } r + \sigma^2.$$

4. Tail probability for geometric Brownian motion

Given the process

$$d\log \mathcal{P}^S = \alpha\, dt + \sigma\, dB, \qquad \text{where } \alpha \text{ is } \left(r + \tfrac{1}{2}\sigma^2\right) \text{ or } \left(r - \tfrac{1}{2}\sigma^2\right),$$

we have that

$$\text{prob}^{\mathbb{S}}\left(\mathcal{P}_T^S > K\right) = N(d_1), \qquad d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-rT}K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}};$$

$$\text{prob}^{\mathbb{Q}}\left(\mathcal{P}_T^S > K\right) = N(d_2), \qquad d_2 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-rT}K}\right) - \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$= d_1 - \sigma\sqrt{T}.$$

7.5 Black and Scholes Option Pricing Model

1. The Black-Scholes value of a European call option is

$$\begin{aligned}\mathcal{P}_0^C &= \mathbb{E}^{\mathbb{P}}\left[M_T\,\mathcal{P}_T^S\,1_A\right] - K\,\mathbb{E}^{\mathbb{P}}\left[M_T\,1_A\right]\\ &= \mathcal{P}_0^S\,\text{prob}^{\mathbb{S}}(A) - e^{-r\,T}\,K\,\text{prob}^{\mathbb{Q}}(A)\\ &= \mathcal{P}_0^S\,N(d_1) - e^{-r\,T}\,K\,N(d_2),\end{aligned}$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}\,K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}};$$

$$d_2 = d_1 - \sigma\,\sqrt{T}.$$

2. The price today of a European call option

$$\mathcal{P}_0^C = \mathcal{P}_0^S\,N(d_1) - e^{-r\,T}\,K\,N(d_2),$$

$$d_1 = \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}\,K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}}; \qquad d_2 = d_1 - \sigma\,\sqrt{T},$$

can be interpreted as the difference between the

- present value of what you will receive, $\mathcal{P}_0^S$, multiplied by the appropriate probability of getting this, $N(d_1) = \text{prob}^{\mathbb{S}}(\mathcal{P}_T^S \geq K)$, and;
- present value of what you will deliver, $e^{-r\,T}\,K$, multiplied by the appropriate probability of paying this, $N(d_2) = \text{prob}^{\mathbb{Q}}(\mathcal{P}_T^S \geq K)$.
- In d_1, the argument of log has the ratio of the present values of the quantities to be received and paid, $\mathcal{P}_0^S/(e^{-r\,T}\,K)$; and,
- The difference between d_1 and d_2, which is $\sigma\,\sqrt{T}$, is the difference in the volatility of the quantities to be received and to be paid.

3. To interpret the Black-Scholes formula for a European call option in another way, recall that the forward price for a stock is:

$$\mathcal{P}_0^{Fwd}(T) = e^{r\,T} \times \mathcal{P}_0^S \qquad \Longleftrightarrow \qquad e^{-r\,T}\mathcal{P}_0^{Fwd}(T) = \mathcal{P}_0^S.$$

Now, we substitute this into the Black-Scholes formula:

$$\begin{aligned}\mathcal{P}_0^C &= \overbrace{\mathcal{P}_0^S}^{\text{PV of asset}}\; N(d_1) - \overbrace{e^{-r\,T}\,K}^{\text{PV of strike}}\; N(d_2) \\ &= e^{-r\,T}\mathcal{P}_0^{Fwd}(T)\,N(d_1) - e^{-r\,T}\,K\,N(d_2) \\ &= e^{-r\,T}\left[\mathcal{P}_0^{Fwd}(T)\,N(d_1) - K\,N(d_2)\right].\end{aligned}$$

Thus, the value of a European call option is the discounted value of the difference between

- the amount you will receive at T, the forward price, multiplied by the probability of receiving this under the appropriate probability; and,
- the amount you will pay at T, the strike, multiplied by the probability of receiving this under the appropriate probability;

Similarly, we can express d_1 and d_2 in terms of the forward price:

$$\begin{aligned}d_1 &= \frac{\log\left(\frac{\mathcal{P}_0^S}{e^{-r\,T}K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}} \\ &= \frac{\log\left(\frac{e^{r\,T}\,\mathcal{P}_0^S}{K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}} \\ &= \frac{\log\left(\frac{\mathcal{P}_0^{Fwd}(T)}{K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}},\end{aligned}$$

and

$$d_2 = d_1 - \sigma\,\sqrt{T}.$$

4. The fundamental PDE that all derivatives (on a non-dividend paying underlying asset) must satisfy is:

$$V_t + r\,\mathcal{P}^S\,V_{\mathcal{P}S} + \frac{1}{2}\sigma^2\,(\mathcal{P}^S)^2\,V_{\mathcal{P}S\mathcal{P}S} \;= r\,V.$$

5. The expected return on the stock, μ,

- does not appear explicitly in the Black-Scholes formula, but
- its effect is present via the stock price (or via the SDF).

7.6 Extensions of the Black and Scholes Model

1. The Black-Scholes formula for European call options can be extended to allow for dividends, and can then be interpreted as:

$$\text{Value of call option} \;=\; \overbrace{e^{-q\,T}\,\mathcal{P}_0^S}^{\text{PV of receipt}}\;N(d_1) - \overbrace{e^{-r\,T}\,K}^{\text{PV of payment}}\;N(d_2)$$

$$\text{where} \qquad d_1 \;=\; \frac{\log\left(\frac{e^{-q\,T}\mathcal{P}_0^S}{e^{-r\,T}K}\right) + \frac{1}{2}\sigma^2\,T}{\sigma\sqrt{T}}$$

$$d_2 \;=\; d_1 - \sigma\,\sqrt{T}.$$

2. The option to exchange one risky asset for another can be valued by changing the numeraire to one of the risky assets.

Margrabe's formula for exchange options is:

$$\text{Value of exchange option} = e^{-q_1 T}\,\mathcal{P}_0^{S_1}\,N(d_1) - e^{-q_2 T}\,\mathcal{P}_0^{S_2}\,N(d_2)$$

$$\text{where} \qquad d_1 = \frac{\log\left(\frac{e^{-q_1 T}\,\mathcal{P}_0^{S_1}}{e^{-q_2 T}\,\mathcal{P}_0^{S_2}}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}.$$

where σ is the constant volatility of the ratio of the two asset prices.

3. Interpreting the option on forwards in terms of Margrabe's formula leads to Black's formula for the price of European options on forwards:

$$\text{Call Price} = \mathcal{P}^Z(0,T')\left[\mathcal{P}^{Fwd}(0,T')\,N(d_1) - K\,N(d_2)\right],$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^{Fwd}(0,T')}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T},$$

and σ is the volatility of the forward contract.

4. Using Covered Interest Parity and noting that at maturity the value of a forward contract and the spot are the same, we get Merton's formula for options, which does not require the risk-free rate to be constant:

$$\text{Call} = e^{-q\,T}\,\mathcal{P}_0^S\,N(d_1) - \mathcal{P}^Z(0,T)\,K\,N(d_2),$$

where

$$d_1 = \frac{\log\left(\frac{e^{-q\,T}\,\mathcal{P}_0^S}{\mathcal{P}^Z(0,T)\,K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\,\sqrt{T}}$$

$$d_2 = d_1 - \sigma\,\sqrt{T},$$

where σ is the constant volatility of the forward price.

5. Using Margrabe's formula and Covered Interest Parity, we obtain the the value of a deferred exchange option

$$\text{Value} = e^{-q_1 T'}\mathcal{P}_0^{S_1}N(d_1) - e^{-q_2 T'}\mathcal{P}_0^{S_2}N(d_2),$$

$$\text{where} \quad d_1 = \frac{\log\left(\frac{e^{-q_1\,T'}\,\mathcal{P}_0^{S_1}}{e^{-q_2\,T'}\,\mathcal{P}_0^{S_2}}\right) + \frac{1}{2}\sigma^2}{\sigma\,\sqrt{T}}$$

$$d_2 = d_1 - \sigma\,\sqrt{T}.$$

where

- σ is the volatility of the ratio of the prices of the two assets being exchanged,
- the option matures at date T, and
- the exchange takes place at $T' > T$.

6. The value of options on foreign currency can be obtained using the Black-Scholes model with the following inputs:

- Initial asset price $= \mathcal{P}_0^X$;
- Dividend-yield $= r^*$; that is, the foreign interest rate;
- Risk-free rate $= r$;
- Volatility $= \sigma_x$; that is, the volatility of the spot exchange rate.

7. Alternatively, using Covered Interest Parity and Merton's formula, we can obtain the value of currency options without assuming that domestic and foreign risk-free interest rates are constant:

$$\text{Call} = \mathcal{P}^{Z^*}(0,T)\,\mathcal{P}_0^X\,N(d_1) - \mathcal{P}^Z(0,T)\,K\,N(d_2),$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^{Z^*}(0,T)\,\mathcal{P}_0^X}{\mathcal{P}^Z(0,T)}\frac{1}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} = \frac{\log\left(\frac{\mathcal{P}^{Fwd}(0,T)}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}.$$

where σ now denotes the constant volatility of the forward exchange rate.

8. All of the inputs into the option pricing formula are observable, except for the volatility coefficient, σ.

- We can estimate volatility either
  - ⋆ from historical data (standard deviation), or
  - ⋆ from the prices of other options (implied volatility).

- The Black-Scholes formula is still valid even if σ_t is a not constant,
  - ⋆ as long as it is a deterministic (non-random) function of time.
  - ⋆ In this case, we need to replace in the formula σ by $\sigma^2_{\text{avg}} = \frac{1}{T}\int_0^T \sigma_t^2\, dt$.

9. If we compute implied volatilities for options with the same maturity but different strikes, we obtain different implied volatilities rather than the same.

- This suggests that the return distribution is fat tailed and skewed, rather than being normal, and that we should consider a model that allows for stochastic volatility.

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Lecture Notes for 2025

Continuous Time Financial Economics

Appendix A

Practice Questions for Part 1

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EDHEC Business School

Enclosed below are some practice questions that will help you improve your understanding of the material covered in class; your answers to these questions do **not** have to be submitted to me (and the solutions are included below).

1. Market completeness

1.1. What do we mean when we say "financial markets are complete?"

1.2. In the real world, are financial markets complete or incomplete?

1.3. For valuing financial assets, is it reasonable to assume markets are complete?

1.4. When markets are complete, what method of valuation gives answers that are close to the price of the security? Why?

1.5. When markets are not complete, how can we value securities?

Solution:

1a We say "financial markets are complete" when one can generate any desired pattern of cashflows. In a model where from each node there are two possible future outcomes (that is, a binomial model), we need two securities (usually the stock and bond) to complete the market. In a more general model, if from each node there are M possible outcomes, then for markets to be complete we need M securities (whose payoffs are independent).

1b In the real world, markets are unlikely to be complete because

- ⋆ there are a very large number of possible outcomes from any current state of the economy;
- ⋆ the number of traded securities is limited;
- ⋆ there are frictions on trading, such as transactions costs and taxes, that will make it difficult to replicate exactly the desired pattern of cashflows.

1c When valuing financial assets, in most cases it is reasonable to assume that markets are complete. The reason for this is that it is usually possible to find traded assets (such as stocks, bonds, futures, forwards etc.) that allow us to generate/replicate the pattern of cashflows we want. Transactions costs are small in financial markets and may be ignored as a first approximation.

1d When markets are complete, using the no-arbitrage (that is, replicating or exact hedging) approach to value financial assets gives us prices that are close to the prices observed in the market. That is, the gap between the model price and the actual price is relatively small.

1e When markets are not complete, then we cannot reply on no-arbitrage arguments to value financial assets. In this case, we have to rely on equilibrium arguments to value securities. The gap between the observed market price and the model price obtained using equilibrium arguments can be quite large.

2. Suppose that at date 0 you are asked to value a stream of fixed (risk-free) cashflows for the next 10 years; that is, $\{C(1), C(2), \ldots, C(10)\}$. You are also given the prices of zero-coupon bonds for maturities 1 to 10 years; let us denote these prices as $\{Z(0,1),\ Z(0,2),\ \ldots,\ Z(0,10)\}$.

2.1. What is the value at date 0 of receiving one unit of the cashflow in date t?

2.2. How would you value the stream of ten cashflows described above?

Solution:

2a The value at date 0 of receiving one unit of cashflow in date t is given by the price of the zero coupon bond for that maturity, $Z(0,t)$.

2b Each zero-coupon bond gives us the value at date 0 of receiving one unit at t, the maturity date of that bond. Thus, the value at date 0 of a cashflow C_t is simply the product of this cashflow with the zero-coupon bond price; that is, $Z(0,t) \times C_t$. To find the value of the entire stream of cashflows, we need to add up these products over the ten years:

$$\text{Value of the 10 fixed cashflows} = \sum_{t=1}^{10} Z(0,t) \times C(t).$$

3. Suppose that you are the treasurer of a Japanese firm called Aiwa. You have decided to raise new financing for your firm in the eurodollar bond market by issuing one-year discount bonds. The face value of each bond that you issue is USD1000 and the bond gives an effective return of 5% at the end of the year; that is, $e^{r_{USD}} = 1.05$. The current exchange rate is JPY 200/USD and the yen bond gives an effective return of 1% at the end of the year; that is, $e^{r_{JPY}} = 1.01$. At maturity, the principal amount of the bonds may be redeemed, at the owner's option, for dollars or for yen at an exchange rate of JPY 200/USD. Assume that one year from now the spot rate can be either JPY 225/USD or JPY175/USD; these are the only two values of the future exchange rate that you need to consider (that is, consider a binomial setting).

3.1. Explain how to decompose the security Aiwa is issuing into simpler financial contracts.

3.2. Find the value today of the security Aiwa is issuing using the risk-neutral valuation approach.

3.3. Find the value today of the security Aiwa is issuing using the replication approach.

3.4. You, the treasurer of Aiwa, wish to be fully hedged against changes in the spot exchange rate. That is, you wish to know exactly how many dollars you will need at the end of the year to pay off the buyer of the bond-option. What is your exposure to USD after issuing each of these "bonds." How can you hedge your position using forward contracts?

Solution:

3a The security that Aiwa is issuing can be interpreted as a package of two assets: (i) a straight USD bond with a 5% annual return; and, (ii) an option to sell USD 1000 for JPY (a put option on USD) at a strike price of JPY 200/USD. Note that for from Aiwa's point of view, the home currency is the JPY and the foreign currency is the USD.

3b We can either value the package as a whole, or we can value the two contracts separately. Let us value each contract separately so that the calculations are more transparent.

$$\text{Value today of the straight USD bond} = \frac{\text{USD } 1000}{1.05} = \text{USD } 952.38.$$

Now, we need to find the value of the put option. To use the risk-neutral approach, we use the fact that the forward rate is the certainty equivalent value of the future spot rate. Thus, denoting by p_u and p_d the risk-neutral probabilities:

$$\begin{aligned} \text{Forward rate} &= p_u \text{Spot}_u + p_d \text{Spot}_d \\ 200 \times \frac{1.01}{1.05} &= 192.38 = p_u 225 + (1 - p_u)175 \\ p_u &= 0.3476. \end{aligned}$$

Thus, the risk-neutral payoff to the put option in JPY/USD is

$$\begin{aligned} \text{Expected payoff to put} &= [p_u \times P_u] + [(1 - p_u) \times P_d] \\ &= p_u \max(0, K - S_u) + (1 - p_u) \max(0, K - S_d) \\ &= 0.3476 \max(0, 200 - 225) + (1 - 0.3476)(0, 200 - 175) \\ &= 16.3095. \end{aligned}$$

Because each bond is for USD 1000, the expected payoff to the option on the bond is JPY 16,309.50. The value of the option today is equal to the present value of this payoff:

$$\text{Value of put in JPY} = \frac{16,309.50}{1.01} = 16,148.00.$$

Finally, we convert this into USD using the spot exchange rate:

$$\text{Value of put in USD} = \frac{16,148}{200} = \text{USD } 80.74.$$

So, the total value of the security is:

$$\text{Value of packaged security} = 952.38 + 80.74 = \text{USD } 1,033.12.$$

Of course, one could do this entire calculation in one step by noting that in state S_u it is optimal not to exercise the option and so the USD 1000 payment is worth JPY 225,000; in state S_d it will be optimal to exercise the option and so the USD 1,000 payment is worth JPY 200,000 (instead of JPY 175,000 without the option). So, the USD value of this payoff is:

$$\text{Value of packaged security} = \frac{\frac{p_u 225{,}000 + p_d 200{,}000}{1.01}}{200} = \text{USD } 1033.12.$$

3c Suppose we replicate this packaged payoff (that is, the bond *and* the option combined together) using a JPY bond and a forward contract on USD. Let a denote the number of JPY bonds held in the replicating portfolio, and b the number of forward contracts. Furthermore, we assume that the value of the bonds at date 0 is JPY 1 (slightly different

from the assumption in the lecture slides), so that the value at maturity is JPY 1.01. Then, the two replicating equations are:

$$a\,\text{Bond}_u + b\,(S_u - F_0) \;=\; \text{Cashflow}_u,$$

$$a\,\text{Bond}_d + b\,(S_d - F_0) \;=\; \text{Cashflow}_d.$$

Substituting in the numerical values, we get:

$$a\,1.01 + b\,(225 - 192.381) \;=\; 225,000,$$

$$a\,1.01 + b\,(175 - 192.381) \;=\; 200,000.$$

Solving this system of simultaneous equations, we get:

$$a = 206,624; \quad \text{and} \quad b = 500.$$

Note that no payment is made at date 0 for the forward contracts; so the cost of the replicating portfolio is just the cost of the bonds, which is JPY 206,624, which in USD terms is equal to USD 206,624/200 = USD 1033.12.

3d From the replicating equations above, we see that the bond-option package is equivalent to a position in JPY bonds and a position in 500 forward contracts. The Aiwa treasurer has sold this security, so the treasurer is short 500 forward contracts. Thus, to hedge, the treasurer should go long (buy) 500 forward contracts.

4. Consider a two date (single-period) binomial model where the current price of the stock is $S(0)$ and it can take only one of two possible values at T, S_u and S_d. There is a bond in this economy whose current price is $R(0) = 1$ and at T its price will be $R(T) = e^{r\,T}$ in both the up and down state. Suppose we wish to value a put option with strike price K in this model.

4.1. Write down the two equations that show how to determine the portfolio of stock and bond that replicates the payoff to the put option.

4.2. From these two replicating equations, identify the state prices. Find the value of the put option using these state prices. Show that the stock and bond price can also be valued using the state prices.

4.3. Find the risk-neutral probabilities from the state prices. Find the value of the put option in terms of the risk-neutral probabilities. Show that the stock can also be valued using the risk-neutral probabilities.

4.4. Now, change the numeraire from cash to the underlying stock. Find the probabilities that correspond to this numeraire. Find the value of the put option and the bond in terms of this numeraire using martingale pricing.

Solution:

4a Let a denote the number of bonds held and b the number of shares. Then two equations that show how the portfolio of bonds and shares will replicate the payoff to the put option in the up and down state are:

$$\begin{aligned} a\,R(T) + b\,S_u &= P_u \\ a\,R(T) + b\,S_d &= P_d. \end{aligned}$$

4b Solving these two equations gives:

$$b = \frac{P_u - P_d}{S_u - S_d}, \qquad \text{and} \qquad a = \frac{1}{R(T)}\left(P_u - b\,S_u\right).$$

To preclude arbitrage, the value of the put option must be the same as the value of the replicating portfolio, which is:

$$\text{Value of put} = P_0 = a\,R(0) + b\,S(0) = \left[\frac{1}{R(T)}\left(P_u - b\,S_u\right)\right]1 + \left[\frac{P_u - P_d}{S_u - S_d}\right]S(0).$$

Simplifying the above by collecting the terms that multiply P_u and P_d, we get that

$$P_0 = \pi_u\,P_u + \pi_d\,P_d,$$

where the state prices π_u and π_d are:

$$\pi_u = \left(\frac{S(0) - e^{-r\,T}S_d}{S_u - S_d}\right) \quad \text{and} \quad \pi_d = \left(\frac{e^{-r\,T}S_u - S(0)}{S_u - S_d}\right).$$

To show that the state prices can be used to value the bond, we need to show that

$$\begin{aligned} R(0) = 1 &= \pi_u\,R(T) + \pi_d\,R(T) \\ &= \left(\frac{S(0) - e^{-r\,T}S_d}{S_u - S_d}\right)e^{r\,T} + \left(\frac{e^{-r\,T}S_u - S(0)}{S_u - S_d}\right)e^{r\,T} \\ &= \frac{e^{r\,T}S(0) - S_d + S_u - e^{r\,T}S(0)}{S_u - S_d} \\ &= \frac{S_u - S_d}{S_u - S_d} \\ &= 1. \end{aligned}$$

Similarly, one can show that

$$\begin{aligned} S(0) &= \pi_u\,S_u + \pi_d\,S_d \\ &= \left(\frac{S(0) - e^{-r\,T}S_d}{S_u - S_d}\right)S_u + \left(\frac{e^{-r\,T}S_u - S(0)}{S_u - S_d}\right)S_d \\ &= S(0)\left(\frac{S_u - S_d}{S_u - S_d}\right) \\ &= S(0). \end{aligned}$$

4c The risk-neutral probabilities, p_u and p_d are

$$p_u = \pi_u\frac{R(T)}{R(0)} = \pi_u\frac{e^{r\,T}}{1} \quad \text{and} \quad p_d = \pi_d\frac{R(T)}{R(0)} = \pi_d\frac{e^{r\,T}}{1}.$$

Therefore, the price of the put in terms of these risk-neutral probabilities is

$$P(0) = e^{-r\,T}\left[p_u\,P_u + p_d\,P_d\right].$$

Similarly, the price of the stock is:

$$S(0) = e^{-r\,T}\left[p_u\,S_u + p_d\,S_d\right].$$

4d Denote by q_u and q_d the probabilities when the stock is the numeraire. For these probabilities, it must be the case that

$$\frac{S(0)}{S(0)} = 1 = \left[\pi_u \frac{S_u}{S(0)} + \pi_d \frac{S_d}{S(0)}\right],$$

which means that

$$q_u = \pi_u \frac{S_u}{S(0)} \quad \text{and} \quad q_d = \pi_d \frac{S_d}{S(0)}.$$

Now, let us find the value of the put using these probabilities (and remembering to use S as the numeraire):

$$\frac{P(0)}{S(0)} = q_u \left(\frac{P_u}{S_u}\right) + q_d \left(\frac{P_d}{S_d}\right).$$

After substituting in the values for q_u and q_d, the above equation simplifies to

$$P_0 = \pi_u P_u + \pi_d P_d,$$

which we already proved is the value of the put option using state prices.

The same analysis can be repeated for the price of the bond, again remembering to use S as the numeraire.

5. A stock price is currently \$40. At the end of the month it will either be \$42 or \$38. The risk-free rate is 8% p.a. with continuous compounding.

5.1. What are the risk-neutral probabilities in this model?

5.2. What are the state prices (prices of Arrow-Debreu securities) in this model?

5.3. Suppose the numeraire is the stock instead of the bond. What are the probabilities that correspond to this numeraire?

5.4. What is the value of a one-month European call option with strike price \$40?

5.5. What is the value of a one-month European put option with strike price \$40?

5.6. Verify that the call and put prices satisfy the put-call parity relation.

5.7. What is the value of a one-month option that pays out $\log S_u$ in the up state and $\log S_d$ in the down state?

5.8. What is the value of a one-month option that pays out the square of the stock price at maturity; that is, S_u^2 in the up state and S_d^2 in the down state?

Solution:

5a One can derive the risk-neutral probabilities, p_u and p_d, from the equation:

$$S(0) = e^{-rT}\left[p_u S_u + p_d S_d\right].$$

After making the substitution $p_d = 1 - p_u$, we have one linear equation in one unknown, p_u:

$$S(0) = e^{-rT}\left[p_u S_u + (1 - p_u) S_d\right].$$

We can solve this equation to get:

$$p_u = \frac{e^{rT} S(0) - S_d}{S_u - S_d}.$$

Plugging in the values, we get:

$$p_u = \frac{e^{0.08\times(1/12)}\, 40 - 38}{42 - 38} = 0.5669,$$

implying that $p_d = 1 - p_u = 0.4331$.

5b In **Q**4b, we have already worked out the expressions for the state prices.

$$\pi_u = \frac{S(0) - e^{-rT} S_d}{S_u - S_d} = \frac{40 - e^{-0.08\times(1/12)}\, 38}{42 - 38} = 0.5631,$$

and

$$\pi_d = \frac{e^{-rT} S_u - S(0)}{S_u - S_d} = \frac{e^{-0.08\times(1/12)}\, 42 - 40}{42 - 38} = 0.4302.$$

Alternatively, we know that the state prices are the discounted values of the risk-neutral probabilities, so

$$\pi_u = e^{-rT} p_u = 0.5631; \quad \text{and} \quad \pi_d = e^{-rT} p_d = 0.4302.$$

5c From **Q**4d we know that when the numeraire is the stock, then the probabilities are:

$$q_u = \pi_u \frac{S_u}{S(0)} = 0.5631 \frac{42}{40} = 0.5913; \text{ and } q_d = \pi_d \frac{S_d}{S(0)} = 0.4302 \frac{38}{40} = 0.4087.$$

5d The value of a call option can be obtained using any of the quantities we have computed above. In terms of state prices:

$$\begin{aligned} \text{Value of call} &= \pi_u C_u + \pi_d C_d \\ &= \pi_u \max(0, 42 - 40) + \pi_d \max(0, 38 - 40) \\ &= [\pi_u \times 2] + [\pi_d \times 0] \\ &= \$1.1262, \quad \text{where } \pi_u = 0.5631. \end{aligned}$$

5e Just as above, one can work out that the value of the put option is:

$$\text{Value of put} = \pi_u P_u + \pi_d P_d = [\pi_u \times 0] + [\pi_d \times 2] = 0.8604.$$

5f The put-call parity relation says:

$$S(0) = C(0) + e^{-rT} K - P(0).$$

Substituting into the right-hand side of this equation the values of the call and put that we have derived above, along with the discounted value of the strike price K, gives us \$40, which is indeed the current stock price, $S(0)$.

5g The value of an option that pays off $\log(T)$ at maturity is

$$\text{Value of option} = [\pi_u \times \log S_u] + [\pi_d \times \log S_d] = \$3.6698.$$

5h The value of an option that pays off $S^2(T)$ at maturity is

$$\text{Value of option} = [\pi_u \times S_u^2] + [\pi_d \times S_d^2] = \$1614.60.$$

6. Consider a two-date (one period) *trinomial* model, where the stock price at T can take one of three values. One can also invest in a risk-free bond. We wish to value a call option in this model.

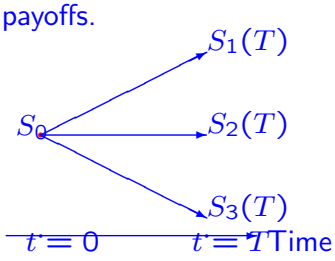
6.1. Explain why one cannot value the call option if at T the stock can take one of *three* values.

6.2. Explain how you could change the model so that even if the stock can take one of three values at T it is still possible to value the option. That is, what would you have to allow an investor to do so that it becomes possible to hedge/replicate this option exactly, and hence, value it using our standard arguments. This is an important question, so please give it serious thought before looking at the solution.

Solution:

6a One way to think of no-arbitrage valuation is that if we wish to find the value of a particular option (in our example, a call option), we then use other traded securities (in our example, the bond and the underlying stock) to replicate exactly the payoff to the option. No-arbitrage then requires the value of the option to match exactly the value of the replicating portfolio.

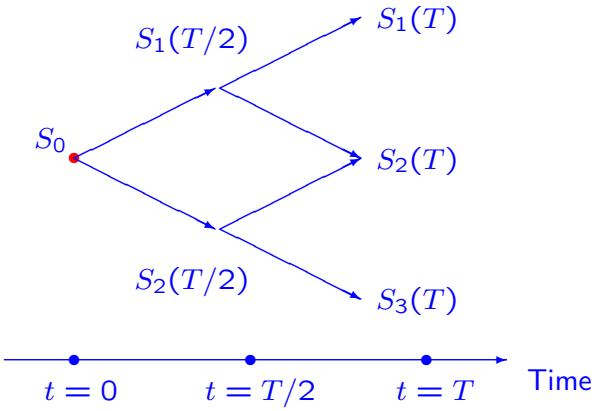
In this question, we are told that the stock can take one of three values at T; thus, one cannot construct a replicating portfolio using only the two assets — the stock and the bond — that will match all three payoffs.



6b What you need to do is to allow the investor to trade between dates 0 and T; that is, one needs to change the model so that there is an additional trading date between 0 and T; let us call this intermediate trading date $T/2$.

Once we introduce this additional trading date, we need to write the stock price process so that there are *only two possible outcomes for the stock price* at $T/2$, and a further two outcomes from each node going at $T/2$ to date T (see the figure).

With this change, the investor can now replicate exactly the payoff to the option: that is, the additional trading opportunity at $T/2$ allows one to replicate the payoff to the option for all three stock-price outcomes at date T.



7. Consider a three-date (two-period) binomial model. The three dates are $t = \{0, 1, 2\}$. At each future date (after $t = 0$), the stock price can either go up or go down. Also available to the investor is a constant risk-free rate. Suppose the investor is allowed to trade only at $t = 0$ and $T = 2$.

7.1. Can the investor value a call option using our standard approach? Explain why.

Solution:

7a This situation described in this question is very similar to that in the last question. Because the investor is not allowed to trade between 0 and T, even though the stock price process is binomial, for the investor it is essentially trinomial. Thus, we will not be able to value the option using our standard approach that replies on exact replication/hedging arguments.

8. We wish to value an option in a model where at T the stock can take a continuum of values. In addition to the stock, there is also available a bond with a risk-free rate that is constant between date 0 and T.

8.1. Suppose that there are also a continuum of state-securities (corresponding to each of the states) that are available. In this case, how would you find the value of the option?

8.2. Now suppose that only the stock and bond are available for trading, and state-securities are *not* available. Is it possible to find the value of the option? If not, what additional assumption do you need to make so that it becomes possible to value the option using our standard approach. Think carefully — this is the principal insight of the Black-Scholes-Merton model.

Solution:

8a If there are a continuum of state securities that are available, then the value of the option is simply the product of the payoff to the option in each state multiplied by the price of the state security in that state.

8b If the only securities available for trading are the stock and the bond and there are a continuum of states at T, then the only way to value an option using our standard arguments is to have a model where one can *trade continuously in time*. Continuous trading will allow us to have essentially a binomial setting over each small time interval, Δt. Thus, by dividing the time interval $[0, T]$ into an infinite number of small periods, we will be able to use the same arguments for valuing options that we have used in the two-date binomial model.

9. One of the simplest Ito processes is the **arithmetic** Brownian motion:

$$dX = \mu\, dt + \sigma\, dB,$$

where B is a Brownian motion under the actual probability measure, and μ and σ are constants.

9.1. Show that the solution to the above stochastic differential equation is

$$X(t) = X(0) + \mu\, t + \sigma\, B(t).$$

9.2. Show that $X(t)$ has a Normal distribution with mean $X(0) + \mu\, t$ and variance equal to $\sigma^2\, t$.

9.3. What happens to the variance of $X(t)$ as time goes to infinity?

9.4. Is this a good process to use for modeling stock prices? Explain your answer.

Solution:

9a Integrating the equation for the arithmetic Brownian motion, we get:

$$X(t) = X(0) + \mu\, t + \sigma B(t), \quad \text{where} B(0) = 0.$$

9b The right-hand side of the equation above has only one random variable, which is $B(t)$. We know from the properties of a Brownian motion that, given $B(0) = 0$, the mean of $B(t)$ is 0, the variance is t, and the distribution of $B(t)$ is Normal. Thus, the mean of $X(t)$ is $X(0) + \mu\, t$, the variance is $\sigma^2\, t$, and the distribution is Normal.

9c Because the variance of $X(t)$ is equal to $\sigma^2\, t$, when $t \to \infty$, the variance $\to \infty$.

9d The arithmetic Brownian motion is **not** a very good process for modeling stock prices. There are several reasons for this. The most important is that $X(t)$ can become negative under this process, which is ruled out for stocks because of limited liability. The other less-than-attractive properties of this process are that the expected increase in the stock price, $\mu\, t$, and the variance of the stock price, $\sigma^2\, t$, do not depend on the stock price; that is, these moments are the same (in dollar terms) no matter whether the stock price is high or low.

10. (From Hull) A company's cash position, measured in millions of dollars, follows an arithmetic Brownian motion with $\mu = 0.10$ *per month* and $\sigma^2 = 0.16$ *per month*. The initial cash position is 2 million.

10.1. What is the probability distributions of the cash position after 6 months?

10.2. What is the probability of a negative cash position after 6 months?

Solution:

10a Given the process for an arithmetic Brownian motion, after 6 months, that is, $t = 6$, we have:

$$X(t) = X(0) + \mu\, t + \sigma\, B(t) = 2 + (0.10 \times 6) + \sqrt{0.16 \times 6}.$$

implying that the distribution is Normal with mean $\mu_x = 2 + [0.10 \times 6] = 2.6$ and variance $= \sigma_x^2 = 0.16 \times 6 = 0.96$.

10b We wish to find the probability that the cashflow in 6 months time, $X(6)$, is less than 0:

$$\begin{aligned}
\text{Prob}\left[X(6) < 0\right] &\iff \text{Prob}\left[\frac{X(6) - \mu_x}{\sigma_x} < \frac{0 - \mu_x}{\sigma_x}\right] \\
&\iff \text{Prob}\left[\frac{X(6) - 2.6}{\sqrt{0.96}} < \frac{0 - 2.6}{\sqrt{0.96}}\right] \\
&\iff N\left[\frac{-2.6}{\sqrt{0.96}}\right] \\
&\iff N\left[-2.65361\right] \\
&= 0.00398174.
\end{aligned}$$

11. Let $dX = \mu\, dt + \sigma\, dB$. Consider a function $Y = f(X)$. Using Ito's formula, compute dY for the following functions:

11.1. $Y = X^{\alpha}$, where α is a constant.

11.2. $Y = \log(X)$.

11.3. $Y = \exp(X)$.

Solution:

11a Given $Y = f(X) = X^{\alpha}$:

$$\begin{aligned} dY &= f_x\, dX + (1/2) f_{xx}\, (dX)^2 \\ &= \alpha X^{\alpha-1}\, dX + (1/2)\alpha(\alpha-1)X^{\alpha-2}(dX)^2 \\ &= \alpha X^{\alpha-1}\left[\mu\, dt + \sigma\, dB\right] + (1/2)\alpha(\alpha-1)X^{\alpha-2}(\sigma^2\, dt) \\ &= \left(\alpha\mu X^{\alpha-1} + \frac{1}{2}\alpha(\alpha-1)X^{\alpha-2}\sigma^2\right)dt + \alpha X^{\alpha-1}\sigma dB. \end{aligned}$$

11b Given $Y = f(X) = \log X$:

$$\begin{aligned} dY &= f_x\, dX + (1/2) f_{xx}\, (dX)^2 \\ &= \frac{1}{X}\, dX - (1/2)\frac{1}{X^2}\, (dX)^2 \\ &= \left(\frac{\mu}{X} - \frac{\sigma^2}{2X^2}\right)dt + \frac{\sigma}{X} dB. \end{aligned}$$

11c Given $Y = f(X) = e^X$:

$$\begin{aligned} dY &= f_x\,dX + (1/2) f_{xx}\,(dX)^2 \\ &= e^X\,dX + (1/2) e^X\,(dX)^2 \\ &= \left(\mu + \frac{1}{2}\sigma^2\right) Y\,dt + \sigma Y\,dB \\ \frac{dY}{Y} &= \left(\mu + \frac{1}{2}\sigma^2\right) dt + \sigma\,dB. \end{aligned}$$

12. We studied in class the **geometric** Brownian motion, which will be the process used in the Black-Scholes model:

$$dX = \mu\,X\,dt + \sigma\,X\,dB, \quad \text{where } \mu \text{ and } \sigma \text{ are constants,}$$

and where B is a Brownian motion under the actual probability measure.

12.1. Show that the solution to the above equation is

$$X(t) = X(0) \exp\left(\mu\,t - \frac{1}{2}\sigma^2\,t + \sigma\,B(t)\right).$$

12.2. Show that now it is $\log X(t)$ that follows a Normal distribution with mean $\log X(0) + \left(\mu - \frac{1}{2}\sigma^2\right) t$ and variance $\sigma^2\,t$.

12.3. What happens to the variance of $\log X(t)$ as time goes to infinity?

12.4. Is this a good process to use to model stock prices? Why?

Solution:

12a There are two possible approaches.

- ⋆ The first is to start with the solution that is given, and show that dX leads to the geometric Brownian motion.
- ⋆ The second approach is to start with the process for dX and show that its solution is the one that is given.

The first approach is exactly like the one considered in the previous question, where we need to note that $X(t) = f(t, B) = X(0)\exp\left(\mu\, t - \frac{1}{2}\sigma^2\, t + \sigma\, B(t)\right)$, and then

$$\begin{aligned} dX &= f_t dt + f_B(dB) + (1/2) f_{BB}(dB)^2 \\ &= X\left(\mu - \frac{1}{2}\sigma^2\right)dt + X\sigma\,(dB) + \frac{1}{2}X\,\sigma^2 (dB)^2 \\ &= X\left(\mu - \frac{1}{2}\sigma^2\right)dt + X\sigma\,(dB) + \frac{1}{2}X\,\sigma^2\, dt \\ \frac{dX}{X} &= \mu\, dt + \sigma\, dB. \end{aligned}$$

Under the second approach, consider a function $f(X) = \log(X)$, and compute:

$$\begin{aligned} df &= f_x\, dX + (1/2) f_{xx}\,(dX)^2 \\ &= \frac{1}{X} dX - \frac{1}{2}\frac{1}{X^2}(dX)^2 \\ &= \frac{1}{X}\left[\mu\, X\, dt + \sigma\, X\, dB\right] - \frac{1}{2}\frac{1}{X^2}\left[X^2\,\sigma^2\, dt\right] \\ d\log(X) &= \left[\mu - (1/2)\sigma^2\right] dt + [\sigma\, dB] \\ \log\left[\frac{X(t)}{X(0)}\right] &= \left[\mu - (1/2)\sigma^2\right] t + [\sigma\, B] \\ \left[\frac{X(t)}{X(0)}\right] &= \exp\left(\left[\mu - (1/2)\sigma^2\right] t + [\sigma\, B]\right) \\ X(t) &= X(0)\exp\left(\left[\mu - (1/2)\sigma^2\right] t + \sigma\, B\right), \end{aligned}$$

which is the solution we wanted to verify.

12b We can write the solution as

$$\log[X(t)] = \log[X(0)] + \left(\left[\mu - (1/2)\sigma^2\right] t + \sigma\, B\right).$$

From the above, we conclude that the distribution of $X(t)$ is lognormal; that is, $\log X(t)$ follows a Normal distribution with mean $\log[X(0)] + \left[\mu - (1/2)\sigma^2\right] t$ and variance $\sigma^2 t$.

12c Just like for the arithmetic Brownian motion, as $t \to \infty$ the variance goes to infinity.

12d The geometric Brownian motion is a good process for modeling stock returns. First, under this process the stock price is always positive – the reason being that as $S(t)$ decreases, the volatility of the process also decreases. Secondly, this process has the attractive property that the expected change and variance of the *percentage change* in the price (that is, returns) are constant.

13. (From Hull) A stock price follows a geometric Brownian motion process with current price 50. Assume that $\mu = 16\%$ p.a. and $\sigma = 30\%$ p.a.

13.1. What is the probability distribution for the log of the stock price in two years? What are the mean and standard deviation of the log of the stock price in two years.

13.2. Determine the 95% confidence interval for the stock price in two years.

Solution:

13a If the stock price follows a geometric Brownian motion process, then we know (see the previous question) that $\log S(t)$ follows a Normal distribution with mean $\log S(0) + \left(\mu - \frac{1}{2}\sigma^2\right) t$ and variance $\sigma^2 t$.
So, the mean of $\log S(2)$ is

$$\log S(0) + \left(\mu - \frac{1}{2}\sigma^2\right) t = \log 50 + \left(0.16 - \frac{1}{2}(0.30)^2\right) 2 = 4.1420,$$

and the variance is

$$\sigma^2 t = (0.30)^2 2 = 0.18,$$

so that its volatility is

$$\sqrt{\sigma^2 t} = \sqrt{(0.30)^2 2} = \sqrt{0.18} = 0.4243.$$

13b The 95% confidence interval for $\log S(2)$ is given by

$$\text{mean} \pm (2 \times \text{volatility}) = 4.1420 \pm (2 \times 0.4243) = \{3.2934, 4.9906\};$$

that is,

$$3.2934 < \log S(2) < 4.9906.$$

Thus, the 95% confidence interval for $S(2)$ is given by

$$e^{3.2934} < S(2) < e^{4.9906} \iff 26.9343 < S(2) < 147.0250.$$

14. Suppose that a stock price S, follows a geometric Brownian motion:

$$dS = \mu S\, dt + \sigma S\, dB, \quad \text{where } \mu \text{ and } \sigma \text{ are constants,}$$

and where B is a Brownian motion under the actual probability measure. Consider a function $Y = f(S)$. Compute dY for the following functions, and explain if the process for Y is a geometric Brownian motion:

14.1. $Y = S^\alpha$, where α is a constant.

14.2. $Y = \log(S)$.

14.3. $Y = \exp(S)$.

Solution:

14a Given $Y = f(S) = S^{\alpha}$:

$$\begin{aligned} dY &= f_s\,dS + (1/2) f_{ss}\,(dS)^2 \\ &= \alpha S^{\alpha-1}\,dS + (1/2)\alpha(\alpha-1)S^{\alpha-2}(dS)^2 \\ &= \alpha S^{\alpha-1}\left[\mu\,S\,dt + \sigma\,S\,dB\right] + (1/2)\alpha(\alpha-1)S^{\alpha-2}\left(S^2\,\sigma^2\,dt\right) \\ &= S^{\alpha}\big(\alpha\mu + \frac{1}{2}\alpha(\alpha-1)\sigma^2\big)dt + S^{\alpha}\alpha\,\sigma\,dB \\ \frac{dY}{Y} &= \big(\alpha\mu + \frac{1}{2}\alpha(\alpha-1)\sigma^2\big)dt + \alpha\,\sigma\,dB. \end{aligned}$$

The above process is a geometric Brownian motion because the proportional change in Y has a constant mean $\left(\alpha\mu + \frac{1}{2}\alpha(\alpha-1)\sigma^2\right)dt$ and a constant variance $(\alpha\sigma)^2 dt$.

14b Given $Y = f(S) = \log S$:

$$\begin{aligned} dY &= f_s\,dS + (1/2) f_{ss}\,(dS)^2 \\ &= \frac{1}{S}\,dS - \frac{1}{2}\frac{1}{S^2}\,(dS)^2 \\ &= \frac{1}{S}\left[\mu\,S\,dt + \sigma\,S\,dB\right] - \frac{1}{2}\frac{1}{S^2}\,\left(S^2\,\sigma^2\,dt\right) \\ &= \big(\mu - \frac{1}{2}\sigma^2\big)dt + \sigma\,dB, \end{aligned}$$

so the process for $Y = \log S$ is an arithmetic Brownian motion.

14c Given $Y = f(S) = e^S$:

$$\begin{aligned} dY &= f_s\,dS + (1/2)\,f_s\,(dS)^2 = e^S\,dS + (1/2)\,e^S\,(dS)^2 \\ &= e^S\left[\mu\,S\,dt + \sigma\,S\,dB\right] + (1/2)\,e^S\,\left(S^2\,\sigma^2\,dt\right) \\ &= \big(\mu\,S + \frac{1}{2}\sigma^2\,S^2\big)Y\,dt + \sigma Y\,S\,dB \\ \frac{dY}{Y} &= \big(\mu\,S + \frac{1}{2}\sigma^2\,S^2\big)dt + \sigma S\,dB. \end{aligned}$$

This process is not a geometric Brownian motion because the mean and volatility of the proportional change in Y contain the term S, which is not constant.

15. Find the differential equation that a contract must satisfy if the payoff to the contract depends on the prices of two non-dividend paying traded securities. Assume that the price of each of the underlying securities follows a geometric Brownian motion and that the correlation between the Brownian motions for the two prices has a correlation of ρ.

Solution:

- We are given that the prices for the two risky assets are given by

$$\frac{dS_i}{S_i} = \mu_i\, dt + \sigma_i\, dB_i,$$

where each B_i is a Brownian motion under the actual probability measure, and the correlation between them is ρ.

- We wish to find the PDE satisfied by the value of a contract whose final payoff depends on the prices of both risky assets, that is, $C(T, S_1(T), S_2(T))$.

- Let us denote the value of this contract at date t by $C(t, S_1, S_2)$.

- Using Ito's formula, we get the following equation for the change in the value of this contract:

$$\begin{aligned} dC &= C_t\,dt + C_{s_1}\,dS_1 + C_{s_2}\,dS_2 \\ &\quad + \frac{1}{2}C_{s_1s_1}\,(dS_1)^2 + \frac{1}{2}C_{s_2s_2}\,(dS_2)^2 + C_{s_1s_2}\,(dS_1)(dS_2) \\ &= C_t\,dt + C_{s_1}\,[\mu_1\,S_1\,dt + \sigma_1\,S_1\,dB_1] + C_{s_2}\,[\mu_2\,S_2\,dt + \sigma_2\,S_2\,dB_2] \\ &\quad + \frac{1}{2}C_{s_1s_1}\,(\sigma_1\,S_1)^2dt + \frac{1}{2}C_{s_2s_2}\,(\sigma_2\,S_2)^2dt + C_{s_1s_2}\,\rho(\sigma_1\,S_1)(\sigma_2\,S_2)dt \\ &= \Big(C_t + \mu_1 S_1\,C_{s_1} + \mu_2 S_2\,C_{s_2} \\ &\quad + \frac{1}{2}C_{s_1s_1}\,(\sigma_1\,S_1)^2 + \frac{1}{2}C_{s_2s_2}\,(\sigma_2\,S_2)^2 + \rho\,\sigma_1\,\sigma_2\,S_1\,S_2\,C_{s_1s_2}\Big)\,dt \\ &\quad + \sigma_1\,S_1\,C_{s_1}dB_1 + \sigma_2\,S_2\,C_{s_2}dB_2 \\ &= \Big(\mu_{\text{misc}} + \mu_1\,S_1\,C_{s_1} + \mu_2\,S_2\,C_{s_2}\Big)\,dt + \sigma_1\,S_1\,C_{s_1}dB_1 + \sigma_2\,S_2\,C_{s_2}dB_2, \end{aligned}$$

  where

$$\mu_{\text{misc}} = C_t + \frac{1}{2}C_{s_1s_1}\,(\sigma_1\,S_1)^2 + \frac{1}{2}C_{s_2s_2}\,(\sigma_2\,S_2)^2 + \rho\,\sigma_1\,\sigma_2\,S_1\,S_2\,C_{s_1s_2}.$$

- To replicate this payoff (or to hedge it), we need to hold the bond and the two underlying risky assets.
- Denote the investment in the bond by $b(t)$ and denote the value of the bond by $R(t)$, where we know that $dR(t) = r\,R(t)\,dt$.

- Denote the investment in the risky assets by $a_1(t)$ and $a_2(t)$, and the value of this portfolio by $W(t)$:

  $W(t) = a_1(t)S_1(t) + a_2(t)S_2(t) + b(t)\,R(t).$

  At the maturity date T, we would like the value of this portfolio to match that of the contract we wish to value:

  $W(T) = C(T, S_1(T), S_2(T)).$

- Using Ito's formula, the change in the value of the replicating portfolio is:

$$\begin{aligned} dW &= a_1\,dS_1 + a_2\,dS_2 + b(t)\,dR(t) \\ &= a_1\,[\mu_1\,S_1\,dt + \sigma_1\,S_1\,dB_1] + a_2\,[\mu_2\,S_2\,dt + \sigma_2\,S_2\,dB_2] + b\,r\,R. \\ &= (a_1\,\mu_1\,S_1 + a_2\,\mu_2\,S_2 + b\,r\,R)\,dt + a_1\,\sigma_1\,S_1dB_1 + a_2\,\sigma_2\,S_2dB_2. \end{aligned}$$

- Matching the coefficients of dB_i in the process for dC and dW, we see that

$$\begin{aligned} a_1 &= C_{s_1}, \\ a_2 &= C_{s_2}. \end{aligned}$$

- Matching the coefficient of dt in the process for dC and dW, and using the expressions for a_i obtained just above, we get:

$$\begin{aligned} \mu_{\text{misc}} + \mu_1\,S_1\,C_{s_1} + \mu_2\,S_2\,C_{s_2} &= \mu_1\,S_1\,C_{s_1} + \mu_2\,S_2\,C_{s_2} + b\,r\,R \\ \mu_{\text{misc}} &= b\,r\,R, \end{aligned}$$

which implies that

$$b = \frac{\mu_{\text{misc}}}{r\,R}.$$

So, now we have identified a_1, a_2, and b.

- The no-arbitrage condition then requires that the value of the claim must satisfy:

$$\begin{aligned} C(t,S_1,S_2) &= W(t,S_1,S_2) \\ &= a_1 S_1 + a_2 S_2 + b\,R \\ &= C_{s_1} S_1 + C_{s_2} S_2(t) + \frac{\mu_{\text{misc}}}{r\,R} R \\ &= C_{s_1} S_1 + C_{s_2} S_2(t) + \frac{\mu_{\text{misc}}}{r}, \end{aligned}$$

where we have used our solutions for a_i and b to get the third equality above.

- Multiplying both sides of the above equation by r, and then substituting back the expression for μ_{misc}, we get the PDE that this claim must satisfy:

$$\begin{aligned} r\,C &= \mu_{\text{misc}} + C_{s_1}\,r\,S_1 + C_{s_2}\,r\,S_2(t) \\ 0 &= \mu_{\text{misc}} + C_{s_1}\,r\,S_1 + C_{s_2}\,r\,S_2(t) - r\,C \\ 0 &= \left(C_t + \frac{1}{2} C_{s_1 s_1} (\sigma_1\,S_1)^2 + \frac{1}{2} C_{s_2 s_2} (\sigma_2\,S_2)^2 + \rho\,\sigma_1\,\sigma_2\,S_1\,S_2\,C_{s_1 s_2} \right) \\ &\quad + C_{s_1}\,r\,S_1 + C_{s_2}\,r\,S_2(t) - r\,C. \end{aligned}$$

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Appendix B

Homework Questions for Part 1 . . . To be handed in

Raman Uppal

EDHEC Business School

Instructions for Homework Questions:

To get credit for your homework, you need to follow these instructions.

- You need to submit your solutions to only the **five** questions given below. Most of these questions are from previous exams, so working on these questions will give you an idea of what to expect on the exam. (Of course, the questions that appear on your exam will *appear* to you to be much more difficult!)
- Each question has several parts that are designed to help you get to the final answer.
- Each question is worth 10 points, and the total homework is worth 50 points.
- If you think that there is any error in the questions, please send me an email.
- You should work on these questions independently; this is *not* group work. You are welcome to consult the class notes (lecture slides) and any text book, but please do not search for solutions to these questions on the internet.
- When submitting your work, please submit only a PDF file (please do not send .doc files, because they may appear differently on my computer compared to what they appear on your computer; in any case, it is best to use LaTeX).
- Please name your file using the following convention: `LASTNAME-Firstname-HW1.pdf`. For example, if I were submitting a file it would be named: `UPPAL-Raman-HW1.pdf`.

- Please type your answers legibly and neatly – if I cannot read your work, I will not be able to give you credit for it. This is a good opportunity to learn LaTeX.
- Please use a font size of 12 points and margins of 1.5 inches.
- Please start each question on a new page.
- Your homework should be emailed to julie.smadja@edhec.edu. Please do NOT copy me on your email.
- The deadline for submitting your answers to these questions is 7 April 2025 (the deadline for the homework for the course on *Financial Econometrics* may be different).
- Assignments will not be accepted (or graded) after the deadline and a grade of 0 will be assigned in that case.
- The LaTeX file with these questions is available from Julie; please email her if you do not find it in the course folder on Blackboard. You can use this file to start your homework; that way, you will not need to retype the questions.

1. Numeraires and Probabilities

- We wish to understand how the distribution of a stock (that is, its probability measure) changes as we change the numeraire.

- In this question, we wish to consider as numeraire an *arbitrary risky asset (denoted Y) that does not pay dividends*. (*Hint:* In class, we already studied two numeraires: (1) the risk-free bank deposit, and (2) a dividend-paying stock; the classnotes will help you solve this problem.)

- Let $r(t)$ denote the instantaneous risk-free rate at date t, and let $R(t) = \exp(\int_0^t r(u)du)$, so that:
$$\frac{dR}{R} = r(t)\,dt.$$

- Let S be the price of an asset that has a constant dividend yield q. Let $V(t) = e^{q\,t}S(t)$ be the *value* of the portfolio in which all dividends are reinvested, so that:
$$\frac{dV}{V} = \underbrace{q\,dt}_{\text{dividends}} + \underbrace{\frac{dS}{S}}_{\text{capital gains}},$$
with
$$\frac{dS}{S} = \mu_s\,dt + \sigma_s\,dB_s.$$

- Finally, let Y be the price of another asset that does not pay any dividends:
$$\frac{dY}{Y} = \mu_y\,dt + \sigma_y\,dB_y,$$
where B_s and B_y are Brownian motions under the physical (actual) probability measure with correlation ρ, and $\{\mu_s, \mu_y, \sigma_s, \sigma_y,\ \rho\}$ are general random processes.

Please answer the following questions.

1.1. Show that the drift of the process dY/Y when Y is the numeraire equals $r + \sigma_y^2$.
Hint: To do this, identify the drift of the process dC/C where $C(t) = R(t)/Y(t)$.

1.2. Now, using the result in the previous part, identify the drift of dS/S by finding the drift of dC/C, where $C(t) = V(t)/Y(t)$.

1.3. Finally, show that the results derived in class are special cases of the result derived above. That is, show that if Y is the price of instantaneously risk-free asset so that $\sigma_y = 0$, then your answer to the previous question implies that the drift of dS/S is $r - q$. And, if Y is the price of the risky stock S, then the drift of dS/S equals $r - q + \sigma_s^2$ (where q denotes the dividend yield).

2. Option valuation using the CAPM

- In the original Black and Scholes (1973) paper, there are *two* derivations of the option-pricing formula. One approach relies on constructing an option position that is delta-hedged with a stock, and because this position has zero risk, it is argued that it must earn just the risk-free rate of return; this argument is then used to derive the fundamental partial differential equation.
- The second approach for deriving the value of options relies on the Capital Asset Pricing Model (CAPM), and this question asks you to use that approach to derive the value of a call option. Recall that the CAPM states that the relation between the risk and return of a security is a linear function of its beta (β), where the β denotes the covariance between the return on the asset and the return on the market portfolio, divided by the variance of the return on the market portfolio.
- Assume the setting of a standard Black-Scholes economy with a risk-free bank account with price $R(t)$ that earns a constant risk-free rate r, and stock price $S(t)$ that follows a Geometric Brownian Motion with constant volatility σ_S and constant drift μ_S:
$$\frac{dS(t)}{S(t)} = \mu_S\, dt + \sigma_S\, dB_S(t).$$
- Finally, assume that the *return* on the market portfolio is given by:
$$dR_M(t) = \mu_M\, dt + \sigma_M\, dB_M(t),$$

where μ_M and σ_M are constants, and $dB_S(t) \times dB_M(t) = \rho_{MS}$.

- Now, consider a call option with strike price K and maturity T. Denote the price today of this option by $C(S_t, t)$, where S_t is the price of the stock on the current date t.
- Let β_{MC} denote the beta of the call option with respect to the return on the market, and let β_{MS} denote the beta of the stock underlying this call option.

Please answer the following questions,

2.1. Write down (a) the instantaneous expected return on the call option according to the CAPM, and (b) use Itô's formula to identify the process for the return on the call option.

2.2. Find the covariance between the return on the call option and the return on the market portfolio. Then, use this result to identify the relation between the beta on the call, β_{MC}, and the beta on the stock, β_{MS}.

2.3. Use the results of the previous parts to derive the fundamental partial differential equation that the price of the call option must satisfy, which is:
$$r\, C = C_t + r\, C_S\, S + \frac{1}{2}\sigma_S^2\, C_{SS}\, S^2.$$

2.4. Explain which approach for deriving the option price is better: the approach based on constructing a risk-free hedge portfolio, or the approach based on the CAPM. Why?

3. Options and hedging

- In class, we studied the Black-Scholes model of option pricing that assumes that the interest rate is constant. We also studied Merton's model of option pricing that does not require the assumption of a constant interest rate; the Merton model also allows for volatility that is changing over time, as long as the changes are deterministic (non-stochastic).
- Suppose the price $P^S(t)$ of a non-dividend paying stock has a constant volatility σ_S.
- Let the price of a discount bond be denoted by $P^D(t,T)$.
- Let the constant $\kappa > 0$ denote the rate of mean reversion of the short rate process and the constant $\phi > 0$ the (absolute) volatility of the short rate process.
- Assume that that volatility at date t of a discount bond maturing at $T > t$ is

  $$\sigma_D(t) = \frac{\phi}{\kappa}\left(1 - e^{-\kappa(T-t)}\right),$$

  and the return on the stock and bond have constant correlation ρ.

Please answer the following questions,

3.1. Find the volatility at t of $\frac{P^S(t)}{P^D(t,T)}$; that is, the volatility term of the process

$$\frac{d\left(\frac{P^S(t)}{P^D(t,T)}\right)}{\frac{P^S(t)}{P^D(t,T)}}.$$

Denote this volatility of $\frac{P^S(t)}{P^D(t,T)}$ by $\boldsymbol{\sigma}(t)$.

3.2. Define

$$\boldsymbol{\sigma}_{avg} = \sqrt{\frac{1}{T}\int_0^T [\boldsymbol{\sigma}(t)]^2\, dt}.$$

Show that

$$\boldsymbol{\sigma}_{avg}^2 = \sigma_S^2 + \frac{1}{\kappa^2}\left[\phi^2 - 2\kappa\rho\sigma_S\phi - \left(2\phi^2 - 2\kappa\rho\sigma_S\phi\right)\left(\frac{1-e^{-\kappa T}}{\kappa T}\right) + \phi^2\left(\frac{1-e^{-2\kappa T}}{2\kappa T}\right)\right].$$

3.3. For small T, use l'Hôpital's rule to show that $\boldsymbol{\sigma}_{avg} \approx \sigma_S$.

3.4. For large T, show that $\boldsymbol{\sigma}_{avg} > \sigma_S$ if ρ is sufficiently small.

4. Option valuation and hedging

- We now study the problem of an option dealer who has sold a particular option whose payoff depends on the prices of two stocks and the dealer wishes to hedge her position.
- Suppose that the price of the first stock, $\mathcal{P}^{S_1}$ follows the process
$$\frac{d\mathcal{P}^{S_1}(t)}{\mathcal{P}^{S_1}(t)} = \mu_1 dt + \sigma_1 dB_1(t),$$
and that the price of the second stock, $\mathcal{P}^{S_2}$ follows the process
$$\frac{d\mathcal{P}^{S_2}(t)}{\mathcal{P}^{S_2}(t)} = \mu_2 dt + \sigma_2 dB_2(t),$$
where μ_1, μ_2, σ_1, and σ_2 are constants, $dB_1(t)$ and $dB_2(t)$ are Brownian motions, with the correlation ρ between the two Brownian motions also being constant: $dB_1(t) \times dB_2(t) = \rho\, dt$.
- Let $D(t)$ denote the price of a bank-deposit that earns the constant risk-free rate of return:
$$\frac{dD(t)}{D(t)} = r\, dt.$$
- Let $C(t)$ denote the value at t of a European option written on the difference between the prices of these two stocks. Specifically, at the maturity date T of the option, the value of the option is
$$C(T) = \max\Big(0, \mathcal{P}^{S_1}(T) - \mathcal{P}^{S_2}(T)\Big),$$

Please answer the following questions,

4.1. Using Itô's formula, derive the process followed by the value of this option.

4.2. Suppose that you are an option dealer who has just sold (written) one of these options for a customer. You now wish to form a hedge portfolio composed of your unit short position in the option and positions in the two stocks. Let $H(t)$ denote the date t value of this hedge portfolio. Let $n_1(t)$ and $n_2(t)$ denote the number of shares in each of these two stocks at t. Write down the process for $H(t)$.

4.3. Identify the number of shares to hold in each of the two stocks so that the hedge portfolio $H(t)$ is riskless.

4.4. In the absence of arbitrage, derive the partial differential equation that must be satisfied by the option value $C(t)$. Does the risk-free rate appear in this differential equation? Explain why or why not?

5. Valuation of a forward-start option

- We wish to value a *forward-start option*. A forward-start option is an option for which the strike price is set equal to the stock price at some later date. In essence, it is an option that is issued at a later date with the strike set so that the option is at-the-money on the issue date. Often, such options are promised to corporate executives.

- Let us denote by T^{mat} the maturity date of the option, and by T^{set} the date on which the strike price of the option is set. Thus, we have that $T^{set} < T^{mat}$.



- Assume that the setting is of a standard Black-Scholes economy where there is a bond with price $R(t)$ that earns a constant risk-free rate r, and the stock price $S(t)$ follows a Geometric Brownian Motion with constant volatility σ and constant dividend yield q. That is:

$$\frac{dR}{R} = r\,dt,$$

$$\frac{dS}{S} = \mu\,dt + \sigma\,dB.$$

Please answer the following questions,

5.1. Write down the process for the price of the stock under the risk-neutral measure. Explain, in words, the logic underlying the drift and volatility of this process. Then, write the payoff to a forward-start call option on the maturity date T^{mat}.

5.2. Define an indicator variable x whose value is equal to 1 if the option is exercised, and is equal to 0 if the option is not exercised. Choose as your numeraire $V(t) = e^{q\,t}S(t)$. Find the value at $t = 0$ of the payoff at maturity of $H(T^{mat}) = x\,S(T^{mat})$.

5.3. Find the value at $t = 0$ of the payoff at maturity of $G(T^{mat}) = x\,S(T^{set})$. To value this, note that for $t < T^{set}$, $S(T^{set})$ is random, but once the strike price is fixed to be $S(T^{set})$ at $t = T^{set}$, then it is no longer random.
Hint: To value this payoff, choose as the numeraire the following *portfolio*, whose value we will denote as $Z(t)$:

i. Purchase $e^{-q\,T^{set}}$ shares of the stock at $t = 0$ and reinvest the dividends until date T^{set}, which will result in the ownership of one share worth $S(T^{set})$ at the date T^{set};

ii. At date T^{set}, sell the share and invest the proceeds in the risk-free asset until T^{mat}, whose value at $t = T^{mat}$ is: $S(T^{set})\,e^{r(T^{mat}-T^{set})}$.

5.4. Next, we need to calculate the probability of exercising the option under the two numeraires. So, find the probability of $S(T^{mat}) > S(T^{set})$ when $V(t)$ is used as the numeraire; that is, find $\text{prob}^V\left[S(T^{mat}) > S(T^{set})\right]$.

Similarly, find the probability of $S(T^{mat}) > S(T^{set})$ when $Z(t)$ is used as the numeraire; that is, find $\text{prob}^Z\left[S(T^{mat}) > S(T^{set})\right]$. *Hint:* Note that between T^{set} and T^{mat} the portfolio $Z(t)$ earns the risk-free rate.

5.5. Combining the answers to the questions above, please give the final expression for the value at $t = 0$ of a forward-start call option.

Then, provide a simple interpretation in plain English of the expression you have obtained for the value of the forward-start call option in the previous question.

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Appendix C

Information about the Final Exam

Raman Uppal

EDHEC Business School

C.1 Information about the exam

- The exam tests your understanding of the material covered in class
  - ⋆ The exam does this by asking you to apply the material covered in class to *new problems* that were not covered in class.
- There are **two questions** on the exam.
  - ⋆ The first question is on the first part of the course
    - ∗ Material on valuation and hedging of options (options on equities, exchange rates, interest rates, swaps, etc.).
  - ⋆ The second question is on the second part of the course
    - ∗ Material on portfolio selection and equilibrium asset pricing.
- Each question consists of several parts (usually three or four); the parts are designed to guide you to the final answer.
- The two questions have equal weight on the exam.

C.2 Rules for the exam

- You have two hours to work on the exam.
- The exam is closed book (no notes allowed).
- During the exam, you are not allowed to have the following with you:
  - ⋆ Any sheet with notes on the course material;
  - ⋆ The slides (lecture notes) for the course;
  - ⋆ Any textbook;
  - ⋆ A mobile phone.

C.3 Suggestions on how to prepare for the exam

- The exam is designed to test your understanding of the material that we covered in class;
  - ⋆ The exam does this by asking you to apply the material covered in class to two new problems;
- The above implies that
  1. You must understand very well the material covered in class
  2. You must be well rested so that you can think clearly on how to apply the material covered in class to the new problems.
- So, in my view, the best way to prepare for the exam is to:
  - ⋆ Review carefully the material covered in the course slides and homeworks;
  - ⋆ Sleep well the night before the exam and stay relaxed during the exam so that you can think.

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Appendix D

Sample Final Exam

Raman Uppal

EDHEC Business School

Instructions for the final exam

1. Start by putting your name on the first page of both handouts that you have received—without that, it will not be possible to give you credit for the exam.

2. You must return **all** the material you have received (questions + answers) to get credit.

3. Credit will be given for providing the logic underlying your answer, so please explain clearly the reasoning behind your answer.

4. I can only give points for material that I can read, understand, and interpret. So, it is to your advantage to express your ideas clearly, and to highlight key arguments and quantities that you would like me to focus on when reading the exam.

5. Please put your answer in the boxed space provided on the answer sheet—this is where I will be looking for your response. There is space below the boxes to show your work.

6. There is no reward for long answers, and incorrect/illogical statements will be penalized. So, please keep your answers brief and to the point.

7. If something is not clear, please make a reasonable assumption and continue. Please highlight your assumptions so that I can identify them when grading the exam.

Rules for the exam

1. You have **two hours** to work on the exam.
2. You are **not allowed** to have the following with you during the exam:
   - Any sheet with handwritten or typed notes;
   - Any slides (lecture notes) for the course;
   - Any books;
   - A mobile phone.

Q1. Options and hedging

- In class, we studied the Black-Scholes model of option pricing that assumes that the interest rate is constant. We also studied Merton's model of option pricing that does not require the assumption of a constant interest rate; the Merton model also allows for volatility that is changing over time, as long as the changes are deterministic (non-stochastic).
- This question is about *implementing* Merton's formula when the discount bond price is given by Vasicek's model (that is, we analyze just the expression for volatility for implementing this option-pricing model).
- Suppose the price $P^S(t)$ of a non-dividend paying stock has a constant volatility σ_S.
- Let the price of a discount bond be denoted by $P^D(t,T)$.
- Let the constant $\kappa > 0$ denote the rate of mean reversion of the short rate process and the constant $\phi > 0$ the (absolute) volatility of the short rate process.
- Assume that that volatility at date t of a discount bond maturing at $T > t$ is
$$\sigma_D(t) = \frac{\phi}{\kappa}\left(1 - e^{-\kappa(T-t)}\right).$$
- Assume that the return on the stock and bond have constant correlation denoted by ρ.

Please answer the following questions, each of which is worth 2.5 points.

Q1.1 Find the volatility at t of $\frac{P^S(t)}{P^D(t,T)}$; that is, the volatility term of the process
$$\frac{d\left(\frac{P^S(t)}{P^D(t,T)}\right)}{\frac{P^S(t)}{P^D(t,T)}}.$$
Denote this volatility of $\frac{P^S(t)}{P^D(t,T)}$ by $\boldsymbol{\sigma}(t)$.

Q1.2 Define
$$\boldsymbol{\sigma}_{avg} = \sqrt{\frac{1}{T}\int_0^T [\boldsymbol{\sigma}(t)]^2\, dt}.$$
Show that
$$\boldsymbol{\sigma}_{avg}^2 = \sigma_S^2 + \frac{1}{\kappa^2}\left[\phi^2 - 2\kappa\rho\sigma_S\phi - \left(2\phi^2 - 2\kappa\rho\sigma_S\phi\right)\left(\frac{1-e^{-\kappa T}}{\kappa T}\right) + \phi^2\left(\frac{1-e^{-2\kappa T}}{2\,\kappa T}\right)\right].$$

Q1.3 For small T, use l'Hôpital's rule to show that $\boldsymbol{\sigma}_{avg} \approx \sigma_S$.

Q1.4 For large T, show that $\boldsymbol{\sigma}_{avg} > \sigma_S$ if ρ is sufficiently small.

END of first question ♠

Q2. Portfolio choice and general equilibrium asset pricing

- Consider an infinite-horizon representative-agent economy (that is, all agents are identical) with time-separable power utility function:

$$U(C,t) = e^{-\delta t}\frac{C_t^{1-\gamma}}{1-\gamma}.$$

- In our course, we studied an "endowment" economy, where the amount of output available for consumption was given *exogenously*.
- In this question, we study a "production" economy, where there is a single good (such as "corn") that can either be consumed or invested in the production process. The representative investor chooses how much to consume and how much to invest each instant; the investment this instant then determines the output next instant. Thus, in this economy output is determined *endogenously*.
- We assume that the production sector has constant returns to scale. The output each instant, denoted by $Y(t)$, depends on the capital stock, $K(t)$, as described by $\frac{dY(t)}{Y(t)} = \mu\, K(t)\, dt + \sigma\, K(t)\, dB_t$, where μ, σ are constants; $B(t)$ is a Brownian motion.
- The change in the capital stock is the change in the aggregate output, less consumption:

$$\frac{dK(t)}{K(t)} = \frac{dY_t}{Y_t} - C(t)\, dt.$$

- Each investor in this economy can choose to invest in this production process or to borrow/lend at the risk-free rate $r(t)$, which will be determined endogenously in equilibrium.
- Each investor also needs to decide how much to consume each period (with the remainder of the aggregate output reinvested).

Based on the above, please answer the following questions, each of which is worth 2.5 points.

Q2.1 Write the optimization problem of the representative agent along with the dynamic budget constraint that this optimization problem must satisfy.

Q2.2 Denote the value function by V. What are the state variables for the optimization problem of the representative investor? What is the Bellman equation?

Q2.3 Write the first-order conditions that the optimal consumption policy and portfolio policy of the representative investor must satisfy.

Q2.4 Use the market-clearing condition (that in equilibrium aggregate borrowing must equal zero) to obtain the risk-free rate in equilibrium. Use this to obtain the expression for the risk premium. Then, provide an economic interpretation for the risk-premium.

END of second question ♠

Lecture Notes for 2025

Continuous Time Financial Economics

Appendix E

Guide To Notation

Raman Uppal

EDHEC Business School

E.1 Roman letters

Symbol	Definition	Appears for first time in Chapter
1_A	random variable whose value is 1 if the event (set of states) A occurs, 0 otherwise	1
$B(t)$	Brownian motion's level at t	2
c or C	Consumption	1
Cov$[x, y]$	Covariance between x and y	3
E	Expectation operator (unconditional, at $t = 0$)	1
E_t	Conditional expectation operator at t	1
$\mathcal{E}$	Elasticity of an option	1
f	Exercise boundary for an American option	5
K	Strike price of an option	1
m	Stochastic discount factor (SDF) or state price density in a single-period (two-date) model	1
M_t	Stochastic discount factor (SDF) process	3
$\mathcal{P}$	Price of an asset	1
$\mathcal{P}^{A_k}$	Price of Arrow-Debreu state security paying one unit in state k	1
$\mathcal{P}^B$	Price of a coupon bond	6
$\mathcal{P}^C$	Price of a call option	1
$\mathcal{P}^D$	Price of bank deposit or money-market account; $\mathcal{P}_0^D = 1$	1
$\mathcal{P}^{Fwd}$	Price of forward contract	6
$\mathcal{P}^S$	Price of stock	1
$\mathcal{P}^X$	Price of foreign currency; that is, the spot exchange rate	6
$\mathcal{P}^Z$	Price of zero-coupon discount bond	6
$\mathcal{P}^{Z^*}$	Price of zero-coupon bond in foreign currency	6
$\mathbb{P}$	Physical probability measure (could be objective or subjective)	1

prob$^{\mathbb{P}}$	Physical probability under the measure $\mathbb{P}$, with the measure $\mathbb{P}$ written explicitly	1
prob$^{\mathbb{Q}}$	Risk-neutral probability (RNP) under the measure $\mathbb{Q}$; prob$^{\mathbb{Q}}(A) = E^{\mathbb{P}}[\xi_T \, 1_A]$	1
prob$^{\mathbb{S}}$	Probability when the numeraire is the stock, so under the measure $\mathbb{S}$	1
q	Dividend yield on an asset	2
$\mathbb{Q}$	Risk-neutral probability measure	1
r	Risk-free interest rate	1
r^*	Risk-free interest rate in a foreign currency	6
R_f	Gross risk-free interest rate ($R_f = 1 + r$) or ($R_f = e^r$)	1
R_i	Gross rate of return on asset i	1
R_{mkt}	Gross rate of return on market portfolio	1
R_z	Gross rate of return on zero-beta portfolio	1
RNP	Abbreviation for "risk-neutral probability"	1
$\mathbb{S}$	Probability measure when stock is the numeraire	1
SDF	Abbreviation for "stochastic discount factor"	1
t	Time, usually today (current date)	1
T	Terminal date	1
T'	Used to refer to dates after the terminal date; that is, $T' > T$	2
$\top$	Transpose operator (for example, to change row vectors into column vectors)	3
$\mathcal{U}$	Utility, which is a function of lifetime consumption	3
U	Utility, which is a function of one period consumption	3
u	Utility, which is a function of one period consumption, with $U(C_t) = e^{-\delta t} C_t$	3
$V(t, \mathcal{P}_t^S)$	Value of an option at date t	5
Var$[x]$	Variance of x	3
w_n	Portfolio weight; proportion of wealth invested in asset n; also denoted by π_n	1
W	Wealth	2
y	Yield on a bond	6
y^*	Yield on a bond in foreign currency	6

E.2 Greek letters

Symbol	Definition	Appears for first time in Chapter
Δ	Delta of an option, measures change in option price as price of underlying asset changes	5
Δt	Change in t	5
$\Delta Y(t)$	Change in $Y(t)$	5
Γ	Gamma of an option, measures change in option delta as stock price changes	5
λ	Factor risk premium; or, market price(s) of risk; also known as Sharpe ratio	3
λ	Factor risk premium; or, market price(s) of risk; also known as Sharpe ratio	3
μ	Drift of an Itô process	2
π_n	Portfolio weight; proportion of wealth invested in asset n; also denoted by w_n	5
ρ	Correlation coefficient, correlation between two Brownian motions	2
ρ	Rho of an option, measures change in option value as interest rate changes	5
σ	Volatility (scalar or matrix); diffusion coefficient of an Itô process	2
Σ	Variance-covariance matrix, also denoted by V	3
τ	Stopping time (the first time a particular condition is satisfied)	2
Θ	Theta, measures change in option value as t increases (time-to-maturity, T, decreases)	2
$\mathcal{V}$	Vega of an option, measures change in option value as volatility changes	5
ψ	Factor risk premium divided by variance of the factor, $\psi = \lambda/\text{Var}[f]$ or $\psi = \Sigma_F^{-1}/\lambda$	3
ξ_T	Radon-Nikodym derivative, $\xi_T = \frac{d\mathbb{Q}}{d\mathbb{P}} = \frac{d\text{prob}^{\mathbb{Q}}}{d\text{prob}^{\mathbb{P}}} = M_T \mathcal{P}_T^D$	4
ξ_t	Conditional expectation of ξ_T; that is, $\xi_t = E_t[\xi_T]$	4

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Lecture Notes for 2025

Continuous Time Financial Economics

Appendix F

List of Ph.D. Elective Courses Offered by EDHEC

Raman Uppal

EDHEC Business School

F.1 List of Ph.D. Courses Offered in the Past

Year	Instructor	Affiliation	Course title
2009	Lionel Martellini	EDHEC	Dynamic Asset Allocation Decision
2009	Pietro Veronesi	U. of Chicago	Recent Developments in Portfolio Allocation and Macro-Finance Model
2010	Rama Cont	Columbia	Monte Carlo Methods in Finance
2010	Tim Bollerslev	Duke	Volatility Modelling
2010	Harrison Hong	Princeton	Behavioural Finance
2010	Ekkehart Boehmer	EDHEC	Microstructure
2010	Francis Diebold	U. of Penn.	The Yield Curve
2011	Peter Christoffersen	Toronto	Risk Management and Extreme Risks
2011	Yacine Aït-Sahalia	Princeton	Financial Econometrics of Jumps
2011	Antonio Mello	UW-Madison	Advanced Corporate Finance
2011	Jerome Detemple	Boston	Derivaties Securities
2011	Nicholas Polson	Chicago	Bayesian Methods in Financial Econometrics
2011	Allan Timmermann	UC San Diego	Predictive Modeling and Forecast Evaluation in Financial Markets
2011	Michael Brandt	Duke	Portfolio Choice
2011	Ravi Bansal	Duke	Long-Run Risks in Asset Prices
2011	Tarun Ramadorai	Oxford	Risk and Return in Hedge Funds
2012	Mikhail Chernov	LSE	Empirical Option Pricing
2012	Rama Cont	Columbia	Credit Risk Modelling
2012	Ekkehart Boehmer	EDHEC	Microstructure
2012	Federico Bandi	Johns Hopkins	The Econometrics of Continuous-Time Models
2012	Harrison Hong	Princeton	Behavioural Finance
2012	Tim Bollerslev	Duke	Volatility Modelling
2012	Lionel Martellini	EDHEC	Dynamic Asset Allocation-Long-term Investing with Short-term Constraints
2012	Pietro Veronesi	U. of Chicago	Macroeconomic and Political Uncertainty and the Dynamics of Asset Prices

Year	Instructor	Affiliation	Course title
2012	Jianqing Fan	Princeton	Portfolio Problems with High-Dimension Covariance Matrices
2012	Sanjiv Das	Santa Clara	Credit Risk Modelling
2013	Michael Brandt	Duke	Portfolio Allocation
2013	Tarun Ramadorai	Oxford	International Finance
2013	Jerome Detemple	Boston	Strategic Asset Allocation
2013	Torben Andersen	Northwestern	High-Frequency Asset Pricing
2013	Vikas Agarwal	Georgia State	Hedge Funds
2013	Allan Timmermann	UC San Diego	Predictive Modeling in Financial Markets
2013	Federico Bandi	Johns Hopkins	Estimation of Continuous-time Models
2013	Harrison Hong	Princeton	Behavioural Finance
2013	Peter Christoffersen	Toronto	Risk Management and Extreme Risks
2013	Rama Cont	Imperial	Monte-Carlo Methods in Finance
2014	Allan Timmermann	UC San Diego	Markov Switching and Time-varying Parameter Models in Finance
2014	Tim Bollerslev	Duke	Volatility Modelling
2014	Ekkehart Boehmer	EDHEC	Microstructure
2014	Yacine Aït-Sahalia	Princeton	An Introduction to High Frequency Financial Econometrics
2014	Mikhail Chernov	LSE	Empirical Option Pricing
2014	Sanjiv Das	Santa Clara	Data Science, Analytics, Algorithms
2014	Lionel Martellini	EDHEC	Dynamic Risk and Asset Allocation Decisions
2014	Ludovic Phalippou	Oxford	Private Equity
2014	Pietro Veronesi	U. of Chicago	Macroeconomic and Political Uncertainty and Dynamics of Asset Prices
2015	Federico Bandi	Johns Hopkins	The Econometrics of Continuous-Time Models
2015	Torben Andersen	Northwestern	High-Frequency Asset Pricing
2015	Peter Christoffersen	Toronto	Risk Management and Extreme Risks
2015	Harrison Hong	Princeton	Behavioral Finance and Asset Management
2016	Jerome Detemple	Boston	Dynamic Asset Allocation
2016	Vikas Agarwal	Georgia State	Asset Management (Mutual Funds & Hedge Funds)

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Year	Instructor	Affiliation	Course title
2016	Francis Diebold	U. of Penn.	Aspects of Real-Time Measurement and Monitoring in Financial Markets, Macro Fundamentals and the Interface
2016	Tim Bollerslev	Duke	Volatility Modeling
2016	Ludovic Phalippou	Oxford	Private Equity
2016	Pietro Veronesi	U. of Chicago	Macroeconomic and Political Uncertainty and the Dynamics of Asset Prices
2017	Rama Cont	Imperial	Liquidity Risk
2017	Riccardo Rebonato	EDHEC	Term Structure Modelling in the P and Q measures
2017	Sanjiv Das	Santa Clara	Data Science, Analytics, Algorithms
2017	Harjoat Bhamra	Imperial College	Monetary Policy and Asset Prices
2017	Marcel Rindisbacher	Boston Univ.	Monte Carlo Methods
2017	Harrison Hong	Columbia Univ.	Behavioral Finance and Asset Management
2017	Federico Bandi	Johns Hopkins	Continuous-time Financial Econometrics
2018	Michael Chernov	LSE, now UCLA	Macro-Based Valuation Across Assets and Horizons
2018	Jeffrey M. Wooldridge	Michigan State Univ.	A Course in Microeconometrics
2018	Yacine Aït-Sahalia	Princeton	Topics in Financial Econometrics
2018	Tarun Ramadorai	Imperial College	International Finance
2018	Allan Timmermann	UC San Diego	Predictive Modeling and Forecast Evaluation in Financial Markets
2019	Bryan Kelly	Yale	Empirical Asset Pricing
2019	José Luis Peydró	Pompeu Fabra Univ.	Systemic Risk and Macroprudential Policy
2019	Toni Whited	Michigan	Dynamic Corporate Finance
2019	Michael Brandt	Duke	Macroeconomic Nowcasting
2019	Jesus Fernandez-Villaverde	Wharton	Numerical Solutions and Bayesian Estimation Methods for General Equilibrium Asset Pricing Models
2020	Jeffrey M. Wooldridge	Michigan State Univ.	Panel Data Analysis
2020	Albert Menkveld	Vrije U. of Amsterdam	Market Microstructure
2020	Jaume Ventura	Barcelona	The Macroeconomics of Credit and Asset Bubbles
2020	Federico Bandi	Johns Hopkins	Nonparametric Methods in Finance

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Year	Instructor	Affiliation	Course title
2021	Yacine Aït-Sahalia	Princeton	Topics in Financial Econometrics
2021	CEPR Conference	EDHEC	Household Finance
2021	Pietro Veronesi	U. of of Chicago	Fixed Income Securities
2021	Sanjiv Das	Santa Clara	Data Science, Analytics, Algorithms
2022	Stefano Giglio	Yale University	Empirical Asset Pricing
2022	Andrew Patton	Duke University	Financial Econometrics
2022	Vikas Agarwal	Georgia State University	Asset Management: Mutual Funds and Hedge Funds
2022	Harrison Hong	Columbia University	Behavioral Finance and Asset Management
2022	Marcin Kacperczyk	Imperial College	Climate/Sustainable Finance
2023	Gordon Phillips	Dartmouth University	Finance and Industrial Organization
2023	Jeffrey Wooldridge	Michigan State	Topics in Microeconometrics
2023	Yacine Aït-Sahalia	Princeton University	Machine Learning with Applications in Finance
2023	Ludovic Phalippou	Oxford University	Private Equity
2023	Harjoat Bhamra	Imperial College	Asset Pricing and Monetary Policy
2024	Frederico Belo	INSEAD	Production-based Asset Pricing: Theory and Empirics
2024	Federico Bandi	Johns Hopkins	Spectral Asset Pricing
2024	Lukas Schmid	Univ. of Southern California	Models and Methods in Macro-Finance
2024	Pietro Veronesi	Chicago Booth	Fixed Income
2024	Tarun Ramadorai	Imperial College	Household Finance
2025	Robert Kosowski	Imperial College	Alternative Risk Premia and Replication of Financial Anomalies
2025	Albert Menkveld	Univ. of Amsterdam	Market Microstructure

F.2 List of Ph.D. Courses Scheduled (subject to change)

Year	Instructor	Affiliation	Course title
2025	Ayako Yasuda	Berkeley	Sustainable Finance and Impact Investing

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Continuous Time Financial Economics